CLARUS GC/MS



Environmental Tutorial
Basic Principles

PerkinElmer®

Clarus SQ8 GC/MS
Environmental Tutorial

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Table Of Contents

Chapter 1. Introduction 7
About this Tutorial 9
Prerequisites 9
About the Clarus GC/MS System 10
Basic Operating Summary 11
About Sample Analysis 13
Conventions Used in This Manual 14
Clarus GC 15
Touch Screen 15
Chapter 2. How to Build a Quantification Method for Reporting 17
Quantification Methods 18
About Internal and External Standards 19
Internal Standards 19
External Standards 19
Terminology 20
Creating the Quantify Method 22
Loading the Tutorial_VOA / Tutorial_SVOA Project 23
Editing a Quantify Method 29
Setting Environmental Parameters 35
Entering Environmental Parameter Values 40
Entering Target Compound Information 41
Entering Surrogate Compound Information 43
Entering Spike Compound Information 44
Setting QA/QC Limits 45
Entering Parameters into the Quantify Method (For Expert Users) 46
Creating a GC and MS Method 46
Chapter 3. How to Set Up Submitter/Task Data 49
About the Submitter/Task Information 49
To access the Submitter/Task Data window: 50
To add a submitter: 50
To Import a Custom Compound List (.CCL) 51
To Define Report Methods 53
Custom Compound List 55
Proper Reporting Limit (RL) Setup 57
Chapter 4. How to use Generic TIC Names 59
About Generic TIC Names 61
Chapter 5. How to Build a Sample List using the Wizard 63
About a Sample List 65
Building a Sample List Using the Wizard 66
Adding Samples to the Sample List 67
Sample Injection Information 68
Sample Prep Information 69
Lab Information 70
Editing Decimal Places in the Sample List 72
Chapter 6. How to Process Quantification Results 73
Processing Quantification Results 75
Processing Samples 76
Chapter 7. How to Generate Environmental Reports 81
About Environmental Reports 83
Environmental Reports 84
Before Generating Environmental Reports 87

Copy and Paste *.RME files to your new Project 87
Copy the Qualitative Database File to your new Project 88
About Quantitative and Qualitative Processing 89
Steps for Producing Environmental Forms 90
Step 1 – Generating a Sample List with the Required Fields 91
Step 2 – Creating a Quantify Method 95
Step 3 – Running Samples, QC, Tune, Initial, Calibration, and Continuing Calibration 95
Step 4 – Integrating/Calibrating/Quantifying Initial Calibration Files 95
Step 5 – Integrating/Quantifying Sample files 96
Step 6 – Qualitatively Processing Samples 97
Step 7 – Producing Environmental Reports via the Environmental Forms Dialog 98
Form 1 – Organics Analysis Data Sheet 101
Form 1TIC – Tentatively Identified Compounds(TICs) 109
Form 2 – SMC/Surrogate Compound Recovery 116
Form 3 - Matrix Spike/Matrix Duplicate Recovery 119
Form 4 - Method Blank Summary 123
Form 5 - Instrument Performance Check 126
Form 6 – Initial Calibration Data 132
Form 7 – Continuing Calibration Check 135
Form 8 – Internal Standard Area and RT Summary 138
PKIEnvQuant Template Form Generation 141
LCS_Voa and LCS_SV Template Form Generation 147
Form1_CSV – Organics Analysis Data Sheet 150
Form5_CSV - Instrument Performance Check (CSV format) 153
Form6_CSV – Initial Calibration Data (CSV format) 156
SpectrumQuant Report_Env Template Form Generation 159
Chapter 8. How to Save your Data to LIMS 163
Exporting Your Environmental Data to LIMS 165
Appendix 1. Environmental Reporting Calculations 169
On-column Amount Calculations 171
Average Relative Response Factor (RRF) 171
Linear Regression 172
Quadratic 174
Target Calculations 176
Volatile Water Target 177
Volatile Low Level Soil Target 179
Volatile Medium Level Soil Target 181
Semi-Volatile Water Target 184
Semi-Volatile Soil Target 186
Tentatively Identified Compounds Calculations 188
Volatile Water Tentatively Identified Compound 188
Volatile Low Level Soil Tentatively Identified Compound 190
Semi-Volatile Water Tentatively Identified Compound 195
Semi-Volatile Soil Tentatively Identified Compound 197
Surrogate Recovery Calculations 199
Volatile Surrogate Recovery 199
Semi-Volatile Surrogate Recovery 201
Spike Recovery Calculations 203
Volatile Spike Recovery 203
Semi-Volatile Spike Recovery 205
Concentrations for Spike Added Calculations 207
Spike Added Volatile Water Concentration 207
Spike Added Volatile Low Level Soil Concentration 208
Spike Added Volatile Medium Level Soil Concentration 210
Spike Added Semi-Volatile Water Concentration 212

Spike Added Semi-Volatile Soil Concentration 213
Relative Percent Difference (RPD) Spike Calculations 215
RPD by Concentration 215
RPD by Recovery 216
Reporting Limit Calculations (RL or MDL) 217
Volatile Water Reporting Limit using RL or MDL 217
Volatile Low Level Soil Reporting Limit (RL or MDL) 218
Volatile Medium Level Soil Reporting Limit (RL or MDL) 219
Semi-Volatile Water Reporting Limit (RL or MDL) 221
Semi-Volatile Soil Reporting Limit (RL or MDL) 223
Appendix 2. Error Messages and Warnings 225
About the Error Messages and Warnings 227
Error Messages 227
Warnings 228
Form Specific Checks 229
Form 1 229
Form 2 229
Form 3 230
Form 4 231
Form 5 231
Form 6 231
Form 7 232
Form 8 232
Index 233



Introduction 1



About this Tutorial

The Clarus GC/ MS Environmental Tutorial is your guide to using the TurboMass environmental reporting software. This tutorial is designed to teach you the basic operating procedures using data provided in Tutorial Projects (both Volatile Organic Analysis-VOA and Semi-Volatile Organic Analysis-SVOA) to produce environmental reports. More specifically, this tutorial begins by showing you how to build a quantification method. Next, it shows you how to enter Submitter/Task data, how to use generic TIC names, create a Sample List, and generate environmental reports.

This tutorial is intended to supplement the Clarus GC/MS Tutorial. Before using this tutorial, make sure that you are familiar with the procedures described in the Clarus GC/MS Tutorial; for example, setting up the instrument, starting, leak checking, tuning, acquiring data, etc.

After completing this tutorial, you will have the basic background information required to explore and use all of the features necessary to successfully produce environmental reports. For additional details and instructions about this product, contact PerkinElmer to enroll in a customer training course.

Prerequisites

Before operating the Clarus MS system you should:

- Thoroughly understand the recommended safety practices. Read the Warnings and Safety Information section in the Clarus SQ 8 MS Hardware Guide (P/N 09931017).

- Have a basic understanding of how to use the Clarus GC, computer, and Microsoft Windows. For details, refer to the corresponding manuals provided with each product. Also have a basic understanding of how the TurboMass software works, more specifically review the Clarus GC/MS Tutorial (P/N 09931018).

- This Clarus GC/MS Environmental Tutorial is presented with the understanding that you have a basic familiarity with Microsoft Windows operations. If you need a refresher about the basic principles of Microsoft Windows, such as opening and closing files, adjusting the window size or position, printing, or using the Windows Explorer, refer to your printed Microsoft Windows documentation, or online Help files, for details.

About the Clarus GC/MS System

The Clarus Gas Chromatograph/Mass Spectrometer (GC/MS) is a sophisticated benchtop mass spectrometric detector that provides you with the simple tools needed to generate environmental reports using gas chromatography/mass spectrometry analyses as well as the sophisticated tools needed to perform the more complex analyses. Clarus MS runs analyses that best characterize your sample by using either the electron ionization (EI) mode or chemical ionization (CI) mode. Designed as a detector for the Clarus GC, this system produces positive identification and quantification of compounds separated by the Clarus GC, even those complex compounds that coelute.

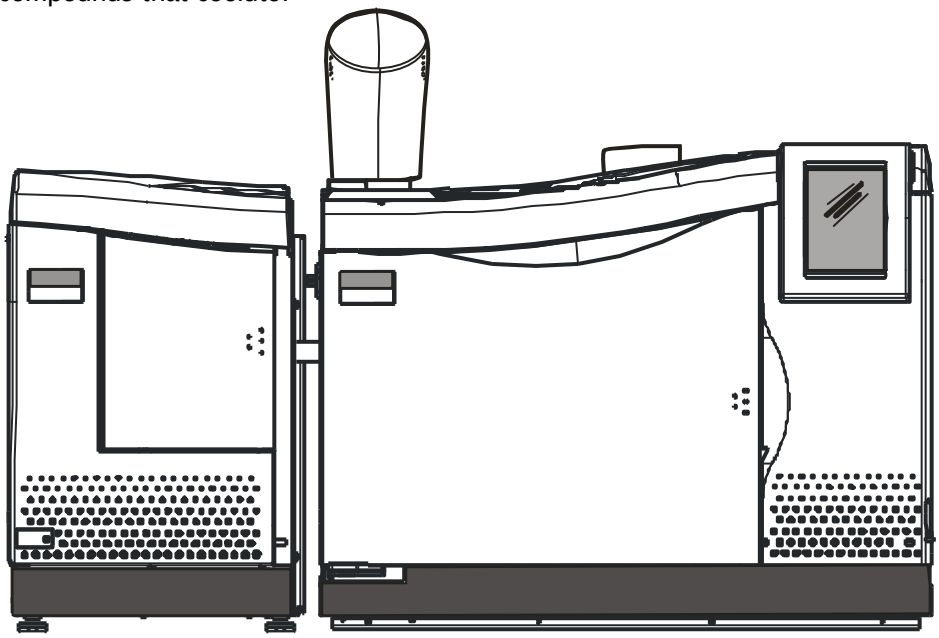


Figure 1 The Clarus GC/MS System

The Clarus GC/MS is controlled by a personal computer (PC) based data system using the Microsoft Windows operating environment. The user interface contains color graphics and provides full user interaction by using either the keyboard or the mouse. The TurboMass software completely controls the GC/MS system from tuning and data acquisition (scanning or selected ion recording mode) through quantifying your results. Complete operating instructions of all controls are in the *TurboMass Software User's Guide* (P/N 09931016), supplied with your system.

Basic Operating Summary

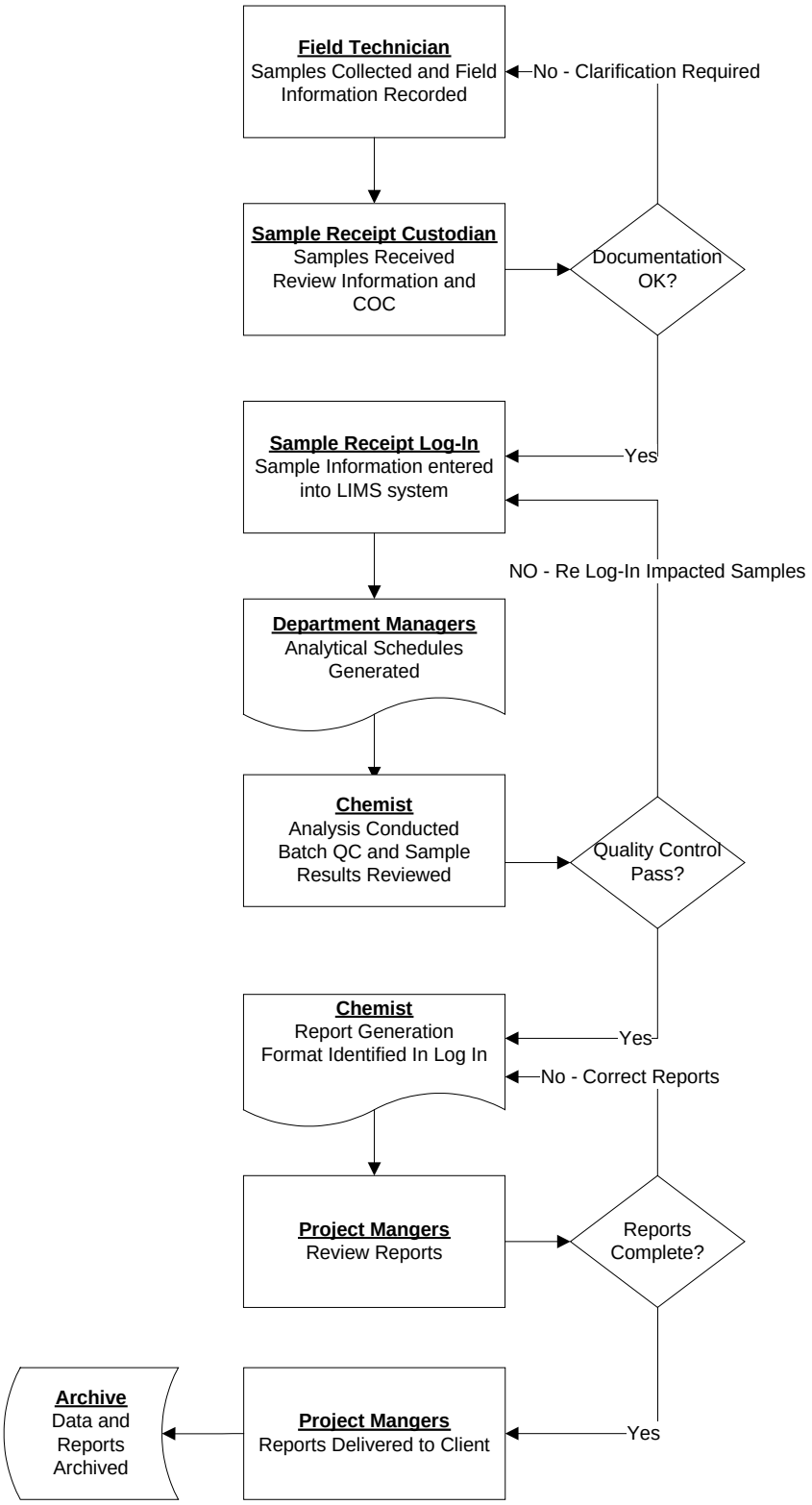
The Environmental Testing Laboratory is a complex business with many methods being performed on various sample matrices. There are critical check points whereby information must be analyzed prior to performing the next step in the process.

The ability to quickly evaluate data for accuracy and completeness is essential to producing a legally defensible report. The software used to process the data must be able to scrutinize the data to insure the quality control parameters in the methods are being achieved. Not only should the data processing system generate the necessary forms but it should also easily review the sample quality control data (i.e. indicate out of control parameters) and the batch quality control information to insure compliance.

The samples must also be collected using the proper bottles (container size and material). For example, samples to be collected for heavy metal analysis must be stored in plastic containers to prevent contamination by inorganic parameters such as boron and silica, which are found in glass containers. Organic semi-volatile analysis (methods SW8270, 625, 525) must be stored in darkened glass since many of these compounds are susceptible to degradation by ultra violet light. Volatile organic analysis must be collected in preserved; Teflon capped vials that are filled so that no headspace is available for the volatile organics to escape.

The samples must be labeled with indelible ink to prevent it from being washed away due to exposure to the elements or wiped off. The samples must be packed in ice or a substance that allows specified temperature limits to be achieved. A custody seal must be secured around the outside of the container to prevent tampering and to insure the integrity of the samples is not compromised.

The following diagram shows the environmental analysis work flow:



About Sample Analysis

There are a variety of different analyses that take place within an environmental laboratory. For the purposes of this analysis we will focus on organic methods utilizing GC/MS. Samples are analyzed via GC/MS for volatile organic, semi-volatile organic and water-soluble constituents. There are several types of samples that are analyzed by the laboratory.

Sample Types

- **Field Blank** – A sample of water that is exposed to the environment of the site and sent to the laboratory for analysis. It is used to identify the presence of contamination at the sight. It is analyzed in the same manner as a field sample.

- **Trip Blank** – A sample of water that resides with the volatile vials of the field samples. The trip blank is used to identify the presence of volatile contamination during the sampling event or shipment to/from the laboratory. It is analyzed in the same manner as a field sample.

- **System Blank** – An aliquot of water generated at the laboratory that is analyzed to assure that there is no contamination with the GC/MS system. It is analyzed in the same manner as a field sample but the system needs to identify it as a System Blank.

- **Method Blank** – An aliquot of water generated at the laboratory that is subjected to the same sample preparation procedures as the field samples. It is used to determine if contamination was introduced during preparation. It is analyzed in the same manner as a field sample but the system needs to identify it as a Method Blank.

- **LCS** – Laboratory Control Sample. The LCS consists of an aliquot of a clean (control) matrix similar to the sample matrix. The LCS is spiked with the same analytes at the same concentrations as the matrix spike. It is analyzed in the same manner as a field sample but the system needs to identify it as the LCS

- **Initial Calibration Standards** – Standards used to calibrate the instrument. The system needs to identify each level of the initial calibration standards.

- **Continuing Calibration Standard** – A standard that is analyzed periodically (according to the method) to assure that the instrument has maintained calibration. The system must identify the continuing calibration standard.

- **Matrix Spike/Matrix Spike Duplicate** – A field sample is spiked with a known concentration of analytes in duplicate. The system must identify the original sample, the spike and spike duplicate so that the proper calculation can be performed.

Conventions Used in This Manual

Normal text is used to provide information and instructions.

Bold text refers to text that is displayed on the screen.

UPPERCASE text, for example ENTER or ALT, refers to keys on the PC keyboard. '+' is used to show that you have to press two keys at the same time, for example ALT + F.

Three terms are used in the text. Each one implies a particular level of observation or action as follows:

***Note**: A note indicates additional, significant information that is provided with some procedures.*

CAUTION	**A caution indicates an operation that could cause instrument damage if precautions are not followed.**
ATTENTION	**Une précaution indique une opération qui pourrait causer des dommages aux instruments si les précautions ne sont pas respectées.**



WARNING



AVERTISSEMENT

A warning indicates an operation that could cause personal injury if precautions are not followed.

Un avertissement indique une opération qui pourrait causer des blessures si les précautions ne sont pas respectées.

Clarus GC

You can operate the Clarus GC using the touch screen or through the TurboMass software. The touch screen is the interface between you and the instrument, enabling you to control the Clarus GC in order for you to perform your analyses easily and conveniently. The touch screen contains active areas that you touch to perform the required action.

You may find it more convenient to use a stylus or you may like to touch the active areas of the touch screen. **Do not** use sharp, pointed objects. A light touch is all that is needed. You do not need to press forcefully on an active area.

Please note that when you touch an active area, various processes are started in the instrument. Some of these processes may take several seconds. Be patient and wait for the process to complete before touching another active area on the touch screen.

Touch Screen

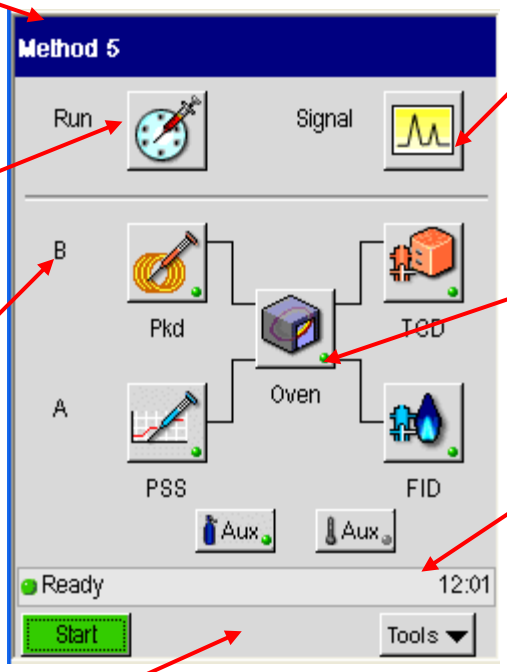
The main displays on the touch screen are termed screens. You use various screens to set up your Clarus GC to perform the analyses. These screens contain:

- **Entry fields** which allow you to make entries,
- **Buttons** that you touch to start or stop actions or to display a dialog,
- **Option Buttons** that you select an option from a list and
- **Boxes** that you use to switch functions on or off.

The Clarus GC is controlled by a collection of operating parameters called the Active Method. You can prepare and save up to five methods and make any one of them the Active Method. However, the fifth method is reserved for TotalChrom or TurboMass and may be overwritten.

The following Status screen shows the different sections of the user interface.

The Status screen displays icons that provide quick access to major areas of the system. The injector and detector buttons show graphic representations of the devices for each channel. If an auxiliary zone is configured, the Aux button appears below the Oven button. The icon buttons that represent the heated zones (injectors, detectors, oven, and Aux if configured) include a light to indicate the ready/not ready status. A red blinking light indicates not ready and a steady green light indicates ready status.



The title bar displays the name of the active method.

The Run button provides access to the Autosampler and Manual Inject

Channel B appears on top of Channel A to emulate the order on top of the GC

Click here to view the Signal. If a method is not running the screen will display a flat baseline.

Icon buttons provide quick access to all configured areas of the instrument. An indicator light shows the ready/not ready status of each heated zone.

The status bar displays the overall ready/not ready status as well as text message and the real time clock/date display.

The bottom bar displays the Tools pop up menu. When the system is running, the Stop button also appears.

2 How to Build a Quantification Method for Reporting

Quantification Methods

Since Quantification is analyzing data to determine the concentration of each sample using an internal or external standard, you need to acquire data from more than one sample. This quantification example uses a data set of previously obtained data stored in the Tutorial_VOA.pro Project. You will find this project in the **C:\TurboMass\Tutorial_VOA.PRO** directory on your hard drive.

This chapter describes how to use TurboMass to perform quantitative assays for environmental analysis.

When using the TurboMass for environmental analysis you will enter parameters into many additional fields, depending on your needs. This tutorial will show you how to work with a typical environmental quantitation method. Standard EPA method parameters are used whenever possible.

For additional details on building a basic quantification method, see the Clarus GC/MS Tutorial.

About Internal and External Standards

Before you begin, you need to decide whether you will use internal or external standards for your quantitative method. This section provides information on both.

Internal Standards

Typical environmental analysis requires an internal standard calculation.

In an **internal standard method**, a known and constant quantity of a compound that is not one of the analytes is added to the sample; this is the internal standard. The ratio of its retention time to the retention times of the analytes has been established. The ratio of its peak area to the peak areas of the analytes is determined for various concentrations of analytes. The unknown concentrations of the analytes in samples are then calculated, using the area of the internal standard peak as a reference. Response factors are included in the calculation to compensate for differences in the sensitivity of the detector to different analytes.

Note: *When you are creating your Quantify method, put all of the internal standards into the method first, before you enter your surrogates, targets, and spikes,*
Analysis by an internal standard is the preferred technique whenever practical because it corrects for errors in sample preparation and variations in the amount of sample injected.

The concentrations reported for the peaks of interest are affected only by the quantities of the various components and the quantity of internal standard added.

Internal standards should be of the same family as the target compounds (for example, phenols) but their retention times should generally not overlap those of the targets. An exception is isotopically labeled internal standards.

External Standards

In an **external standard method**, known amounts of analytes are run in separate analyses (the standard runs), and the resulting peak areas are used to obtain calibrated response factors. In subsequent analyses of samples with unknown concentrations, the concentrations of the analytes are calculated by applying the response factors obtained from the standard runs.

Since, in an external standard method, there is no standard peak whose area changes with variations in injection size, the sample injection size must be reproducible from run to run. Therefore, external standard methods are best used with an autosampler. They are not recommended for manual injection.

Terminology

The software and the procedures in this chapter frequently use the following words and expressions:

Target compound
An analyte whose concentration is to be determined.

Quantification
Quantitative analysis is determining the concentration of a target compound from mass spectral data.

Internal Standard
An internal standard is a compound that is added, in a known and constant concentration, to the sample. The ratio of its retention time to the retention times of the target compounds must be established, and the ratio of its peak area to the peak areas of target compounds is determined for various concentrations of the target compounds. The software uses this information to determine the unknown concentrations of the target compounds in the sample. Multiple internal standards may be present in a chromatogram.

External Standard
Known amounts of analytes are run in a separate analysis, a standard run, and the resulting peak areas are used to obtain calibrated response factors that are stored in a calibration library. In later runs, these response factors are used to calculate analyte concentrations.

Response Factor Curve
This is a plot of peak area versus concentration for a given target compound. The software uses response curves to compensate for differences in the sensitivity of the detector to different compounds.

Calibration
In the Quantification software, the process of generating points on the response curve using scan files in which the analytes and internal standards (if any) are present in known concentrations.

Calibration Library
A file that contains retention times, mass peak intensities, and concentrations for the internal standards, if any, and the target compounds. This file constitutes, in effect, a method for quantitative analysis.

Minimum Detection Limit (MDL)
For the purpose of environmental reports the term MDL is used to indicate the threshold value for the "U" qualifier flag. Values below this threshold value will flag the compound with a "U" in the Form 1 report.

Qualifiers (Q-Flags)
In addition to the concentration of a compound the Form 1 also contains a column labeled "Q" for qualifier. A qualifier provides additional information about the compound.

Surrogate

Surrogate is a compound that is added to every standard, blank, matrix spike, matrix spike duplicate, and unknown sample before sample preparation. The surrogate is added at an exact, known concentration. It is used to determine the efficiency of the sample preparation process. Surrogates should possess chemical properties similar to those of the target compounds but should not be found in a real sample matrix. Deuterated compounds are frequently used as surrogate compounds.

Matrix

Matrix is the predominant material in the sample to be analyzed, usually water or soil.

Matrix Spike

Matrix spike is an aliquot of sample to which has been added known quantities of specific target compounds (matrix spike compounds). The matrix spike is subjected to the entire analytical procedure so that recovery of the matrix spike compounds can be determined.

Creating the Quantify Method

Before performing Integration or Quantification, you will create a Quantify Method using the Quantify Method Editor. By selecting a method from within the Method Editor this method becomes the current system method file and is used when performing Quantify operations. Changes made to the method are not permanent until they have been saved to your hard disk. Consequently, you must save the method before it can be used to perform quantification by selecting **Save** from the **File** menu to update the current method file, or **Save As** from the **File** menu to save to a new method file.

This tutorial takes you through the steps necessary to build a **new environmental Quantify method** and it will also use a **previously built environmental example** to demonstrate how an environmental method should look.

Note: *Each Sample List should have only one Quantify method.*

The Quantify method describes how a data file is processed to produce calibration curves and quantitative information. Details must be entered into the method for each of the compounds being used in the analysis.

The Quantify Method specifies information for performing the following tasks:

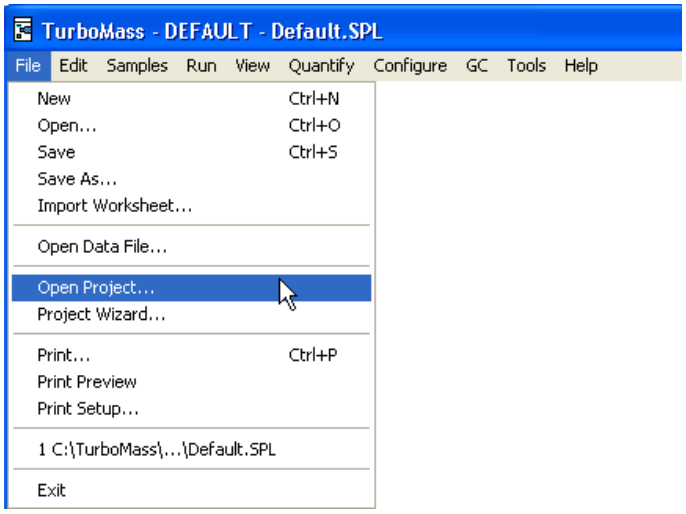
- Integration of a chromatogram trace to obtain peak information.
- Location of the chromatogram peak relating to a specific compound from the list of detected peaks.
- Calculation of a response factor for the located peak.
- Formation of a Quantify calibration curve.
- QA/QC Limits – A single set of values that will be applied to all compounds in the method.
- Environmental Parameters that are defined independently for each compound in the method.
- Peak matching
- Concentration designation

Loading the Tutorial_VOA / Tutorial_SVOA Project

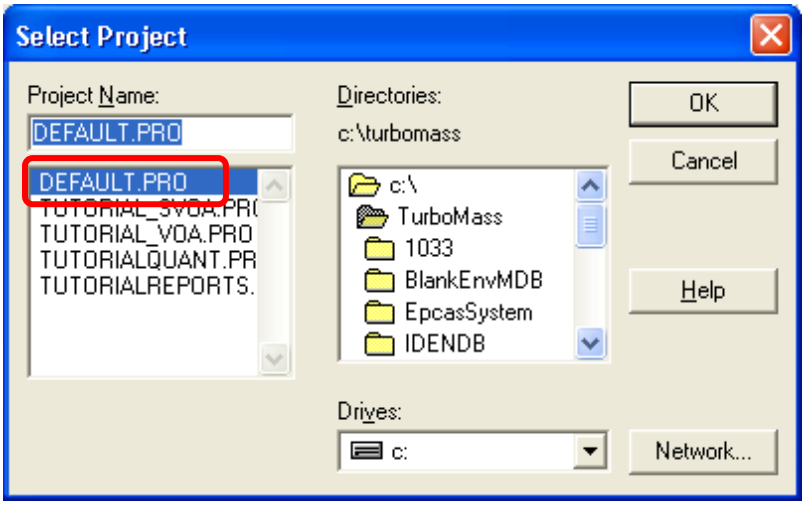
TurboMass ships with two projects (Tutorial_VOA and Tutorial_SVOA) that contain example data to help you understand environmental reporting. This Tutorial primarily uses the Tutorial_VOA.PRO project to help illustrate the software.

To open the Tutorial_VOA project:

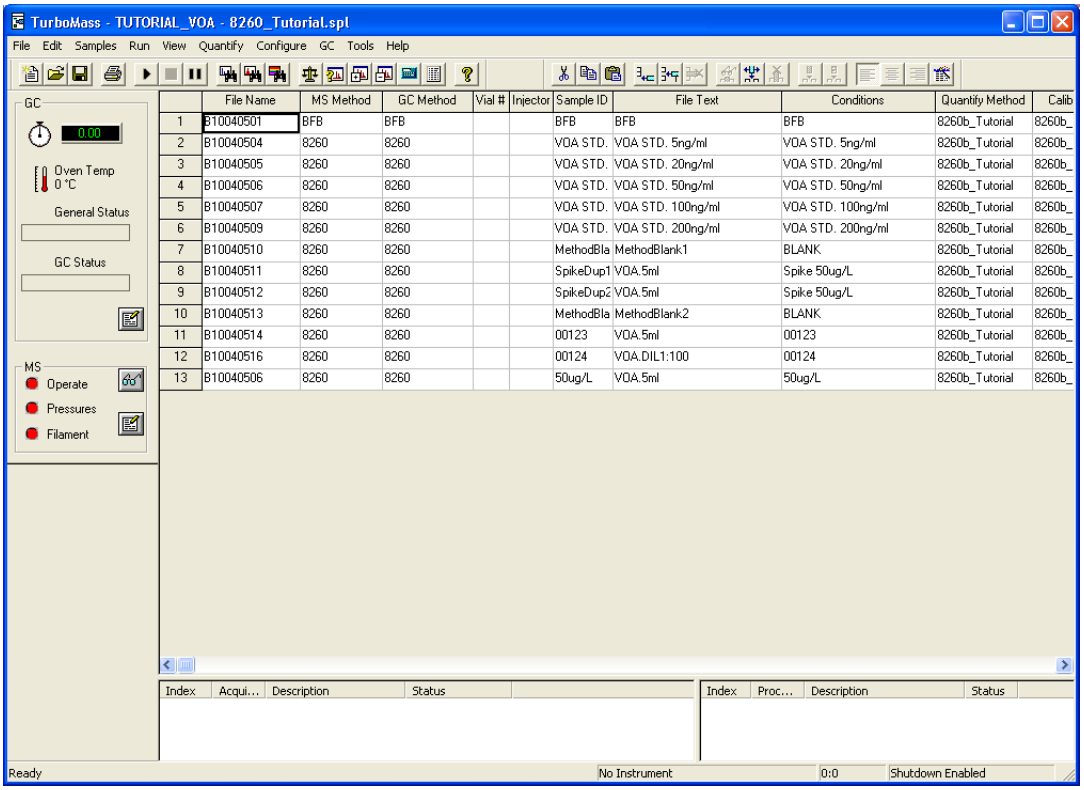
1. Select **Open Project...** from the **File** menu.



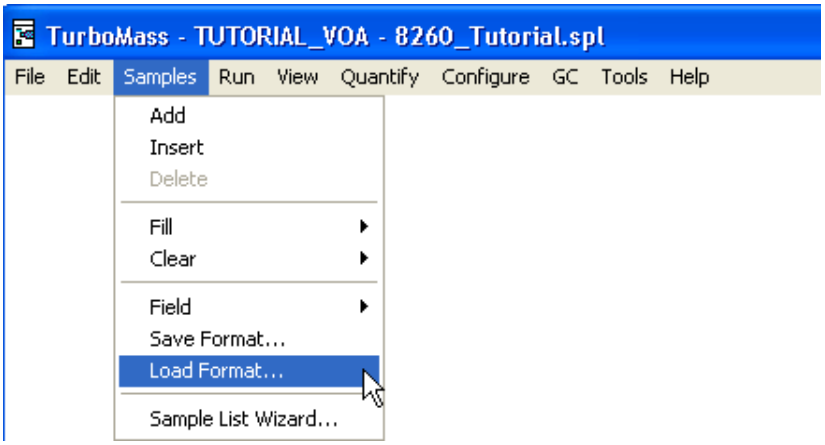
The Select Project dialog appears with DEFAULT.PRO selected:



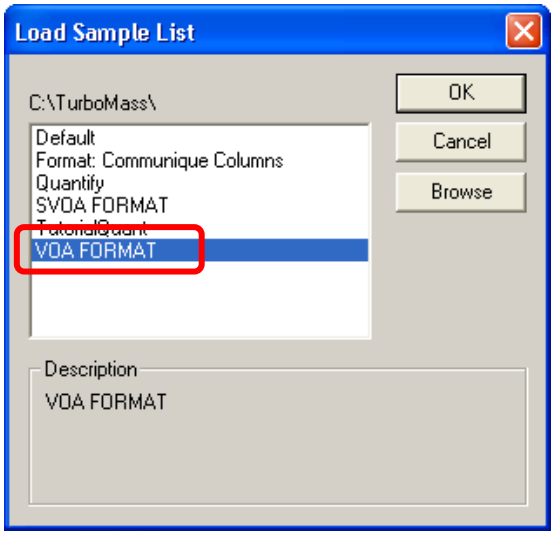
2. Select **TUTORIAL_VOA.PRO** from the list then click the **OK** button.
   The **TUTORIAL_VOA.PRO** project opens as shown below on the Sample List page.



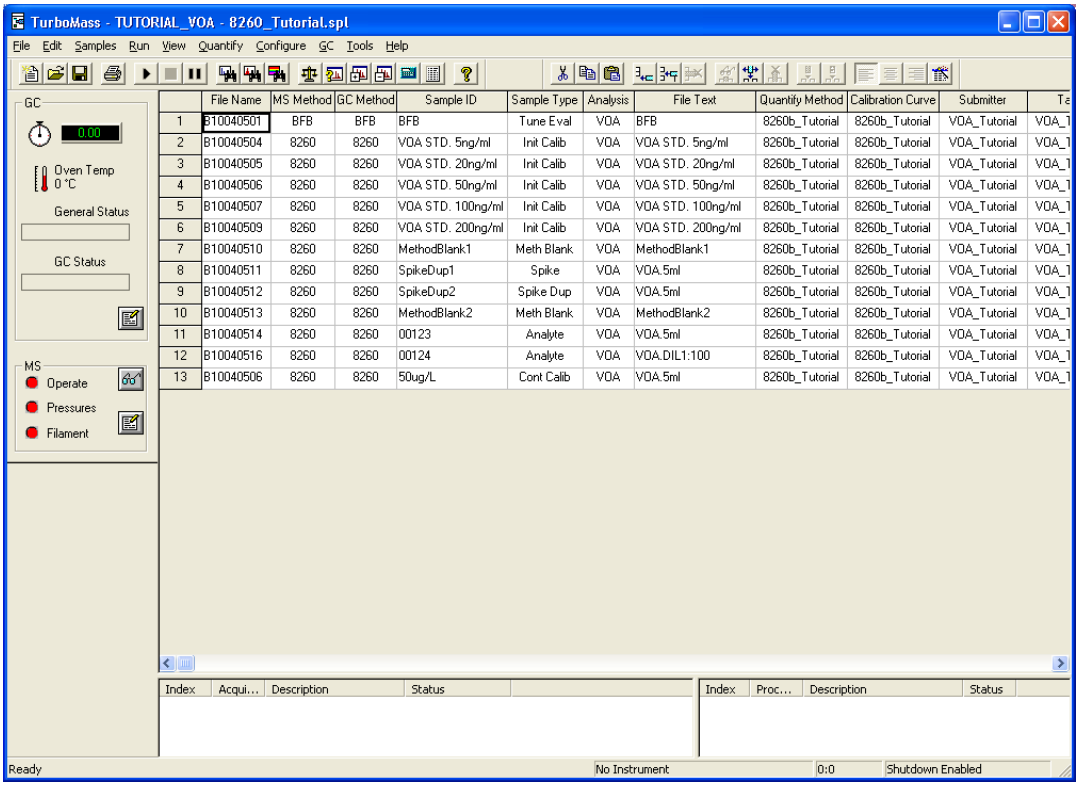
3. To ensure that all of the fields required for sample processing are included in the Sample List, select **Load Format...** from the **Samples** menu.



   The following dialog appears:

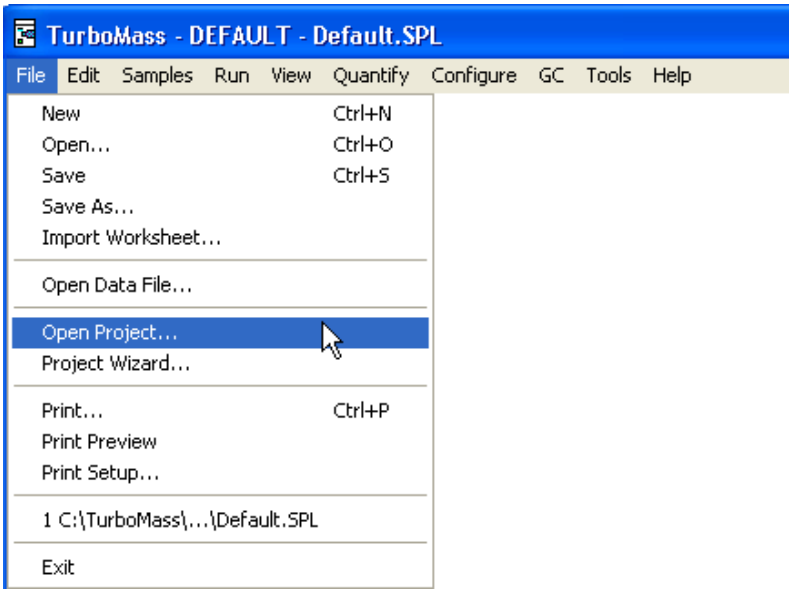


4. Select **VOA FORMAT** from the **Project Name** list and click the **OK** button.
   Now the Sample List contains the required fields and columns for processing the data.

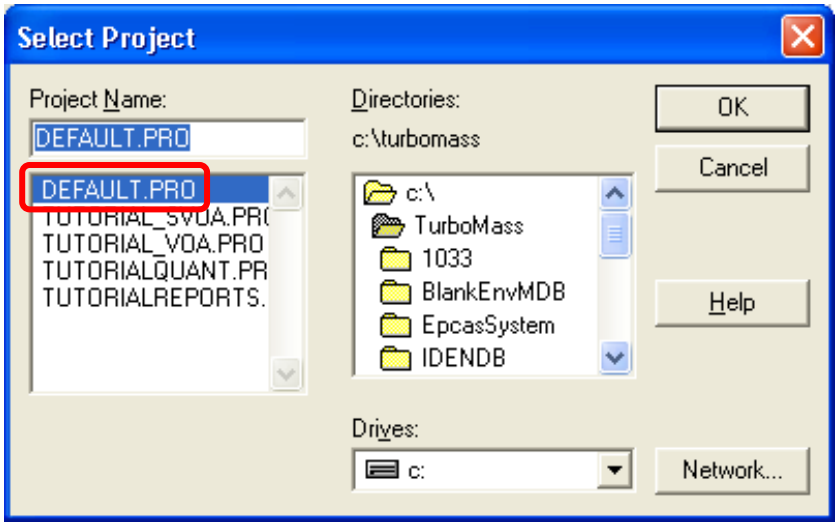


To open the Tutorial_SVOA project:

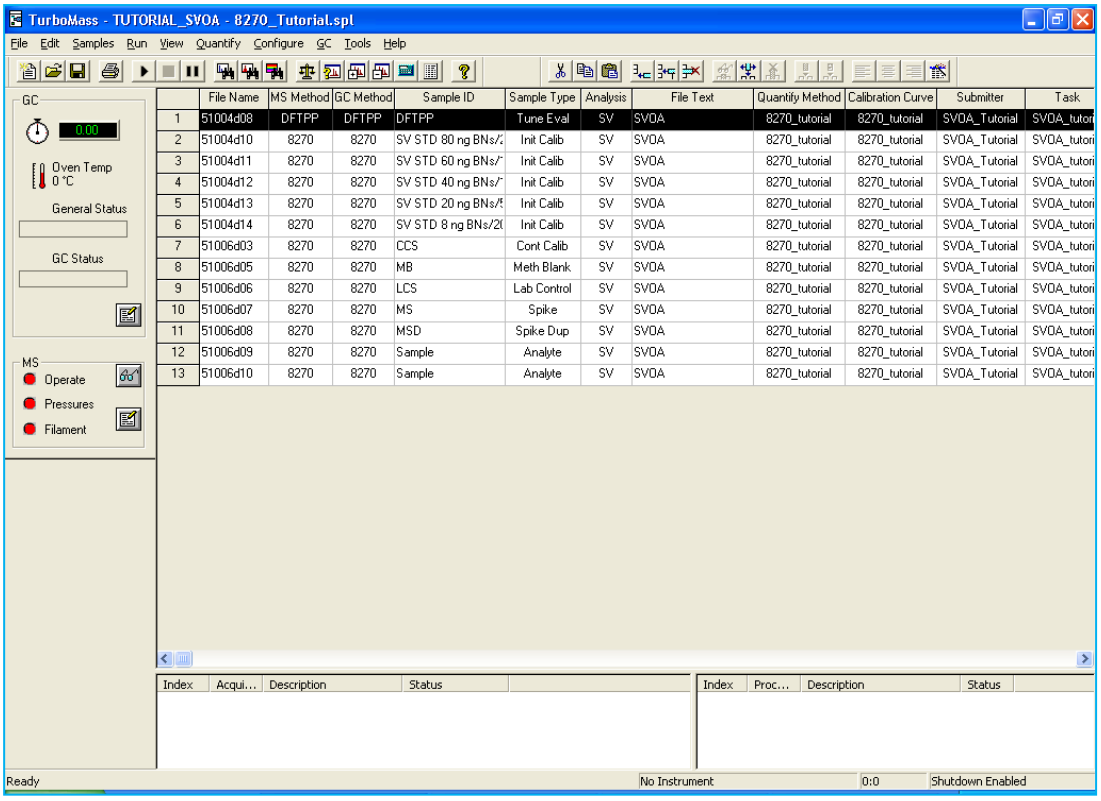
1. Select **Open Project…** from the **File** menu.



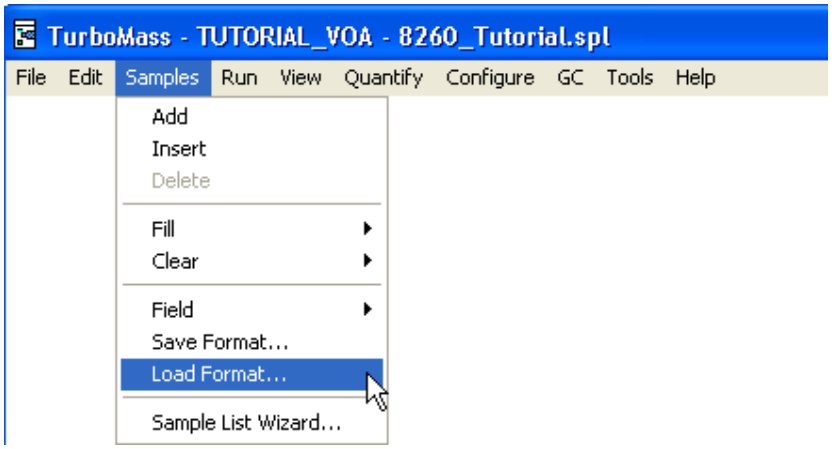
The Select Project dialog appears with **DEFAULT.PRO** selected:



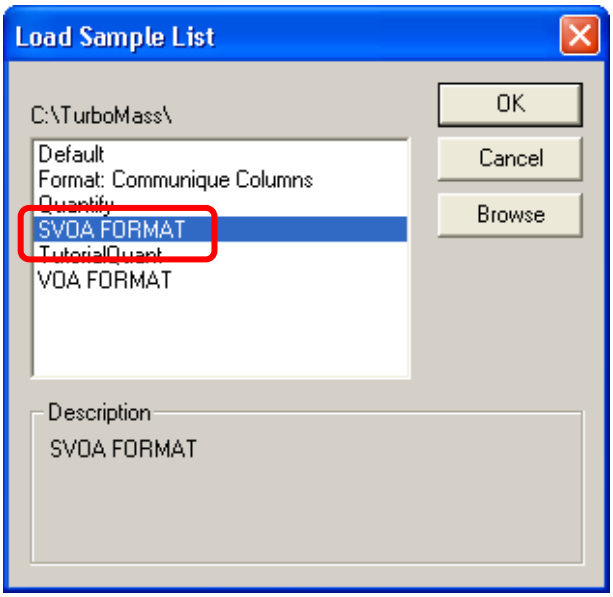
2. Select **TUTORIAL_SVOA.PRO** from the **Project Name** list then click the **OK** button. The **TUTORIAL_SVOA.PRO** project opens as shown below on the Sample List page.



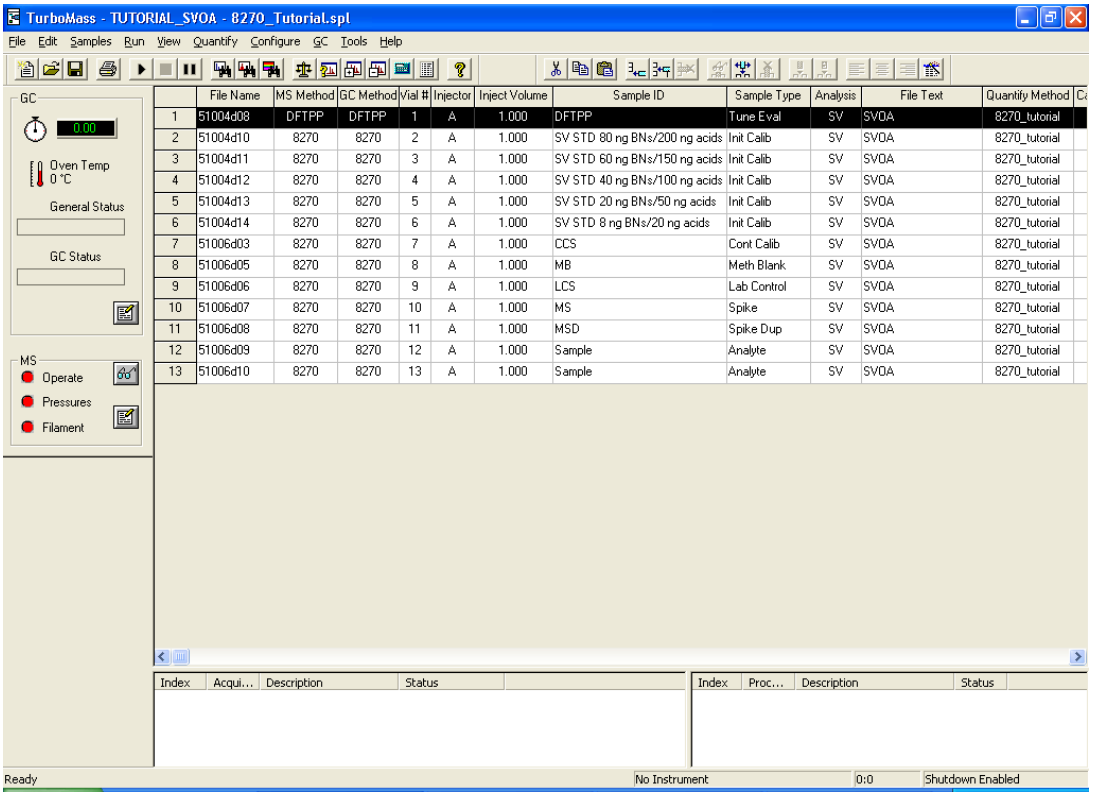
3. To ensure that all of the fields required for sample processing are included in the Sample List, select **Load Format…** from the **Samples** menu.



The following dialog appears:



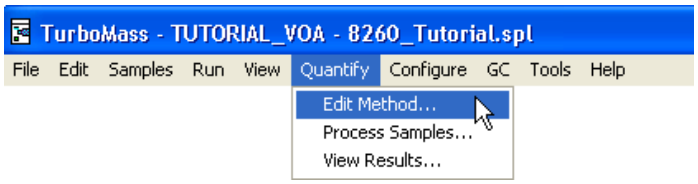
4. Select **SVOA FORMAT** from the list and click the **OK** button.
   Now the Sample List contains the required fields and columns for processing the data.



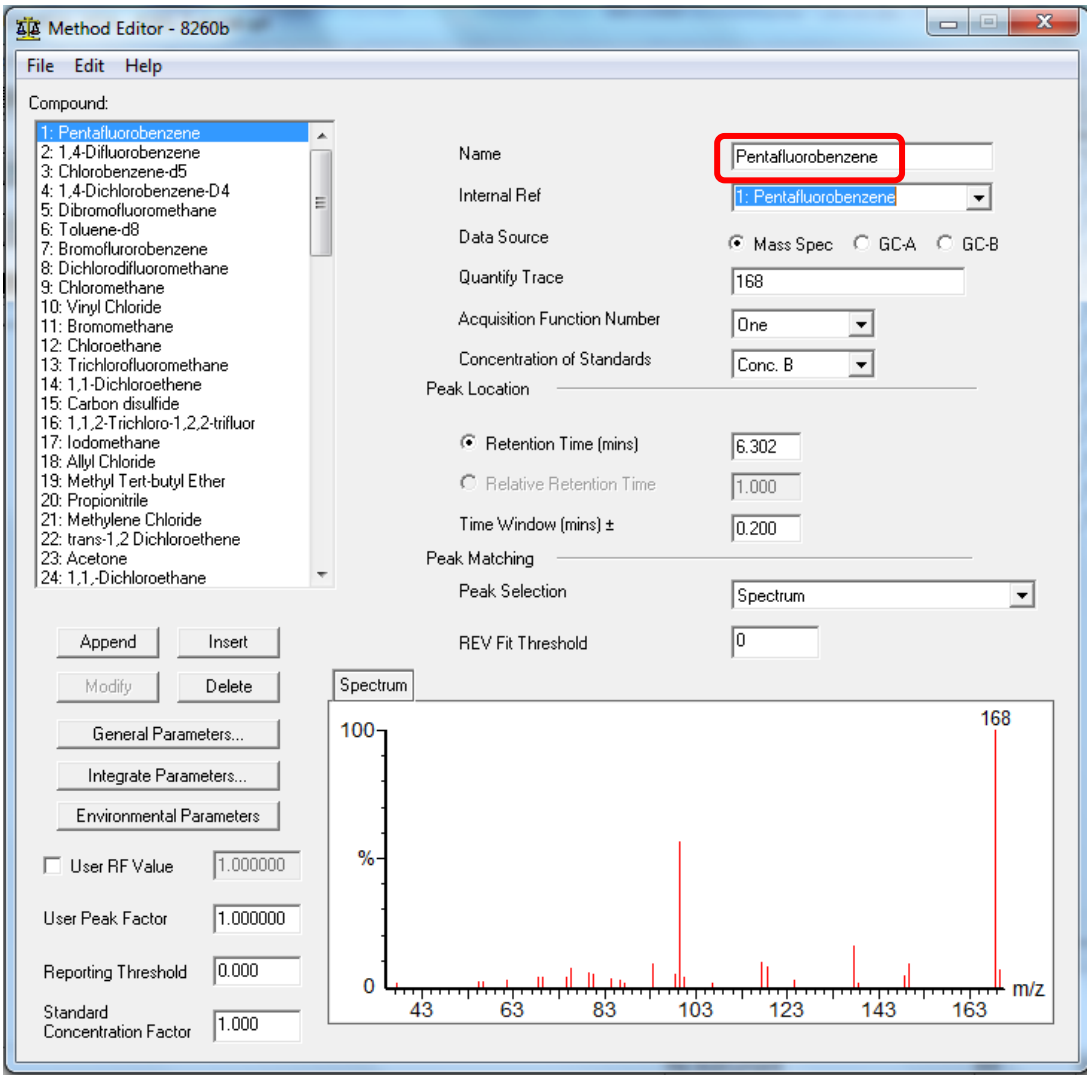
Editing a Quantify Method

We will continue to work with the **Tutorial_VOA.PRO** supplied with your software.

1. From the **Quantify** menu, select **Edit Method...**



The Quantify Method Editor dialog appears:

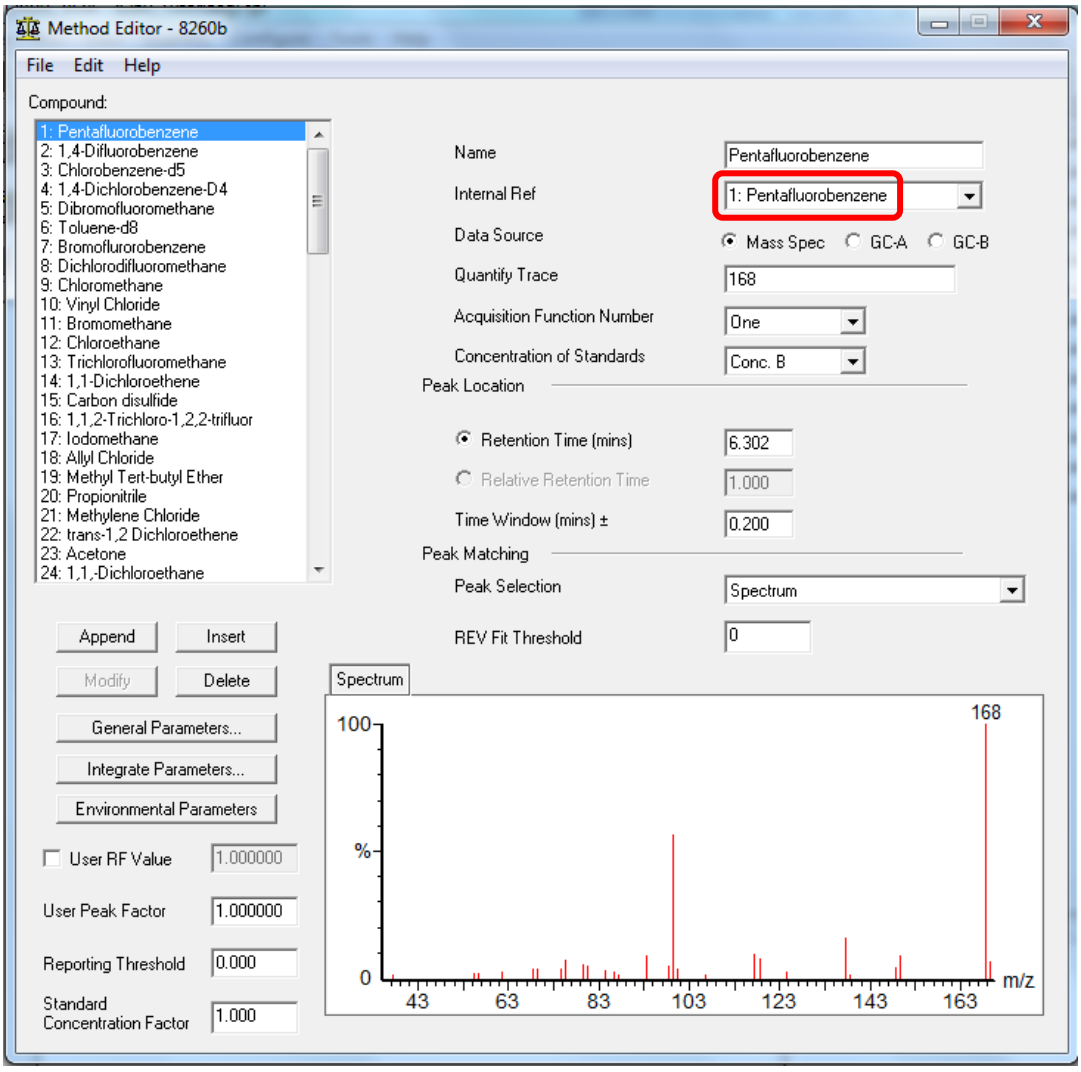


The compound **Name** is the name of the compound as it appears in the **Compound** list. This dialog contains all of the method parameters associated with the compound.

If you are using internal standards (Internal ref), then enter your internal standards before entering other compounds. This is shown in the following dialog to make it easy for you to select the internal standard you wish to associate with the compound.

In this example, select **1: Pentafluorobenzene** as the **Internal Ref**.

When using a Compound as an Internal Ref do not select [None] from the dropdown but select the compound so that it references itself.



The Internal Reference compound is the internal standard or retention reference compound displayed in the **Internal Ref** text box. The first four compounds are internal standards: 1:Pentafluorobenzene; 2: 1,4-Difluorobenzene; and 3: Chlorobenzene-d5; 1,4-Dichlorobenzene-D4.

2. Select **Mass Spec** as the **Data Source** of the peak for the selected compound.

3. Set **Quantify Trace** to the major ion descriptor being used to quantify the compound. In this example we are using a mass of 168.

4. The Quantify Trace parameter specifies an ion to be integrated when TurboMass is performing automatic peak detection, and is used during the locate phase when TurboMass is matching peak list entries against method compounds.

5. Quantification on a selected mass chromatogram is usually preferred for sensitivity and selectivity. Ideally, the ion should be:
    - Characteristic of the compound
    - Have a high relative intensity
    - A low background level
    - Unique to that compound – not overlapped by another compound with the same ion.

Note: *TurboMass enters this value automatically if you use the mouse to enter the Peak Location parameters. Be sure to verify that it is the best choice for avoiding co-eluting compounds.*

The function number in the **Acquisition Function Number** text box used to quantify all of the compounds in this example is **One**. This function number is identified by the mass spectrometer acquisition method.

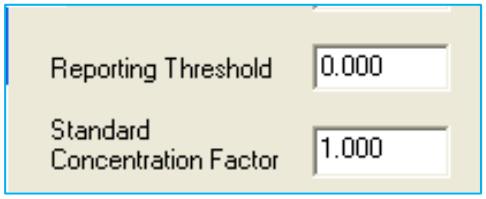
Note: *A function number will refer to Scan or SIR mode. There may be more than one function per method.*

6. Set the **Concentration of Standards** parameter to the Sample List column that contains the compound's concentration level within each Standard or QC sample. In this example use **Conc A** for the **Target** and **Spike** compounds, **Conc B** for the **Surrogates**, and **Conc C** for the **Internal Standards**.

    The software allows up to 10 concentration levels within a single sample. For example, if one group of compounds is initially at 50 ng, a second group is at 100 ng, and a third is at 400 ng, these three concentration levels can be defined as Conc. A, Conc. B, and Conc. C. As the standard is serial-diluted, change the values assigned to Conc. A, B, and C in the Sample List to reflect the new concentrations.

Note: *If additional fields are required, right click in the Sample List, select Customize Display, then scroll down the list to find additional concentration levels to add to your Sample List.*

A second option used to enter concentration levels is the **Standard Concentration Factor** (shown in the bottom left of the Quantify method screen. This is a multiplier of the concentration designated for that compound.



7. Set the **Peak Location** Parameters. The Peak Location parameters determine how a peak within a peak list is identified as matching a method compound. A peak can be classified as a match according to its Retention Time or Relative Retention Time, whether it falls within the specified Time Window, and whether it satisfies the Peak Selection criterion.
    - **Retention Time:** If selected, a peak within a peak list is identified as a match if it elutes at the Retention Time specified and within the Time Window specified.
    - **Relative Retention Time:** If selected, a peak within a peak list is identified as a match if it elutes at the time at which the compound is expected to elute relative to the compound specified in the Internal Ref text box

    Set the **Retention Time** and **Time Window** parameters in one of the following ways: ***Using the mouse*** or ***Using the keyboard.***

    Set the **Time Window** to specify by how much the compound elution time may vary. The Time Window is applied either side of the predicted retention time to give a valid window. The Time Window also defines the chromatogram range that will be integrated. This example uses 0.200 minutes.

8. In the **Peak Matching** section of the method, set **Peak Selection** to specify which peak should be located when more than one peak is detected within the time window. By default, the peak Nearest to the specified retention time will be selected. Other options that can be selected are Largest peak and First peak or Last peak in the specified time window.

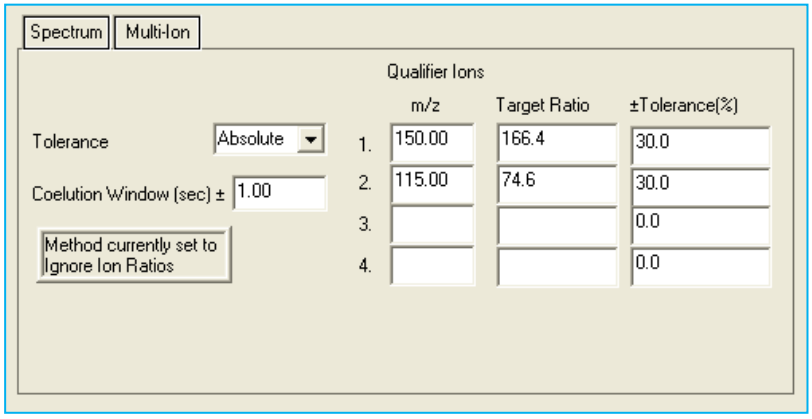
Note: *The setting for the Peak Selection control will determine the appearance of the lower part of the window.*

- **Spectrum** is selected for the first two Internal Standards (1: Pentafluorobenzene and 2: 1,4-Difluorobenzene) . Only peaks above the specified **Rev Fit Threshold** value are considered for the match. The REV Fit Threshold control will be enabled and the spectrum display will be visible. This example uses a **Rev Fit Threshold** value of 0.

Note: *Quantify does not perform background subtraction or Auto refine during Spectral peak matching. For example, when using spectrum data objects in a Communiqué report template, check the Spectrum Properties to make sure the Background treatment is set to* ***None****.*

- **Select Multiple Ion Ratio – to Quantify Trace**. When selected the spectrum control is hidden. The Qualifier Ion grid, the Tolerance control and the Coelution window control will be visible. The REV Fit Threshold control will be enabled. In this example set the following:

Setting	**Value**
Tolerance	Absolute
Coelution Window (sec) ±	1.00
± Tolerance	30%



The majority of the compounds in the example tutorial use multiple ion ratio. Scroll down through the list in the Method Editor to see example of this.

9. In this example do not specify the **User RF Value**. The User RF Value is used in cases where there are no calibration standards to plot a calibration curve. It represents the gradient of a curve and is used as a multiplication factor that will be applied to peak responses for the current compound to determine concentrations.

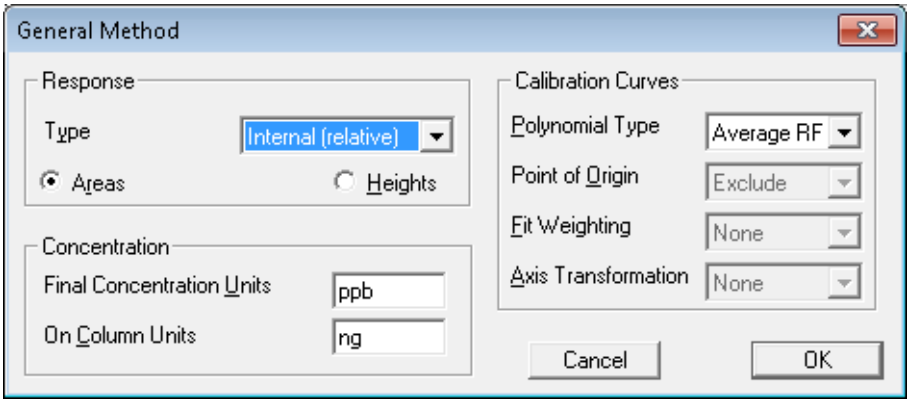
10. Set the **User Peak Factor**. This value is a multiplication factor that will be applied to all calculated concentrations for the current compound. If the User Peak Value is left at zero or set to 1, the concentration values will not be changed. In this example, the value is set to 1.000000

11. Set the **Reporting Threshold**. An edit box that defines a value to be passed to environmental reporting (Communiqué) via the data source for use when filtering quantitative results in reports. In this example the value is set to 0.000. Values must be entered as **On-column** amounts in nanograms (ng). Equivalent to values entered into the Sample List in the Conc fields for the initial calibration compound amounts for lowest level being reported.

IMPORTANT: *The **Reporting Threshold value** will be used as the **Reporting Limit** for the purpose of setting flags on Form 1 and determining what Compounds to show for Compound on the general environmental Quantitative Report (PKIEnvQuant template) if no Custom Compound List (which includes Reporting Limits) is defined.*

12. Set the Standard Concentration Factor (**Std Conc Factor**). Set this parameter on a per compound basis and it is used by Quantify to adjust the concentration values in 'standard' samples in the Sample List (including **Init Calib** and **Cont Calib**) prior to calibration of the compound. In this example the value is set to 1.000

13. Set the **General Parameters** by clicking the button.
    The General Parameters dialog appears:



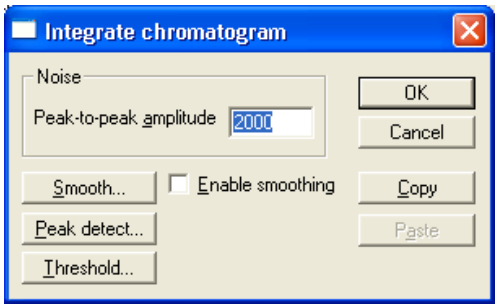
In this example the General Method Parameter settings are the same for all compounds as show in the above dialog box. Make sure the Response Type is set to **Internal** (relative).

Final Concentration Units: This value is used to define final concentration units in the quantify method. This value is used for the adjusted concentration units (environmental users) and for concentration units for non-environmental users.

On-Column units: This is the value defined in the quantify method for on-column amounts – this parameter is for environmental users who are required to enter ng into sample list for calculations to work correctly. This allows the quantify/view results to display the correct units (on-column ng) and to still have final concentration units available for reports.

Note: *The Concentrations defined in the Sample List **must ALL** be entered as **NG on column** values for both volatile and semi-volatile analysis. This applies to both the Internal Standard and individual target compounds.*

14. Click the **Integrate Parameter**s button and set the parameters. In this example they are set the same for all compounds.



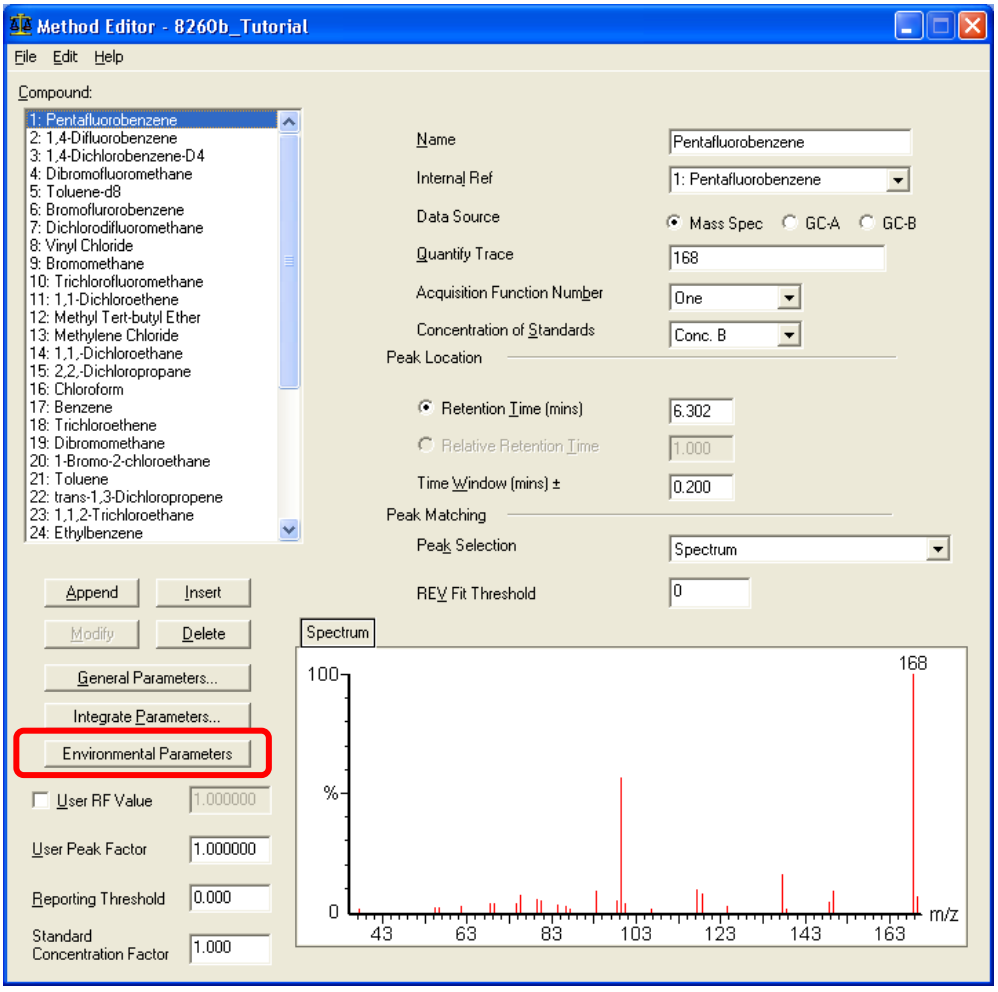
Next look at parameters for Compound 2: 1,4-Difluorobenzene. Continue until each Compound has the correct settings.

Setting Environmental Parameters

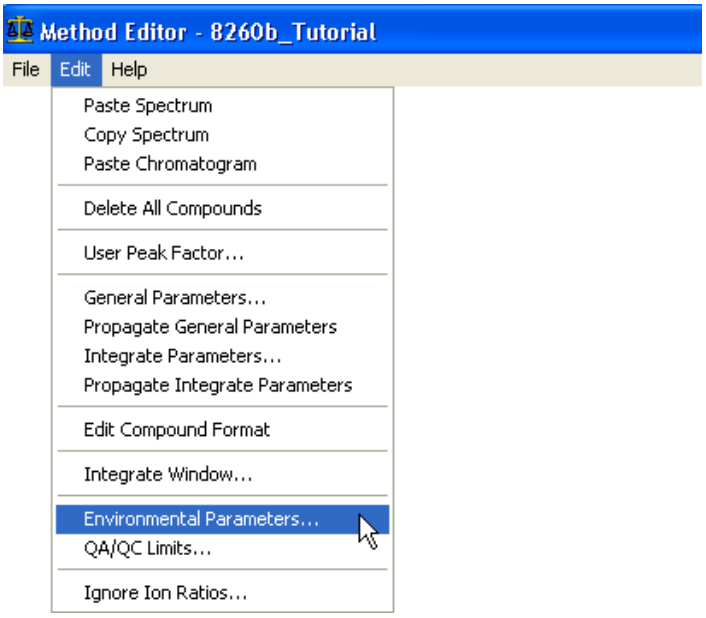
The environmental parameters will not have to be changed frequently, however when changes are required it involves reviewing a large number of compounds. Using the Environmental Parameters dialogs makes interaction more streamlined than that required by the main dialog of the Quantify Method Editor (i.e., the need to click Modify after editing each compound before selecting the next one).

The Environmental Parameters dialog displays the parameters based on the type of the currently selected compound; **Target Compound**, **Surrogate Compound**, **Spike Compound**, or **Internal Standard Compound**

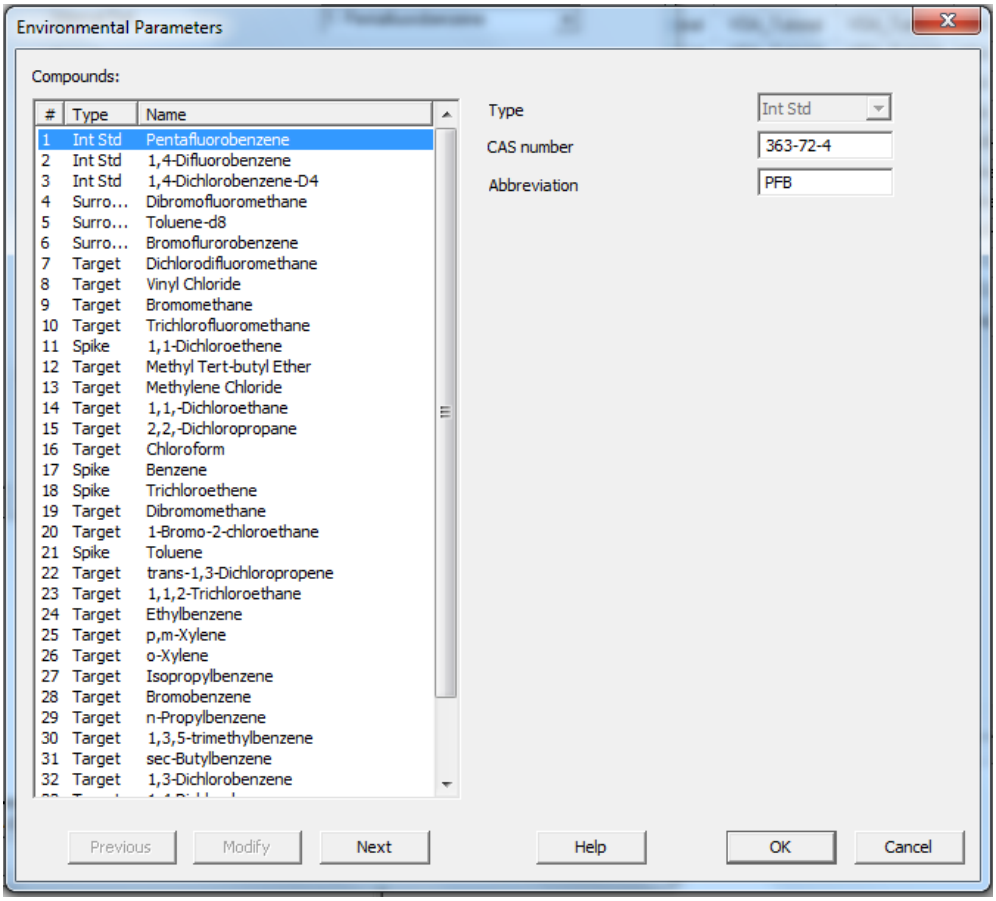
Note: *To ensure that environmental calculations function correctly, please take note that modifying any part of the Quantify Method may cause these calculations to be invalid. If a change to the Quantify Method is required, a complete integration and processing of the applicable data is required.*



➢ Click the **Environmental Parameters** button in the main Quantify Method Editor window (or choose the **Environmental Parameters...** command in the **Edit** menu) to display the Environmental Parameters dialog.



The Environmental parameters dialog appears. This dialog displays a list view of all **Compounds** defined in the Quantify Method.



The currently selected compound row appears highlighted. The list displays:

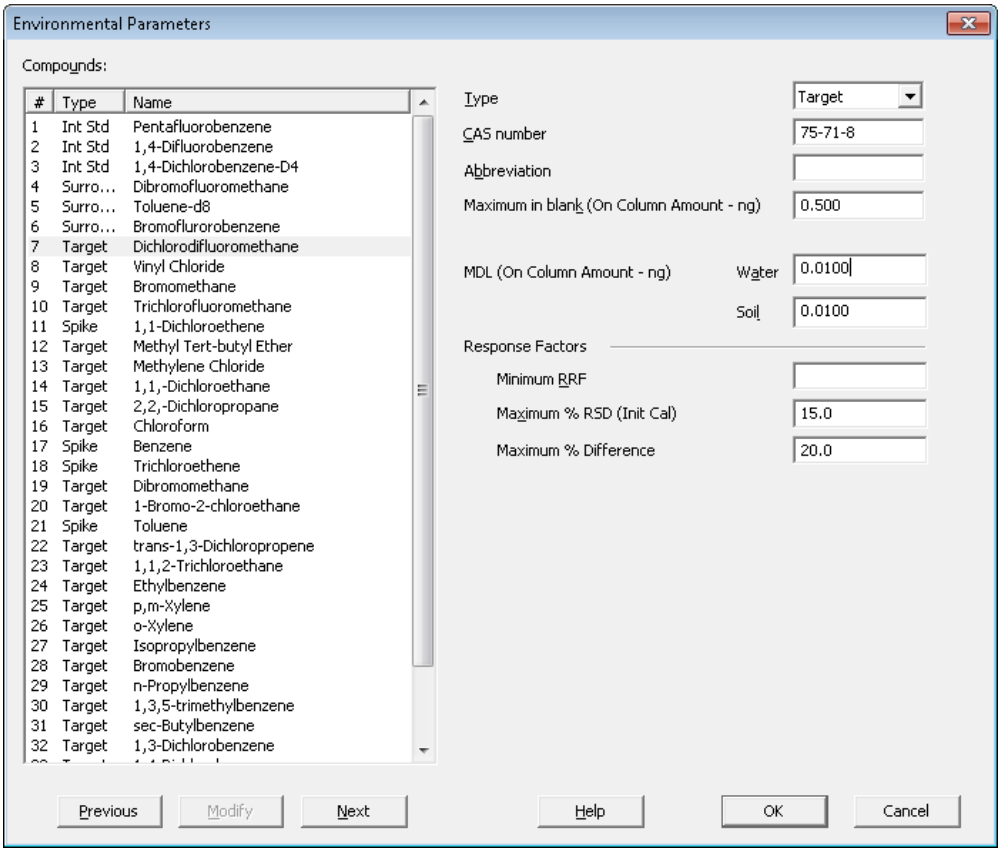
- The **compound number #** – This is the same number displayed in the main Quantify Method Editor window). Clicking on # sorts the compounds in the order in which they appear in the Quantify Method main window (or reverse order if clicked again).

- The compound **Type** – Target, Spike, Surrogate or Internal Standard. (see also Type control). Clicking on Type sorts the compounds by type in alphabetical (or reverse alphabetical order if clicked a second time).

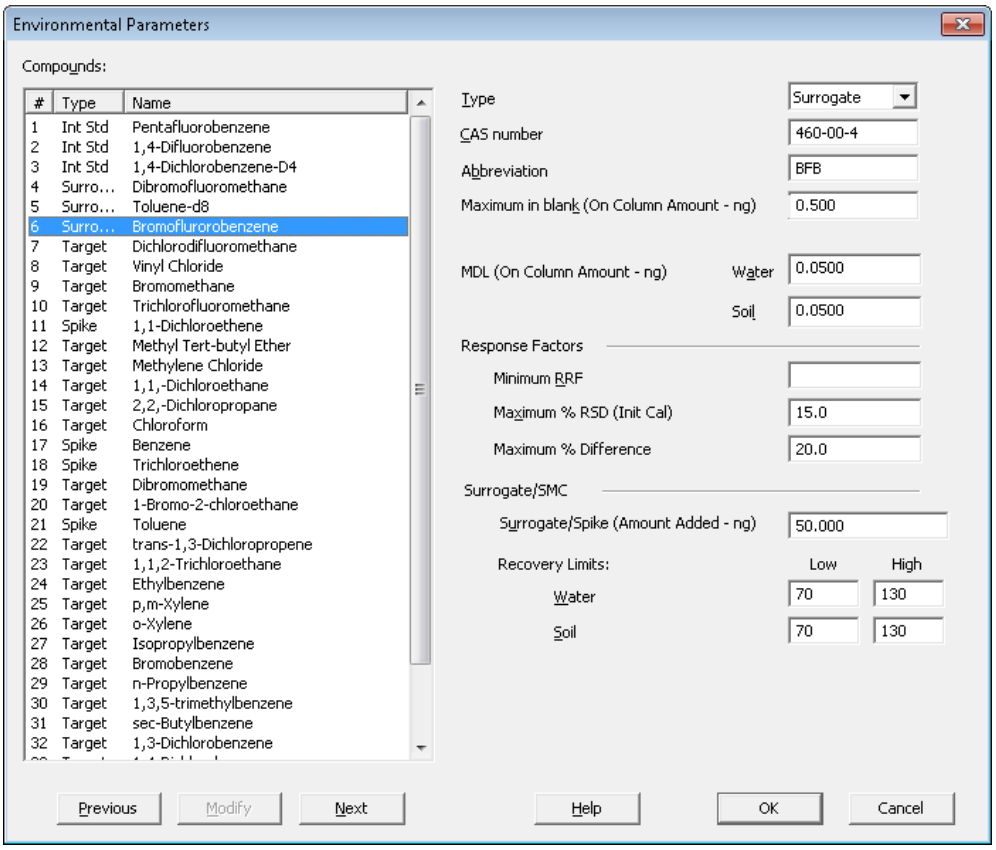
- The compound **Name**, as it appears in the main Quantify Method Editor window. Clicking on Name sorts the compounds by name in alphabetical (or reverse alphabetical order if clicked a second time).

The display of the Environmental Parameters dialog depends on the type of the currently selected compound. If the Compound Type is **Internal Standard**, then the following parameters are displayed: **Type**, **CAS number** and **Abbreviation**.

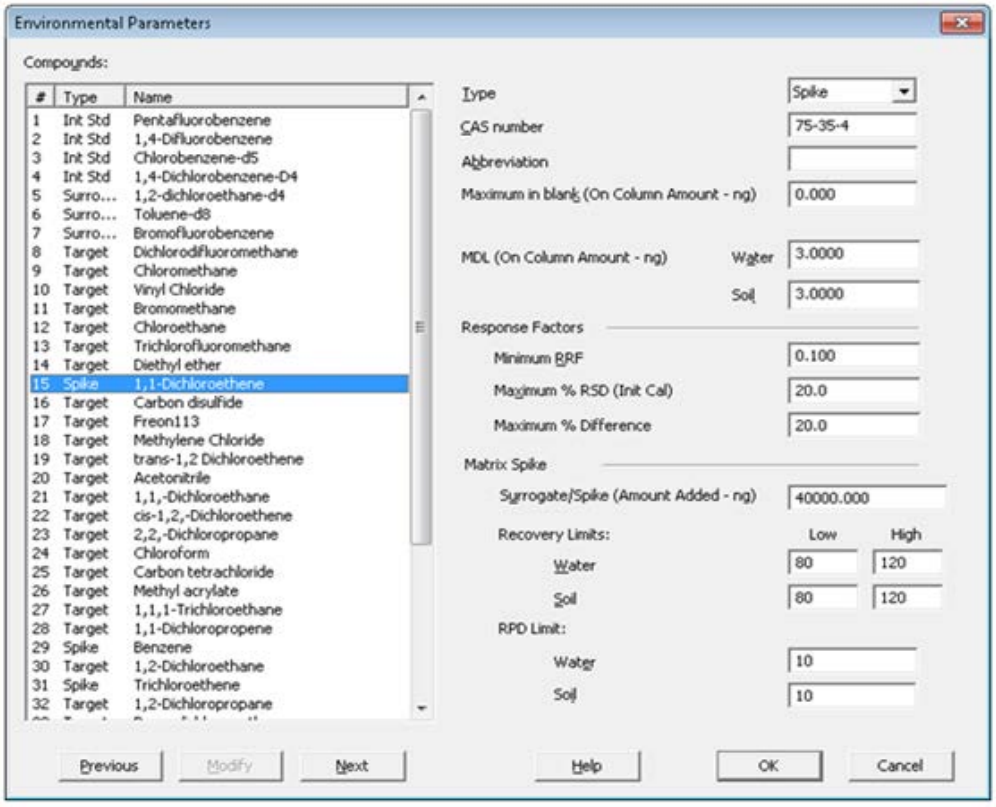
If the Compound Type is **Target** the following dialog appears:



If the Compound Type is **Surrogate**, the following dialog appears:



If the Compound Type is **Spike**, the following dialog appears:



Entering Environmental Parameter Values

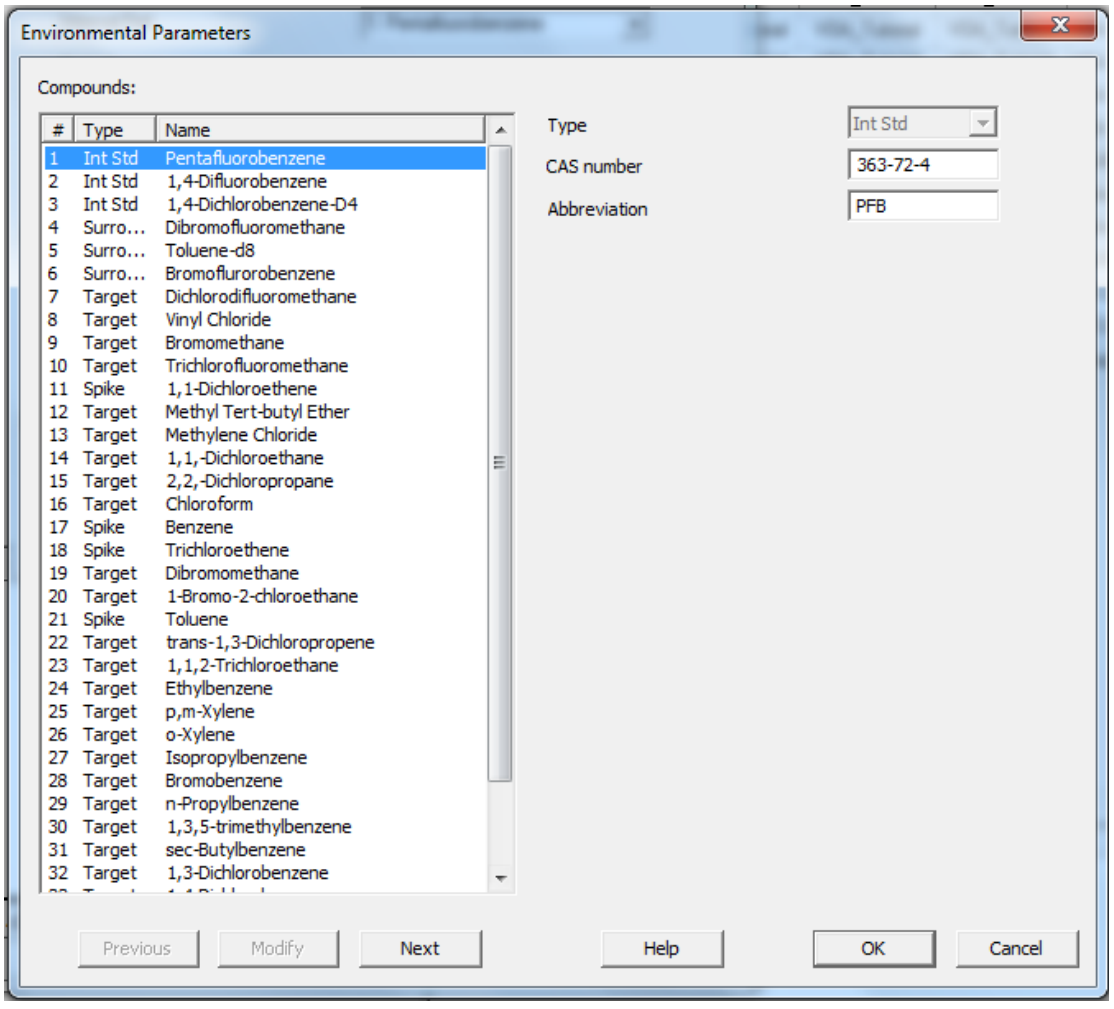
This section provides you with information the various parameters that are available on the Environmental Parameters dialog.

Type – A drop–down list that indicates the nature of the compound: a **Target**, a **SMC/Surrogate**, a **Spike**, or an **Internal Standard**.

Note: *A compound cannot be designated as an internal standard within this dialog. The use of a compound as an Internal Reference within the main Quantify Method Editor window defines it as an internal standard and will cause the Int Std selection to be displayed in this control as read–only.*

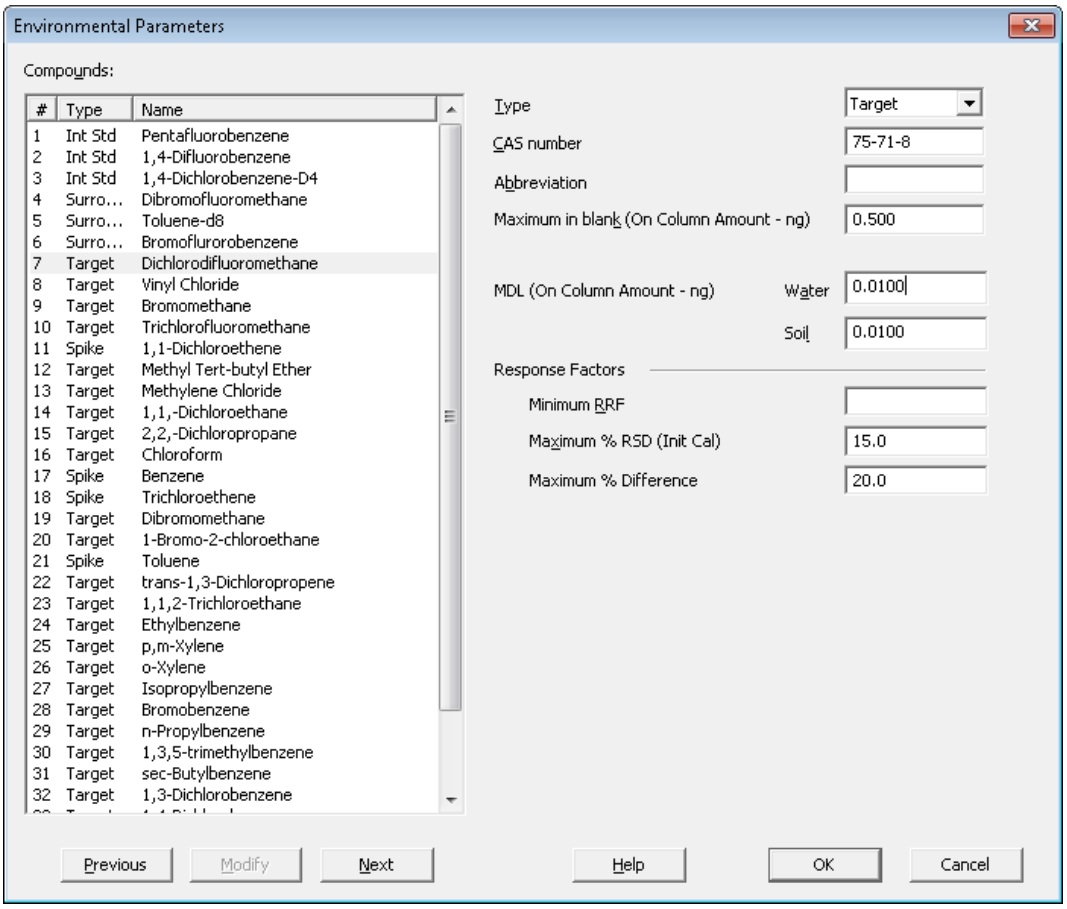
CAS number – An edit box that specifies the Chemical Abstracts Number for the compound.

Abbreviation – An edit box that specifies the abbreviation to be used for the compound on Forms 2 and 8 (i.e., this is only used for Surrogates and Int. Stds.)



Entering Target Compound Information

This section provides you with information on the parameters that are specific to a Target Compound.



Maximum in blank (On column Amount ng) - An edit box that indicates the maximum amount permitted (0.0000 to 9999.9999 or left empty) in the method blank before a 'B' flag will be assigned.

Note: *"B" flags (indicating blank contamination) should not be indicated on a Form 1 when the sample concentration is less than the adjusted minimum detection limit. The "B" flag should only show for a positive result (greater than the minimum detection limit). To do this, set the Maximum in blank value (ng) equal to the on-column minimum detection level (MDL)*

If the Maximum in blank value (ng) is set lower than the on-column MDL, a compound detected in the sample and blank that are above the maximum in blank value but below the MDL will be flagged with a "UB".

Method Detection Limit Values (MDL) – On column Amount - ng

- Applies to Voa waters, Voa low level soils, Voa Med level soils, Svoa Water, Svoa Soils
- Must be entered as 'on-column' amounts in nanograms (ng).
- This value is typically determined statistically by analyzing 7 replicate samples.

Note: *Volatile Medium level soil values must be entered into the Soil field.*

MDL: Water – An edit box that is used to indicate the minimum detection limit (MDL) for the selected compound in water samples. This control is not visible if the selected compound is an internal standard. This control is enabled for targets, spikes and surrogates. This value you enter in this field is used to trigger the "J" flag in the CLP-Like forms when a compound is above the MDL but below the RL.

MDL: Soil – An edit box that is used to indicate the minimum detection limit (MDL) for the selected compound in soil samples. This control is not visible if the selected compound is an internal standard. This control is enabled for targets, spikes and surrogates.

***Note**: For the purpose of environmental reports the term MDL is used to indicate the threshold value for the "U" qualifier flag. Values below this threshold value will flag the compound with a "U" in the Form 1 report and no concentration values will be printed.*

Response Factors

For each target and surrogate compound you can define:

Minimum RRF - An edit box that defines the minimum acceptable RRF for this compound (0.0000 to 9999.9999 or left empty) in initial and continuing calibrations.

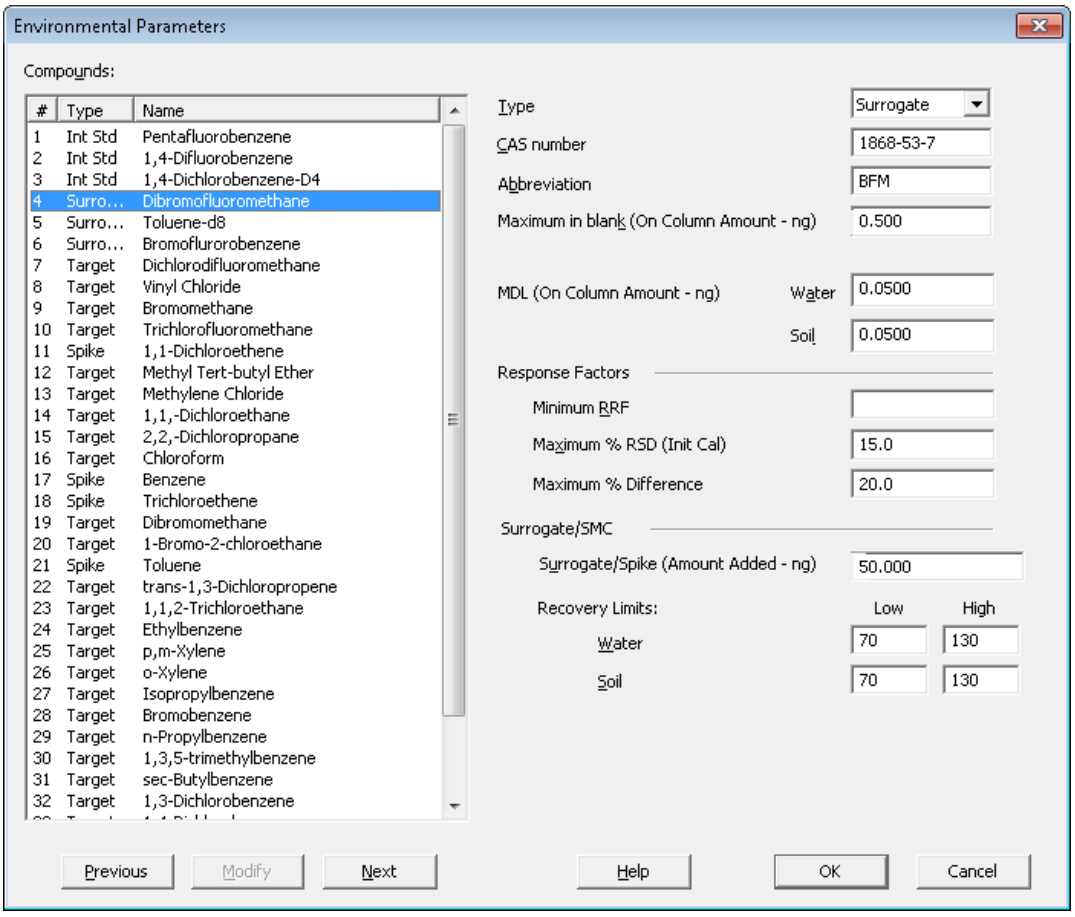
***IMPORTANT:** If you change the Minimum RRF value, Maximum % RSD value and/or Maximum % Difference value, you must reprocess (recalibrate) the Sample List for this new value to be used in the calibration acceptance testing.*

Maximum % RSD (Init Cal) – An edit box that defines the maximum acceptable percentage relative standard deviation (0.0 to 100.0 or left empty) between response factors calculated for each concentration level of the initial calibration.

Maximum % Difference [Maximum % Drift For compounds using curve fit] – An edit box that defines the maximum acceptable percentage difference between the RRF calculated for this compound from the continuing calibration and the average RRF from the initial calibration. For compounds using curve fit this becomes the 'Maximum % Drift' – the acceptable difference between the concentration calculated for the compound in the continuing calibration standard using the calibration equation and the known concentration.

Entering Surrogate Compound Information

This section provides you with information on the parameters that are specific to a Surrogate Compound.



Surrogate/SMC

Environmental Parameters for Surrogate Compounds contains all of the parameters described in the previous sections plus the following:

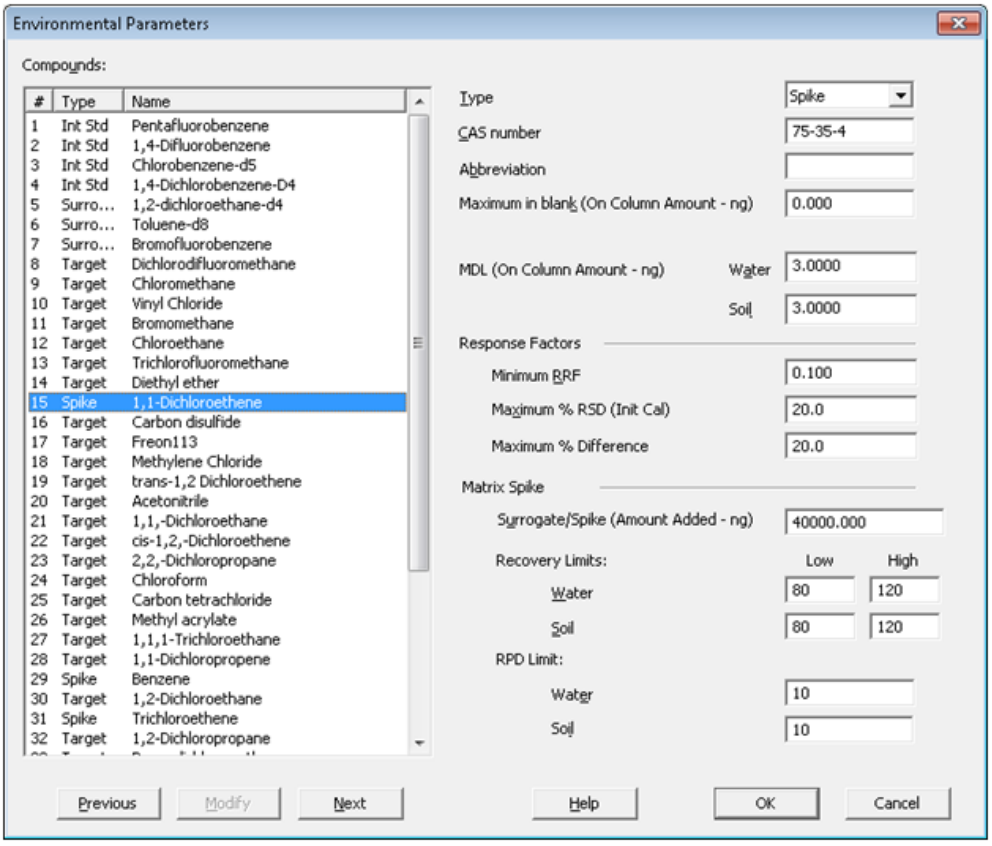
Surrogate/Spike (Amount-Added – ng) – An edit box that defines the amount of the compound used to spike the sample. This should be entered as the amount of spike added as described on pages 199-201 in **Appendix 1 Environmental Reporting Calculations**.

Low Recovery limits: Water and Soil – An edit box that defines the minimum acceptable recovery percentage for the compound (spike or surrogate) in a Water sample and a Soil sample. (Form 2)

High Recovery limits: Water and Soil – An edit box that defines the maximum acceptable recovery percentage for the compound (spike or surrogate) in a Water sample and a Soil sample. (Form 2)

Entering Spike Compound Information

This section provides you with information on the parameters that are specific to a Spike Compound.



Matrix Spike

This dialog contains all of the parameters described in the previous sections plus the following:

In the **Surrogate/Spike (Amount-Added—ng)** field enter the Amount Added in nanograms (ng). The amount added is defined in the Spike and Surrogate Recovery sections for volatile and semi-volatile (refer to **Appendix 1 Environmental Reporting Calculations**).

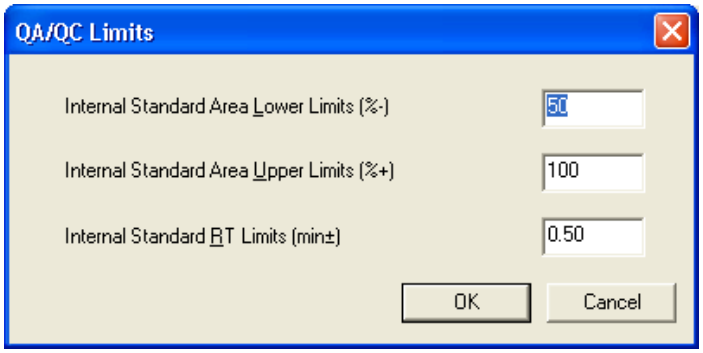
RPD Limit for Water – An edit box that defines the maximum acceptable relative percent difference (RPD) value between the matrix spike and matrix spike duplicate recoveries, for water samples.

RPD Limit for Soil – An edit box that defines the maximum acceptable relative percent difference value between the matrix spike and matrix spike duplicate recoveries, for soil samples.

Setting QA/QC Limits

The QA/QC limits are a set of values that apply to all compounds (global parameters) in the method. For this reason the QA/QC Limits dialog is accessed from the **Method Editor Edit** menu rather than from a button in the compound list section of the Method Editor since all the buttons on this window relate to values than can be compound–specific; the QA/QC limits cannot be.

The environmental QA/QC limits have also been kept separate from the **Ignore ion ratios** when matching peaks option (the only other global parameter) because the QA/QC Limit values are used for different applications and combining them might be confusing.



Internal Standard Area Lower Limit (% -) – Enter a value (from 0 to 100) that defines the acceptable lower limit of the area measured for each internal standard peak in a sample, compared to that in the most recent continuing calibration, or mid–level of the initial calibration if no continuing calibration has yet been performed.

Internal Standard Area Upper Limit (% +) – Enter a value (from 0 to 100) that defines the acceptable upper limit of the area measured for each internal standard peak in a sample, compared to that in the most recent continuing calibration, or mid–level of the initial calibration if no continuing calibration has yet been performed.

Internal Standard RT Limits (min ±) – Enter a value (from 0.00 to 999.99) that defines the acceptable limits for the actual retention time of each internal standard peak in a sample, compared to that in the most recent continuing calibration, or mid–level of the initial calibration if no continuing calibration has yet been performed.

1. When done, select **Save As** from the File menu and name the Quantify method Tutorial.

2. Close all windows except the Sample List.

Note: *After entering all of the Quantitation Method parameters, this quantitation method must be placed in and saved in the Sample List for the correct processing of environmental reports..*

Entering Parameters into the Quantify Method (For Expert Users)

For complete details, refer to the TurboMass GC/MS Tutorial. The following is a summary of the procedure:

1. Open the chromatogram of the calibration run of interest. Have the method editor open at the same time.
2. In the chromatogram screen, select the peak of interest and click to get the spectra window open.
3. In the spectra window, double-click the mass you want to quantify on. This will open another chromatogram with that extracted mass of interest.
4. Right-click and drag a retention time window in the extracted mass chromatogram, and notice that in the message bar it say "retention time window XX where XX is minutes.
5. Then right click on the extracted ion peak of interest.
6. Finally copy the spectrum list from the spectra window.
7. Toggle back to the method, and observe that all the values are in the right place.
8. Paste the spectrum, add the name, and append

Creating a GC and MS Method

If you wish to run your own analysis, create and save a method to control the GC for your analysis, and create and save a method to control the mass spectrometer. The method shown in this tutorial was created specifically to analyze EPA Method 8260 using a Headspace trap with the GC/MS. You will need to create methods in a similar way for your own quantification analysis. Refer to the *Clarus GC/MS Tutorial* for details on setting up you GC and MS acquisition methods.

Note: *The example in this tutorial assumes you have run an analysis and collected data. To illustrate this, we will use a sample data set in the VOA_Tutorial project containing mass chromatograms.*

3 How to Set Up Submitter/Task Data



About the Submitter/Task Information

The TurboMass software provides a simple dialog (Submitter/Task Data from Environmental Configuration in the Tools menu.) that allows you to easily assign report templates to forms you may customize to meet the needs of your clients. A report template designed to capture certain evaluation criteria may also be substituted for another one by an administrator or another with the designated security permissions.

This dialog allows you to optionally predefine which customized report and associated component list is generated on a per-client and per-task basis. This eliminates the possibility of generating the wrong customized form for the client. In addition, when reporting to LIMS it allows on the necessary data to be reported to LIMS.

You can customize your reports to include your laboratory's or organization's logo, chromatograms, spectra, calibration plots, a quantification report, library-search reports, page header and footer and more. You can plot the total ion chromatogram and/or selected ions and overlay target and qualifier multiple ion traces.

Additionally, reports may include items such as initial calibration response factors, concentrations of the target compounds, and/or TICs.

The Submitter/Task Data Window combines several related functions:

- Maintenance of Submitter/Task hierarchy
- Maintenance of Custom Compound Lists
- Mapping of Forms to Report Methods

These functions are combined in *one Submitter/Task Data window* since both Custom Compound Lists and mapping of Forms to Report Methods are specific to a Task.

The Custom Compound Lists are displayed so an environmental laboratory will be able to define subsets of the compounds in a quantify method to be reported for a specified project (task). This task may be one of the multiple tasks for an overall submitter (client). TurboMass allows for multiple submitters and multiple tasks for each submitter. The compound subset (or full list) for each task will also set the (concentration) reporting levels of each of these compounds uniquely for each client's (Submitter's) projects.

The mapping of Forms to Report Methods allows Form customization by providing a translation between the EPA Forms and Communiqué templates (referenced in TurboMass report methods). This translation enables the environmental report generation process, which is driven from the selection of the Forms to be printed.

This window is a tree view showing clients (Submitters) and their projects (Tasks) displayed in alphabetical order. The tree view can be expanded and collapsed; but you cannot drag–and–drop items (nodes) in the tree view. Renaming of nodes is possible, by setting the label into edit mode by right-clicking at the end of it then selecting Rename from the context menu. The currently selected node will remain highlighted when the list does not have focus.

You can also associate a Custom Compound List with a particular Task. (A Custom Compound List is never directly associated with a Submitter.)

Note: *If you do not choose a Submitter and Task then the Default is selected.*

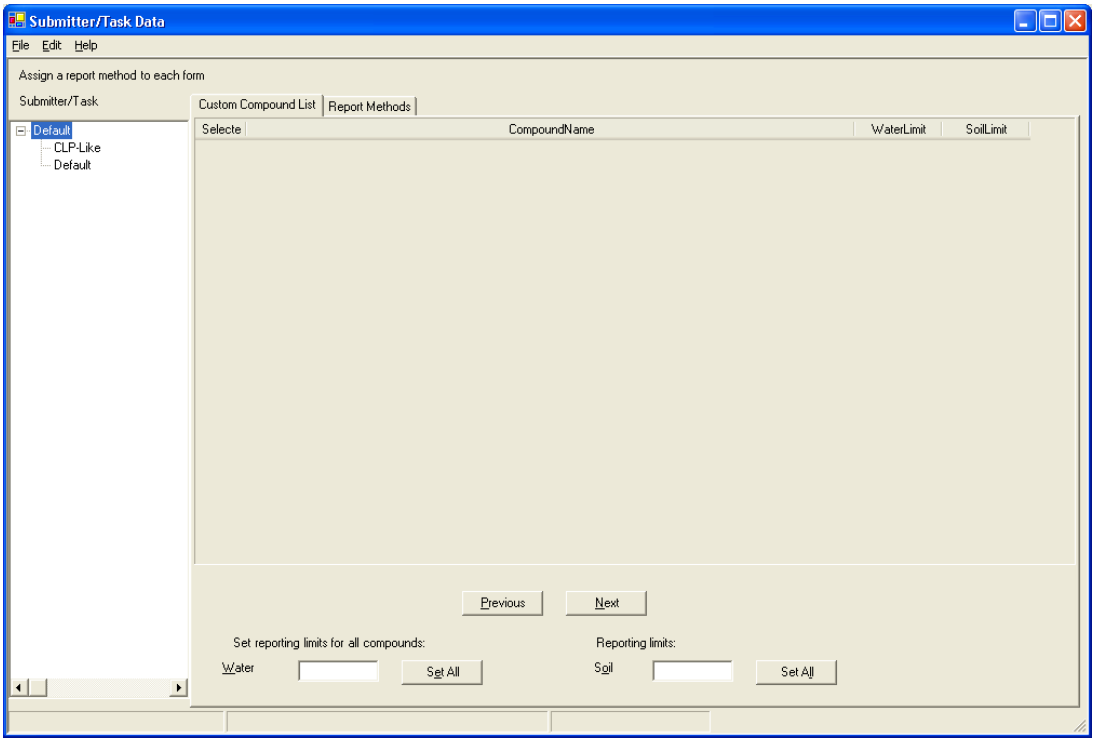
To access the Submitter/Task Data window:

- From the TurboMass **Sample List** window select Tools > Environmental Configuration > **Submitter/Task Data...**

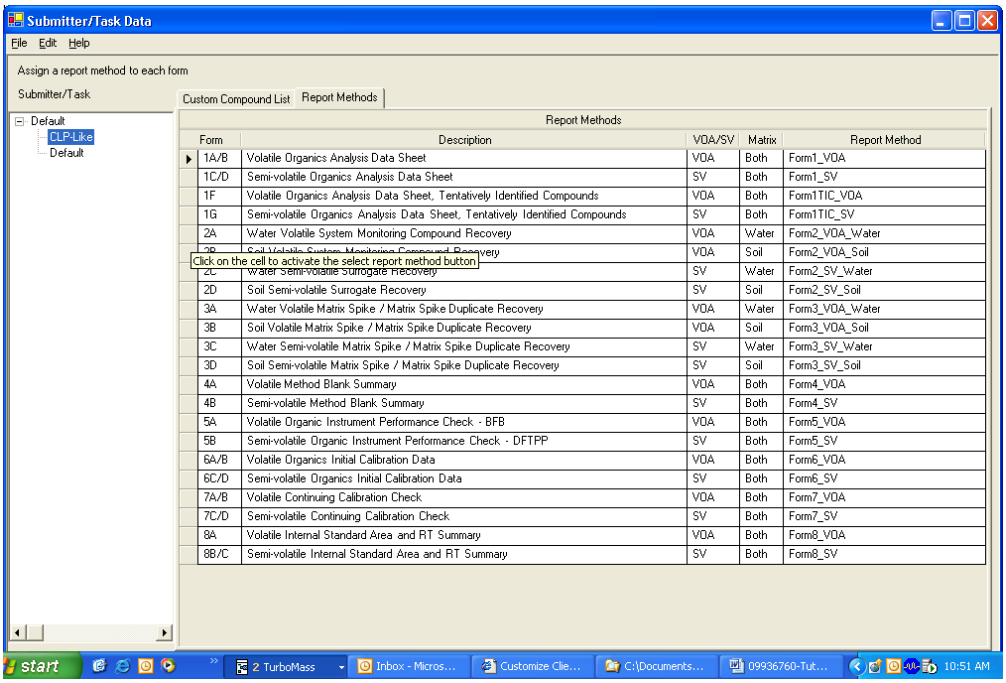
To add a submitter:

1. From the TurboMass Sample List window select **Tools > Environmental Configuration > Submitter/Task Data...**

   The **Submitter/Task Data** window is displayed:



2. Select the **Report Methods** tab. This provides an independent mapping of Forms to report methods for each Submitter and Task. This mapping is global and it will be up to the user to ensure the appropriate report methods are in the project (this occurs automatically when a new project is created via the Project Wizard).



Form	Description	VOA/SV	Matrix	Report Method
1A/B	Volatile Organics Analysis Data Sheet	VOA	Both	Form1_VOA
1C/D	Semi-volatile Organics Analysis Data Sheet	SV	Both	Form1_SV
1F	Volatile Organics Analysis Data Sheet, Tentatively Identified Compounds	VOA	Both	Form1TIC_VOA
1G	Semi-volatile Organics Analysis Data Sheet, Tentatively Identified Compounds	SV	Both	Form1TIC_SV
2A	Water Volatile System Monitoring Compound Recovery	VOA	Water	Form2_VOA_Water
2B	Soil Volatile System Monitoring Compound Recovery	VOA	Soil	Form2_VOA_Soil
2C	Water Semi-volatile Surrogate Recovery	SV	Water	Form2_SV_Water
2D	Soil Semi-volatile Surrogate Recovery	SV	Soil	Form2_SV_Soil
3A	Water Volatile Matrix Spike / Matrix Spike Duplicate Recovery	VOA	Water	Form3_VOA_Water
3B	Soil Volatile Matrix Spike / Matrix Spike Duplicate Recovery	VOA	Soil	Form3_VOA_Soil
3C	Water Semi-volatile Matrix Spike / Matrix Spike Duplicate Recovery	SV	Water	Form3_SV_Water
3D	Soil Semi-volatile Matrix Spike / Matrix Spike Duplicate Recovery	SV	Soil	Form3_SV_Soil
4A	Volatile Method Blank Summary	VOA	Both	Form4_VOA
4B	Semi-volatile Method Blank Summary	SV	Both	Form4_SV
5A	Volatile Organic Instrument Performance Check - BFB	VOA	Both	Form5_VOA
5B	Semi-volatile Organic Instrument Performance Check - DFTPP	SV	Both	Form5_SV
6A/B	Volatile Organics Initial Calibration Data	VOA	Both	Form6_VOA
6C/D	Semi-volatile Organics Initial Calibration Data	SV	Both	Form6_SV
7A/B	Volatile Continuing Calibration Check	VOA	Both	Form7_VOA
7C/D	Semi-volatile Continuing Calibration Check	SV	Both	Form7_SV
8A	Volatile Internal Standard Area and RT Summary	VOA	Both	Form8_VOA
8B/C	Semi-volatile Internal Standard Area and RT Summary	SV	Both	Form8_SV

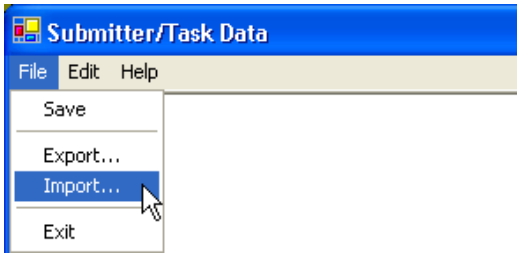
There are two types of Tasks under the Default Submitter; **CLP-Like** and **Default.**

- The **CLP-Like** displays the corresponding Report Methods based on the Contract Laboratory Program (CLP) of the EPA as close as possible.

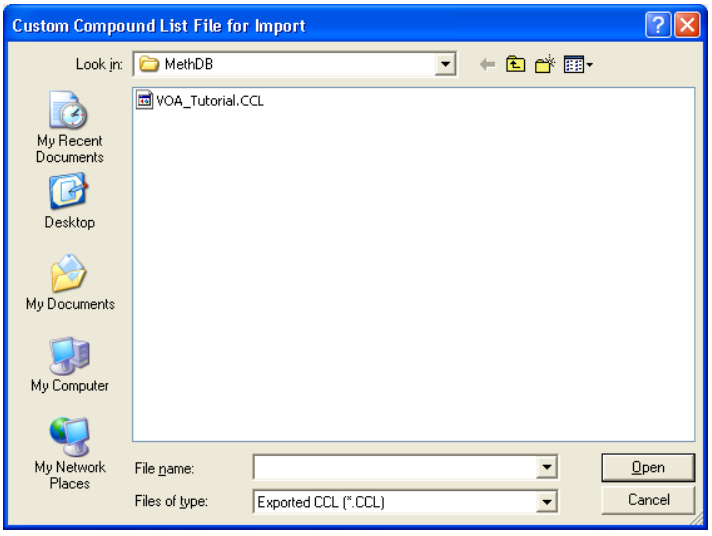
- The **Default** displays the corresponding PerkinElmer version of the forms (PKI) which provides the same results as the CLP-Like with expanded headers to display more information.

To Import a Custom Compound List (.CCL)

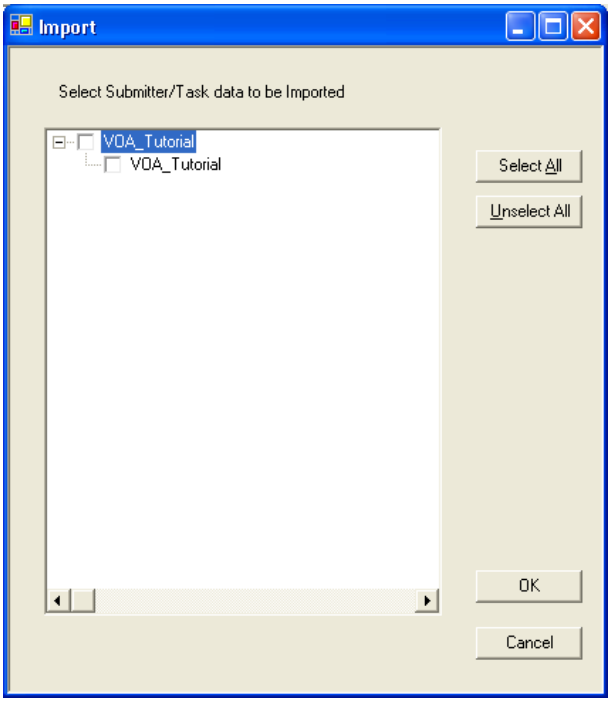
1. From the Submitter/Task Data window select **File > Import**.



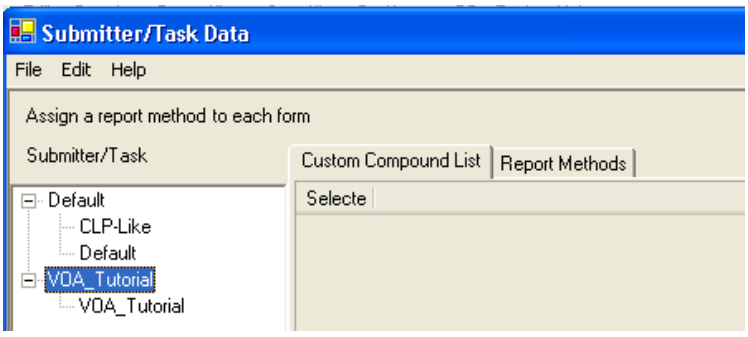
The following dialog appears for the VOA_Tutorial:



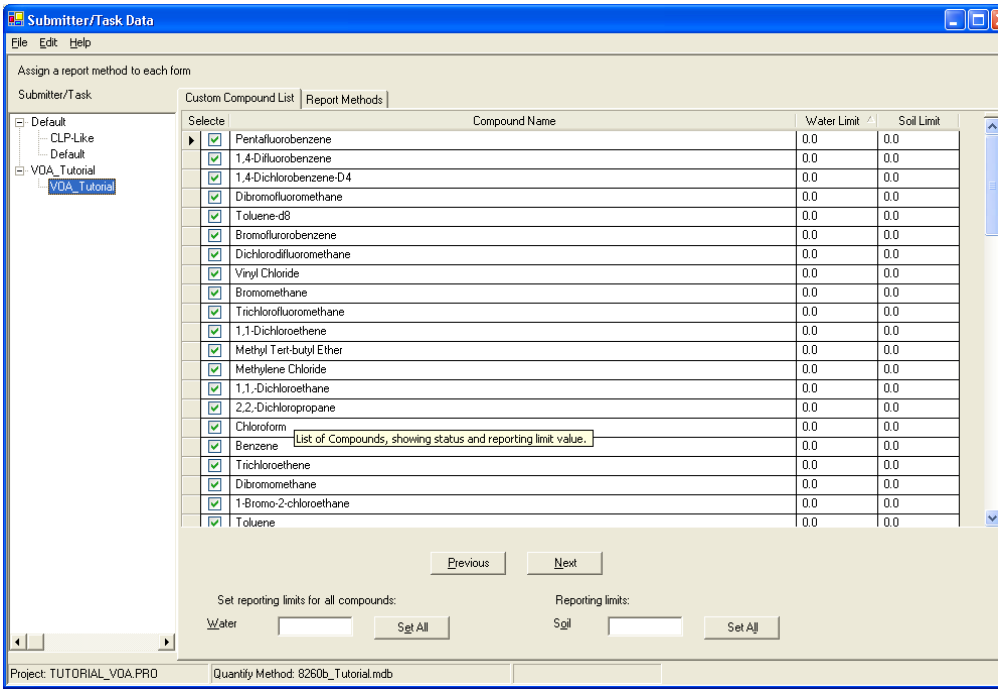
2. Select **VOA_Tutorial.CCL** and click the **Open** button.



3. Select the Submitter/Task to be **Imported** and click the **OK** button. The Submitter and Tasks are now displayed.



4. Click on the **VOA_Tutorial task** to display the Custom Compound List.



5. Close the Submitter/Task window by selecting **File > Exit.**

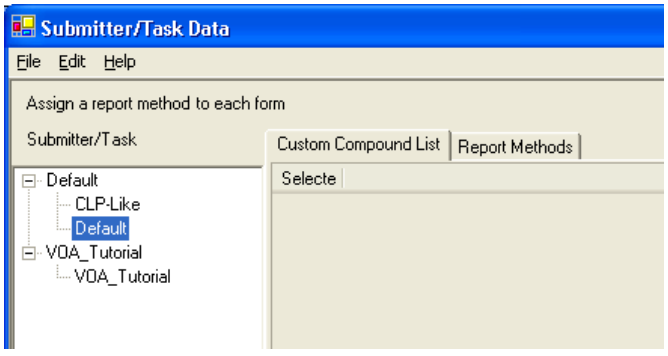
To Define Report Methods

To define the report methods (and hence Communiqué templates) to be used in generating the EPA Forms for an existing Submitter/Task:

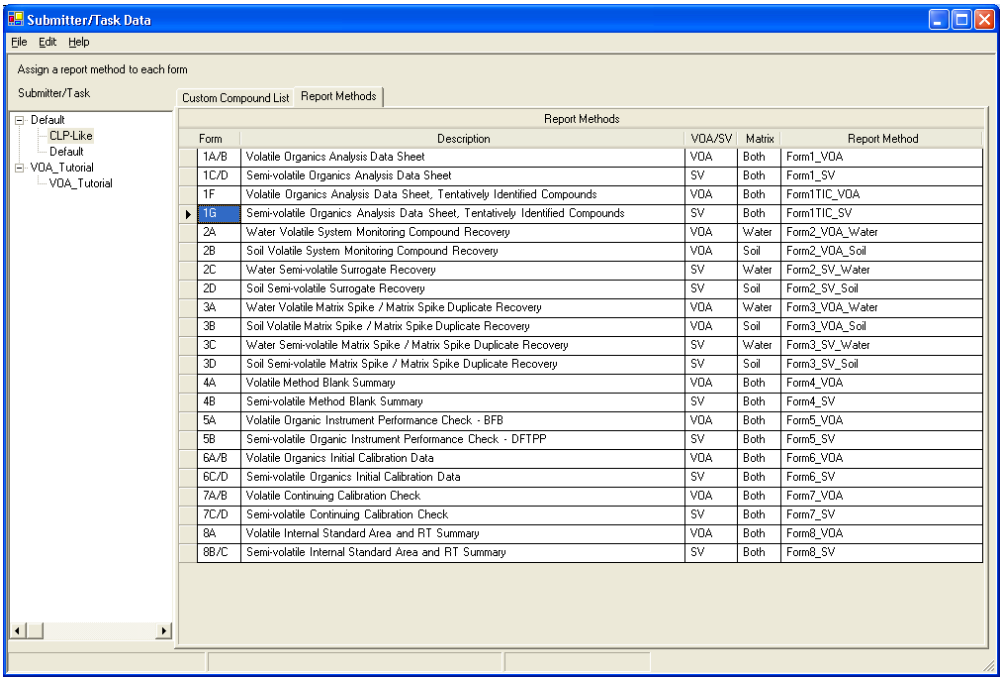
1. From the TurboMass Sample List window select **Tools > Environmental Configuration > Submitter/Task Data...**

   The Submitter/Task Data window displays.

2. Select the appropriate **Task** for the Submitter.



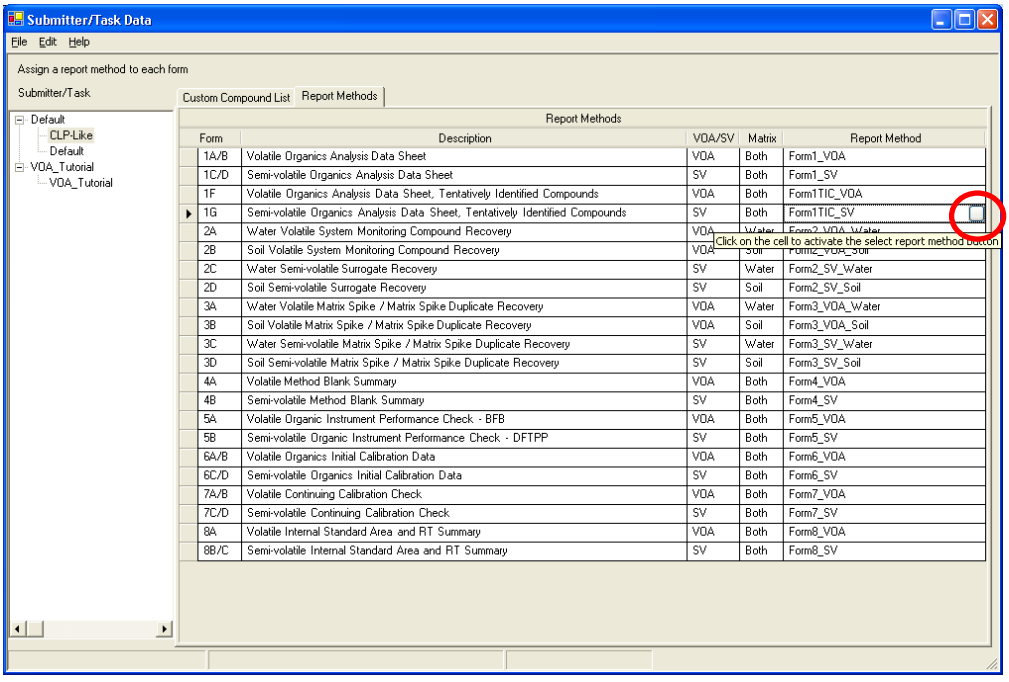
3. Select the **Report Methods** tab.



4. Review the table displaying Forms 1 – 8, and their variants, together with the associated report method.

Note: *For more information on producing Forms 1 – 8 refer to Steps for Producing Environmental Forms on page 90.*

5. Where necessary, select the appropriate report method to be associated with each Form by positioning the cursor in the Report Method cell.



6. Click the button that appears on the far right side to display the Report Method data files. The Report Method Data File window displays.

7. Select the method file you want and click the **Open** button.

   In this Report Method Data File list: Forms 1 – 8 are volatile and semi-volatile CLP-Like: there are also PKI versions of the Forms 1 – 8; and LCS (Laboratory Control Sample) and ICV (Initial Calibration Verification) Forms.

8. When the mapping of Forms to report methods is correct, choose the **File / Save** command.

Custom Compound List

When a **Custom Compound List** is displayed the status bar displays the exact origin of the custom compound list: the Project name is displayed in the left–hand segment and the Quantify Method name, along with the time and date the compound list was imported, is displayed in the right–hand segment. At all other times the status bar will be left empty.

Although defining a custom list will not be an everyday operation it nevertheless needs to be a simple and efficient process. Since it may involve changing a single value (reporting level) for a whole series of compounds the editing procedure should be more streamlined than that required by the main dialog of the Quantify Method Editor (i.e., clicking Modify after editing each compound before selecting the next one), while not being so different that it is confusing.

When a Task is selected and a custom compound list is displayed, the status bar shows the Project and Quantify Method from which the compound list was originally extracted.

To Create a Custom Compound List

To create a custom compound list for a new task of an existing submitter:

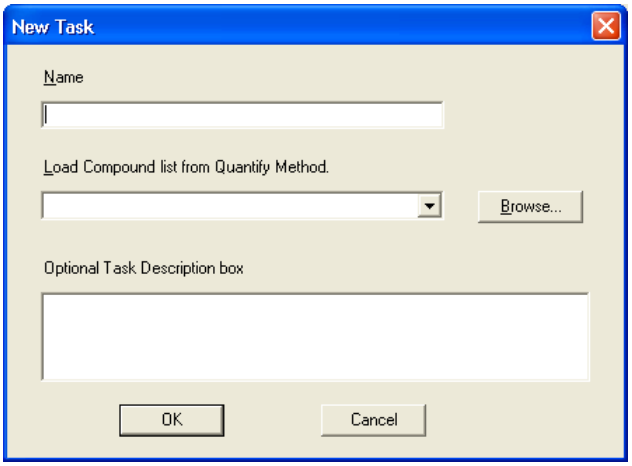
1. From the TurboMass Sample List window select **Tools > Environmental Configuration > Submitter / Task Data...**

   The Submitter/Task Data window displays.

2. In the Submitter/Task Data window select the submitter for whom the new task is to be generated.

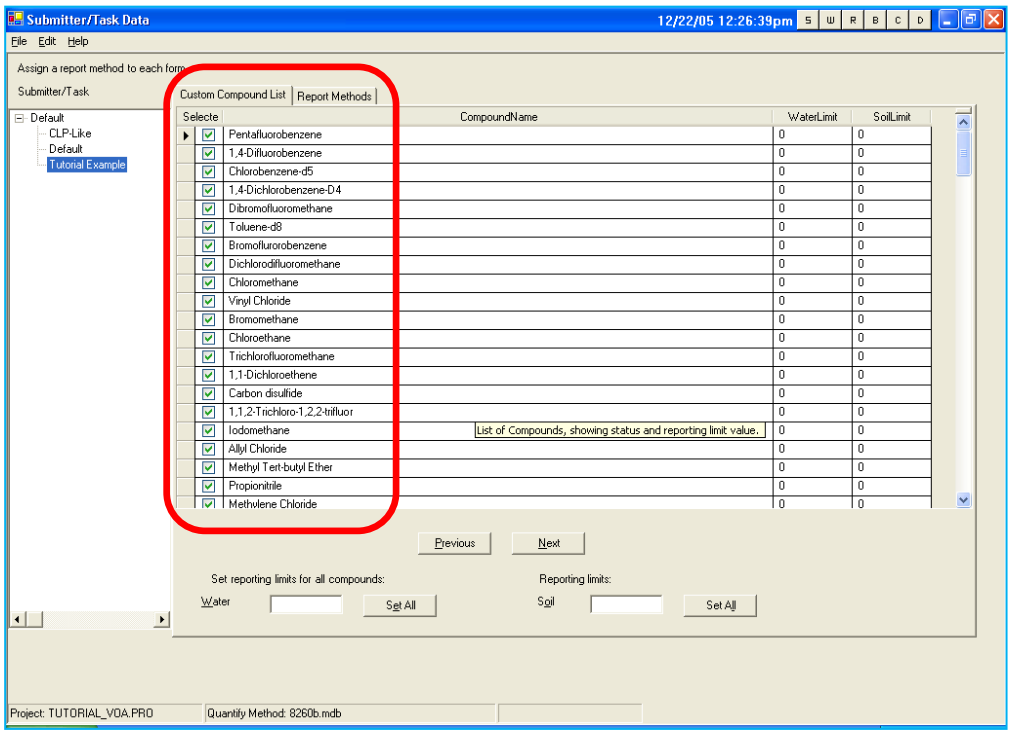
3. Choose **New Task** from the **Edit** menu (or the context menu for the tree list).

   The **New Task** dialog displays.



4. Enter a **Name** for the new task and select the **Quantify Method** from which the Compound List is to be imported. The compounds are displayed and initially each one is checked.

5. Click the **OK** button.
   The Compound List is displayed.



6. Click on the check box to unselected a **Compound Name** that is not required to be reported for this submitter/task. Repeat as required.

7. Select a compound for which a different **Reporting Limit** (Water or Soil) is required.

8. Enter the new value in the appropriate edit field below the list and click **Save** from the File menu to save the change or **Next** to save the change and select the next compound (or select Previous to save the change and select the previous compound).

Note: *The Reporting Limits entry in this dialog supersedes the Reporting Threshold Value in the Quantify Method.*

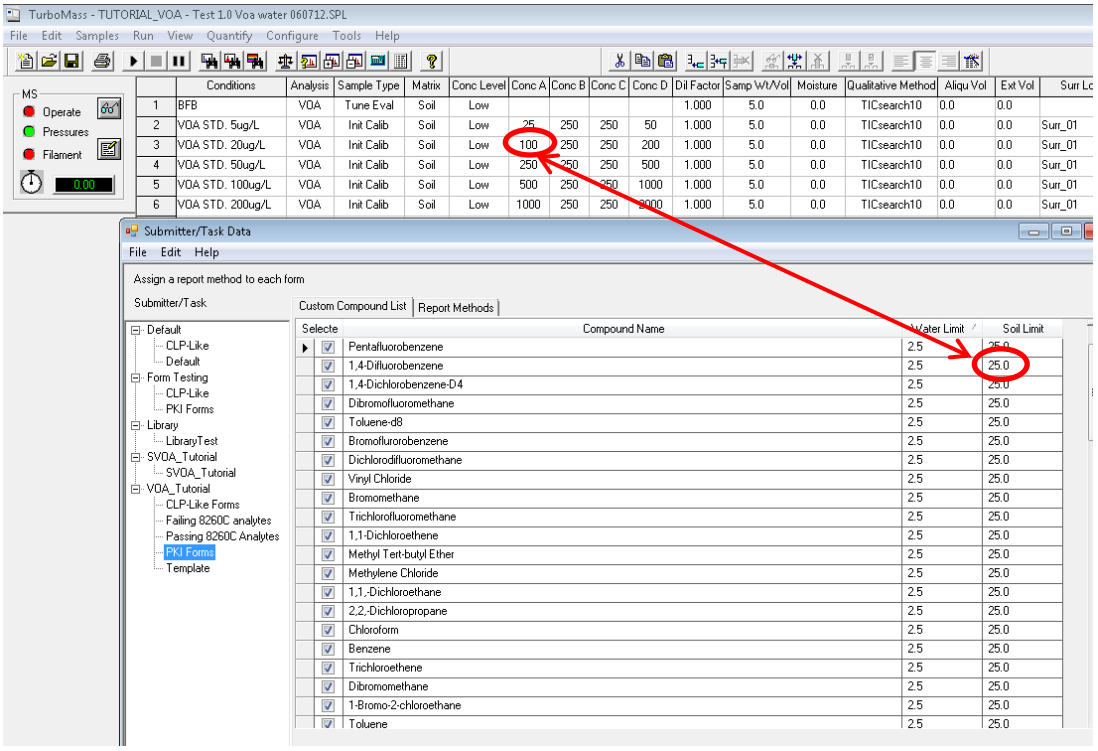
The Reporting Limits value should be entered as on column units in nanograms (ng). Refer to the following section for information on how to properly set these values.

9. When done, choose **Save** from the **File** menu to save the new Custom Compound List.

Proper Reporting Limit (RL) Setup

- Applies to Voa waters, Voa low level soils, Voa Med level soils, Svoa Water, Svoa Soils
- Must be entered as 'on-column' amounts in nanograms (ng). Equivalent to values entered into Sample List in the Conc fields for the initial calibration compound amounts for lowest level being reported.
- This value is typically determined by the lowest level that is used in the initial calibration curve.

Note: *Volatile Medium Level Soil values must be entered into the* ***Soil Limit*** *field.*





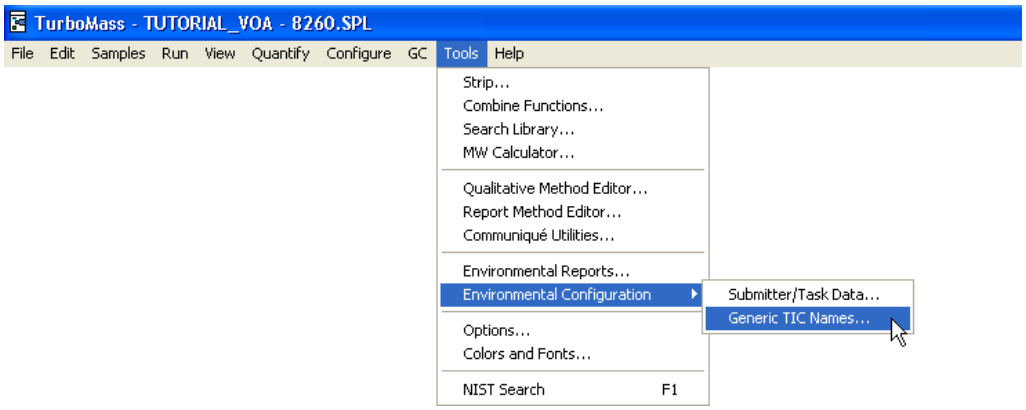
4 How to use Generic TIC Names



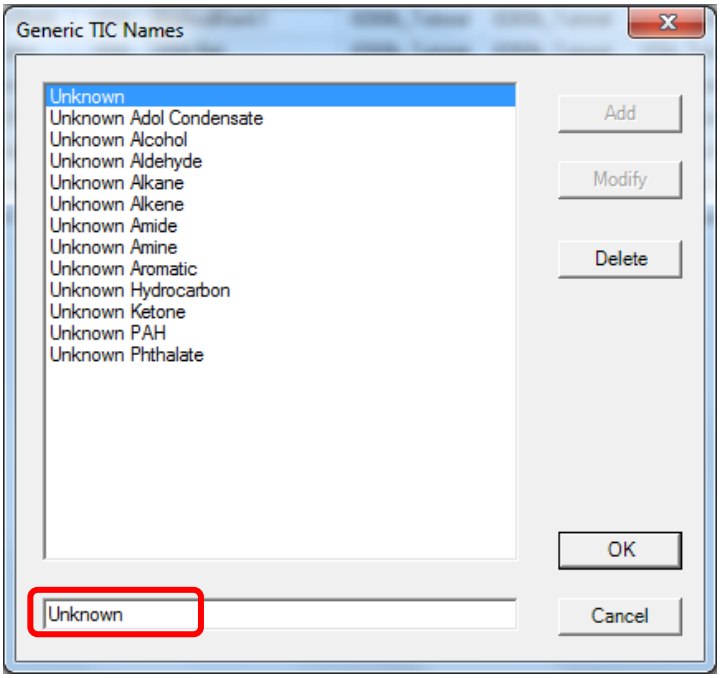
About Generic TIC Names

Administrators' or other people with appropriate access permission can edit the list of generic compound names that appear in the Tentatively Identified Compound dialog.

1. To access this dialog select **Tools > Environmental Configuration > Generic TIC Names...** from the TurboMass Sample List window.



The **Generic TIC Name** dialog appears:



Generating a Tentatively Identified Compound report requires identifying unknowns. Setting the generic TIC names will allow a customization of identifying your unknown compounds.

2. Type a generic name into the edit box and click the **Add** button.

3. Repeat step 2 for each generic name required (the list is sorted alphabetically as each new name is added).

4. Click **OK**.

To edit the list of generic names:

1. From the From the **Tools** menu select **Environmental Configuration > Generic TIC Names...**
2. Edit an existing name by selecting that name, editing as required and clicking the **Modify** button.
3. Add a new name by typing the name into the edit box and clicking the **Add** button.
4. Delete an existing name by selecting the name and clicking the **Delete** button.
5. Click **OK**.

5 How to Build a Sample List using the Wizard



About a Sample List

The TurboMass Sample List window is in a spreadsheet format, which allows easy editing of multiple samples but becomes unwieldy when there is a lot of information required for each sample. The Sample List Wizard is a forms-format equivalent display. It allows easier editing of the large amount of per-sample information required of environmental and QA/QC samples.

The Sample List Wizard enables you to select an existing Sample List or create a new one. Only the parameters required for the current type of analysis (VOA, SV, or QA/QC) and current matrix (water/soil) are displayed. The controls are grouped in a way to allow efficient entry of sample-specific data. It also provides the ability to propagate changes made to one row to subsequent rows; a command to update vial numbers; a command to update sample IDs. Several fields have automatic incrementing of numeric values, and there are commands to insert, delete, and add sample rows.

Note: *All environmental reporting users should use the Sample List Wizard to create a sample list with the majority of required fields for their analysis and matrix type. You MUST verify that all of the correct fields are available by reviewing the table on page 92.*

You can enter or modify all of your environmental and QA/QC Sample List parameters using the Sample List Wizard. You can still edit directly from the Sample List.

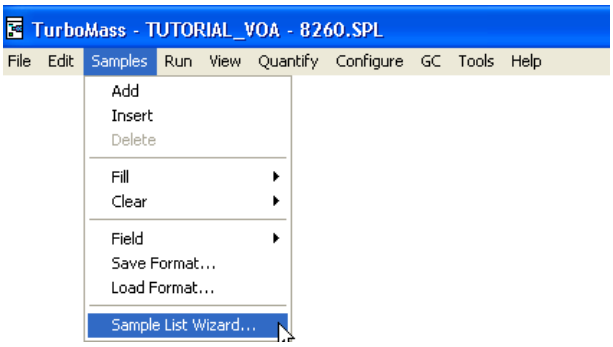
The recommended approach to using the Sample List Wizard is to create your original "template" Sample List so that all the per-sample information is correctly filled in. When you run later sets of samples which follow a similar sequence you may then read in the original "template" Sample List, rename it, and make your per-sample changes in the Wizard or Sample List window spreadsheet environment – which ever you find most convenient for your current task

Building a Sample List Using the Wizard

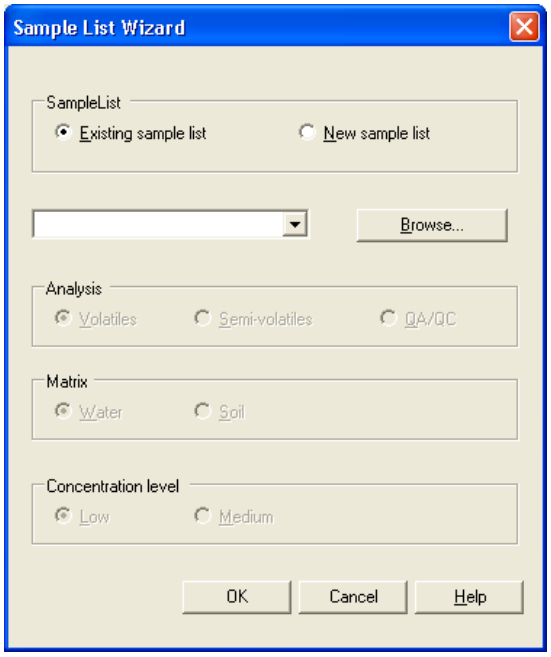
1. You can build a Sample List manually or use the Sample List Wizard. This procedure describes how to use the Sample List Wizard to build your Sample List.

2. In this example we will show how to use the Sample List Wizard to build a Sample List.

3. From the **Samples** menu select **Sample List Wizard...**



The first dialog opens.



You can modify an existing Sample List or create a new Sample List.

4. Select the **Existing Sample List** radio button.

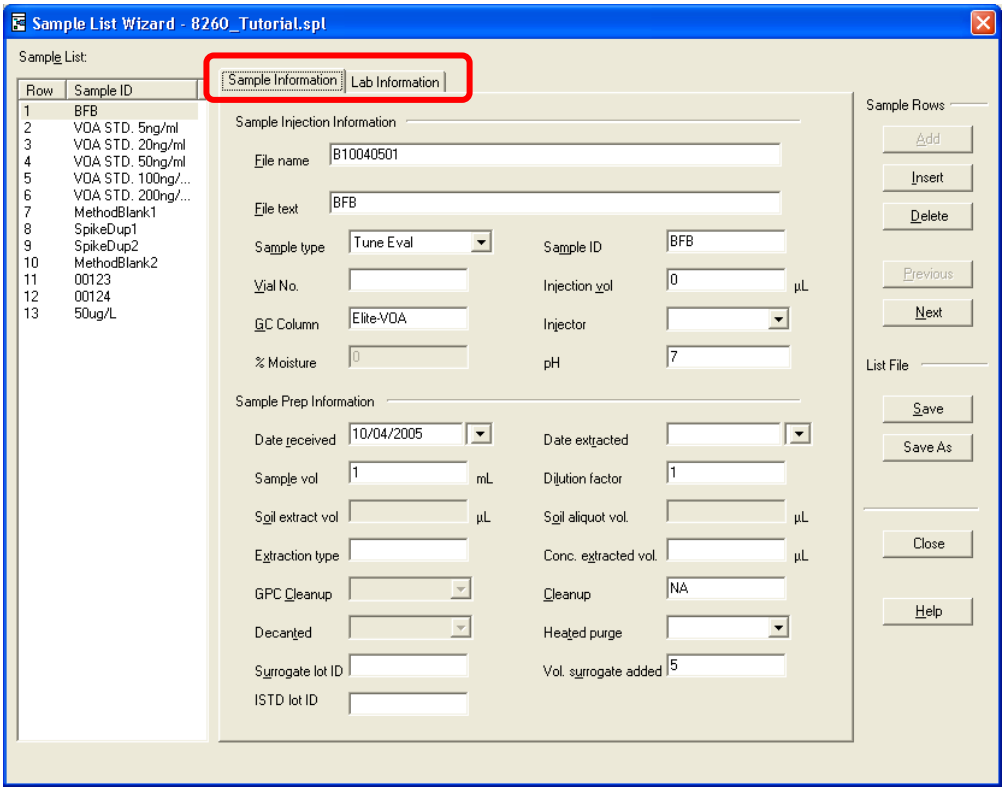
5. Select the type of **Analysis**: **Volatiles**, **Semi-volatiles**, or **QA/QC**.
   For example Volatiles.

6. Select the **Matrix: Water** or **Soil**.
   For example Soil.

7. When **Soil** is selected you must also select a Concentration level of either **Low** or **Medium**.

8. Click **OK**.

   The next dialog appears and allows you to enter the **Sample Information** and **Lab Information** for each of the Samples.



Adding Samples to the Sample List

The Sample List window is a listing of the **Row** numbers and **Sample ID**s for the samples in the current Sample List. As long at least one sample exists in the list, a sample is always selected. The currently selected sample is indicated with a highlight. You can change the selected row using the up and down arrow keys or with the mouse. If the **Ctrl** key is held down, the up arrow will move the currently selected sample row up one position in the list. Similarly, with the Ctrl key held down, the down arrow will move the currently selected sample row down one position in the list.

- Enter the **Sample Injection** Information.

  This window contains many fields required for environmental and QA/QC data. Not all fields need to be entered since they are not used for calculations but only displayed on the reports. They can be ignored or added later on a LIMS system.

Sample Injection Information

File name – Enter the raw data file name.

File text – Enter additional comments about the sample.

Sample Type – Select the type of sample from a drop-down box. The following Sample Types are displayed in the order in which they are to appear in the drop–down list:

- **Analyte** – Analytical sample with unknown concentrations of target compounds.
- **Blank** – An analytical blank. For environmental and QA/QC, use Meth Blank instead.
- **QC** – Quality Control sample.
- **Standard Concentration calibration standard** – For environmental and QA/QC, use Init Calib or Cont Calib instead.
- **Analyte Dup** – Reinjection of an Analyte (Duplicate).
- **Tune Eval** –DFTPP or BFB tuning check.
- **Init Calib** – Initial calibration standard (e.g., one level of a 5-level calibration).
- **Cont Calib** – Continuing Calibration standard (injected periodically to validate the initial calibration curve).
- **Meth Blank** – Analytical Method Blank. Contains all Internal Standards and Surrogates.
- **Lab Control** – Laboratory Control Sample (LCS), typically a Cont Calib prepared from a different stock solution to validate the Init Calib and Cont Calib concentrations.
- **Spike** – Matrix Spike sample.
- **Spike Dup** – Matrix Spike Duplicate sample.
- **Dilution** – Dilution of an Analyte. Sample List "Conc" values for internal standards and Surrogates will need to be adjusted if they are diluted.
- **Re-Extract** – Re-extracted sample.

Sample ID – Enter this primary sample descriptor for the environmental reporting software. This descriptor appears in the Sample List column in this window.

Vial No. – The sample position in the autosampler from which the injection will be made.

Injection vol – The amount of sample injected into the GC. **This value does not control the injection volume (that value is in the GC Method) but it is used in calculations.**

GC column – Enter information about the GC column used for the analysis. It is enabled if a GC is configured on the system.

Injector – Select from the drop-down list injection port (A or B) into which the sample will be injected (this controls the Clarus GC autosampler).

% Moisture – Enter the moisture content determined for a soil sample. This field is enabled only if the Matrix is Soil.

pH – Enter the pH of the sample.

Sample Prep Information

Date received – Date the sample was received in the lab. The date is displayed in the short format defined in the Windows Regional settings. Clicking the down arrow command button displays a calendar control, to enable you to select the Date received.

Dated extracted – Date the sample was extracted for analysis. The date will be displayed in the short format defined in the Windows Regional settings.

- Clicking the down arrow command button displays a calendar control, to enable you to select the Date extracted.

Sample wt (Sample vol) – The weight or volume (depending on sample matrix) taken for analysis. This field is enabled if the Analysis type is Volatiles or Semi-volatiles.

- If the Matrix setting is Soil the caption is Sample wt and the units displayed are grams (g).
- If the Matrix setting is Water the caption is Sample vol the units displayed are milliters (mL).

Dilution factor – Dilution factor applied to the sample. (Undiluted is 1)

Soil extract vol – The total volume of the methanol extract. Enabled only if the Analysis type is Volatiles, the Matrix is Soil and the Concentration level is Medium.

Soil aliquot vol – Volume of the aliquot of the sample methanol extract. Enabled only if the Analysis type is Volatiles, the Matrix is Soil and the Concentration level is Medium.

Extraction type – Text field that indicates how the sample was extracted.

Conc. extract vol. – The concentrated extract volume. Enabled if the Analysis type is Semi-volatiles.

GPC Cleanup – A drop-down box that indicates whether or not the sample was subject to a GPC (gel permeation chromatography) cleanup procedure. Enabled if the Analysis type is Semi-volatiles.

Cleanup – A text field that describes any (non-GPC) cleanup procedure used. Enabled if the Analysis type is Volatiles or Semi-volatiles.

VOA Moisture Calc – When this option is set to **Yes**, then when Volatile Soil Samples (Medium Level Concentration) are calculated, an additional calculation is performed to account for any possible dilution occurring on the extract.

The following parameters are informational only used for reports:

Decanted – A drop-down list that indicates whether or not the sample was decanted. Enabled only if the Analysis type is Semi-volatiles and Matrix is Soil.

Heated purge – A drop-down box that indicates whether or not a heated purge was used. Enabled if the Analysis type is Volatiles

Surrogate lot ID. – A text field allowing identification of the lot number of the surrogate standard compounds mix. This field is enabled if the Analysis type is Volatiles or Semi-volatiles.

Vol surrogate added – An edit box that indicates the volume of surrogate standard added to the sample. This field is enabled if the Analysis type is Volatiles or Semi-volatiles.

ISTD lot no. – A text field allowing identification of the lot number of the internal standard compounds mix. It is Enabled if the Analysis type is Volatiles or Semi-volatiles. All concentration calculations for target compounds are made relative to an internal standard.

Lab Information

- Click the **Lab Information** tab and enter the Lab Information for this sample.

  Lab information displays for each Sample ID in the Sample List. These values typically do not change on a per-sample basis, so have been put in this second tab to simplify the user interface.

Concentrations Display – The concentrations grid (with a horizontal scroll bar) contains cells for you to enter standard Concentrations A to T (identical with the Conc A to T values shown in the Sample List spreadsheet display). These concentrations levels are defined for the standard compounds in the Quantify method associated with your samples.

Sample Tracking

Submitter – Drop–down list of the Submitters previously entered in the Submitter/Task Data window.

Task – Drop–down list of the Tasks previously entered in Submitter/Task Data window. Optionally defines the list of compounds to be reported, reporting limits, and printouts (report methods).

Job – Text field describing the job the sample is associated with.

Contract – Text field describing the contract under which the sample is being analyzed.

EPA sample No. – Text field defining the EPA sample number.

Case No. – Text field defining the case number.

SAS No. – Text field defining the Special Analytical Services (SAS) number.

SDG No. – Text field defining the Sample Delivery Group (SDG) number.

Lab Code –Text field containing the laboratory code.

Files

The following files must be generated in TurboMass before you can enter them in your Sample List.

GC Method – A drop-down list enabling you to select the GC Method to be used for the analysis.

MS Method – A drop-own list enabling you to select the MS Method to be used for the analysis.

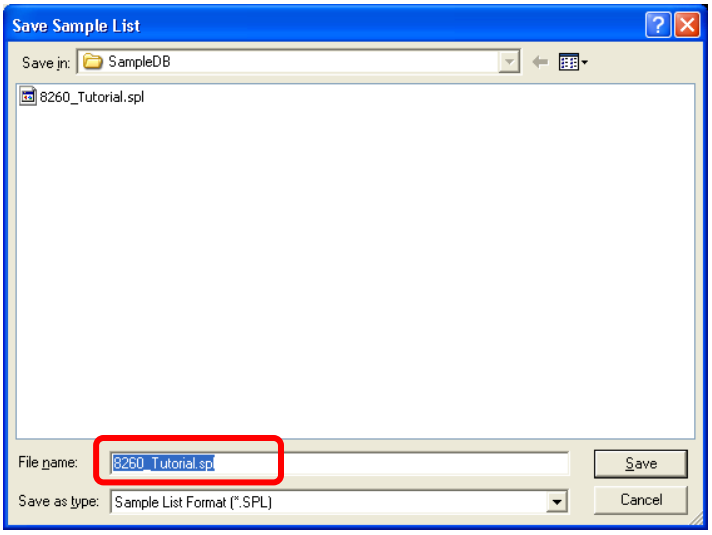
Quantify Method – A drop-down list enabling you to select the Quantify Method to be used for the analysis.

Calibration Curve – A drop-down list enabling you to select the calibration file to be used for the forms and required calculations. See Chapter 6.

Qualitative Method – A drop-down list enabling you to select the Qualitative Method (typically the TIC library search) to be used for the analysis. This is required for Form 1 TIC.

1. Click the **Save** button.
   The **Save Sample List** dialog appears.

2. Type a name for this Sample List and make sure that it is saved in the correct Project directory. The *.spl extension is added to the name and the file is stored in the Project directory.



Note: *After making a change in the Sample List Wizard you must save the changed file, then reopen it even if the file is currently open in the Sample List.*

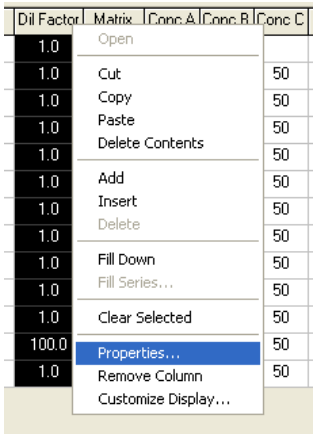
Note: *The Sample List is saved with the .RAW data file when data is acquired. If you change the Sample ID field in the Sample List and try to print Environmental Reports, the Sample ID reverts back to the original setting when the files were acquired. If you make a change to the sample ID field this must be done before the acquisition starts. The sample ID is required to change at the same time the file name change changes.*

Editing Decimal Places in the Sample List

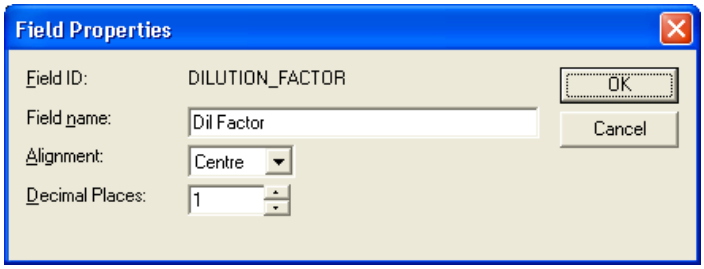
The Sample List Wizard handles decimal places differently from the Sample List. For example, if you enter values with two decimal places in the Sample List Wizard, you may see that value displayed in the Sample List with one decimal place. You can change the Sample List to display decimal places as follows.

1. Click on a **Sample List** heading (for example, Dil Factor).

2. Right-click the mouse and choose **Properties** from the menu.



The Field Properties dialog appears:



3. Set your **Decimal Places** and click the **OK** button.

6 How to Process Quantification Results



Processing Quantification Results

After acquiring your data and creating your quantification method, your next step is to process and view the results. This chapter shows how to process results, view calibration curves, and print reports.

The following steps summarize the procedure for processing quantification results.

1. Highlight the samples you wish to process in the Sample List.
2. Select **Process Samples** from the Quantify menu.
3. Verify the proper Quantify Method file.
4. Create or select the calibration curve file.
5. Verify the samples you wish to process.
6. Run **Quantify**.

Note: *This may take a few minutes. Processing is dependent on integration parameters, number of target compounds, and the number of samples processed.*

7. Select **View Results** from the **Quantify** menu.
   The Quantify window appears and the View Results window also displays on the bottom of the screen.
8. Select **View** from the **Display** menu to set the way you want your results displayed.
9. In the Summary dialog, double-click on a component.
   The Edit Quantify Peak dialog appears along with the chromatogram, spectra, and multiple ion ratio windows.
10. At this point you may want to manually integrate the peak. Use the little black boxes displayed on either side of the peak near the base.

Note: *For more information refer to the Manual Peak Integration topic in the TurboMass Help file or in the TurboMass Software User's Guide.*

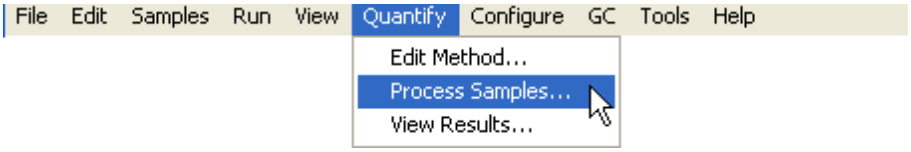
11. Click **OK** to accept the integrated peak.
12. View the integrated peak.
    The concentration shown is related to the processed curve and quantify method.

Note: *All environmental parameters entered into the Sample List will be taken into account in the final environmental report. The results displayed in the final environmental reports may be different than the results you see in this raw quantify display and review. This quantify results view shows the On-column amount values based on the processed curve. For example, you will not observe the dilution factor change until you process your results in the environmental reports environment.*

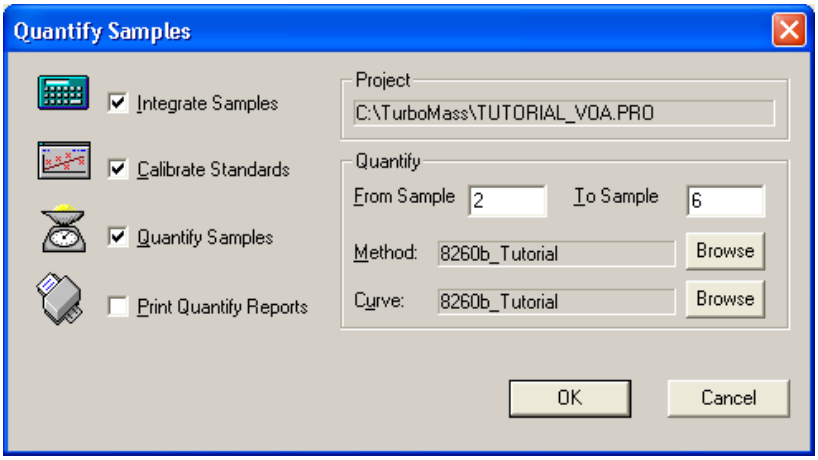
Processing Samples

Note: *Checking **Integrate Samples** only will perform peak detection but will not actually integrate the peak and produce areas. **Quantify Samples** must also be checked for the peak to produce areas.*

1. From the **Quantify** menu select **Process Samples**.

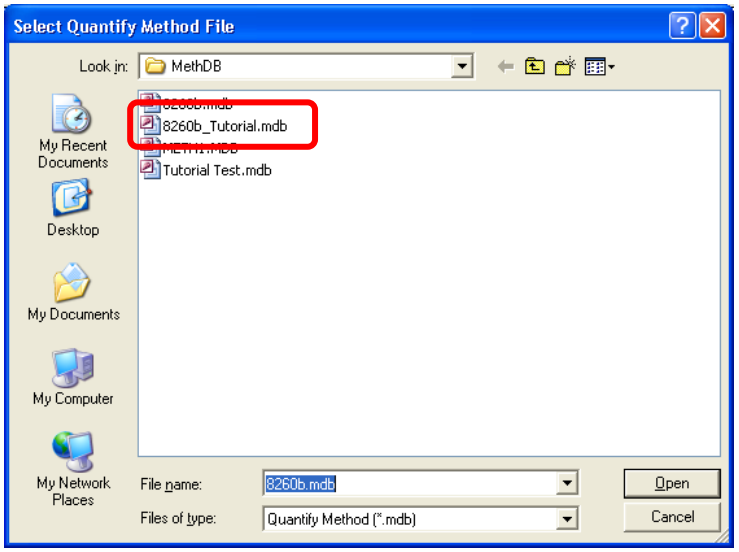


The **Quantify Samples** dialog is displayed.



2. Make sure that the proper boxes are checked, that you are quantifying the proper samples (in this example, **From Sample 2 To Sample 6**), and that you are quantifying with the proper **Method** and calibration **Curve**.

3. Verify the proper Quantify Method file by clicking **Browse** next to the Method text box.
   The **Select Quantify Method** File dialog is displayed.

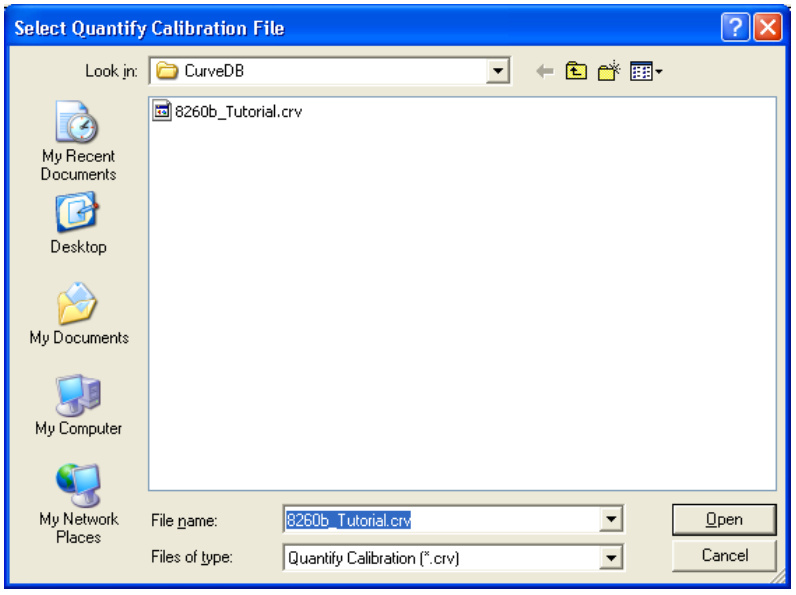


The **Quantify Method** file in this example is **8260b_Tutorial.mdb**.

4. If necessary, select **8260b_Tutorial.mdb** and click **Open**.

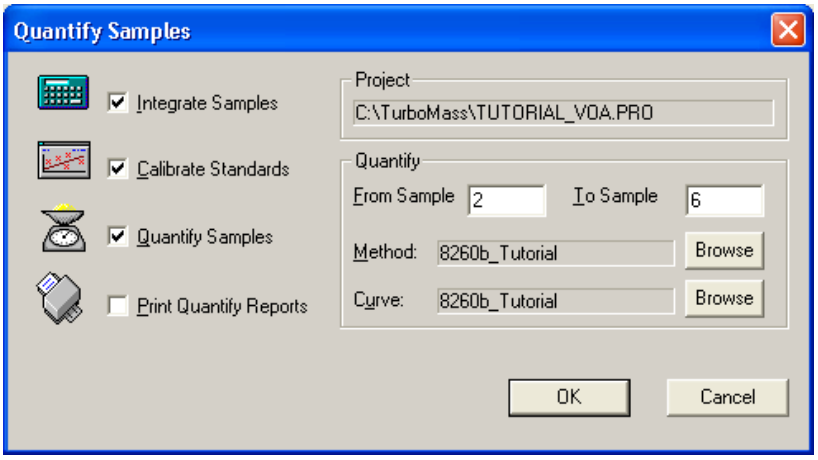
Calibrate the method by creating or using an existing Curve file as follows:

1. Click **Browse** next to the **Curve** text box.
   The **Select Quantify Calibration File** dialog is displayed.

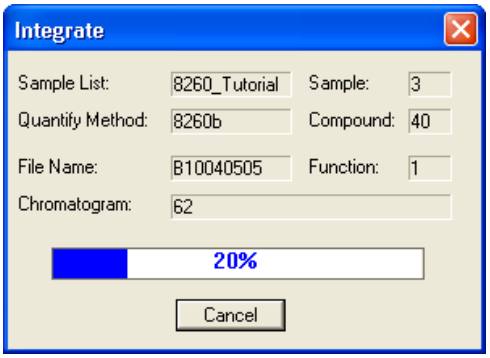


2. Enter a unique **File name**.
   In this example a unique file name is 8260b_tutorial.crv.

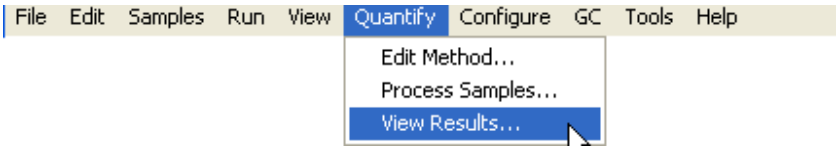
3. Click **Open**.
   The Quantify Samples dialog is displayed.



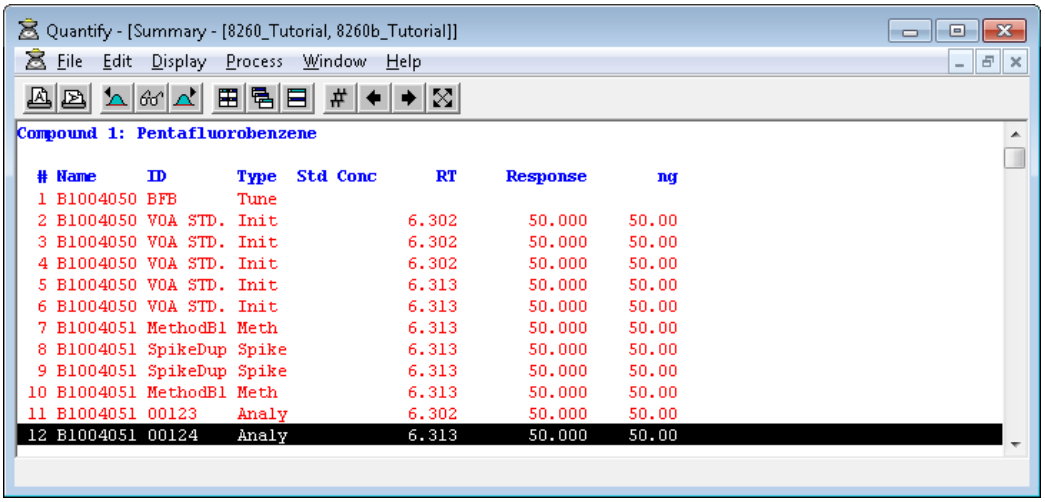
4. Click **OK** to quantify.
   The following dialog is displayed as the quantification process progresses.



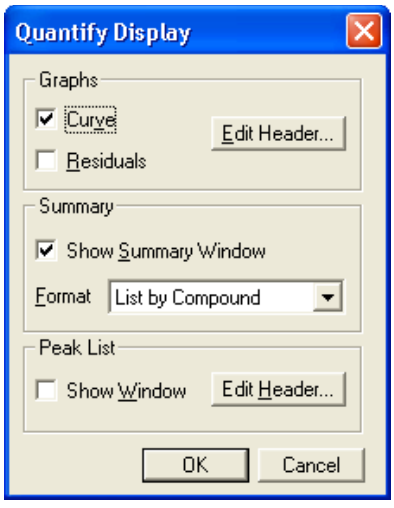
5. Select **View Results** from the **Quantify** menu (or open Quantify Results that is minimized in the bottom window).



The following **Quantify** window is displayed.



6. To set the way results are displayed, select **View** from the **Display** menu. The **Quantify Display** dialog is displayed.



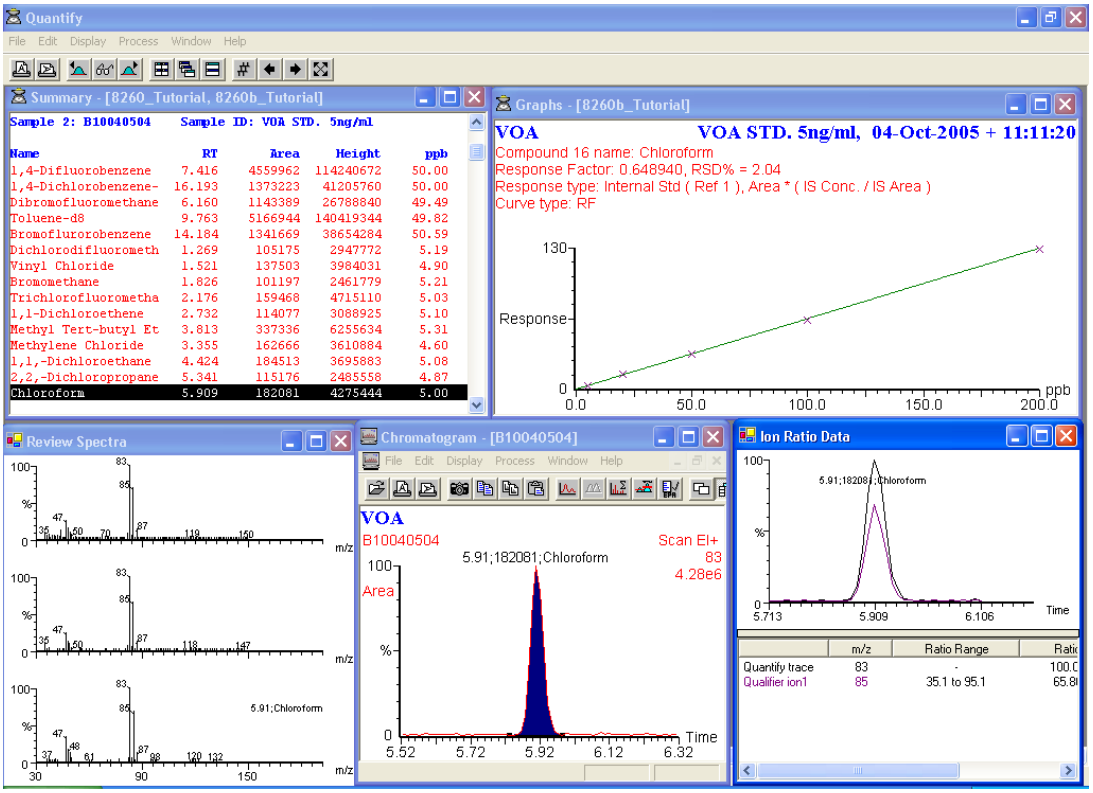
This dialog enables you to set the way you want your results displayed. You can select **List by Compound** to show the results for all chromatograms in the *Sample List* for that compound, or *List by Sample* to show all of the compounds for that sample.

This example shows that **List by Compound** was selected.

7. View the integrated peak by double-clicking on the compound name in the *Summary List* or double-clicking on the plotted point in the graph.

   You can interactively review the data using a view similar to the following:

Note: *The different windows appear and you will have to arrange them for your particular need as shown below. You need to shrink the chromatogram window to fit.*



If the peak isn't scaled properly, click  . You can also drag the small black boxes at the base of the peak to perform manual integration of that peak.

8. To view data for the next (or previous) peak click the toolbar button:  (or ). To view data for the same peak but in the next sample, click on the arrow buttons.

Manual Peak Integration

If the automated peak detection is not determining peak baselines satisfactorily, it is possible to define the baselines manually. This can be achieved by modifying the peak information held in the Peak Lists or by creating them from scratch.

1. To display an integrated peak in Chromatogram, select the desired entry in the Summary window or the Peak List window entry.
2. To display calibration standard peaks, double-click on the desired calibration point in the Calibration Curve window.
   TurboMass opens the Chromatogram window with the relevant peak displayed. Also, the Edit Quantify Peak dialog box is automatically opened that displays detailed peak information and from which you can manually adjust the baseline.
3. To modify the peak baseline, select the handles that appear at either side of the baseline, and adjust the baseline as required. TurboMass will update the Peak Information displayed in the Edit Quantify Peak dialog box.

4. When you are satisfied with the manual integration, choose OK to save the new peak integration information.

5. Optionally, add a comment to be stored in the peak list for the selected peak. The comment can be included in the printed report.

6. If no peak was detected, the chromatogram, which should have contained the peak, can be displayed by using the mouse to select the appropriate Summary window entry.

7. To add a baseline, right-click at one end of the chromatogram region of interest, and drag the mouse horizontally to the other end. As you drag the mouse TurboMass indicates the selected range. When you release the mouse, a baseline will be drawn.

8. To delete the current peak, choose **Delete** and then **OK** in the **Edit Quantify** dialog box.
   The peak list and associated windows will be updated. If the peak is a calibration standard, you will be asked if you want to recalculate the calibration curve. If a new curve is calculated, all compounds will be re-quantified.

The Summary window can be formatted to include the Detection Flags for each peak. The Detection Flags give information about the start and end points of the peak and can have the following values:

b peak baseline starts/ends on chromatogram curve.

v peak baseline starts/end as valley dropline between two peaks.

s peak baseline starts/ends as shoulder dropline between two peaks.

d peak starts or ends on a dropline.

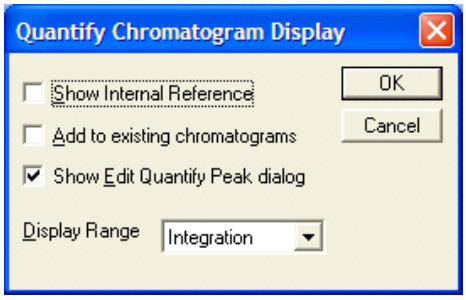
M peak start or end point has been manually assigned.

X calibration point has been excluded from calibration curve.

To modify peak data:

1. Double-click on the compound name in the Summary window you wish to manually integrate.

2. If the **Edit Quantify Peak** dialog is not displayed, select Chromatogram from the Display menu in Quantify and make sure the Show Edit Quantify Peak dialog is selected. Then try step 1 again.



3. Perform manual reintegration, if necessary.

4. Click **OK** to accept the changes made and have the results display updated.
   At this point your curve is processed. Process and review the rest of the samples in the list.

7 How to Generate Environmental Reports



About Environmental Reports

TurboMass Environmental Reporting software generates reports based on lists of PerkinElmer TurboMass and Clarus GC/MS samples (Sample List). These reports, while based on U.S. Environmental Protection Agency (US EPA) requirements, are designed to be flexible and customizable to support worldwide environmental and QA/QC reporting requirements governed by NELAC and ISO 17025.

You can add samples to the list or create new lists using either the Sample List or Sample List Wizard, as described earlier in this tutorial. While the Sample List is typically created prior to data acquisition, it can be created and edited any time prior to environmental report generation.

In the Report Generation window (accessed from the Sample List window by selecting **Tools > Environmental Reports...**) you can select the Forms you want printed and the data set to be used as input. Also, the selection of Tentatively Identified Compounds (TIC) from the initial library search qualitative processing is made within this environment.

IMPORTANT: *If you change the quantitative results at the report generation time (for example, you select a different method) you must reprocess, otherwise the qualifier flag assignments may be invalid.*

The exact format of the report you generate and the calculations that occur depends upon several selections you make within the Sample List or Sample List Wizard environments. These include the following:

- Sample Type
- Volatiles (VOA), Semi-volatiles (SV), or QA/QC
- Matrix: Water or Soil
- Concentration Level (Volatile or Semi-Volatile)

PerkinElmer provides two sets of reports, one based on the US EPA OLM04.2 specification ("CLP-Like), and the other (the default) a format enhanced with additional laboratory information and with a more attractive and convenient format (PKI format). As described in an earlier chapter, you can choose which format to print on a Form by form basis using the Submitter/Task dialog.

Environmental Reports

Using the Environmental Report Window you can generate the following Environmental Reports:

Form 1 – 8 Types Table

Format	Fraction	Matrix	Report Method *.rme	Contains Template*.tpl
CLP-Like Style	Voa	Water/Soil	Form1_Voa	Form1AB
	SVoa	Water/Soil	Form1_SV	Form1CD
	Voa	Water/Soil	Form1TIC_Voa	Form1F
	SVoa	Water/Soil	Form1TIC_SV	Form1G
	Voa	Water	Form2_Voa_water	Form2A
	Voa	Soil	Form2_Voa_soil	Form2B
	SVoa	Water	Form2_SV_water	Form2C
	SVoa	Soil	Form2_SV_soil	Form2D
	Voa	Water	Form3_Voa_Water	Form3A
	Voa	Soil	Form3_Voa_Soil	Form3B
	Svoa	Water	Form3_SV_Water	Form3C
	Svoa	Soil	Form3_SV_Soil	Form3D
	Voa	Water/Soil	Form4_Voa	Form4A
	Svoa	Water/Soil	Form4_SV	Form4B
	Voa	Water/Soil	Form5_Voa	Form5A
	Svoa	Water/Soil	Form5_SV	Form5B
	Voa	Water/Soil	Form6_Voa	Form6AB
	Svoa	Water/Soil	Form6_SV	Form6CD
	Voa	Water/Soil	Form7_Voa	Form7AB
	Svoa	Water/Soil	Form7_SV	Form7CD
	Voa	Water/Soil	Form8_Voa	Form8Voa
	Svoa	Water/Soil	Form8_SV	Form8SV
PKI Format	Voa	Water/Soil	PKI1_Voa	PKI1AB
	Svoa	Water/Soil	PKI1_SV	PKI1CD
	Voa	Water/Soil	PKI1TIC_Voa	PKI1F
	Svoa	Water/Soil	PKI1TIC_Svoa	PKI1G
	Voa	Water	PKI2_Voa_water	PKI2A
	Voa	Soil	PKI2_Voa_soil	PKI2B
	Svoa	Water	PKI2_SV_water	PKI2C
	Svoa	Soil	PKI2_SV_soil	PKI2D
	Voa	Water	PKI3_Voa_water	PKI3A_8000
	Voa	Soil	PKI3_Voa_soil	PKI3B_8000
	Svoa	Water	PKI3_SV_water	PKI3C_8000
	SVoa	Soil	PKI3_SV_soil	PKI3D_8000
	Voa	Water/Soil	PKI4_Voa	PKI4A
	Svoa	Water/Soil	PKI4_SV	PKI4B
	Voa	Water/Soil	PKI5_Voa	PKI5A
	Svoa	Water/Soil	PKI5_SV	PKI5B
	Voa	Water/Soil	PKI6_Voa	PKI6ABL
	Svoa	Water/Soil	PKI6_SV	PKI6CDL
	Voa	Water/Soil	PKI7_Voa	PKI7AB
	Svoa	Water/Soil	PKI7_SV	PKI7CD
	Voa	Water/Soil	PKI8_Voa	PKI8VOA
	SVoa	Water/Soil	PKI8_SV	PKI8SV

Note**: Sample reports for Forms 1–8 for the volatile and semi-volatile fractions are located in the following directories: **C:\TurboMass\Tutorial_VOA.pro\Data** and C:**TurboMass\Tutorial_SVOA.pro\Data.

Form 1–8 Descriptions

Note: *Form numbers referenced in this section include both PKI style format and the CLP-Like format. This section provides a general description of each form type.*

- **Form 1 types – Organics Data Analysis Sheet**: Form 1 reports the concentrations of target compounds in the sample.

- **Form 1 TIC types – Organics Data Analysis Sheet, Tentatively Identified Compounds**: Form 1 TIC reports show library search results for the largest (typically 10 to 30) non-target peaks in the chromatogram. Each is assigned an estimated concentration based upon the response of the Total Ion Chromatogram Area of the nearest internal standard compound.

- **Form 2 types – System Monitoring Compounds (surrogates):** Form 2 reports the recoveries of the specific analytes defined as System Monitoring compounds/Surrogates. It flags any compound whose recovery is outside of the acceptable limits.

- **Form 3 types – Matrix Spike / Matrix Spike Duplicate Recovery**: Form 3 reports the recoveries of specific analytes defined as spike compounds in the sample; the matrix spike (MS), and the matrix spike duplicate (MSD). It flags any compound whose recovery or relative percent difference (RPD is outside of the acceptable limits.

- **Form 4 types – Method Blank Summary Report**: Form 4 designates which samples were acquired, associated with a particular method blank.

- **Form 5 types Tune – Evaluation**: Form 5 reports the compliance of the tune evaluation sample (instrument performance check), and lists all data files acquired while this tune file was valid.

- **Form 6 types – Initial Calibration Data**: Form 6 reports the initial calibration of the instrument based on a multi-point calibration. The average relative response factor (RRF) and its relative standard deviation (RSD) are reported for all target compounds. It flags any compound whose response factors or relative standard deviations are outside of the acceptable limits.

- **Form 7 types – Continuing Calibration Check**: Form 7 compares the daily calibration (continuing calibration) data against the initial calibration data of the instrument. Compounds whose response factors fall below a minimum value are flagged. It flags any compound whose percent difference or percent drift from the initial calibration is outside of the criteria set in the quantify method.

- **Form 8 types – Internal Standard Area and Retention Time Summary Report**: Form 8 reports the retention time and area reproducibility for the internal standards for all data files in the group of samples; it flags any internal standard area counts that are outside of the established area range based on the selected mid-point calibration analysis.

Additional Environmental Form Types
The table below shows additional environmental forms that are available to you:

Fraction	Matrix	Report Method *.rme	Contains Template*.tpl
Voa/Svoa	Water/Soil	PKIEnvQuant	PKIEnvQuant
Voa	Water/Soil	LCS_VOA	LCS_VOA
Svoa	Water/Soil	LCS_SV	LCS_SV
Voa	Water/Soil	ICV_VOA	ICV_VOA
Svoa	Water/Soil	ICV_SV	ICV_SV
Voa/Svoa	Water/Soil		FORM1_CSV
Voa/Svoa	Water/Soil		FORM5_CSV
Voa/Svoa	Water/Soil		FORM6_CSV
Voa/Svoa	Water/Soil		SpectrumQuant Report_Env

Note: *For the environmental forms with only a template listed, you must create a report method using the template you wish to use.*

Following is a description of the additional environmental forms available to you:

- **PKIEnvQuant – Organics Quantitation Report:** Reports the concentrations of target compounds and their reporting limits in the sample, ISTD amounts and Surrogate recoveries.
- **LCS_Voa – Laboratory Control Sample Recovery Report:** Reports the recoveries in a Laboratory Control Sample (LCS) for the volatile fraction.
- **LCS_SV – Laboratory Control Sample Recovery Report:** Reports the recoveries in a Laboratory Control Sample (LCS) for the semi-volatile fraction.
- **ICV_VOA – Initial Calibration Verification Report (for volatile fraction):** This report compares the initial calibration verification (identified as continuing calibration in the sample list) against the initial calibration data of the instrument. Compounds whose response factors fall below a minimum value are flagged. It flags any compound whose % difference or % drift from the initial calibration is outside of the criteria set in the quantify method.
- **ICV-SV – Initial Calibration Verification Report (for semi-volatile fraction):** This report compares the initial calibration verification (identified as continuing calibration in the sample list) against the initial calibration data of the instrument. Compounds whose response factors fall below a minimum value are flagged. It flags any compound whose % difference or % drift from the initial calibration is outside of the criteria set in the quantify method.
- **Form1_CSV – LIMS report for Targets/TICs in CSV format:** Reports the concentrations of Targets and Tentatively identified compounds (TICs) in a comma-separate format file suitable for transferring data to LABWORKS LIMS.
- **Form5_CSV – Tune Evaluation report in CSV format:** Reports the compliance of the tune evaluation sample (instrument performance check), and lists all data files acquired while this tune file was valid in a comma-separated format (CSV).
- **Form6_CSV – Initial calibration report in CSV format:** Reports the initial calibration of the instrument based on a multi-point calibration. The average relative response factor (RRF) and its relative standard deviation (RSD) are reported for all target compounds.
- **SpectrumQuant Report_Env – Spectrum Quantitation report:** Reports positive target compounds displaying the raw spectrum, background subtracted spectrum and reference spectrum. Quantitated results are displayed along with ion ratio's and the criteria.

Before Generating Environmental Reports

This section provides you with information on what tasks need to be performed prior to generating an Environmental Report.

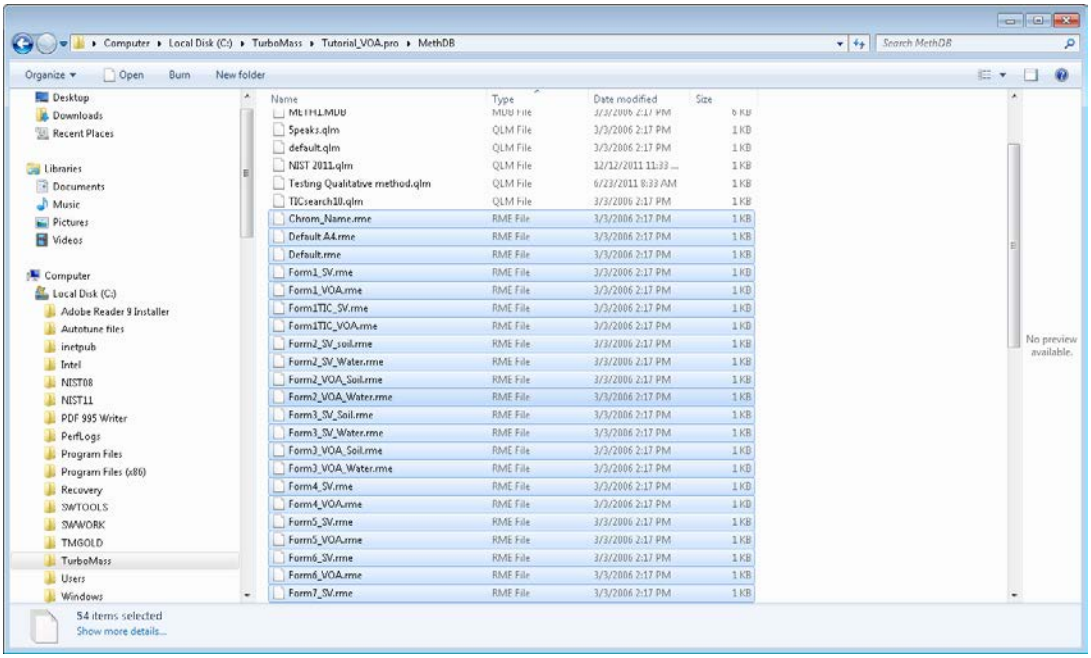
Copy and Paste *.RME files to your new Project

The Environmental forms that ship with TurboMass are not automatically placed into a new or existing project. Initially, the Environmental forms can be found in the VOA_tutorial and the SVOA_tutorial. Therefore, after creating your own project you must copy the *.RME files from the Tutorial_VOA.pro and Tutorial_SVOA.pro folders and paste them into your project\MethodDB directory.

To copy and paste *.rme files to your project directory:

1. Browse to the following directories: C:\TurboMass\Tutorial_VOA.pro\MethDB\
   C:\TurboMass\Tutorial_SVOA.pro\MethDB\

2. Highlight all of the *.rme files and copy and paste them to your project\MethDB\ directory.
   The screen shot below shows all of the *.rme files selected from the **C:\TurboMass\Tutorial_VOA\MethodDB** directory so that these files can then be copied into a new TurboMass project\MethodDB directory.



Copy the Qualitative Database File to your new Project

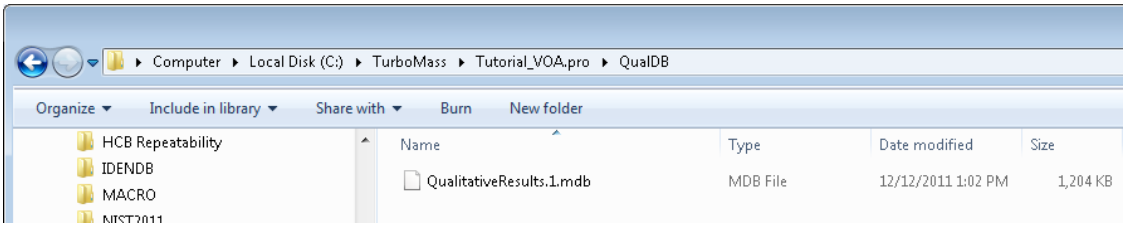
Before generating environmental reports you must copy **the QualitativeResults.1.mdb**, which is located in the following directory: **C:\TurboMass\Tutorial_VOA.pro\QualDB**, and paste it into your current project's QualDB folder. This will allow you to process tentatively identified components within samples and generate a Form 1 TIC report.

To copy and paste the QualitativeResults1.mdb file to your project directory:

1. Browse to the following directory: C:\TurboMass\Tutorial_VOA.pro\QualDB\.

2. Highlight the **QualitativeResults1.mdb** file and copy and paste it into your project\QualDB\ directory.

   The screen shot below shows all of the QualitativeResults1.mdb file selected from the C:]TurboMass\Tutorial_VOA\QualDB directory. Copy and paste this file into you TurboMass project\QualDB directory.



Note: *If you do not copy the QualitativeResults1.mdb file into your project's QualDB folder TurboMass displays the following error message: "**No Qualitative Data Found for some samples."** This message displays within the Environmental Report Window under the Errors/Warnings section when a Form 1 TIC is selected*

About Quantitative and Qualitative Processing

Before you proceed with generating Environmental Forms it is important to understand the purpose of Quantitative Processing and Qualitative Processing. The following section briefly summarizes how Quantitative processing and Qualitative processing can occur.

Quantitative processing can occur in the following ways for Target Compounds:

1. When samples are selected to be quantitatively processed the through the *Quantify* menu when **Process samples** is selected.

2. When the Start Sample List Run is selected from the *Run* Menu and **Start** is selected. ***Autoquantify samples*** must be selected.

3. When the Environmental Reports window is accessed.

Qualitative processing can occur in only the following way for Tentatively Identified Compounds:

- When the Start Sample List Run is selected from the *Run* Menu and **Start** is selected. ***Qualitative calculations*** must be selected.

Steps for Producing Environmental Forms

The process for generating Environmental Forms 1–8 is a multi-step process that involves:

- Step 1 – Generating a Sample List with the required fields
- Step 2 – Creating a Quant Method
- Step 3 – Running Samples, QC, Tune, Initial, Calibration, and Continuing Calibration
- Step 4 – Integrating/Calibrating/Quantifying Initial Calibration Files
- Step 5 – Integrating/Quantifying Sample files
- Step 6 – Qualitatively Processing Samples – For Form 1 Tentatively Identified Compounds only
- Step 7 – Producing Forms 1–8 via the Environmental Forms dialog

This section provides you with step-by step instructions on how to generate Forms 1– 8 for Volatile or Semi-volatile data as well as provides you with critical areas in the software that must be correctly set up for the specific forms.

Step 1 – Generating a Sample List with the Required Fields

The TurboMass software requires a minimum number of Sample List information fields to be filled out in order to produce environmental forms. All forms require the data to be integrated and quantified prior to forms generation.

This section shows you how to load a sample list as well as provides you with information on how to populate your sample list with the information necessary to produce Environmental Forms 1–8.

For in-depth information on the Sample List refer to ***How to Build a Sample List using the Wizard*** *on page* ***63****.*

Load a Sample List

1. From the TurboMass Sample List window select **File > Open Project**...
   The Select Project dialog appears.

2. Select the project you wish to open and click **OK**.

3. From the **Sample** menu select **Load Format**....
   The Load Sample List dialog appears.

4. Open the appropriate Sample List and click **OK**.

5. Refer to the sections below for information on which Sample List fields need to be populated correctly to produce the environmental forms.

Required Sample List Fields for the Following Analysis, Concentration Level, and Matrix Combinations:

		Analysis Type	Concentration Level	Matrix
Volatile Water	=	Voa	'Blank'	Water
Volatile Low Level Soil	=	Voa	Low	Soil
Volatile Medium Level Soil	=	Voa	Med	Soil
Semi-Volatile Water	=	Svoa	Low	Soil
Semi-Volatile Soil	=	Svoa	Low	Soil

Required Field	Voa Water	Voa Low Soil	Voa Med Soil	Svoa Water	Svoa Soil
Analysis	Voa	Voa	Voa	Svoa	Svoa
Matrix	Water	Soil	Soil	Water	Soil
Conc Level	Blank (no text)	Low	Med	Low	Low
Sample Type	Required	Required	Required	Required	Required
Quantify Method	Required	Required	Required	Required	Required
Calibration Curve	Required	Required	Required	Required	Required
Submitter	Required	Required	Required	Required	Required
Task	Required	Required	Required	Required	Required
Conc A-T	Required Units = ng	Required Units = ng	Required Units = ng	Required Units = ng	Required Units = ng
Qualitative Method	Required	Required	Required	Required	Required
Sample Wt/Vol	Required units=mLs	Required units=grams	Required units=grams	Required units=mLs	Required units=grams
Dilution Factor	Required	N/A	Required	Required	Required
% Moisture	N/A	Required Units = percent *	Required Units = percent	N/A	Required Units = percent
Voa Moisture Calc	N/A	N/A	Required Y or N	N/A	N/A
Soil Extract Vol	N/A	N/A	Required Units = uLs	N/A	N/A
Soil Aliquot Vol	N/A	N/A	Required Units = uLs	N/A	N/A
Injection Vol	N/A	N/A	N/A	Required Units = uLs	Required Units = uLs
Conc Ext Vol	N/A	N/A	N/A	Required Units = uLs	Required Units = uLs
GPC Cleanup	N/A	N/A	N/A	Required Y or N	Required Y or N

Moisture * - units are whole percent. Example: A sample that contains 10% moisture by weight, a user would enter in 10 as the value.

Required Sample List Field Definitions

Analysis: The Analysis type must be set to **Volatiles** or **Semi-Volatile**.

Matrix: Must be set to **Water** or **Soil**.

Conc Level: Must be set to **'Blank', Low** or **Medium**

Sample Type: The correct type must be selected for forms to generate.
Valid Environmental sample types are as follows:

- Analyte, Analyte Dup, Tune Eval, Init Calib, Cont Calib, Method Blank, LabControl, Spike, Spike Dup, Dilution and Re-extract.

Note: *Any other sample type that is available are not intended for environmental analysis and should not be used.*

Quantify Method: Must be selected. Refer to *Chapter 2 How to Build a Quantification Method* for information on how to properly set up the Quantify Method.

Calibration Curve: Must be selected, and must be updated with the listed Quantify method using the file(s) that constitute the Initial Calibration.

Submitter/Task: Define a submitter and a task to generate the preferred forms using a defined compound list and defined reporting limit values for both water and soil matrices.

If the submitter and Task fields are left blank, the default reports will generate (PKI format). However, there will be no matrix specific defined reporting limits except for those defined in the Quantify method which accommodates only 1 matrix type per method.

If you wish to print the CLP-Like forms, you can set the **Submitter** field to **Default** and the **Task** field set to **CLP-Like.** Again, there will be no matrix specific defined reporting limits.

*Refer to **Chapter 3 How to Set-Up Submitter/Task Data** for information on how to use the Submitter/Task feature, as well as how to define compound lists and reporting limits.*

Conc A-T: Must be entered as appropriate to the Quantitative method.

- **Concentration fields** in the Sample List must ALL be entered as **ng** (on-column) values for both volatile and semivolatile analysis. This applies to both the Internal Standard and individual target compounds.

Qualitative Method: Must be listed here to produce a Form 1 TIC Report. Refer to ***Form 1TIC – Tentatively Identified Compounds(TICs)*** on page ***109*** for information on correctly setting up Qualitative method parameters to produce a Form 1 TIC.

Sample Wt/Vol: Must be entered for calculations to be correct.

- Water samples must be entered with units in milliliters (mLs)
- Soil samples must be entered with units in grams (g)

Dilution Factor: Must be entered for calculations to be correct.

% Moisture: The moisture content determined for a soil sample must be entered in whole percent value. Example: The sample contained 10% moisture by weight. The value entered into the % Moisture field is 10. This field is only used for soil samples.

VOA Moisture Calc: This field allows the user to apply an additional moisture calculation to account for any dilution that may occur on the extract due to the amount of water content in the sample. This field is only used when the Analysis is VOA, the matrix is Soil, and the concentration level is medium.

- When this option is set to **N** (no), the moisture content of the sample is only used to calculate results based on a dry-weight basis.
- When this option is set to **Y** (yes), the moisture content of the sample is applied to the weight of the sample for dilution effects as well as to calculate the results based on a dry-weight basis.

Note: *Refer to* ***Appendix 1 Environmental Reporting Calculations*** *on page* ***169*** *for detailed calculation information.*

Soil extract vol.: The total volume of the methanol extract must be entered here. This field is only used when the Analysis is Voa, the matrix is soil, and the concentration level is medium.

- This field must be entered in units of microliters (uLs)

Soil aliquot vol.: The volume of the aliquot of the sample methanol extract must be entered here. This field is only used when the Analysis is Voa, the matrix is soil, and the concentration level is medium.

- This field must be entered in units of microliters (uLs)

Injection Vol: This is the amount of sample injected into the GC system. A value must be entered for calculations to be correct. This field is only used when the Analysis is Svoa.

- This field must be entered in units of microliters (uLs)

Conc Ext Vol: This is the volume of the concentrated extract. A value must be entered for calculations to be correct. This field is only used when the Analysis is Svoa.

- This field must be entered in units of microliters (uLs)

GPC Cleanup: The Sample List must indicate whether or not the sample was subject to a GPC (gel permeation chromatography) cleanup procedure. This field is only used when the Analysis is Svoa.

- When this option is set to **N** (no), the default calculation for the analysis/matrix type is applied.
- When this option is set to **Y** (yes), a value of 2 is multiplied to all results and reporting limits.

Note: *Refer to* ***Appendix 1 Environmental Reporting Calculations*** *on page* ***169*** *for detailed calculation information.*

Step 2 – Creating a Quantify Method

Refer to the Chapter 2 How to Build a Quantification Method for Environmental Reporting in the Environmental Tutorial for information on building a Quantify Method.

Note: *Refer to Step 7 of this chapter for information critical to generating correct forms.*

Step 3 – Running Samples, QC, Tune, Initial, Calibration, and Continuing Calibration

Acquire initial data. Refer to the **TurboMass Software Guide** for information on acquiring initial data.

Step 4 – Integrating/Calibrating/Quantifying Initial Calibration Files

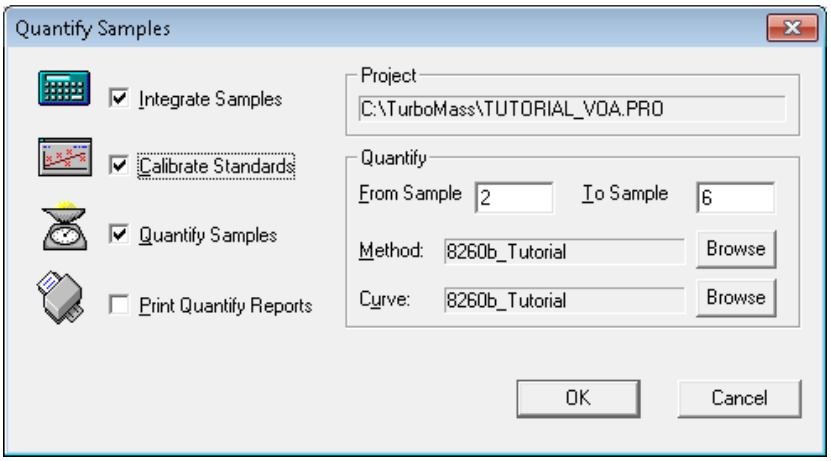
Note: *At least one file must be calibrated with concentrations using the Quant Method and Curve file specified in the currently loaded sample list.*

1. From the TurboMass Sample List view select the Initial Calibration files and then select **Quantify > Process Samples...**

   In the example below the user selects rows 2 – 6, since they are listed as Init Calibration Samples.

	File Name	Quantify Method	Calibration Curve	Sample ID	Analysis	Sample Type	Conc Level	Matrix	Conc A	Conc B	Conc C	Dil Factor	Samp Wt/Vol
1	B10040501	8260b_Tutorial	8260b_Tutorial	BFB	VOA	Tune Eval		Water				1.000	5.0
2	B10040504	8260b_Tutorial	8260b_Tutorial	VOA STD. 5ng/ml	VOA	Init Calib		Water	25	250	250	1.000	5.0
3	B10040505	8260b_Tutorial	8260b_Tutorial	VOA STD. 20ng/ml	VOA	Init Calib		Water	100	250	250	1.000	5.0
4	B10040506	8260b_Tutorial	8260b_Tutorial	VOA STD. 50ng/ml	VOA	Init Calib		Water	250	250	250	1.000	5.0
5	B10040507	8260b_Tutorial	8260b_Tutorial	VOA STD. 100ng/ml	VOA	Init Calib		Water	500	250	250	1.000	5.0
6	B10040509	8260b_Tutorial	8260b_Tutorial	VOA STD. 200ng/ml	VOA	Init Calib		Water	1000	250	250	1.000	5.0
7	B10040510	8260b_Tutorial	8260b_Tutorial	MethodBlank1	VOA	Meth Blank		Water		250	250	1.000	5.0

   The Quantify Samples dialog appears.



2. On the Quantify Samples dialog ensure that **Integrate Samples**, **Calibrate Standards**, and **Quantify Samples** are checked.

Note: ***Integrate Samples*** *should be un-checked if the files were previously integrated and reviewed for valid hits.*

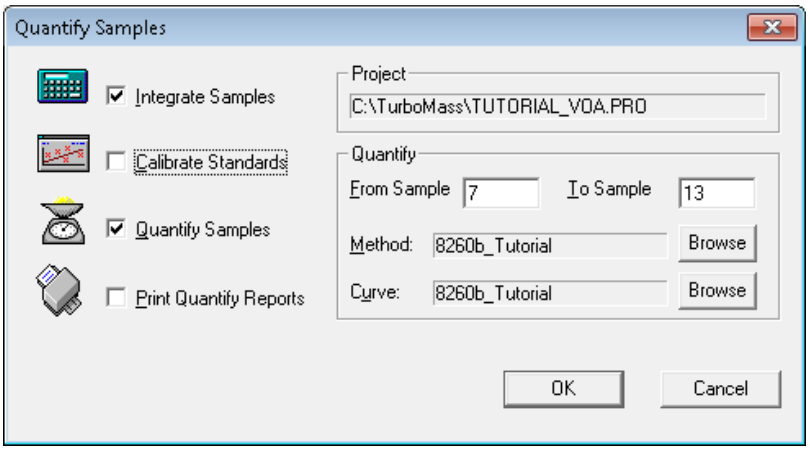
3. Verify the **From** and **To Sample** fields are correct.

4. Verify that the **Method** and **Curve** files are correct.
   If the Method and Curve files are not correct use the Browse button to select a different method/curve.

5. Click **OK**.
   Your Initial Calibration files are processed thereby ensuring that any changes that have been made to the sample list, or to the Quantify method, are fully applied prior to generating a Form.

Step 5 – Integrating/Quantifying Sample files

1. From the TurboMass Sample List view select samples you wish to include on the report and then select **Quantify > Process Samples...**
   The Quantify Samples dialog appears.



2. From the Quantify Samples window, ensure that **Integrate Samples** and **Quantify Samples** are the only options checked.

Note: *The Integrate Samples should be un-checked if the files were previously integrated and reviewed for valid hits.*

Note: *Be very sure to have Calibrate Standards **unselected** when only samples are selected. If **Calibrate Standards** is checked with only samples, the calibration curve will attempt to update and will clear the calibration curve producing incorrect sample results.*

3. Verify the **From** and **To Sample** fields are correct.

4. Verify the **Method and Curve** fields are correct.
   If the Method and Curve files are not correct use the **Browse** button to select a different method/curve.

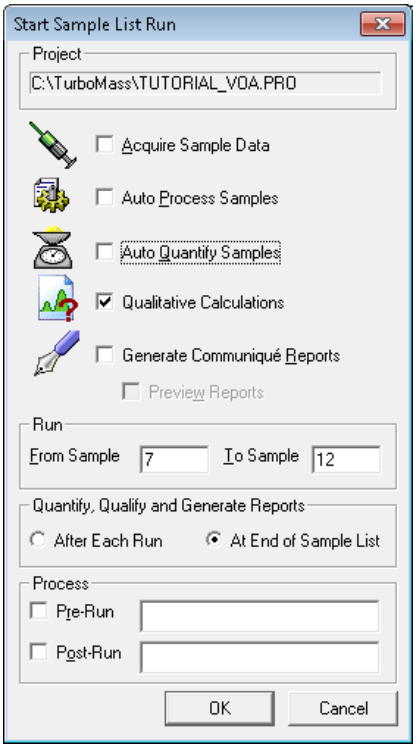
5. Click **OK** to process samples.
   Your samples are processed thereby ensuring that any changes that have been made to the sample list, or to the Quantify method, are fully applied prior to generating environmental reports.

Step 6 – Qualitatively Processing Samples

If you intend to produce a Form 1 Tentatively Identified Compound Report, then you must perform the steps out lined in this section. If you do not intend on generating a Form 1 Tentatively Identified Compound report, then skip to Step 7.

1. From the TurboMass Sample List window select the rows of samples you wish to perform TICs on.
2. From the **Run** menu select **Start** or click on the  the **Start Run** icon.
   The Start Sample List Run window appears.

Note: *When Qualitatively Processing samples the Qualitative method defined in your sample list must be defined.*



3. Check **Qualitative Calculations**.
4. Click **OK**.
   You are now ready to produce Forms 1–8.

Step 7 – Producing Environmental Reports via the Environmental Forms Dialog

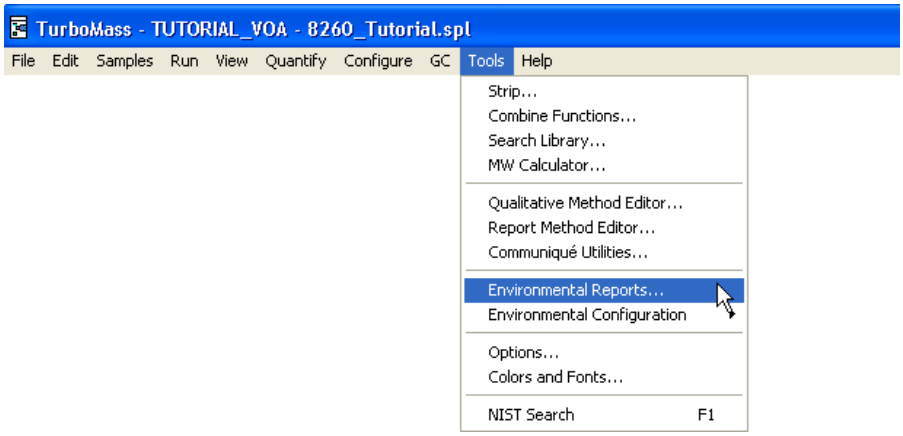
This section provides you with step-by step instructions on generating Environmental Forms 1–8 for Volatile or Semi-volatile data.

Note**: Sample reports for Forms 1–8 for the volatile and semi-volatile fractions are located in the following directories: **C:\TurboMass\Tutorial_VOA.pro\Data** and **C:\TurboMass\Tutorial_SVOA.pro\Data.

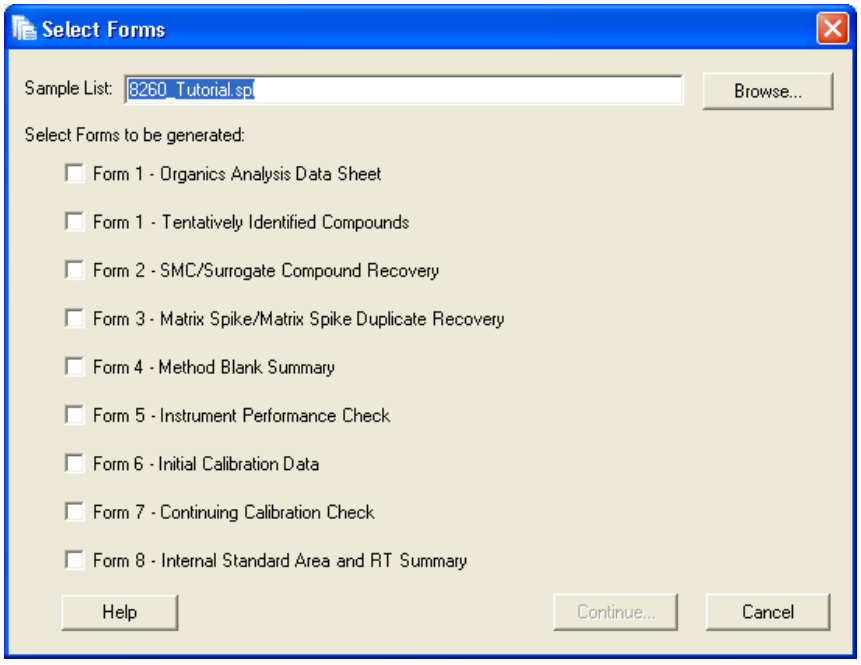
***Note:** Before beginning, make sure the default printer is set up properly or the reports will not print.*

To generate Forms 1–8, follow this procedure:

From the TurboMass Sample List window select **Tools > Environmental Reports...**



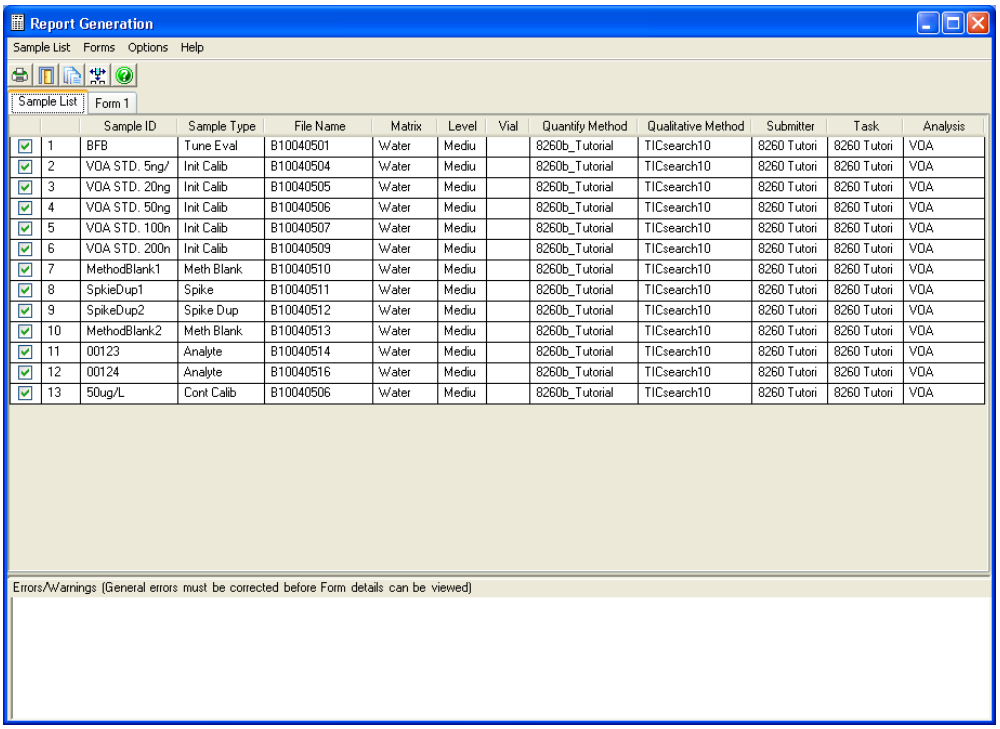
5. The **Select Forms** dialog appears:



***Note**: Make sure any changes made to the Sample List that you are using are saved. If you make changes to the Sample List but do not save the list, the changes will not be used in the generation of the forms.*

6. Accept the displayed Sample List (the one displayed in the TurboMass Sample List window) or select a different one.

7. Select the set of Forms to be generated, and then click the **Continue** button.
The following dialog displays showing the Sample List:



		Sample ID	Sample Type	File Name	Matrix	Level	Vial	Quantify Method	Qualitative Method	Submitter	Task	Analysis
☑	1	BFB	Tune Eval	B10040501	Water	Mediu		8260b_Tutorial	TICsearch10	8260 Tutori	8260 Tutori	VOA
☑	2	VOA STD. 5ng/	Init Calib	B10040504	Water	Mediu		8260b_Tutorial	TICsearch10	8260 Tutori	8260 Tutori	VOA
☑	3	VOA STD. 20ng	Init Calib	B10040505	Water	Mediu		8260b_Tutorial	TICsearch10	8260 Tutori	8260 Tutori	VOA
☑	4	VOA STD. 50ng	Init Calib	B10040506	Water	Mediu		8260b_Tutorial	TICsearch10	8260 Tutori	8260 Tutori	VOA
☑	5	VOA STD. 100n	Init Calib	B10040507	Water	Mediu		8260b_Tutorial	TICsearch10	8260 Tutori	8260 Tutori	VOA
☑	6	VOA STD. 200n	Init Calib	B10040509	Water	Mediu		8260b_Tutorial	TICsearch10	8260 Tutori	8260 Tutori	VOA
☑	7	MethodBlank1	Meth Blank	B10040510	Water	Mediu		8260b_Tutorial	TICsearch10	8260 Tutori	8260 Tutori	VOA
☑	8	SpkieDup1	Spike	B10040511	Water	Mediu		8260b_Tutorial	TICsearch10	8260 Tutori	8260 Tutori	VOA
☑	9	SpikeDup2	Spike Dup	B10040512	Water	Mediu		8260b_Tutorial	TICsearch10	8260 Tutori	8260 Tutori	VOA
☑	10	MethodBlank2	Meth Blank	B10040513	Water	Mediu		8260b_Tutorial	TICsearch10	8260 Tutori	8260 Tutori	VOA
☑	11	00123	Analyte	B10040514	Water	Mediu		8260b_Tutorial	TICsearch10	8260 Tutori	8260 Tutori	VOA
☑	12	00124	Analyte	B10040516	Water	Mediu		8260b_Tutorial	TICsearch10	8260 Tutori	8260 Tutori	VOA
☑	13	50ug/L	Cont Calib	B10040506	Water	Mediu		8260b_Tutorial	TICsearch10	8260 Tutori	8260 Tutori	VOA

8. Review the displayed Sample List for any global errors or form–specific errors identified by the TurboMass software. Refer to ***Appendix 2*** for a list of possible errors.

9. You can right-click in the Report Generation window to display the following menu:

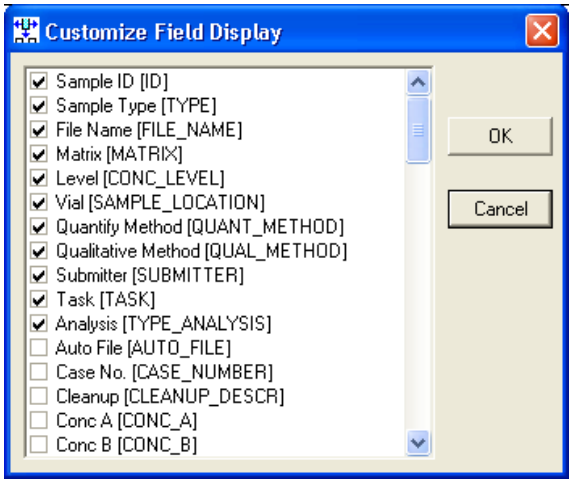


Here you can Select additional forms and De-select or Select All Samples.

10. You can also select **Customize** Display from the **Options** menu.



The following **Customize Field Display** dialog appears:



11. From the Customize Field Display dialog you can add or remove columns to display in the Report Generation window. Click **OK** when finished customizing the display.

Note: *Although the fields are available for display, changes cannot be made to any of the fields.*

12. From the Sample List tab of the Report Generation window select the samples you want to report.

    A check mark appears next to the samples you wish to report.

13. Perform the steps described in the following sections to print Forms 1–8.

Form 1 – Organics Analysis Data Sheet

On a Form 1 report the following information is included:

- The concentrations of target compounds in the sample for volatile or semi-volatile fractions are reported.
- Sample results present below the MDL will display an adjusted reporting limit (derived from sample preparation information) with a "U" flag.
- Sample results equal to or above the MDL but below the reporting limit will display the sample result with a "J" flag.
- Sample results equal to or above the Reporting limit will display the sample result without a flag.

***Note**: Proper MDL and Reporting Limit setup along with detailed calculations are located in **Appendix 1 Environmental Reporting Calculations** on page **169**.*

Below is an example of the PKI style format for a Form 1 report:

Volatile Organics Analysis Data Sheet

Lab Name:	Tutorial	Contract:	CONTRACT_VOA
File Name:	C:\TurboMass\TUTORIAL_VOA.PRO\Data\B10040514.raw		
Sample ID:	00123	Operator:	PerkinElmer
Description:	VOA		
Inject Date/Time:	October 04, 2005 4:47:54 PM	Vial:	0
Instrument:	GC/MS	Tune File:	bbvoa.IPR
GC Method:	AAVOA.mth	MS Method:	BBVOA.EXP
Quantify Method:	8260b_Tutorial_New	Last Updated:	December 18, 2012 4:13:21 PM
Calibration File:	8260b_Tutorial_New	Last Updated:	December 18, 2012 3:58:03 PM
GC Column:	Elite-VOA ID: 180 um		
Matrix:	Water	Date Received:	October 04, 2005
Sample wt/vol:	5.000 mL	Dilution Factor:	1.0

CAS Number	Compound	ppb	Q
75-71-8	Dichlorodifluoromethane	5.00	U
75-01-4	Vinyl Chloride	17.23	
74-83-9	Bromomethane	5.00	U
75-69-4	Trichlorofluoromethane	5.00	U
75-35-4	1,1-Dichloroethene	5.00	U
1634-04-4	Methyl Tert-butyl Ether	5.00	U
75-09-2	Methylene Chloride	5.00	U
75-34-3	1,1,-Dichloroethane	5.05	
594-20-7	2,2,-Dichloropropane	5.00	U
67-66-3	Chloroform	8.73	
71-43-2	Benzene	5.00	U
79-01-6	Trichloroethene	5.00	U
79-95-3	Dibromomethane	5.00	U
107-04-0	1-Bromo-2-chloroethane	5.00	U
108-88-3	Toluene	5.00	U
10061-02-6	trans-1,3-Dichloropropene	5.00	U
79-00-5	1,1,2-Trichloroethane	16.72	
100-41-4	Ethylbenzene	5.00	U
106-42-3/108	p,m-Xylene	5.00	U
95-47-6	o-Xylene	5.00	U
98-82-8	Isopropylbenzene	5.00	U
108-86-1	Bromobenzene	5.00	U
103-65-1	n-Propylbenzene	5.00	U
108-67-8	1,3,5-trimethylbenzene	5.00	U
135-98-8	sec-Butylbenzene	5.00	U
541-73-1	1,3-Dichlorobenzene	5.00	U
106-46-7	1,4-Dichlorobenzene	5.00	U
104-51-8	n-Butylbenzene	5.00	U
95-50-1	1,2-Dichlorobenzene	5.00	U
87-68-3	Hexachlorobutadiene	2.18	BJ

Form 1 Type Report Methods

The following report methods can be used to generate a Form 1 report:

- Form1_Voa.rme
- Form1_SV.rme
- PKI1_Voa.rme
- PKI1_SV.rme

Note: *The Submitter/Task feature can be used to change the report methods selected when generating a Form 1. Refer to* ***Appendix 1 Environmental Reporting Calculations*** *on page* ***225*** *for more information.*

Additional Environmental templates that are acceptable choices and can be produced under similar conditions are as follows:

- **PKIEnvQuant** – for detailed instructions, refer to ***PKIEnvQuant Template Form Generation*** on page ***141.***
- **Form1_CSV –** for detailed instructions, refer to ***Form1_CSV – Organics Analysis Data Sheet*** on page ***150***.
- **SpectrumQuant Report_Env** – for detailed instructions, refer to ***SpectrumQuant Report_Env Template Form Generation*** on page ***159***.

Valid Sample Types for Form 1 Types

A Form 1 report can only be generated using the following valid sample types:

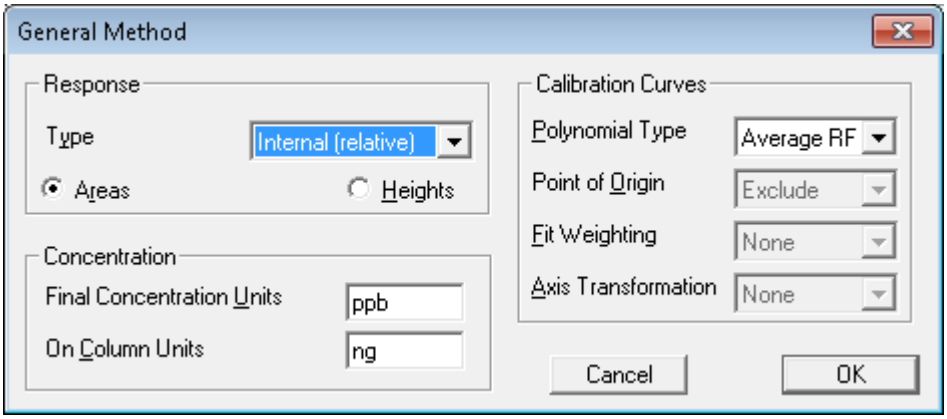
- Analyte
- Analyte Dup
- Method Blank
- Spike
- Spike Dup
- Dilution
- Re-extract
- Lab Control

Critical Quantify Method Information for Form 1 Type Reports

Note: *Detailed set-up of the Quantify method should be done. For more information refer to* ***Chapter 2 How to Build a Quantification Method*** *as well as* ***Appendix 1 Environmental Reporting Calculations*** *on page* ***169*** *for details on correctly entering in the values.*

To generate a Form 1 type report the following parameters in your Quantification Method must be appropriately defined:

- **General Method parameters** – you must specify both the **Final Concentration Units** (value displayed on the form) and the **On Column Units** (value displayed in the Quantify\View Results summary window).



- **Reporting Threshold** – you must define an on column (ng) value *if Submitter/Task will not be used*. This field is applied for both water and soil matrices.

In addition, it is also of critical importance that you accurately set the following within the Environmental Parameters window:

- **Maximum in blank (On column Amount - ng) –**This value determines the "B" flags that are applied to the compounds.

- **MDL (On Column Amount – ng) Water/Soil** – This value determines the "J" flags that are applied to the compounds.

Critical Submitter/Task Information for Form 1 Generation

Note: *Proper set-up of the Submitter/Task should be done. Refer to* ***Chapter 3 How to Set-up Submitter/Task Data*** *as well as* ***Appendix 1 Environmental Reporting Calculations*** *on page* ***169*** *for details on correctly entering in the values.*

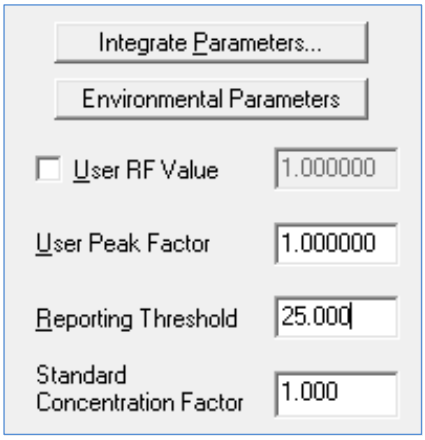
To properly generate a Form 1 type report the following Submitter/Task Information should be appropriately defined:

- **Reporting Limits** – an on column (ng) value must be defined if the Reporting Threshold parameter in your Quantify Method is not to be used. Fields are available for both water and soil matrices on a per compound basis.

Difference between Reporting Threshold versus Reporting Limit

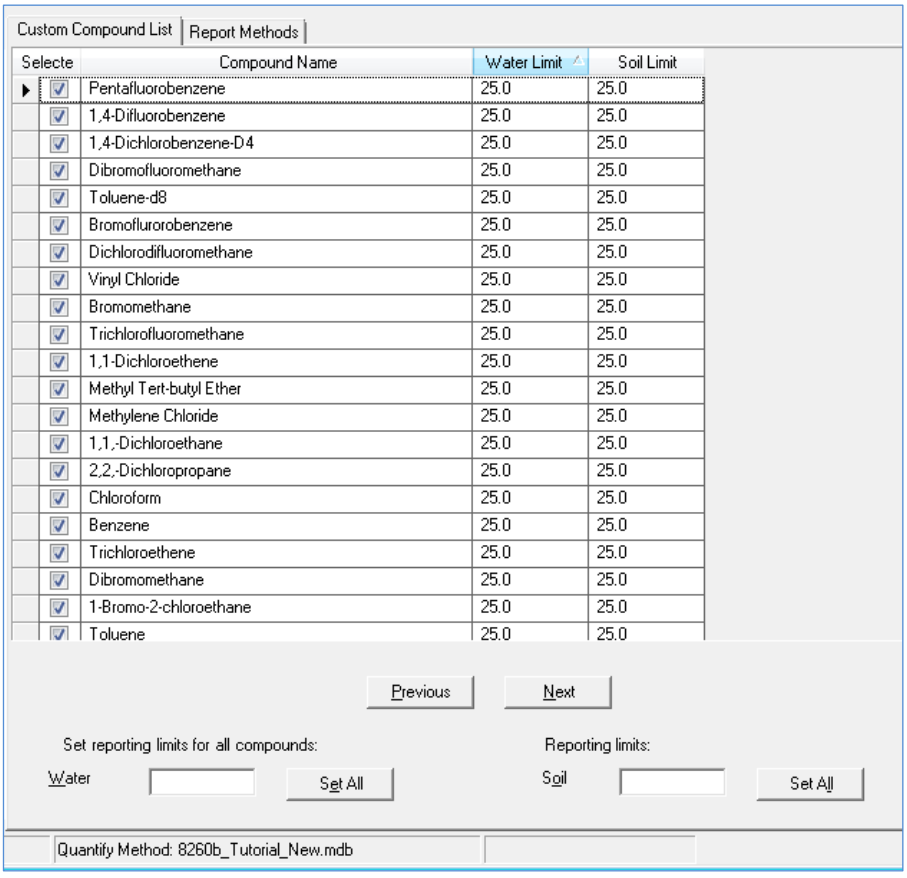
Reporting Threshold and Reporting Limit are essentially the same entity for environmental calculations. The same numerical value defined for a particular compound can be placed in either field.

Reporting Threshold – Designated in the Quantify Method



The **Reporting Threshold** only gives you the ability to enter in one value per compound. This can be used in situations where only one matrix type is analyzed.

Reporting Limit – Designated in Submitter/Task



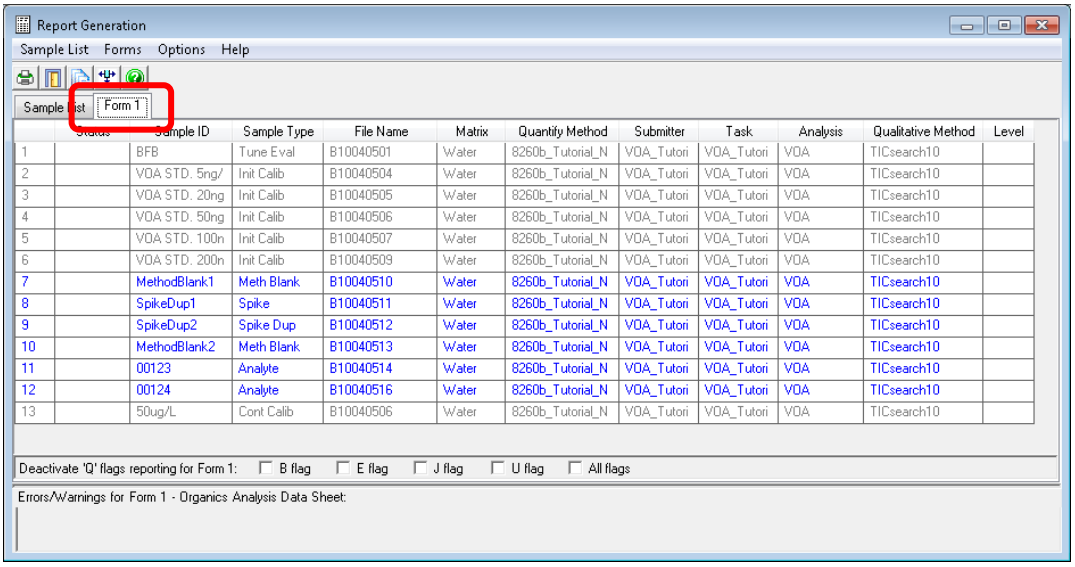
The **Reporting Limits** parameter gives the ability to enter in two values – one for water matrix and one for soil matrix.

- If a **Submitter/Task** list is used, it automatically overrides any value placed in the Reporting threshold field within the Quantify Method for a particular compound.

Form 1 – Generation Steps

1. Verify Steps 1 through 5 have been followed (pages 91-96 of this tutorial).

2. From the **Report Generation** window, click on the **Form 1** tab to display the Form 1 information.



- Correct any global errors (if possible) by changing the rows selected for processing in the Sample List or closing the Report Generation window and adjusting (or adding) parameters in the Sample List..
- Correct any form–specific errors (if possible) by adjusting sample assignments associated with Form.
- From the Report Generation Form 1 tab you can check the Q flags that you do not want to report.

Note: *The Q Flags options determine which of the flags automatically set by the TurboMass software will not be printed in the 'Q' column of Form 1 for any compound.*

3. The samples displayed in blue will be generated for the displayed form.

   In this example, assume you would like to print reports for the analyte samples. To do so you would uncheck samples 7, 8, 9, and 10 from the Sample List tab.

4. To generate Form 1 reports, click **Forms > Print**; otherwise continue to the next section for information on producing Form 1 – Tentatively Identified Compound.

Note: *Printing of forms may take a few minutes. The Report Manager monitors this process and you will be notified of the process in the lower right corner of the computer monitor screen.*

About Qualifiers (Q Flags) – Targets

In addition to the concentration of a compound the Form 1 also contains a column labeled "Q" for qualifier. A qualifier provides additional information about the compound.

IMPORTANT: *If you change the quantitative results at the report generation time (for example, you select a different method) you must reprocess, otherwise the qualifier flag assignments may be invalid.*

The EPA-defined qualifiers are:

U This flag indicates the compound was analyzed for but not detected. The Contract Required Quantitation Limit (CRQL) shall be adjusted accordingly.

NOTE: *For the purpose of environmental reports the term **MDL** is used to indicate the threshold value for the "U" qualifier flag. Values below this threshold value will flag the compound with a "U".*

J This flag indicates an estimated value. This flag is used in the following circumstances:

- When the mass spectral and retention time data indicate the presence of a compound that meets the volatile or semi-volatile GC/MS identification criteria, and the result is greater than or equal to the minimum detection limit (MDL defined in the quantify method) and less than the reporting limit (RL defined in the Submitter\Task).

IMPORTANT: *The "J" flag is set based on both the MDL and the Reporting Threshold/Reporting Limit. When a Submitter/Task list is not defined, the Reporting Threshold from the Quantify method is used to calculate the adjusted Reporting limit based on sample preparation information. If a Submitter/Task list is defined, the Reporting limit from the custom compound list is used to calculate the adjusted Reporting limit based on sample preparation information.*

B This flag is used when the analyte is found in the associated method blank as well as in the sample. It indicates probable blank contamination and warns the data user to take appropriate action. This flag is applied by the software to target compounds under appropriate conditions.

Note: *"B" flags (indicating blank contamination) should not be indicated on a Form 1 when the sample concentration is less than the adjusted minimum detection limit. The "B" flag should only show for a positive result (greater than the minimum detection limit. To do this, set the Maximum in blank value (ng) equal to the on-column minimum detection level (MDL)*

If the Maximum in blank value (ng) is set lower than the on-column MDL, a compound detected in the sample and blank that are above the maximum in blank value but below the MDL will be flagged with a "UB".

E This flag identifies compounds whose concentrations exceed the upper level of the calibration range for that specific analysis. If one or more compounds have a concentration greater than the upper level of the calibration range, the sample or extract should be diluted and reanalyzed.

Note: *For total xylene, where three isomers are quantified as two peaks, the calibration range of each peak shall be considered separately.*

D If a sample or extract is reanalyzed at a higher dilution factor, for example when the concentration of an Analyte exceeds the upper calibration range, the DL suffix is appended by the user to the sample number on Form I for the more diluted sample, and all reported concentrations on that Form I are flagged with the D flag by the software. This flag alerts data users that any discrepancies between the reported concentrations may be due to dilution of the sample or extract. The D flag is not applied to compounds which are not detected in the sample analysis (i.e., compounds reported with the CRQL and the U flag).

X Other specific flags may be required to properly define the results. If used, the flags shall be fully described, with the description attached to the sample data summary package and the Sample Delivery Group (SDG) Narrative. The laboratory-defined flags are limited to X, Y, and Z.

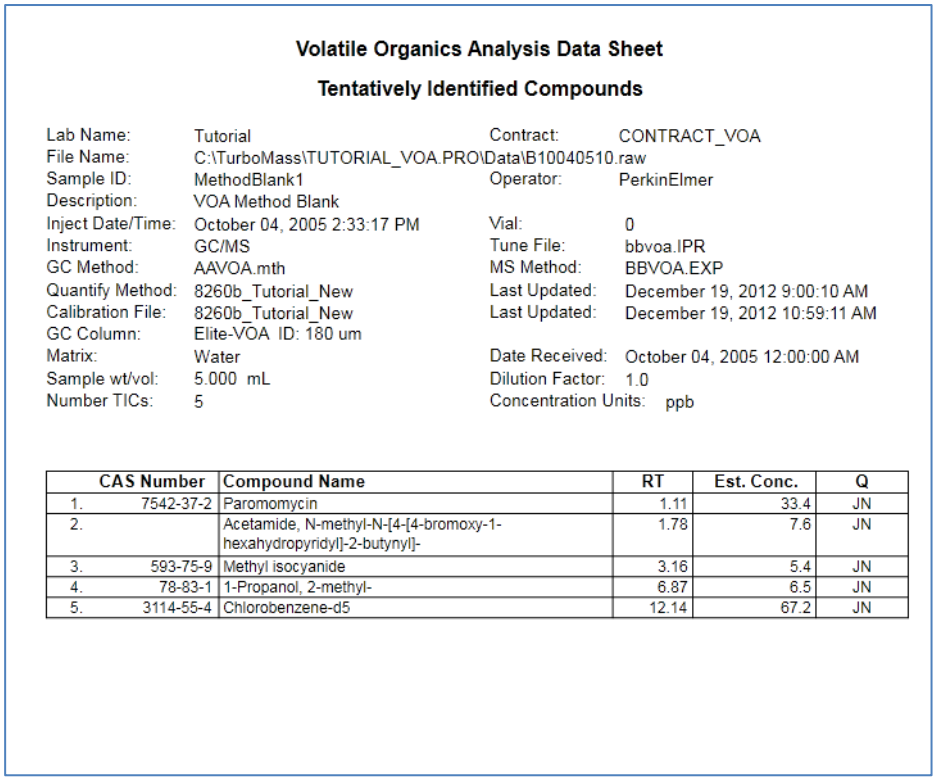
The All flags option will stop any value be populated item in the data source for the Q flag. When All flags is checked all the other flag check boxes are disabled.

Form 1TIC – Tentatively Identified Compounds (TICs)

On a Form 1 TIC the following information is included:

- Library search results for the largest (typically 10 to 30) non-target peaks in the chromatogram.
- Each TIC is assigned an estimated concentration based upon its response compared to the Total Ion Current Chromatogram area of the nearest internal standard compound.

Below is an example of the PKI style format for a Form 1 TIC report:



Volatile Organics Analysis Data Sheet

Tentatively Identified Compounds

Lab Name:	Tutorial	Contract:	CONTRACT_VOA
File Name:	C:\TurboMass\TUTORIAL_VOA.PRO\Data\B10040510.raw		
Sample ID:	MethodBlank1	Operator:	PerkinElmer
Description:	VOA Method Blank		
Inject Date/Time:	October 04, 2005 2:33:17 PM	Vial:	0
Instrument:	GC/MS	Tune File:	bbvoa.IPR
GC Method:	AAVOA.mth	MS Method:	BBVOA.EXP
Quantify Method:	8260b_Tutorial_New	Last Updated:	December 19, 2012 9:00:10 AM
Calibration File:	8260b_Tutorial_New	Last Updated:	December 19, 2012 10:59:11 AM
GC Column:	Elite-VOA ID: 180 um		
Matrix:	Water	Date Received:	October 04, 2005 12:00:00 AM
Sample wt/vol:	5.000 mL	Dilution Factor:	1.0
Number TICs:	5	Concentration Units:	ppb

	CAS Number	Compound Name	RT	Est. Conc.	Q
1.	7542-37-2	Paromomycin	1.11	33.4	JN
2.		Acetamide, N-methyl-N-[4-[4-bromoxy-1-hexahydropyridyl]-2-butynyl]-	1.78	7.6	JN
3.	593-75-9	Methyl isocyanide	3.16	5.4	JN
4.	78-83-1	1-Propanol, 2-methyl-	6.87	6.5	JN
5.	3114-55-4	Chlorobenzene-d5	12.14	67.2	JN

Form 1 TIC Type Report Methods

The following report methods can be generated:

- Form1TIC_Voa.rme
- Form1TIC_SV.rme
- PKI1TIC_Voa.rme
- PKI1TIC_SV.rme

Note: *The Submitter/Task feature can be used to change the report methods selected when generating a Form 1 TIC. Refer to* ***Chapter 3 How to Set-up Submitter/Task Data*** *for more information.*

Additional Environmental templates that are acceptable choices and can be produced under similar conditions are as follows:

- **Form1_CSV** – for detailed instructions, refer to ***Form1_CSV – Organics Analysis Data Sheet*** on page ***150***.

Valid Sample Types for Form 1 TIC Types

A Form 1 TIC report can only be generated using the following valid sample types:

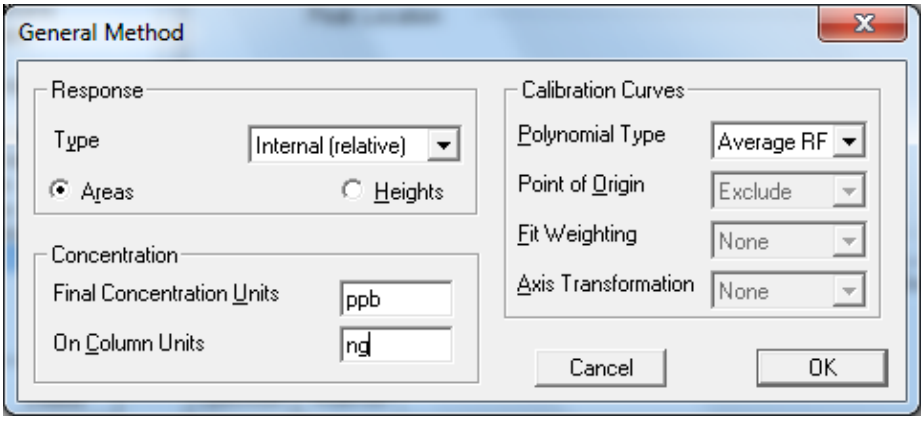
- Analyte
- Analyte Dup
- Method Blank
- Spike
- Spike Dup
- Dilution
- Re-extract
- Lab Control

Critical Quantify Method Information for Form 1 TIC Type Reports

Note: *Detailed set-up of the Quantify method should be done. Refer to* ***Chapter 2 How to Build a Quantification Method*** *as well as* ***Appendix 1 Environmental Reporting Calculations*** *on page* ***169*** *for details on correctly entering in the values.*

To generate a Form 1 TIC type report the following parameters in your Quantification Method must be appropriately defined:

- **General Method** parameters – you must specify the **Final Concentration Units** (value displayed on the form)



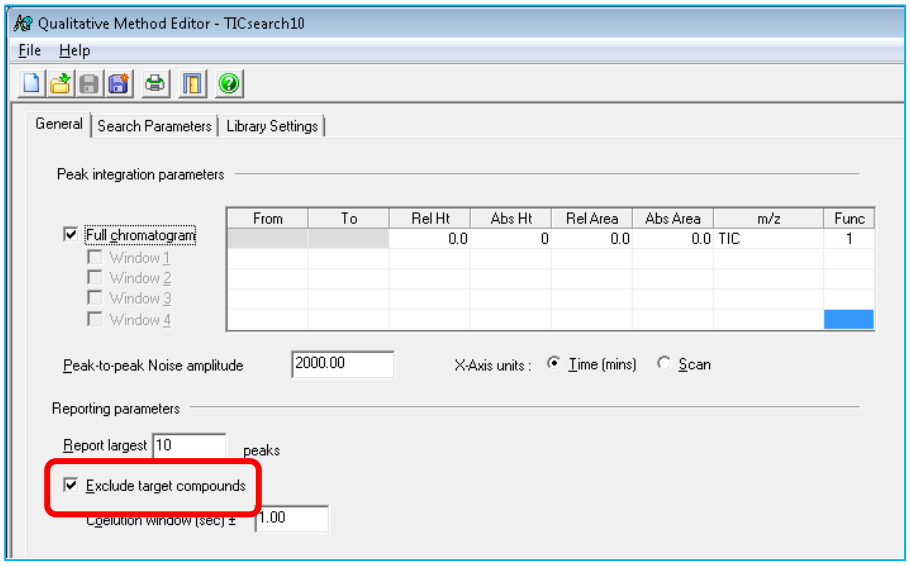
- An **Internal Standard (ISTD)** must be defined in the method in order for the TIC estimated concentrations to be calculated.

Critical Qualitative Method information for Form 1 TIC type reports

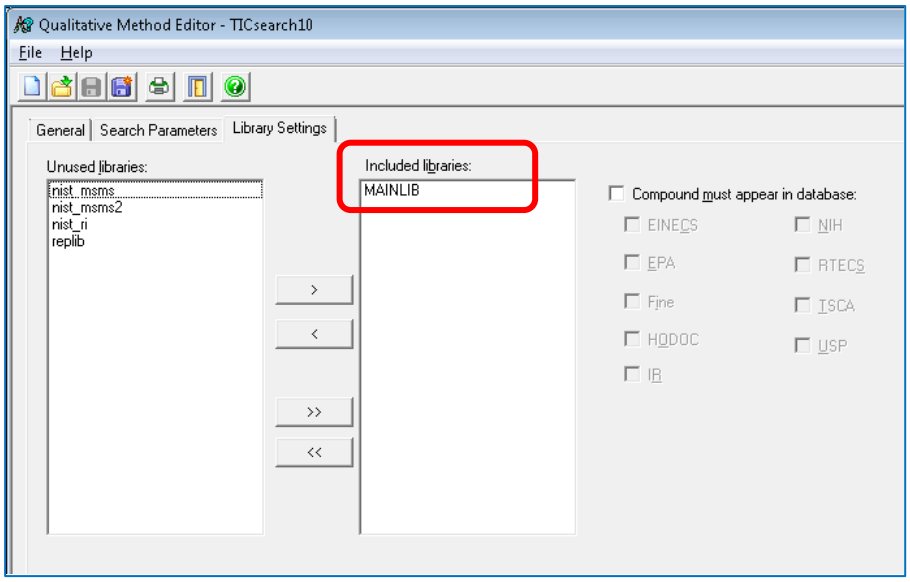
Note: *For detailed set-up of the Qualitative method, refer to* ***Chapter 11 Qualitative Method*** *in the* ***TurboMass Software User's Guide****.*

To generate a Form 1 TIC type report the following parameters in your Qualitative Method must be appropriately defined:

- From the **General tab** of the **Qualitative Method Editor** window make sure that **Exclude target compounds** is checked.



- From the **Library Settings** tab of the **Qualitative Method Editor** window, a library must be present in the **Included librarie**s: section.

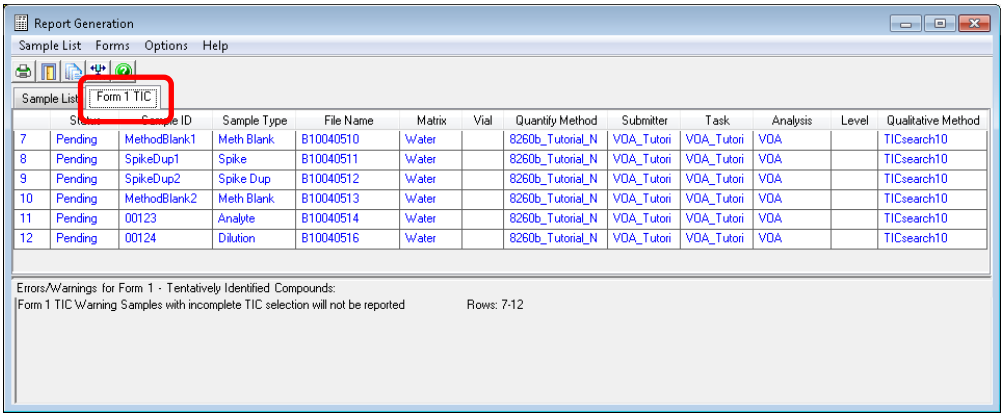


Form 1 TIC Type – Generation Steps

1. Verify the **QualDB** directory exists in your project and the QualitativeResults.1.mdb file resides there. Refer to the section titled ***Copy the Qualitative Database File to your new Project*** on page ***88*** of this manual for details.

2. Verify Steps 1 through 6 have been followed (pages 91-97 of this tutorial).

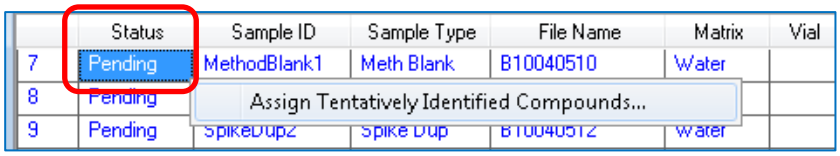
3. From the **Report Generation** window, click on the **Form 1 TIC** tab to display the Form 1 TIC information.



Note: *The Status column of the **Form 1 TIC** (Tentatively Identified Compounds) indicates whether or not the qualitative results have been reviewed and a Tentatively Identified Compound selected for each peak (even if this was just an implicit acceptance of the default top hit). **Pending** indicates that the file has not yet been fully reviewed; **Complete** indicates that TIC selection has been completed for the file.*

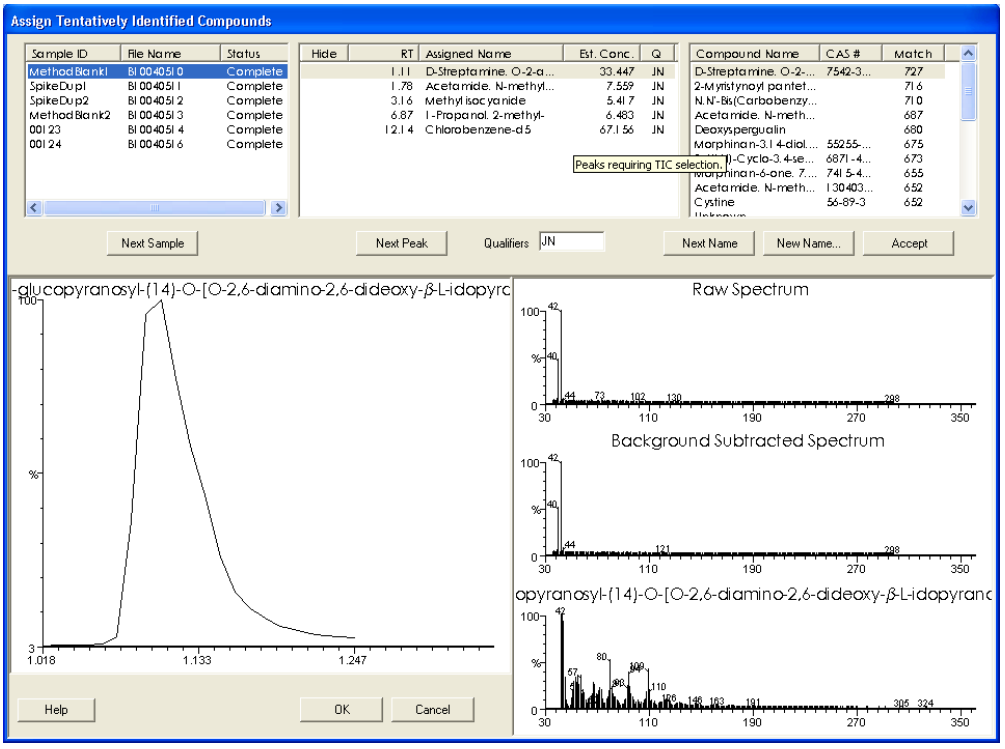
4. Right–click on a **Pending** sample.

   The **Assign Tentatively Identified Compounds...** menu appears.



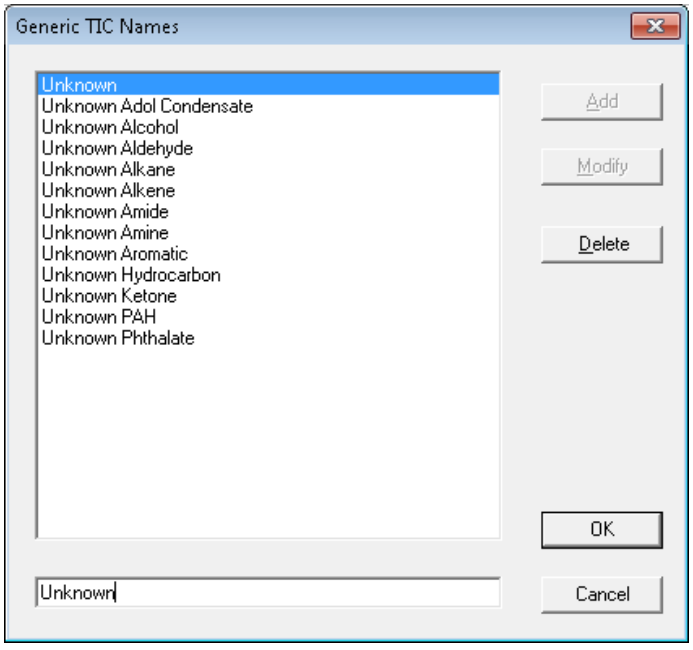
5. Click **Assign Tentatively Identified Compounds...**

   The Assign Tentatively Identified Compounds dialog opens and allows you to match peaks with compound names.



In this dialog you review the hits for the set of unidentified peaks from each appropriate sample and assign a compound name to each peak. In addition to the specific compound names supplied from the NIST library search you have access to a set of generic terms (such as *unknown alkane*, *unknown aromatic*).

Note: *These generic names can be created by selecting "New Name". The following window appears and a new generic TIC name can be added.*



6. Review the displayed peak plot and spectra associated with the first peak in the selected file (and the spectrum for the top hit).

7. Click **Next Name** (or click on another name in the right-hand list) to display the spectrum associated with another plot hit compound.

8. When you have decided which compound name should be assigned to the selected peak, select that name and click **Accept**.

   OR

   Having decided that none of the compound names from the plot hits list can be definitely assigned to the peak, select one of the generic names supplied at the bottom of the Compound Name list.

   OR

   Having decided that none of the generic names is applicable either, click the **New Name** button and enter a new generic name for selection.

   OR

   Right-click on the current peak's retention time and select **Hide/Show**. This will place an "X" under the Del(eted) column and the peak will not be reported.

9. Edit the qualifier codes applied to the peak, if required. (The TurboMass software will automatically change the codes if a generic name is selected.)

10. Click **Next Peak**, or select another peak, or click **Next Sample** to review peaks from the next file.

11. Click **OK** after reviewing the files and changing the TIC assignments as required, to save all assignments and close the dialog. Or you can click **Cancel** to abandon the procedure without saving changes to the data.

12. If the NIST library search has done a good job it may be possible to simply accept all the default compounds names offered (i.e., the highest rated hit). For this reason the software provides for a simultaneous display of the *current tentatively identified compound* setting for each peak in a selected sample. This display also allows reviewing the final set of selections to ensure consistency, e.g., the same compound name has not been used twice).

13. To generate **Form 1 TIC** reports, click **Forms > Print**, otherwise continue to the next section for information on producing Form 2 – SMC/Surrogate Compound Recovery.

About Qualifiers (Q Flags) – Tentatively Identified Compounds (TICs)

In addition to the estimated concentration of a compound the Form 1 TIC also contains a column labeled "Q" for qualifier. A qualifier provides additional information about the compound.

The EPA-defined qualifiers are:

J This flag indicates an estimated value. This flag is used in the following circumstances:

- When estimating a concentration for tentatively identified compounds where a 1:1 response is assumed

N This flag indicates presumptive evidence of a compound. This flag is only used for tentatively identified compounds, where the identification is based on a mass spectral library search. It is applied to all tentatively identified compounds results. For generic characterization of a tentatively identified compound, such as chlorinated hydrocarbon, the N flag is not used.

B This flag is used when the analyte is found in the associated method blank as well as in the sample. It indicates probable blank contamination and warns the data user to take appropriate action. "B" flags must be applied manually by the user for Tentatively identified compounds in the Assign Tentatively Identified Compounds window.

A This flag indicates that a tentatively identified compound is a suspected aldol-condensation product. This flag must be applied manually by the user in the Assign Tentatively Identified Compounds window.

Form 2 – SMC/Surrogate Compound Recovery

A Form 2 reports the recoveries of the specific analytes defined as ***Surrogates***. It flags any compound whose recovery is outside of the acceptable limits.

Below is an example of the PKI style format for a Form 2 report:

Water Volatile System Monitoring Compound Recovery

Lab Name:	Tutorial	Contract:	CONTRACT_VOA
Project Path:	C:\TurboMass\TUTORIAL_VOA.PRO		
Instrument:	GC/MS	Tune File:	bbvoa.IPR
GC Method:	AAVOA.mth	MS Method:	BBVOA.EXP
Quantify Method:	8260b_Tutorial_New	Last Updated:	December 19, 2012 9:00:10 AM
Calibration File:	8260b_Tutorial_New	Last Updated:	December 18, 2012 3:58:03 PM
GC Column:	Elite-VOA ID: 180 um		

	File Name	SMC 1 (BFM) #	SMC 2 (TD8) #	SMC 3 (BFB) #	Tot Out
1	B10040510	99	97	104	0
2	B10040511	100	99	103	0
3	B10040512	100	98	102	0
4	B10040513	98	99	108	0
5	B10040514	98	97	107	0
6	B10040516	96	98	108	0

				QC LIMITS
SMC1	(BFM)	=	Dibromofluoromethane	(70 - 130)
SMC2	(TD8)	=	Toluene-d8	(70 - 130)
SMC3	(BFB)	=	Bromoflurorobenzene	(70 - 130)

Column to be used to flag recovery values

* Values outside of required QC limits

Form 2 Type Report Methods

The following report methods can be generated:

- Form2_Voa_water.rme
- Form2_Voa_soil.rme
- Form2_SV_water.rme
- Form2_SV_soil.rme
- PKI2_Voa_water.rme
- PKI2_Voa_soil.rme
- PKI2_SV_water.rme
- PKI2_SV_soil.rme

Note: *The Submitter/Task feature can be used to change the report methods selected when generating a Form 2. Refer to* ***Chapter 3 How to Set-up Submitter/Task Data*** *for more information.*

Additional Environmental templates that are acceptable choices and can be produced under similar conditions are as follows:

- **PKIEnvQuant** – for detailed instructions, refer to ***PKIEnvQuant Template Form Generation*** on page ***141***.

Valid Sample Types for Form 2 Types

A Form 2 report can only be generated using the following valid sample types:

- Analyte
- Analyte Dup
- Method Blank
- Spike
- Spike Dup
- Dilution
- Re-extract
- Lab Control

Critical Quantify Method Information for Form 2 Type Reports

Note: *Detailed set-up of the Quantify method should be done. Refer to* ***Chapter 2 How to Build a Quantification Method*** *as well as* ***Appendix 1 Environmental Reporting Calculations*** *on page* ***169*** *for details on correctly entering in the values.*

To generate a Form 2 type report the following Environmental Parameters in your Quantification Method must be accurately defined:

- **Surrogate/Spike (Amount added - ng) –** This value is used to calculate the recovery.

IMPORTANT: *Refer to the section Surrogate Recovery Calculations in* ***Appendix 1 Environmental Reporting Calculations*** *for details on how to derive this value.*

Critical Submitter/Task Information for Form 2 Generation

Note: *Proper set-up of the Submitter/Task should be done. For more information, refer to* ***Chapter 3 How to Set-up Submitter/Task Data****.*

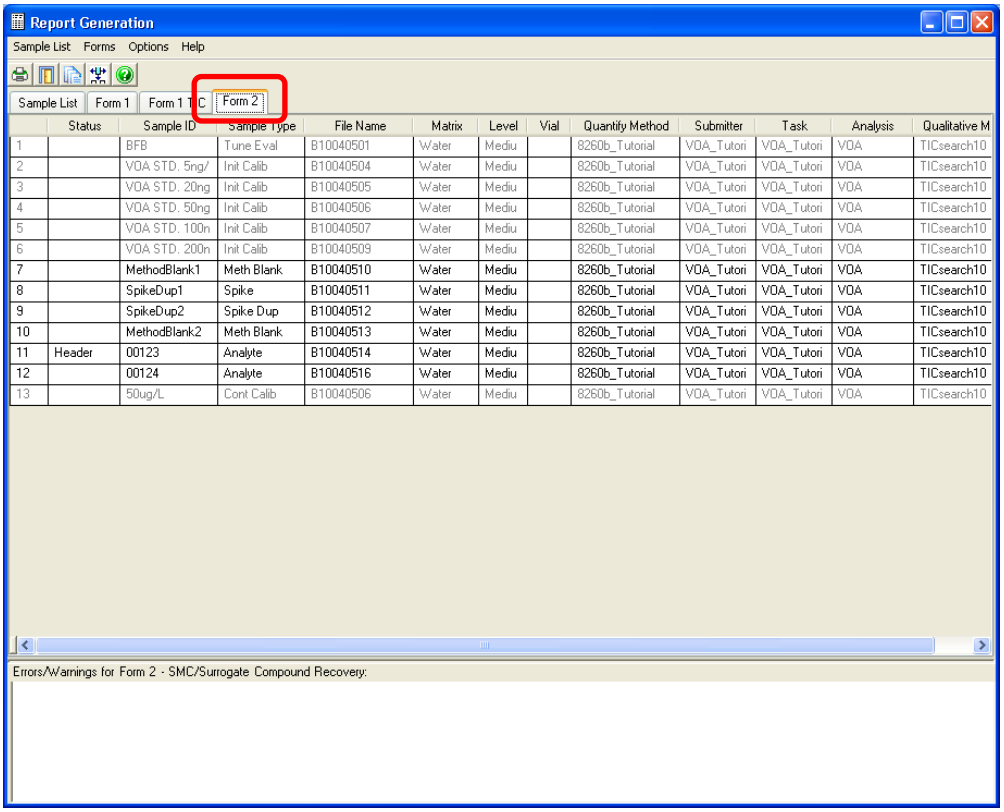
To properly generate a Form 2 type report the following Submitter/Task Information should be appropriately defined:

- If a Submitter/Task is used, the surrogate compounds must be listed in the custom compound list to display on the Form 2.

Form 2 Type – Generation Steps

1. Verify Steps 1 through 5 have been followed (pages 91-96 of this tutorial).

2. From the Report Generation window click on **Form 2**.
   The **Form 2 Tab** appears with the files displayed in the summary data color.



	Status	Sample ID	Sample Type	File Name	Matrix	Level	Vial	Quantify Method	Submitter	Task	Analysis	Qualitative M
1		BFB	Tune Eval	B10040501	Water	Mediu		8260b_Tutorial	VOA_Tutori	VOA_Tutori	VOA	TICsearch10
2		VOA STD. 5ng/	Init Calib	B10040504	Water	Mediu		8260b_Tutorial	VOA_Tutori	VOA_Tutori	VOA	TICsearch10
3		VOA STD. 20ng	Init Calib	B10040505	Water	Mediu		8260b_Tutorial	VOA_Tutori	VOA_Tutori	VOA	TICsearch10
4		VOA STD. 50ng	Init Calib	B10040506	Water	Mediu		8260b_Tutorial	VOA_Tutori	VOA_Tutori	VOA	TICsearch10
5		VOA STD. 100n	Init Calib	B10040507	Water	Mediu		8260b_Tutorial	VOA_Tutori	VOA_Tutori	VOA	TICsearch10
6		VOA STD. 200n	Init Calib	B10040509	Water	Mediu		8260b_Tutorial	VOA_Tutori	VOA_Tutori	VOA	TICsearch10
7		MethodBlank1	Meth Blank	B10040510	Water	Mediu		8260b_Tutorial	VOA_Tutori	VOA_Tutori	VOA	TICsearch10
8		SpikeDup1	Spike	B10040511	Water	Mediu		8260b_Tutorial	VOA_Tutori	VOA_Tutori	VOA	TICsearch10
9		SpikeDup2	Spike Dup	B10040512	Water	Mediu		8260b_Tutorial	VOA_Tutori	VOA_Tutori	VOA	TICsearch10
10		MethodBlank2	Meth Blank	B10040513	Water	Mediu		8260b_Tutorial	VOA_Tutori	VOA_Tutori	VOA	TICsearch10
11	Header	00123	Analyte	B10040514	Water	Mediu		8260b_Tutorial	VOA_Tutori	VOA_Tutori	VOA	TICsearch10
12		00124	Analyte	B10040516	Water	Mediu		8260b_Tutorial	VOA_Tutori	VOA_Tutori	VOA	TICsearch10
13		50ug/L	Cont Calib	B10040506	Water	Mediu		8260b_Tutorial	VOA_Tutori	VOA_Tutori	VOA	TICsearch10

3. Right-click on a row (any cell) of the table and a pop-up menu appears containing the following item: **Assign Header Sample**.
   When you select Assign Header Sample, if the row is an active one, it assigns the selected row as the source of report header information for Form 2. It displays Header in the Status column for that row (and eliminates Header from any other row). Inactive rows (displayed in gray) cannot be selected.

4. To generate a **Form 2** report, click **Forms > Print**, otherwise continue to the next section for information on producing Form 3 – Matrix Spike/Matrix Duplicate Recovery Summary.

Form 3 – Matrix Spike/Matrix Duplicate Recovery

A Form 3 displays the following information:

- The recoveries of specific analytes defined as ***spike*** compounds for the spike (matrix spike-MS) and spike duplicate (matrix spike duplicate–MSD). It flags any compound whose recovery or relative percent difference (RPD) is outside of the acceptable limits.

- Sample results in the un-spiked sample (designated Analyte type in Sample list) that are below the MDL, will not report on this form.

Below is an example of a PKI style format for a Form 3 report:

Water Volatile Matrix Spike/Matrix Spike Duplicate Recovery

Lab Name:	Tutorial	Contract:	CONTRACT_VOA
Project Path:	C:\TurboMass\TUTORIAL_VOA.PRO		
Instrument:	GC/MS	Tune File:	bbvoa.IPR
GC Method:	AAVOA.mth	MS Method:	BBVOA.EXP
Quantify Method:	8260b_Tutorial_New	Last Updated:	December 19, 2012 9:00:10 AM
Calibration File:	8260b_Tutorial_New	Last Updated:	December 18, 2012 3:58:03 PM
GC Column:	Elite-VOA ID: 180 um		

Analyte File Name: B10040514 Sample ID: 00123

Spike File Name: B10040511 Sample ID: SpikeDup1

Compound	Spike Added (ppb)	Sample Concentration (ppb)	MS Concentration (ppb)	MS % Rec	#	QC Limits Rec
1,1-Dichloroethene	20.000	0.000	21.243	106		70 - 130
Benzene	20.000	0.000	20.876	104		70 - 130
Trichloroethene	20.000	0.000	20.438	102		70 - 130
Toluene	20.000	0.000	20.745	104		70 - 130

Spike Duplicate File Name: B10040512 Sample ID: SpikeDup2

Compound	Spike Added (ppb)	MSD Concentration (ppb)	MSD % Rec	#	% RPD	#	RPD	QC Limits Rec
1,1-Dichloroethene	20.000	21.247	106		0		30	70 - 130
Benzene	20.000	20.857	104		0		30	70 - 130
Trichloroethene	20.000	21.075	105		3		30	70 - 130
Toluene	20.000	20.702	104		0		30	70 - 130

Column to be used to flag recovery and RPD values with an asterisk

* Values outside of QC limits

RPD: 0 out of 4 outside limits - Calculated on concentrations

Spike Recovery: 0 out of 8 outside limits

COMMENTS:

Form 3 Type Report Methods

The following report methods can be generated:

- Form3_Voa_water.rme
- Form3_Voa_soil.rme
- Form3_SV_water.rme
- Form3_SV_soil.rme
- PKI3_Voa_water.rme
- PKI3_Voa_soil.rme
- PKI3_SV_water.rme
- PKI3_SV_soil.rme

Note: *The Submitter/Task feature can be used to change the report methods selected when generating a Form 3. Refer to* ***Chapter 3 How to Set-up Submitter/Task Data*** *for more information.*

Additional Environmental templates that are acceptable choices and can be produced under similar conditions are as follows:

Note: *Report methods would need to be created using these templates.*

- **PKI3A.tpl** – this template produces a report similar to PKI3A_8000 template with the exception that the RPD value is calculated using recovery values rather than concentration values.
- **PKI3B.tpl** – this template produces a report similar to PKI3B_8000 with the exception that the RPD value is calculated using recovery values rather than concentration values.
- **PKI3C.tpl** – this template produces a report similar to PKI3C_8000 with the exception that the RPD value is calculated using recovery values rather than concentration values.
- **PKI3D.tpl** – this template produces a report similar to PKI3D_8000 with the exception that the RPD value is calculated using recovery values rather than concentration values.

Note: *The Relative Percent Difference calculation for US EPA GC/MS Form 3 report (Matrix Spike/Matrix Spike Duplicate) can be done in two ways. Refer to the section titled* ***Relative Percent Difference (RPD) Spike Calculations*** *on pages* ***215 - 216*** *for more information.*

Valid Sample Types for Form 3 types

A Form 3 report can only be generated using the following valid sample types:

- Analyte
- Spike
- Spike Dup

Critical Quantify Method Information for Form 3 Type Reports

Note: *Detailed set-up of the Quantify method should be done. Refer to* ***Chapter 2 How to Build a Quantification Method*** *as well as* ***Appendix 1 Environmental Reporting Calculations*** *for detailed calculation information.*

To generate a Form 3 type report the following Environmental Parameters in your Quantification Method must be appropriately defined:

- **Surrogate/Spike (Amount added - ng)** – This value is used to calculate the recovery.
- **MDL (On Column Amount – ng) Water/Soil** – This value will determine if the sample amount is reported and used on the form 3.
- **IMPT** – Refer to the section **Concentrations for Spike Added Calculations** on pages **207-214** for details on how to derive this value.

Critical Submitter/Task information for Form 3 Generation

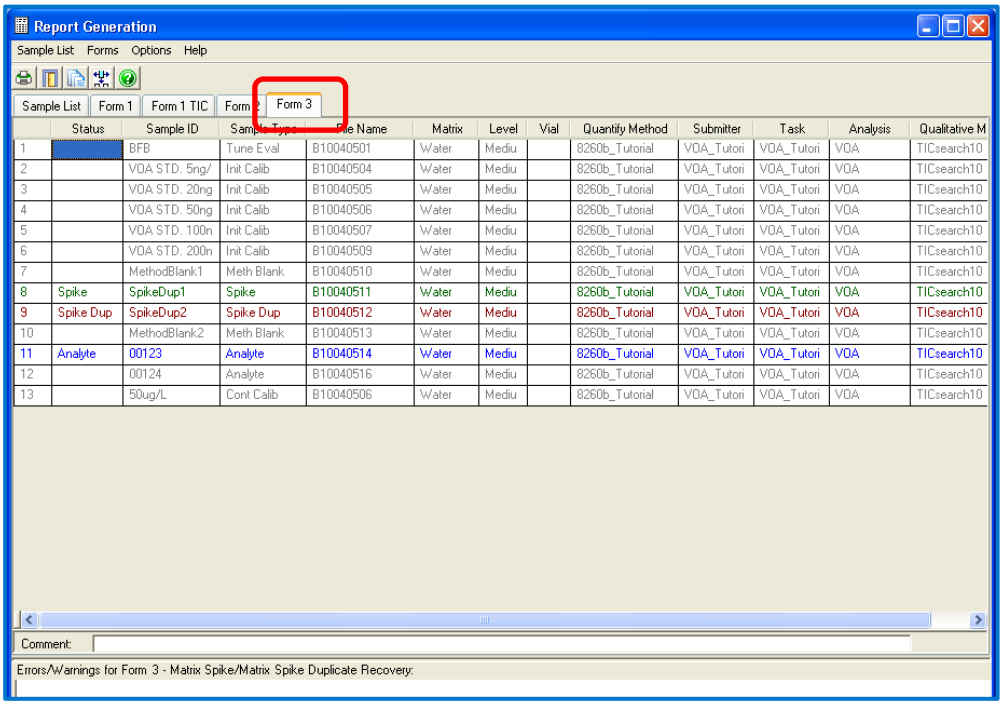
Note: *Proper set-up of the Submitter/Task should be done. Refer to Chapter* ***3 How to Set-up Submitter/Task Data*** *for more information.*

To properly generate a Form 3 type report the following Submitter/Task Information should be appropriately defined:

- If a Submitter/Task is used, the spike compounds must be listed in the custom compound list to display on the Form 3.

Form 3 Type – Generation Steps

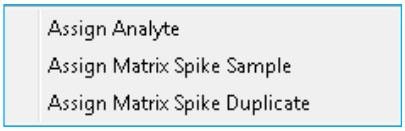
1. Verify Steps 1 through 5 have been followed (pages 91-96 of this tutorial).
2. From the **Report Generation** window click on the **Form 3** tab.



Form 3 shows the primary data file in blue, which represents the unspiked sample analysis. It also uses two unique colors (dark green and dark red) to indicate the **Matrix Spike** sample and the **Matrix Spike Duplicate** samples. All the remaining rows are unused for the report and shown in the grey disregard color.

The Status column indicates which rows are currently flagged as the **Analyte**, **Spike** and **Spike Dup** samples to be used in generating Form 3. This apparent duplication with the Sample Type column is because it will be possible for you to over–ride the Sample List's file Type and assign a different row as one of the three key data files.

A pop-up menu appears when you right–click on a row in the Form 3 Sample List view pane. The pop up menu that appears allow you to reassign the **Matrix**, **Matrix Spike**, and **Matrix Spike Duplicate** samples without returning to the Sample List. (Note that any reassignments made here are lost after the reports are printed.



Assign Analyte: Select this option to assign the selected row as the Unspiked sample from which the MS/MSD sample is prepared. When this option is selected the Status column for the selected row displays "Analyte" and the color for that row becomes blue.

Assign Matrix Spike Sample: Select this option to assign the selected row as the Matrix Spike sample. When this option is selected the Status column for the selected row displays "Spike" and the color for that row becomes dark green.

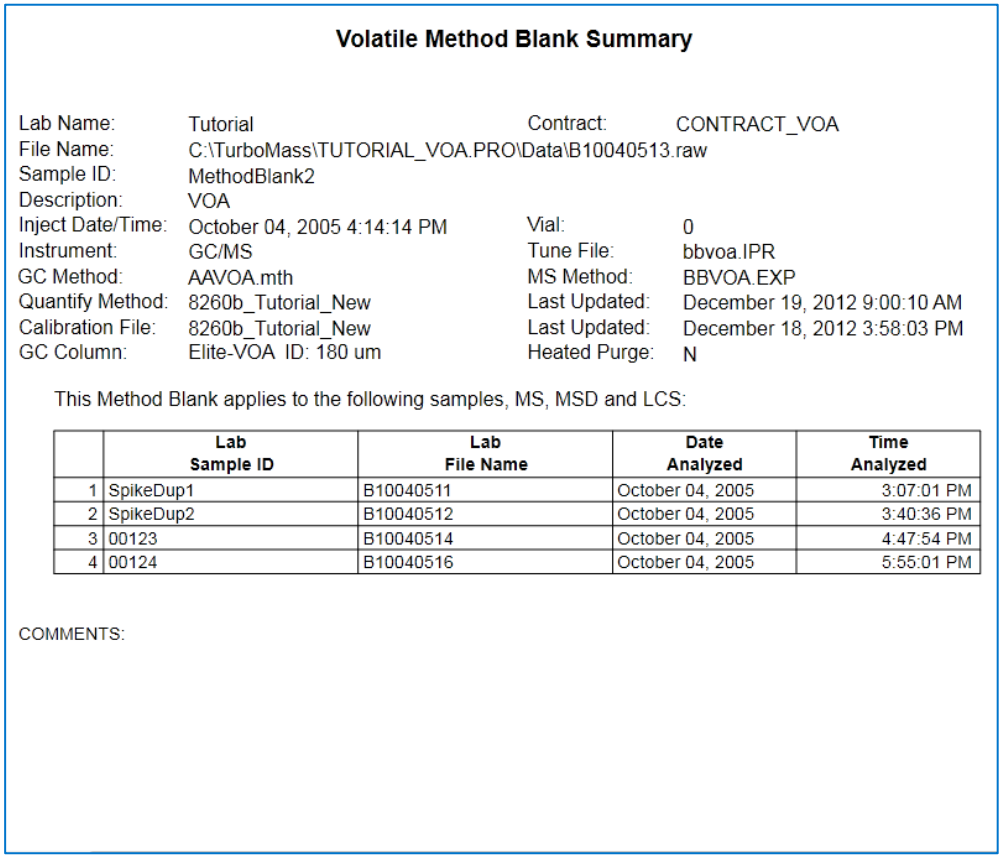
Assign Matrix Spike Duplicate: Select this option to assign the selected row as the Matrix Spike Duplicate sample. When this option is selected the Status column for selected row displays "Spike Dup" and the color for that row becomes dark red.

3. To generate a **Form 3** report, click **Forms > Print**, otherwise continue to the next section for information on producing Form 4 – Method Blank Summary.

Form 4 – Method Blank Summary

This report displays a summary of samples that were acquired associated to a particular method blank.

Below is an example of the PKI style format for a Form 4 report:



Volatile Method Blank Summary

Lab Name:	Tutorial	Contract:	CONTRACT_VOA
File Name:	C:\TurboMass\TUTORIAL_VOA.PRO\Data\B10040513.raw		
Sample ID:	MethodBlank2		
Description:	VOA		
Inject Date/Time:	October 04, 2005 4:14:14 PM	Vial:	0
Instrument:	GC/MS	Tune File:	bbvoa.IPR
GC Method:	AAVOA.mth	MS Method:	BBVOA.EXP
Quantify Method:	8260b_Tutorial_New	Last Updated:	December 19, 2012 9:00:10 AM
Calibration File:	8260b_Tutorial_New	Last Updated:	December 18, 2012 3:58:03 PM
GC Column:	Elite-VOA ID: 180 um	Heated Purge:	N

This Method Blank applies to the following samples, MS, MSD and LCS:

	Lab Sample ID	Lab File Name	Date Analyzed	Time Analyzed
1	SpikeDup1	B10040511	October 04, 2005	3:07:01 PM
2	SpikeDup2	B10040512	October 04, 2005	3:40:36 PM
3	00123	B10040514	October 04, 2005	4:47:54 PM
4	00124	B10040516	October 04, 2005	5:55:01 PM

COMMENTS:

Form 4 Type Report Methods

The following report methods can be generated:

- Form4_Voa.rme
- PKI4_SV.rme

Note: *The Submitter/Task feature can be used to change the report methods selected when generating a Form 4. Refer to* ***Chapter 3 How to Set-up Submitter/ Task Data*** *for more information.*

Valid Sample Types for Form 4 Types

A Form 4 report can only be generated using the following valid sample types:

- Method Blank – used to define the method blank to report

The following types are valid for the summary/table portion of the report:

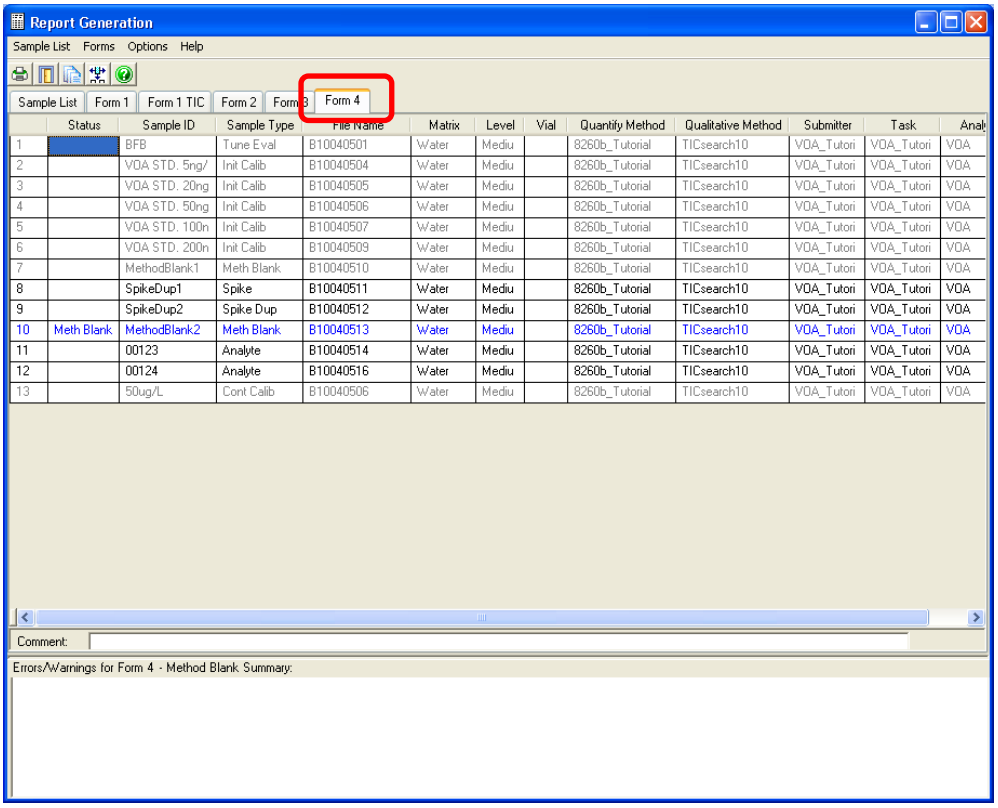
- Analyte
- Analyte Dup

- Spike
- Spike Dup
- Dilution
- Re-extract
- Lab Control

Form 4 Type – Generation Steps

1. Verify Steps 1 through 5 have been followed (pages 91-96 of this tutorial).

2. From the **Report Generation** window click on the **Form 4** tab.

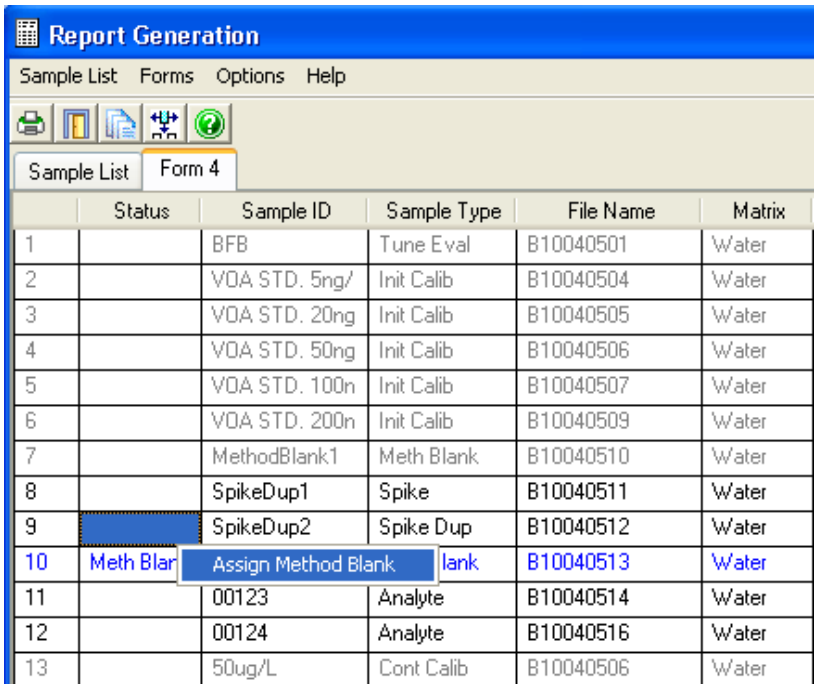


Form 4 uses the primary data file data color (blue) for the method blank (Meth Blank) sample. Any Tune Eval, Cont Calib, Init Calib or additional Meth Blank rows will be shown in the disregard color (gray) since they are not included in Form 4. All remaining rows are shown in the summary color (black), indicating they are included in the summary section of the printed report.

The **Status** column indicates which row is currently flagged as the method blank. This may be used to distinguish between several Meth Blank rows in the Sample List (only one can be treated as the primary data set for Form 4) or to flag a row of another Sample Type as being the method blank.

3. To alter the Method Blank assignments without going back to the Sample List, select the Sample List row which is to become the new primary data set.

4. Right-click on that row and from the pop up menu that appears click **Assign Method Blank.**



Selecting **Assign Method Blank**, assigns the selected row as the Method Blank sample for Form 4. In addition the status column for that row now reads “Meth Blank” and is colored blue.

5. To generate a **Form 4** report, click **Forms > Print**, otherwise continue to the next section for information on producing Form 5 – Instrument Performance Check.

Form 5 – Instrument Performance Check

This report displays the results and compliance of the tune evaluation sample along with a summary of all data files acquired while the tune was valid.

Below is an example of the PKI style format for a Form 5 report:

Volatile Organic Instrument Performance Check

Bromofluorobenzene (BFB)

Lab Name:	Tutorial	Contract:	CONTRACT_VOA
File Name:	C:\TurboMass\TUTORIAL_VOA.PRO\Data\B10040501.raw		
Sample ID:	BFB		
Description:	BFB		
Inject Date/Time:	October 04, 2005 9:39:17 AM		
Instrument:	GC/MS	Tune File:	bbvoa.IPR
GC Method:	AABFB.mth	MS Method:	BBBFB.EXP
GC Column:	Elite-VOA ID: 180 um		
Scans:	COMBINE(304:306)-(292:299)		
Test Name:	BFB 624/8260 TEST	Result:	PASS

m/z	Ion Abundance Criteria	% Relative Abundance	Result
50	15% - 40% of mass 95	16.4	Pass
75	30% - 60% of mass 95	46.8	Pass
95	Base Peak, 100% relative abundance	100.0	Pass
96	5% - 9% of mass 95	6.6	Pass
173	Less than 2% of mass 174	0.5	Pass
174	Greater than 50% of mass 95	80.2	Pass
175	5% - 9% of mass 174	6.5	Pass
176	Greater than 95% but less than 101% of 174	97.9	Pass
177	5% - 9% of mass 176	6.2	Pass

This check applies to the following samples, MS, MSD, blanks and standards:

	Lab Sample ID	Lab File ID	Date Analyzed	Time Analyzed
1	VOA STD. 5ng/ml	B10040504	October 04, 2005	11:11:20 AM
2	VOA STD. 20ng/ml	B10040505	October 04, 2005	11:44:54 AM
3	VOA STD. 50ng/ml	B10040506	October 04, 2005	12:18:34 PM
4	VOA STD. 100ng/ml	B10040507	October 04, 2005	12:52:20 PM
5	VOA STD. 200ng/ml	B10040509	October 04, 2005	01:59:37 PM
6	MethodBlank1	B10040510	October 04, 2005	02:33:17 PM
7	SpikeDup1	B10040511	October 04, 2005	03:07:01 PM
8	SpikeDup2	B10040512	October 04, 2005	03:40:36 PM
9	MethodBlank2	B10040513	October 04, 2005	04:14:14 PM
10	00123	B10040514	October 04, 2005	04:47:54 PM
11	00124	B10040516	October 04, 2005	05:55:01 PM

Form 5 Type Report Methods

The following report methods can be generated:

- Form5_Voa.rme
- Form5_SV.rme
- PKI5_Voa.rme
- PKI5_SV.rme

Note: *The Submitter/Task feature can be used to change the report methods selected when generating a Form 5 report. Refer to* ***Chapter 3 How to Set-up Submitter/Task Data*** *for more information.*

Additional Environmental templates that are acceptable choices and can be produced under similar conditions are as follows:

- **Form5_CSV** – for detailed instructions, refer to ***Form5_CSV - Instrument Performance Check (CSV format)*** on page ***153***.

Valid Sample Types for Form 5 Types

A Form 5 report can only be generated using the following valid sample types:

- Tune Eval – used to define the tune evaluation sample to report

The following types are valid for the summary/table portion of the report

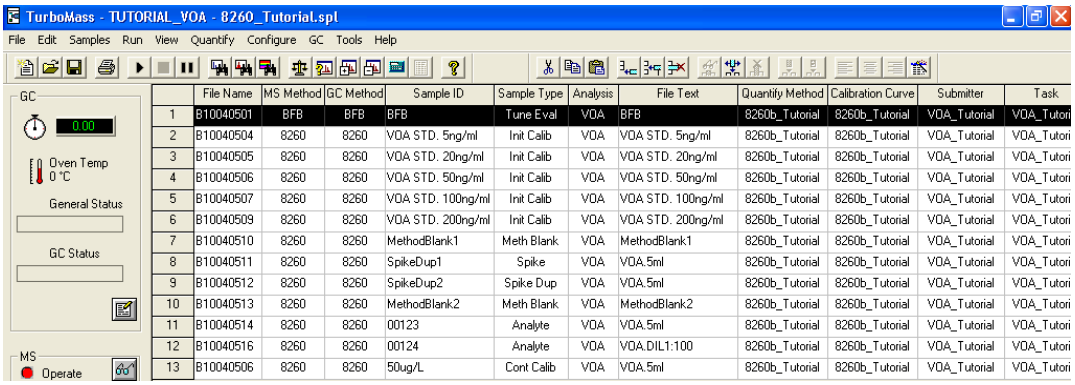
- Analyte
- Analyte Dup
- Init Calib
- Cont Calib
- Meth Blank
- Spike
- Spike Dup
- Dilution
- Re-extract
- Lab Control

Tune Evaluation Processing

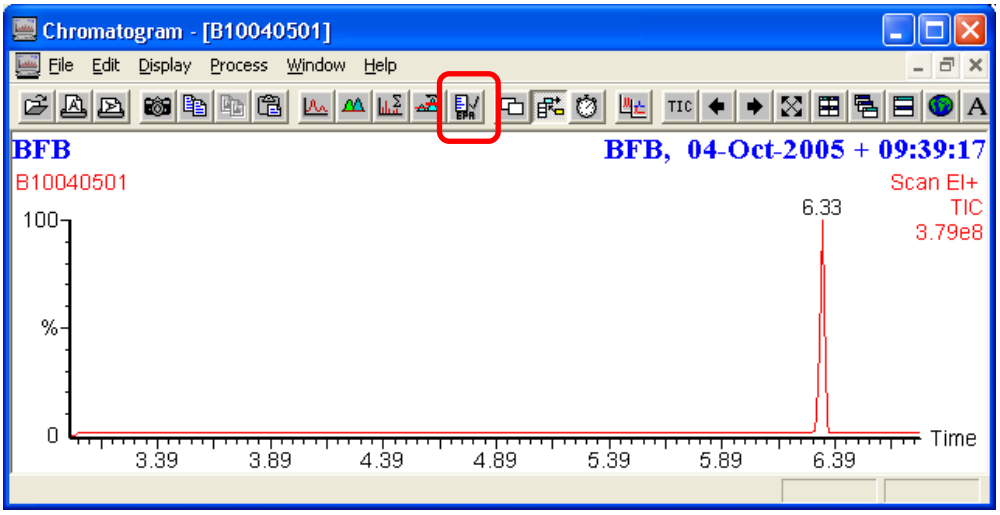
Note: *Before producing a Form 5 the Tune file must be processed using these instructions*

1. From the TurboMass Sample List window, highlight the first row containing BFB.

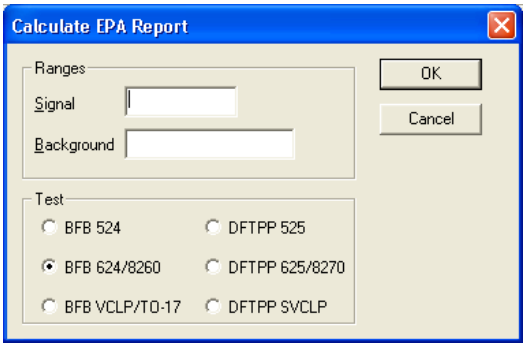
IMPORTANT NOTE: *These instructions can only be performed immediately after opening a session of TurboMass prior to any type of quantitative sample processing. Failure to do this will result in tune evaluation results not being created or updated properly.*



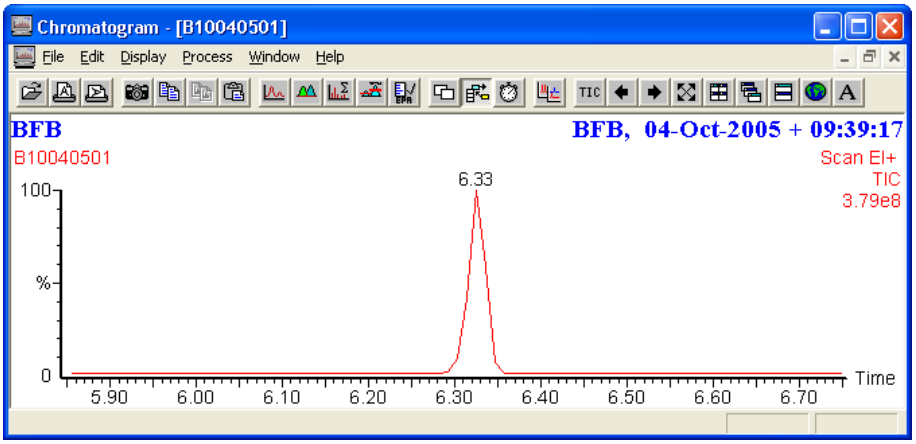
2. From the **View** menu, select **Chromatogram** to open the Chromatogram.



3. Select the EPA button  and the following dialog appears:

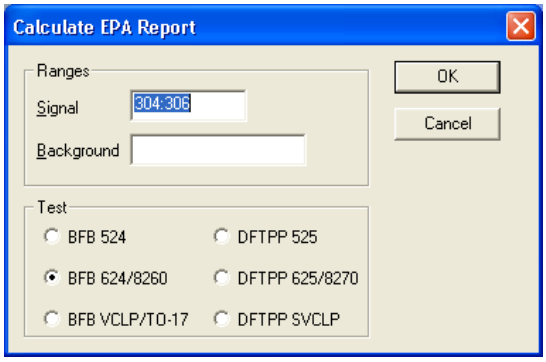


4. Zoom in on the peak of interest (BFB or DFTPP) so that you can easily select the desired scan signal range.



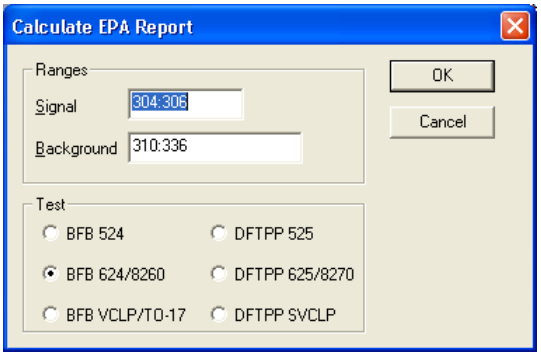
5. To get the signal, right mouse click and hold the mouse button mid-way inside the peak on the left side and drag the mouse to the inside right side and release.

   The signal/signal range selected is displayed in the Calculate EPA Report Signal field:



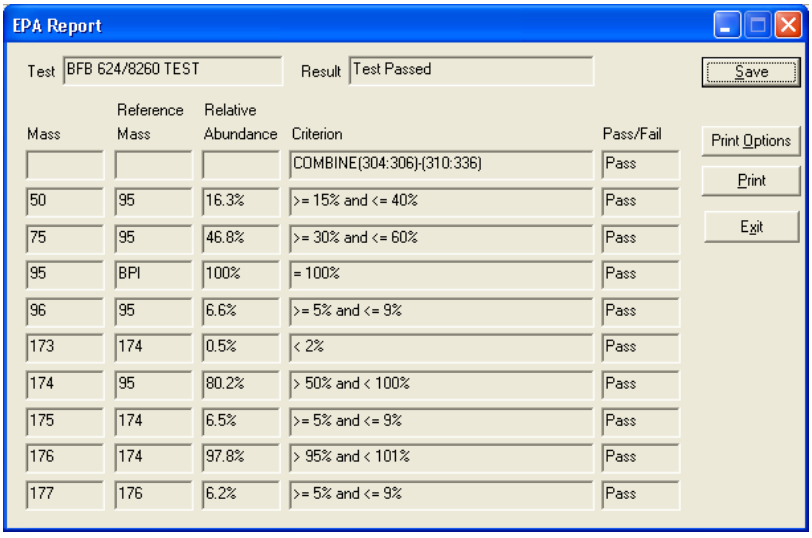
6. To apply a background subtraction, select a scan outside of the peak of interest using a right-mouse click or drag.

   The background subtraction scan can either be a range or a single point.



7. Select the appropriate EPA test.

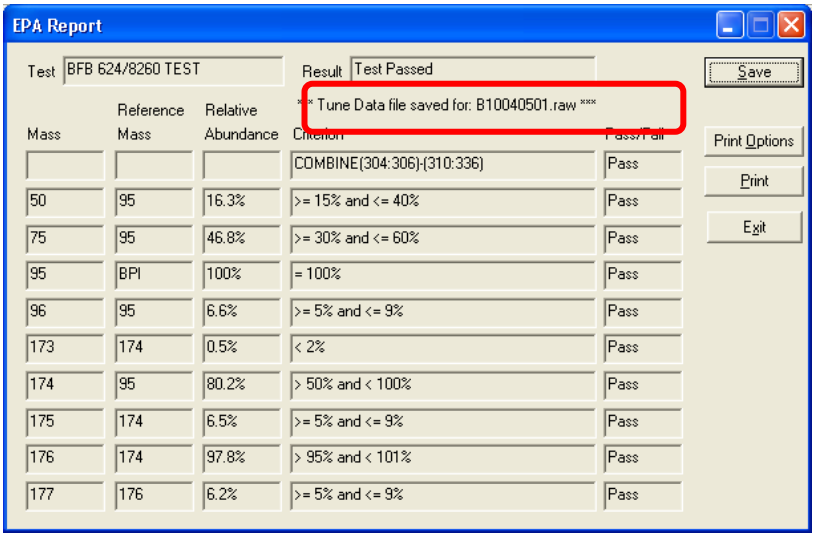
8. Click **OK** and the following **EPA Report** dialog appears:



9. Click **Save** to save the results if the results are acceptable.

   This saves the results to the results database for report processing. This is indicated by the following text appearing:

   *** Tune Data file saved for: B10040501.raw ***



10. Click **Exit** to close the dialog window, then close the chromatogram.

Form 5 Type – Generation Steps

1. Verify Steps 1 through 5 have been followed (pages 91-96 of this tutorial).

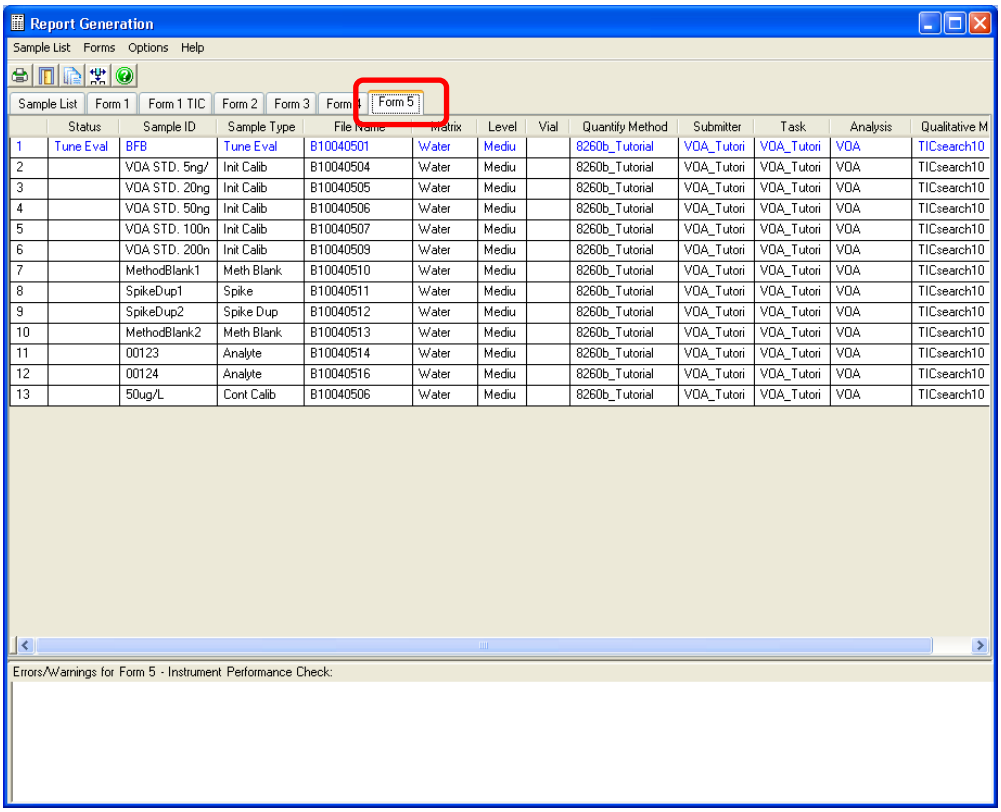
2. Verify the tune evaluation sample has been evaluated and saved within the Chromatogram view.

3. From the Report Generation window click on the **Form 5** tab.

4. From the TurboMass Sample List select **Tools > Environmental Reports.**
   The Select Forms dialog appears.

5. Check the Forms you want to print, including Form 5 and click **Continue**...

6. From the Report Generation window click on the **Form 5** tab.



Form 5 uses the primary data color (blue) to represent the tune evaluation (Tune Eval) sample (BFB/DFTPP). Any additional Tune Eval rows are shown in the disregard color (gray). All other sample type rows will be shown in the summary color (black) and printed on the report.

The Status column indicates which row is currently flagged as the tune evaluation sample. This may be used to distinguish between several Tune Eval rows in the Sample List (although only one can be treated as the primary data set for Form 5) or to flag a row of another Sample Type (e.g. the Continuing Calibration) as being the tune evaluation sample.

7. If the currently assigned Tune Eval file is not the sample you wish to use as the Tune Eval sample, then right-click on the row you want to serve as the Tune Eval.
A popup menu appears.

   From the popup menu that appears select **Assign Tune Evaluation Sample**.

   Select this and assign the selected row as the Tune Evaluation sample for Form 5. The Status column for the selected sample now displays Tune Eval and the row is displayed in the primary data color (blue).

8. To generate a **Form 5** report, click **Forms > Print**, otherwise continue to the next section for information on producing Form 6 – Initial Calibration Data.

Form 6 – Initial Calibration Data

A Form 6 report displays the following information:

- The initial calibration of the instrument based on a multi-point calibration. The average relative response factor (RRF) and the relative standard deviation (RSD). It flags any compound whose response factors or relative standard deviations are outside of the acceptable limits.

- For compounds using a first order curve fit, the correlation coefficient (r^2) is reported. For compounds using a higher order curve fit, the coefficient of determination (CofD) is reported.

Below is an example of the PKI style format for a Form 6 report:

Volatile Organics Initial Calibration Data

Lab Name: Tutorial
Instrument: GC/MS
GC Method: AAVOA.mth
Quantify Method: 8260b_Tutorial_New
Calibration File: 8260b_Tutorial_New
GC Column: Elite-VOA ID: 180 um
Project Path: C:\TurboMass\TUTORIAL_VOA.PRO

Contract: CONTRACT_VOA
Tune File: bbvoa.IPR
MS Method: BBVOA.EXP
Last Updated: December 20, 2012 8:49:05 AM
Last Updated: December 20, 2012 8:49:12 AM
Heated Purge: N

	Concentration	File Name
1	25	B10040504
2	100	B10040505
3	250	B10040506
4	500	B10040507
5	1000	B10040509

Compound	Level 1	Level 2	Level 3	Level 4	Level 5	Avg RRF	% RSD	r²
Dichlorodifluoromethane	0.375	0.377	0.374	0.351	0.327	0.361	6.0	
Vinyl Chloride	0.490	0.529	0.512	0.486	0.482	0.500	4.0	
Bromomethane	0.360	0.343	0.337	0.331	0.357	0.346	3.7	
Trichlorofluoromethane	0.568	0.590	0.568	0.546	0.553	0.565	3.0	
1,1-Dichloroethene	0.406	0.412	0.397	0.386	0.392	0.398	2.7	
Methyl Tert-butyl Ether	1.201	1.225	1.201	1.150	1.197			0.9995
Methylene Chloride	0.579	0.520	0.485	0.460	0.467			0.9998
1,1,-Dichloroethane	0.657	0.666	0.644	0.625	0.642	0.647	2.4	
2,2,-Dichloropropane	0.410	0.437	0.426	0.417	0.418	0.422	2.4	
Chloroform	0.649	0.669	0.651	0.634	0.642	0.649	2.0	
Benzene	0.987	1.019	0.985	0.965	0.985	0.988	2.0	
Trichloroethene	0.229	0.242	0.233	0.233	0.252	0.238	3.8	
Dibromomethane	0.139	0.146	0.144	0.141	0.143	0.143	1.8	
1-Bromo-2-chloroethane	0.311	0.324	0.321	0.316	0.327	0.320	2.0	
Toluene	0.566	0.594	0.578	0.565	0.578	0.576	2.1	
trans-1,3-Dichloropropene	0.262	0.286	0.293	0.304	0.339	0.297	9.5	
1,1,2-Trichloroethane	0.210	0.223	0.222	0.222	0.227	0.221	2.9	
Ethylbenzene	2.850	2.990	2.792	2.746	2.732	2.822	3.7	
p,m-Xylene	2.128	2.278	2.109	2.080	2.070	2.133	4.0	
o-Xylene	2.350	2.531	2.335	2.289	2.293	2.360	4.2	
Isopropylbenzene	2.086	2.255	2.070	2.039	2.028	2.096	4.4	
Bromobenzene	0.753	0.767	0.726	0.703	0.719	0.734	3.6	
n-Propylbenzene	2.245	2.446	2.255	2.206	2.197	2.270	4.5	

December 20, 2012 8:50:17 AM Page 1 of 2

Form 6 Type Report Methods

The following report methods can be generated:

- Form6_Voa.rme
- Form6_SV.rme
- PKI6_Voa.rme
- PKI6_SV.rme

Note: *The Submitter/Task feature can be used to change the report methods selected when generating a Form 6. Refer to* ***Chapter 3 How to Set-up Submitter/Task Data*** *for more information.*

Additional Environmental templates that are acceptable choices and can be produced under similar conditions are as follows:

- **Form6_CSV** – for detailed instructions, refer ***to Form6_CSV – Initial Calibration Data (CSV format)*** on page ***156***.

Valid Sample Types for Form 6 Types

A Form 6 report can only be generated using the following valid sample types:

- Init Calib

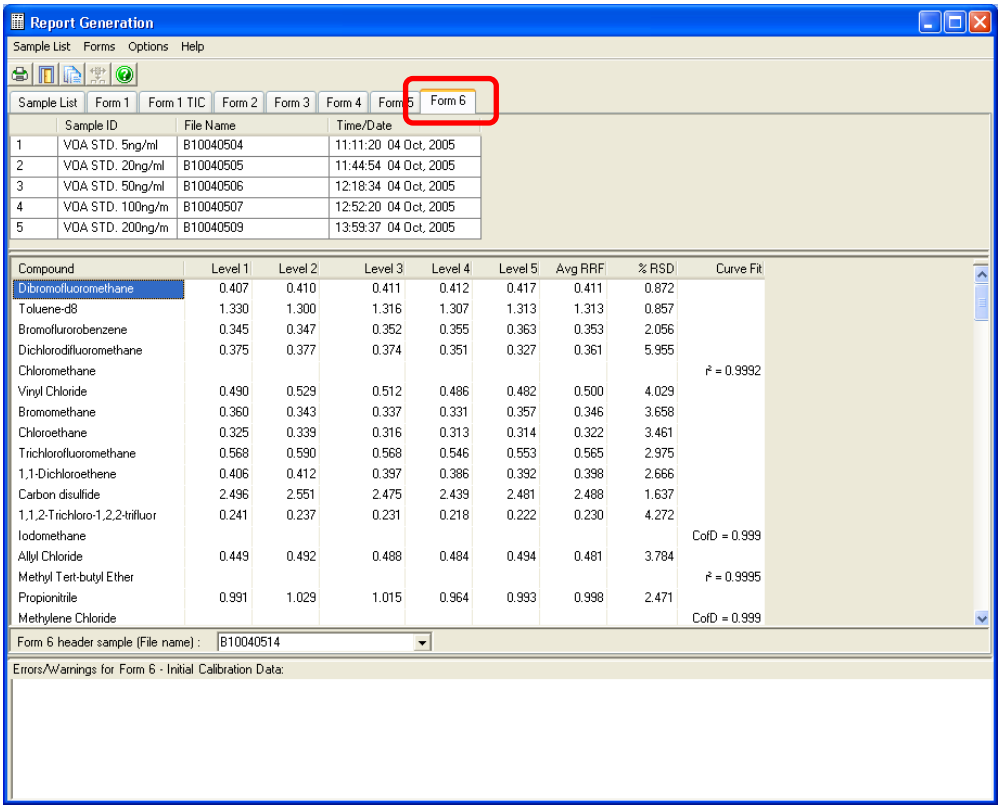
Form 6 Type – Generation Steps

Note: *If you wish to change the number of decimal places that the report template displays for concentration levels, refer to the **Communiqué Template Designer section in the TurboMass Software User's Guide**.*

1. Verify Steps 1 through 4 have been followed (pages 91-95 of this tutorial).

2. From the Report Generation window click on the **Form 6** tab.

Note: *The **Form 6 tab – Initial Calibration Data** has a unique layout among the Forms. It consists of three adjustable–size panes; the calibration sample table, the compound data table and the message window.*



The Form 6 window displays the following information:

Calibration Sample Table: The table lists all the initial calibration data files identified in the calibration file. The table includes columns for:

- Row number
- Sample ID
- File Name (the raw file name)
- Time and date of injection of the standard sample

The data for this table is taken from the first Calibration file referenced by the selected lines of the Sample List.

When a row is selected, the arrow keys can be used to change the selected row and cause the list to scroll. Column widths can be changed in the standard way, by dragging the header divider.

Compound Data Table: The data for this table are also taken from the Calibration file. The table lists all compounds identified in the calibration file and displays the RRF value calculated for each concentration level, plus the average RRF and the percentage relative standard deviation for the RRF values. For compounds using a linear or higher-order curve fit calibration in place of Average RRF, the goodness of fit value (r^2 correlation coefficient for first order, coefficient of determination (CofD) for higher orders).

RRF values flagged as below the minimum required value for that compound (as defined in the Quantify method) will be displayed in red. Similarly if the average RRF value is below the minimum required value or the %RSD value for a compound is greater than the specified maximum value for that compound, they will be displayed in red.

Form 6 Header Sample: This drop-down list enables you to indicate the source of the information that will appear in the Form 6 header. Note that this information is not related to the calibration standard samples but rather the samples analyzed utilizing the calibration. This means that in general the source of the header information will be one of the Analyte samples from the Sample List tab.

The drop-down list contains the file names of all the checked samples from the Sample List tab.

The initial selection will be the first checked Analyte (or Analyte Dup) row from the Sample List tab.

Message Pane: This displays errors and warnings associated with Form 6, in a similar manner to that for all other Forms.

3. To generate a **Form 6** report, click **Forms > Print**, otherwise continue to the next section for information on producing Form 7 – Continuing Calibration Check.

Form 7 – Continuing Calibration Check

Form 7 compares the daily calibration or continuing calibration data against the initial calibration data of the instrument. Compounds whose response factors fall below a minimum value are flagged. It flags any compound whose % difference or % drift deviates outside of the acceptable limits.

Below is an example of the PKI style format for a Form 7 report:

Volatile Continuing Calibration Check

Lab Name:	Tutorial	Contract:	CONTRACT_VOA
File Name:	C:\TurboMass\TUTORIAL_VOA.PRO\Data\B10040506.raw		
Sample ID:	VOA STD. 50ng/ml		
Description:	VOA		
Inject Date/Time:	October 04, 2005 12:18:34 PM	Vial:	0
Instrument:	GC/MS	Tune File:	bbvoa.IPR
GC Method:	AAVOA.mth	MS Method:	BBVOA.EXP
Quantify Method:	8260b_Tutorial_New	Last Updated:	December 20, 2012 8:49:05 AM
Calibration File:	8260b_Tutorial_New	Last Updated:	December 20, 2012 8:49:12 AM
GC Column:	Elite-VOA ID: 180 um	Heated Purge:	N
Init. Calib. Date(s):	Oct. 04, 05 Oct. 04, 05	Init. Calib. Times:	6:11:20 AM 8:59:37 AM

Compound	Avg RRF	RRF Mid	Min RRF	%D	Max %D
Dichlorodifluoromethane	0.361	0.374		3.7	20.0
Vinyl Chloride	0.500	0.512		2.4	20.0
Bromomethane	0.346	0.337		-2.4	20.0
Trichlorofluoromethane	0.565	0.568		0.5	20.0
1,1-Dichloroethene	0.398	0.397		-0.4	20.0
1,1,-Dichloroethane	0.647	0.644	0.100	-0.5	20.0
2,2,-Dichloropropane	0.422	0.426		1.1	20.0
Chloroform	0.649	0.651		0.3	20.0
Benzene	0.988	0.985		-0.3	20.0
Trichloroethene	0.238	0.233		-2.0	20.0
Dibromomethane	0.143	0.144		1.0	20.0
1-Bromo-2-chloroethane	0.320	0.321		0.3	20.0
Toluene	0.576	0.578		0.3	20.0
trans-1,3-Dichloropropene	0.297	0.293		-1.2	20.0
1,1,2-Trichloroethane	0.221	0.222		0.5	20.0
Ethylbenzene	2.822	2.792		-1.1	20.0
p,m-Xylene	2.133	2.109		-1.1	20.0
o-Xylene	2.360	2.335		-1.0	20.0
Isopropylbenzene	2.096	2.070		-1.2	20.0
Bromobenzene	0.734	0.726		-1.0	20.0
n-Propylbenzene	2.270	2.255		-0.6	20.0
1,3,5-trimethylbenzene	1.702	1.703		0.1	20.0
sec-Butylbenzene	1.741	1.719		-1.2	20.0
1,3-Dichlorobenzene	1.047	1.031		-1.5	20.0
1,4-Dichlorobenzene	1.102	1.084		-1.6	20.0
n-Butylbenzene	1.229	1.231		0.2	20.0
1,2-Dichlorobenzene	1.026	1.017		-0.9	20.0
Hexachlorobutadiene	0.192	0.181		-5.5	20.0

Compound	Avg RRF	RRF Mid	Min RRF	%D	Max %D
Dibromofluoromethane	0.411	0.411		-0.1	20.0
Toluene-d8	1.847	1.854		0.4	20.0
Bromoflurorobenzene	0.966	0.956		-1.0	20.0

Number of compounds with RRF below minimum: 0
Number of compounds with % Difference exceeding maximum: 0

Volatile Continuing Calibration Check

Compound	Curve Fit Type	Drift (%)	Max Drift
Methyl Tert-butyl Ether	Linear	1.4	20.0
Methylene Chloride	Linear	1.6	20.0

Number of compounds with % Drift exceeding maximum: 0

Form 7 Type Report Methods

The following report methods can be generated:

- Form7_Voa.rme
- Form7_SV.rme
- PKI7_Voa.rme
- PKI7_SV.rme
- ICV_Voa.rme
- ICV_SV.rme

Note: *The Submitter/Task feature can be used to change the report methods selected when generating a Form 7. Refer to* ***Chapter 3 How to Set-up Submitter/ Task Data*** *for more information.*

Valid Sample Types for Form 7 Types

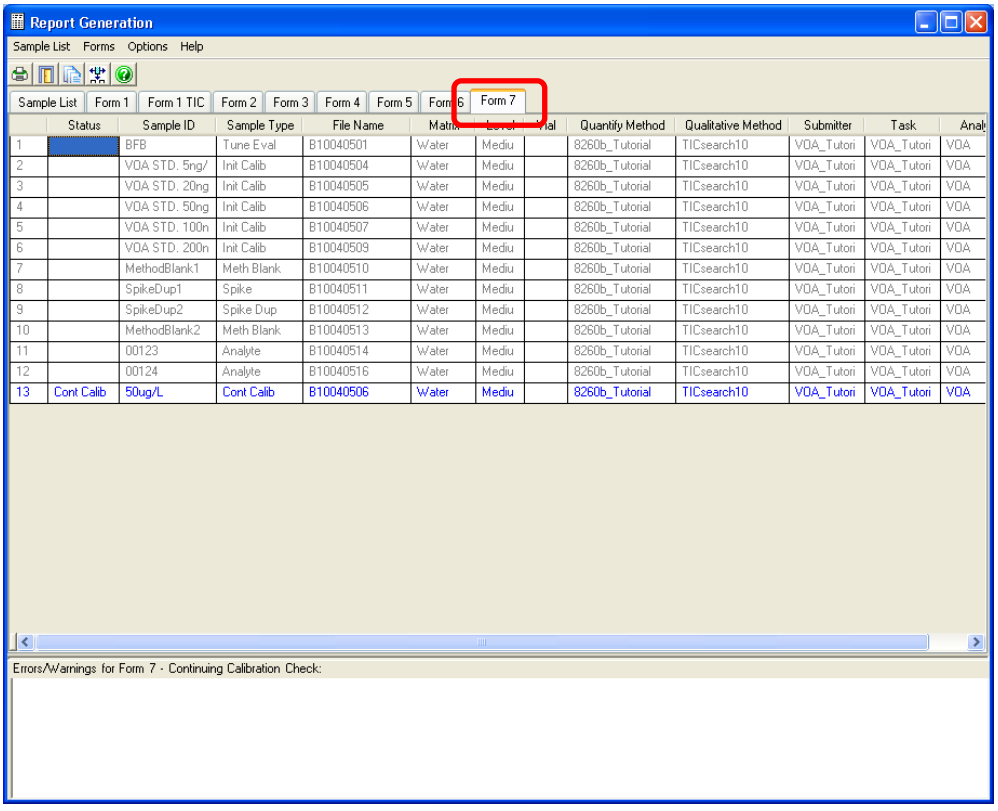
A Form 7 report can only be generated using the following valid sample types:

- Cont Calib

Form 7 Type – Generation Steps

1. Verify Steps 1 through 5 have been followed (pages 91-96 of this tutorial).

2. From the Report Generation window click on the **Form 7** tab.



Note: *If you change the Min RRF value in the Quantify method you must reprocess (recalibrate) the Sample List for this new value to display in the report.*

Form 7 uses a single data file in the primary data color (blue), for the continuing calibration (Cont Calib) sample. All other rows will be shown in the disregard color (gray) and are not used for the report.

The **Status** column indicates which row is currently flagged as the continuing calibration sample. This may be used to distinguish between several Cont Calib rows in the Sample List (although only one can be used for Form 7) or to flag a row of another Sample Type as being the continuing calibration sample.

3. If you want to change which sample is marked as the continuing calibration file you may do so by right mouse clicking on a row in the Form 7 Sample List view and clicking **Assign Continuing Calibration** from the pop up menu that appears.

   This assigns the selected row as the Continuing Calibration (Cont Calib) sample for Form 7. In addition, the Status column for the selected row now displays Cont Calib and row is displayed is now displayed in the primary data color (blue).

4. To generate a **Form 7** report, click **Forms > Print**, otherwise continue to the next section for information on producing Form 8 – Internal Standard Area and RT Summary.

Form 8 – Internal Standard Area and RT Summary

This form reports the retention time and area reproducibility for the internal standards with comparisons to data files in the group of samples. Flags are generated for internal standard areas and retention times that fall outside of the upper and lower limits.

Below is an example of the of PKI style format for a Form 8 report:

Volatile Internal Standard Area and RT Summary

Lab Name:	Tutorial	Contract:	CONTRACT_VOA
File Name (Std):	C:\TurboMass\TUTORIAL_VOA.PRO\Data\B10040506.raw		
Sample ID:	VOA STD. 50ng/ml		
Description:	VOA		
Inject Date/Time:	October 04, 2005 12:18:34 PM	Vial:	0
Instrument:	GC/MS	Tune File:	bbvoa.IPR
GC Method:	AAVOA.mth	MS Method:	BBVOA.EXP
Quantify Method:	8260b_Tutorial_New	Last Updated:	December 20, 2012 8:49:05 AM
Calibration File:	8260b_Tutorial_New	Last Updated:	December 20, 2012 8:49:12 AM
GC Column:	Elite-VOA ID: 180 um	Heated Purge:	N

	IS1 (PFB) Area	RT	IS2 (DFB) Area	RT	IS3 (DCB) Area	RT
12 HOUR STD	2592892	6.30	4215830	7.43	1345867	16.19
UPPER LIMIT	5185785	6.80	8431661	7.93	2691734	16.69
LOWER LIMIT	1296446	5.80	2107915	6.93	672934	15.69

	File Name	Area #	RT #	Area #	RT #	Area #	RT #
1	B10040510	2489168	6.31	4015841	7.43	1176684	16.20
2	B10040511	2407227	6.31	3890673	7.43	1210481	16.20
3	B10040512	2358682	6.31	3806338	7.43	1196931	16.19
4	B10040513	2348216	6.31	3825432	7.43	1086493	16.20
5	B10040514	2277837	6.30	3622894	7.43	1045571	16.20
6	B10040516	2150456	6.31	3488183	7.43	973632	16.19

IS1 (PFB) = Pentafluorobenzene
IS2 (DFB) = 1,4-Difluorobenzene
IS3 (DCB) = 1,4-Dichlorobenzene-D4

AREA UPPER LIMIT = + 100% of internal standard area
AREA LOWER LIMIT = - 50% of internal standard area
RT UPPER LIMIT = + 0.50 minutes of internal standard RT
RT LOWER LIMIT = - 0.50 minutes of internal standard RT

Column used to flag values outside QC limits with an asterisk
* Values outside QC limits

Form 8 Type Report Methods

The following report methods can be generated:

- Form8_Voa.rme
- Form8_SV.rme
- PKI8_Voa.rme
- PKI8_SV.rme

Note: *The Submitter/Task feature can be used to change the report methods selected when generating a Form 8. Refer to* ***Chapter 3 How to Set-up Submitter/Task Data*** *for more information.*

Valid Sample Types for Form 8 Types

One of the following types ***must*** be selected for the file to be used as a comparison to all associated samples:

- **Cont Calib** – This type is selected to define the file used for comparison to the ISTD's in the associated samples
- **Init Calib** – This type is selected to define the file used for comparison to the ISTD's in the associated samples

A Form 8 report can only be generated using the following valid sample types:

- Analyte
- Analyte Dup
- Method Blank
- Spike
- Spike Dup
- Dilution
- Re-extract
- Lab Control

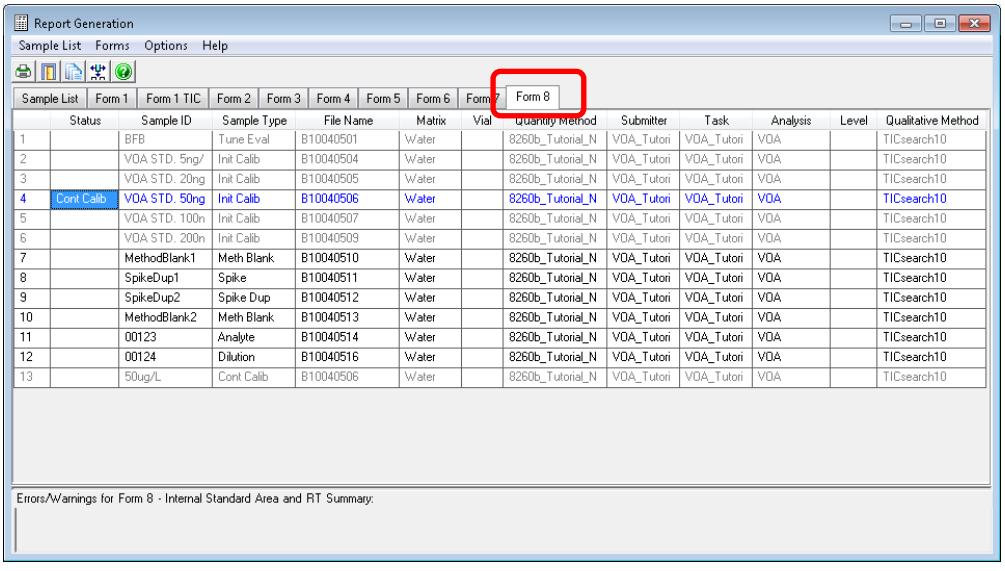
Critical Quantify Method Information for Form 8 Type Reports

Note: *Detailed set-up of the Quantify method should be done. For more information refer to* ***Chapter 2 How to Build a Quantification Method****.*

- At least one compound must be identified as an **Internal Reference**.

Form 8 Type – Generation Steps

1. Verify Steps 1 through 5 have been followed (pages 91-96 of this tutorial).
2. From the Report Generation window, click on the **Form 8** tab to display the Form 8 information.



	Status	Sample ID	Sample Type	File Name	Matrix	Vial	Quantify Method	Submitter	Task	Analysis	Level	Qualitative Method
1		BFB	Tune Eval	B10040501	Water		8260b_Tutorial_N	VOA_Tutori	VOA_Tutori	VOA		TICsearch10
2		VOA STD. 5ng/	Init Calib	B10040504	Water		8260b_Tutorial_N	VOA_Tutori	VOA_Tutori	VOA		TICsearch10
3		VOA STD. 20ng	Init Calib	B10040505	Water		8260b_Tutorial_N	VOA_Tutori	VOA_Tutori	VOA		TICsearch10
4	Cont Calib	VOA STD. 50ng	Init Calib	B10040506	Water		8260b_Tutorial_N	VOA_Tutori	VOA_Tutori	VOA		TICsearch10
5		VOA STD. 100n	Init Calib	B10040507	Water		8260b_Tutorial_N	VOA_Tutori	VOA_Tutori	VOA		TICsearch10
6		VOA STD. 200n	Init Calib	B10040509	Water		8260b_Tutorial_N	VOA_Tutori	VOA_Tutori	VOA		TICsearch10
7		MethodBlank1	Meth Blank	B10040510	Water		8260b_Tutorial_N	VOA_Tutori	VOA_Tutori	VOA		TICsearch10
8		SpikeDup1	Spike	B10040511	Water		8260b_Tutorial_N	VOA_Tutori	VOA_Tutori	VOA		TICsearch10
9		SpikeDup2	Spike Dup	B10040512	Water		8260b_Tutorial_N	VOA_Tutori	VOA_Tutori	VOA		TICsearch10
10		MethodBlank2	Meth Blank	B10040513	Water		8260b_Tutorial_N	VOA_Tutori	VOA_Tutori	VOA		TICsearch10
11		00123	Analyte	B10040514	Water		8260b_Tutorial_N	VOA_Tutori	VOA_Tutori	VOA		TICsearch10
12		00124	Dilution	B10040516	Water		8260b_Tutorial_N	VOA_Tutori	VOA_Tutori	VOA		TICsearch10
13		50ug/L	Cont Calib	B10040506	Water		8260b_Tutorial_N	VOA_Tutori	VOA_Tutori	VOA		TICsearch10

Form 8 uses the primary data color (blue) for the continuing calibration (**Cont Calib**) sample. Any **Tune Eval**, **Init Calib** or additional **Cont Calib** rows will be shown in the disregard color (gray) since they are not included in Form 8.

Note: *An initial Calibration file may be used as the comparison file by selecting it as "Assign Mid-Level Standard" as described below.*

The **Status** column indicates which row is currently flagged as the continuing calibration sample. This may be used to distinguish between several **Cont Calib** rows in the Sample List (only one can be treated as the primary data set for Form 8) or to flag a row of another Sample Type as being the continuing calibration sample.

3. You can alter the Method Blank assignments without going back to the Sample List by right mouse clicking on a row in the Form 8 Sample List and selecting **Assign Mid–Level Standard** from the pop up menu that appears.
   The selected row is assigned as the Mid–Level Standard (labelled as type **Cont Calib**) sample for Form 8. Cont Calib is now displayed in blue in the Status column for that row.

4. To generate a **Form 8** report, click **Forms > Print.**

PKIEnvQuant Template Form Generation

- Reports the concentrations of positively identified target compounds and their reporting limits in the sample. Internal standard amounts and surrogate recoveries are also reported. Compounds that are not detected do not display on this report.
- Sample results present below the Reporting Limit/Reporting Threshold will display a **<RL** in the concentration field.
- Sample results equal to or above the Reporting Limit/Reporting Threshold will display the sample concentration.
- Manual Integrations are noted with an "X" in the MAN column.

Note: *Proper Reporting Limit/Reporting Threshold setup along with detailed calculations are located in* ***Appendix 1 Environmental Reporting Calculations*** *on page* ***169****.*

Below is an example of the PKIEnvQuant template/report:

Quantitation Report

Lab Name:	Tutorial	Contract:	CONTRACT_VOA
File Name:	C:\TurboMass\TUTORIAL_VOA.PRO\Data\B10040514.raw		
Sample ID:	00123	Operator:	PerkinElmer
Description:	VOA		
Inject Date/Time:	October 04, 2005 4:47:54 PM	Vial:	0
Instrument:	GC/MS	Tune File:	bbvoa.IPR
GC Method:	AAVOA.mth	MS Method:	BBVOA.EXP
Quantify Method:	8260b_Tutorial_New	Last Updated:	December 20, 2012 3:39:43 PM
Calibration File:	8260b_Tutorial_New	Last Updated:	December 18, 2012 3:58:03 PM
GC Column:	Elite-VOA ID: 180 um		
Matrix:	Water	Date Received:	October 04, 2005
Sample wt/vol:	5.000 mL	Dilution Factor:	1.0

	Internal Standards	RT	m/z	Area	ng	RT Dev	Man
1	Pentafluorobenzene	6.30	168	2,277,837.2	250.00	0.00	
2	1,4-Difluorobenzene	7.43	114	3,622,894.0	250.00	0.00	
3	1,4-Dichlorobenzene-D4	16.20	152	1,045,571.4	250.00	0.01	

	Surrogate Compounds	RT	m/z	Area	%Recovery	RT Dev	Man
4	Dibromofluoromethane	6.16	113	919,855.6	98	0.00	
5	Toluene-d8	9.76	98	4,089,410.8	97	0.00	
6	Bromoflurorobenzene	14.18	174	1,083,580.9	107	0.00	

	Target Compounds	RT	m/z	Area	ppb	Rpt Limit	Man
7	Dichlorodifluoromethane	1.27	85	2,887.1	<RL	5.00	
8	Vinyl Chloride	1.53	62	392,281.0	17.23	5.00	
9	Bromomethane	1.85	94	3,432.9	<RL	5.00	
10	Trichlorofluoromethane	2.19	101	2,049.7	<RL	5.00	
11	1,1-Dichloroethene	2.73	96	5,692.1	<RL	5.00	
12	Methyl Tert-butyl Ether	3.80	73	2,796.0	<RL	5.00	
13	Methylene Chloride	3.35	84	12,638.9	<RL	5.00	
14	1,1,-Dichloroethane	4.42	63	148,821.4	5.05	5.00	
15	2,2,-Dichloropropane	5.42	77	1,082.4	<RL	5.00	
16	Chloroform	5.91	83	258,118.6	8.73	5.00	
17	Benzene	6.74	78	115,369.7	<RL	5.00	
18	Trichloroethene	7.80	95	18,846.5	<RL	5.00	
19	Dibromomethane	8.32	93	751.3	<RL	5.00	
20	1-Bromo-2-chloroethane	9.06	63	6,614.5	<RL	5.00	
21	Toluene	9.87	92	21,659.3	<RL	5.00	
22	trans-1,3-Dichloropropene	10.29	75	2,597.1	<RL	5.00	

PKIEnvQuant Report Method

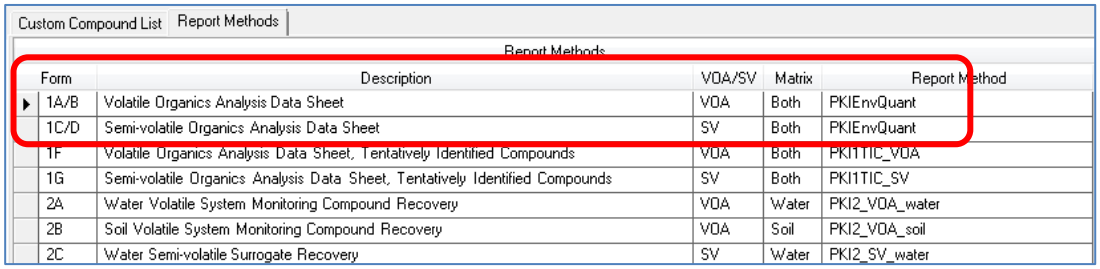
The PKIEnvQuant.rme report method is available in the following directories:

- C:\TurboMass\Tutorial_VOA.pro\MethDB\
- C:\TurboMass\Tutorial_SVOA.pro\MethDB\

Note: *Refer to section on Copy and Paste *.RME files to your new Project on page 87 in this manual to obtain access to this report method.*

Note: *The Submitter/Task feature can be used to change the report methods selected when generating a Form 1 type. For more information, refer to* ***Chapter 3 How to Set-up Submitter/Task Data****.*

- To generate a PKIEnvQuant report, the report method must be placed in the **Form 1A/B row** for **volatile matrices** and in the **Form 1C/D row for semi-volatile** fractions in the **Submitter/Task** list created.



Form	Description	VOA/SV	Matrix	Report Method
1A/B	Volatile Organics Analysis Data Sheet	VOA	Both	PKIEnvQuant
1C/D	Semi-volatile Organics Analysis Data Sheet	SV	Both	PKIEnvQuant
1F	Volatile Organics Analysis Data Sheet, Tentatively Identified Compounds	VOA	Both	PKI1TIC_VOA
1G	Semi-volatile Organics Analysis Data Sheet, Tentatively Identified Compounds	SV	Both	PKI1TIC_SV
2A	Water Volatile System Monitoring Compound Recovery	VOA	Water	PKI2_VOA_water
2B	Soil Volatile System Monitoring Compound Recovery	VOA	Soil	PKI2_VOA_soil
2C	Water Semi-volatile Surrogate Recovery	SV	Water	PKI2_SV_water

Valid Sample Types for PKIEnvQuant

A Form 1 can be generated using the following valid sample types only:

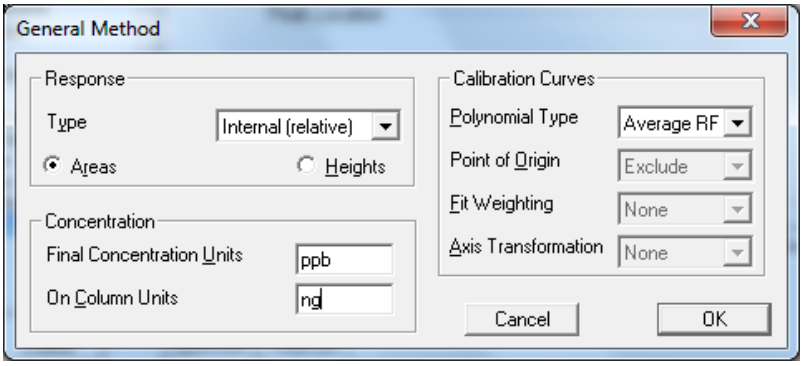
- Analyte
- Analyte Dup
- Method Blank
- Spike
- Spike Dup
- Dilution
- Re-extract
- Lab Control

Critical Quantify Method Information for Form 1 Type Reports

Note: *Detailed set-up of the Quantify method should be done. Refer to* ***Chapter 2 How to Build a Quantification Method for Reporting*** *as well as* ***Appendix 1 Environmental Reporting Calculations*** *on page* ***169*** *for details on correctly entering in the values.*

To generate a Form 1 type report the following parameters in your Quantification Method must be appropriately defined:

- **General Method parameters** – you must specify both the **Final Concentration Units** (value displayed on the form) and the **On Column Units** (value displayed in the Quantify\View Results summary window).



- **Reporting Threshold** – you must define an on column (ng) value *if Submitter/Task will not be used*. This field is applied for both water and soil matrices.

The following Environmental Parameters must be accurately set:

- **Surrogate/Spike (Amount added - ng)** – This value is used to calculate the recovery.

- **IMPT** – Refer to the section Surrogate Recovery Calculations in **Appendix 1 Environmental Reporting Calculations** on page ***169*** for details on how to derive this value.

Critical Submitter/Task Information for PKIEnvQuant Generation

Note: *Proper set-up of the Submitter/Task should be done. For more information refer to* ***Chapter 3 How to Set-up Submitter/Task Data as*** *well as* ***Appendix 1 Environmental Reporting Calculations*** *on page* ***169*** *for details on correctly entering in the values.*

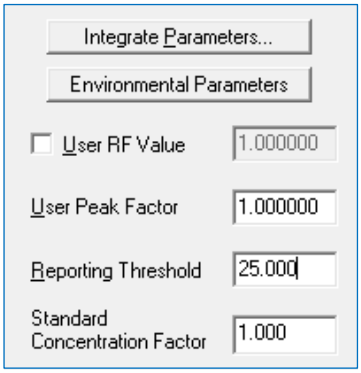
To properly generate this type of report the following Submitter/Task Information should be appropriately defined:

- If a Submitter/Task is used, the surrogate compounds must be listed and selected on in the custom compound list to display on the Form 2.
- Reporting Limits – must have an on column (ng) value defined if Reporting Threshold is not to be used. Fields are available for both water and soil matrices on a per compound basis.
- If a Submitter/Task list is used, it automatically overrides any value placed in the Reporting threshold field within the Quantify Method for a particular compound.

Difference between Reporting Threshold versus Reporting Limit

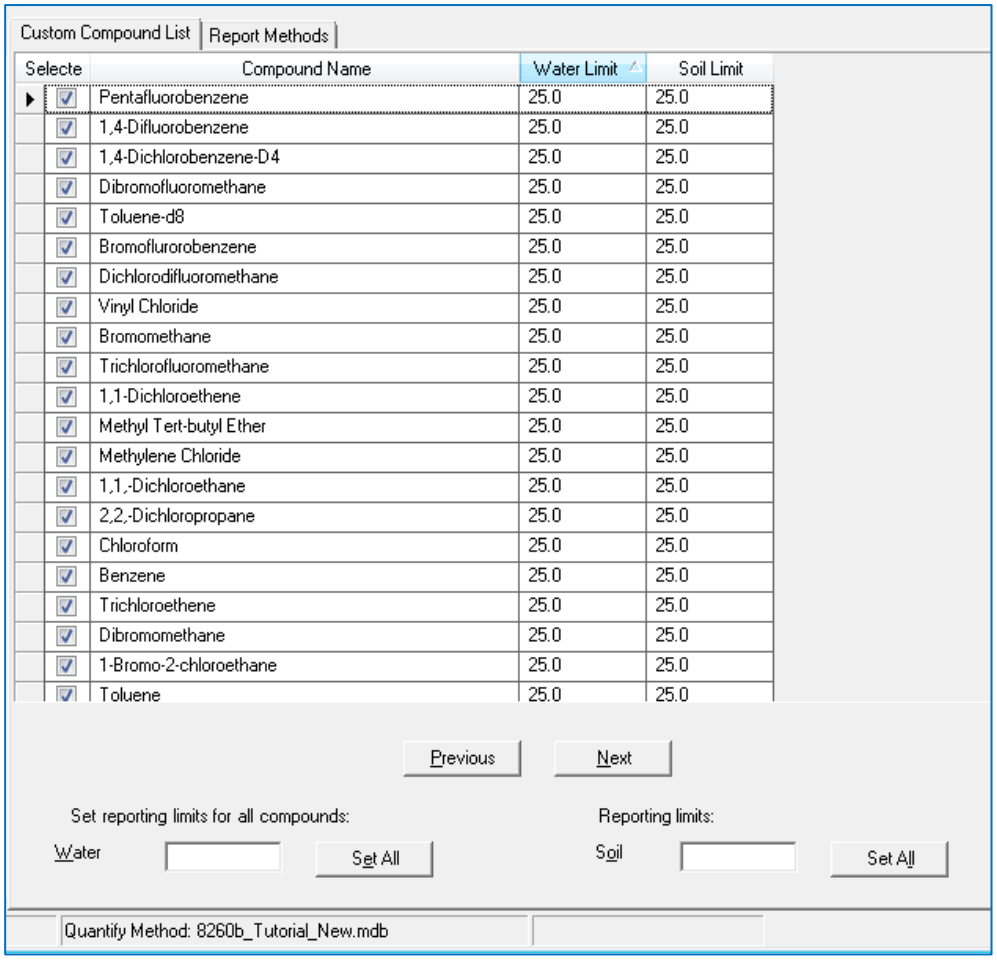
Reporting Threshold and Reporting Limit are essentially the same entity for environmental calculations. The same numerical value defined for a particular compound can be placed in either field.

Reporting Threshold – Designated in the Quantify Method



The **Reporting Threshold** only gives you the ability to enter in one value per compound. This can be used in situations where only one matrix type is analyzed.

Reporting Limit – Designated in Submitter/Task



The **Reporting Limits** parameter gives the ability to enter in two values – one for water matrix and one for soil matrix.

Calculations

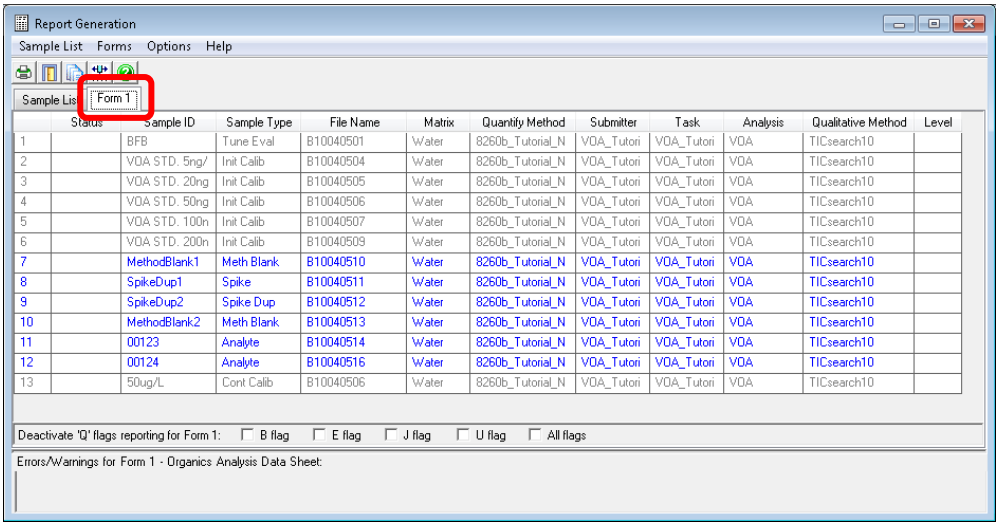
Note: *Refer to* ***Appendix 1 Environmental Reporting Calculations*** *on page* ***169*** *for information on how surrogate recoveries, target concentrations and target reporting limits are calculated based on Analysis type (Voa/Svoa), Matrix (Water/Soil) and concentration Level (low/medium).*

PKIEnvQuant Type – Generation Steps

1. Verify Steps 1 through 5 have been followed (pages 91-96 of this tutorial).

2. Verify the **Submitter/Task** names containing the **PKIEnvQuant** report method are listed in the sample list fields.

3. From the Report Generation window, click on the **Form 1** tab to display the Form 1 information.



Note: *The Deactivate "Q" flags does not apply to the PKIEnvQuant form*

4. The samples displayed in blue will be generated for the displayed form.

   In this example, assume you would like to print reports for the analyte samples. To do so you would uncheck samples 7, 8, 9, and 10 from the Sample List tab.

5. To generate the PKIEnvQuant report, click **Forms > Print**

LCS_Voa and LCS_SV Template Form Generation

An LCS_Voa and LCS_SV template is used to report the recoveries of specific analytes defined as ***spike*** compounds for the volatile analysis or semi-volatile analysis. It flags any compound whose recovery is outside of the acceptable limits.

Note: *If all compounds in a Quantify Method are designated as a spike, the concentration units will not print on the form. At least one compound is required to be designated as a* ***Target*** *for the concentration units to be listed on the form.*

Below is an example of an LCS_Voa report:

Volatile Organics Laboratory Control Sample

Lab Name:	Tutorial	Contract:	CONTRACT_VOA
File Name:	C:\TurboMass\TUTORIAL_VOA.PRO\Data\B10040512.raw		
Sample ID:	SpikeDup2		
Description:	VOA		
Inject Date/Time:	October 04, 2005 3:40:36 PM	Vial:	0
Instrument:	GC/MS	Tune File:	bbvoa.IPR
GC Method:	AAVOA.mth	MS Method:	BBVOA.EXP
Quantify Method:	8260b_Tutorial_New	Last Updated:	December 20, 2012 3:39:43 PM
Calibration File:	8260b_Tutorial_New	Last Updated:	December 18, 2012 3:58:03 PM
GC Column:	Elite-VOA ID: 180 um		
Matrix:	Water		
Sample wt/vol:	5.000 mL		

Compound	Sample Conc. (ppb)	Spike Added (ppb)	Spike Rec. (%)	QC Limits Rec.	(%)
1,1-Dichloroethene	21.25	20.00	106.23	70	130
Benzene	20.86	20.00	104.28	70	130
Trichloroethene	21.07	20.00	105.37	70	130
Toluene	20.70	20.00	103.51	70	130

Report Methods for LCS_Voa and LCS_SV

The LCS_Voa.rme and LCS_SV.rme report methods are available in directories:

- C:\TurboMass\Tutorial_VOA.pro\MethDB\
- C:\TurboMass\Tutorial_SVOA.pro\MethDB\

Note: *Refer to the section titled* ***Copy and Paste *.RME files to your new Project*** *on page* ***87*** *of this manual to obtain access to this report method.*

Valid Sample Type for LCS_Voa and LCS_SV

A LCS_Voa and LCS_SV report can be generated using the following valid sample types only:

- Lab Control Sample

Critical Quantify Method Information for Form 3 Type Reports

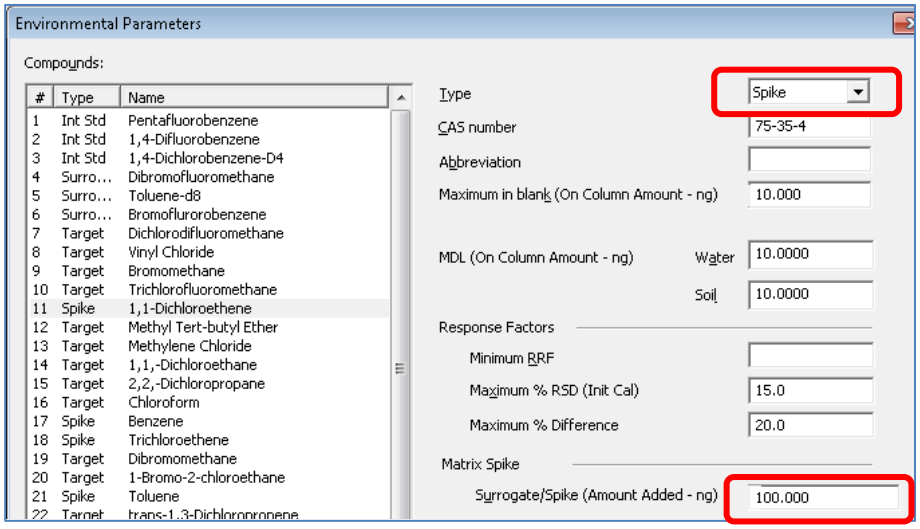
Note: *Detailed set-up of the Quantify method should be done. Refer to* ***Chapter 2 How to Build a Quantification Method*** *as well as* ***Appendix 1 Environmental Reporting Calculations*** *on page* ***169****.for details on correctly entering in the values.*

To generate Form 3 type reports the following parameters in your Quantification Method must be appropriately defined:

- The compound type must be designated as **Spike** for all compounds that are to appear on the spike recovery report.

In addition, the following Environmental Parameters in your Quantification Method must be appropriately defined:

- **Surrogate/Spike (Amount added - ng)** –- This value is used to calculate the recovery.



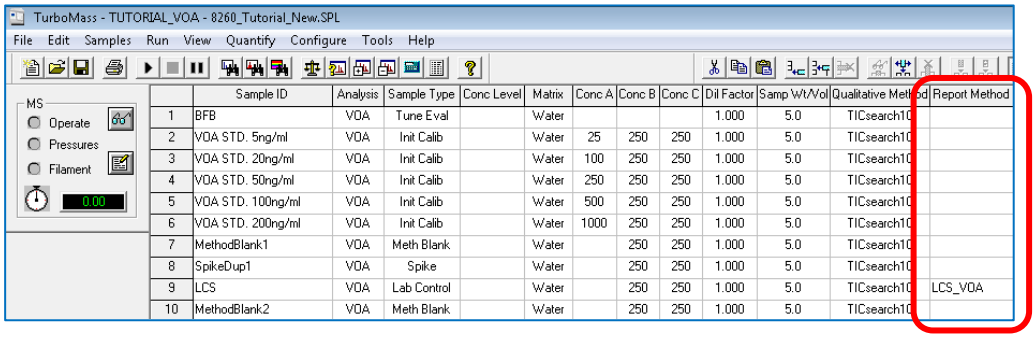
- **IMPT** – Refer to the section **Spike Recovery Calculations** in **Appendix 1 Environmental Reporting Calculations** on pages ***203-206*** for details on how to derive this value.

Note: ***If all compounds in a quantify method are designated as a spike, the concentration units will not print*** *on the form. At least one compound is required to be designated as a target for the concentration units to be listed on the form.*

LCS_Voa or LCS_SV – Generation Steps

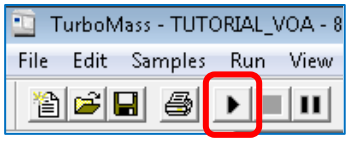
1. Verify Steps 1 through 5 have been followed (pages 91-96 of this tutorial).

2. Select the Report Method (LCS_Voa or LCS_SV) on the appropriate sample line.



	Sample ID	Analysis	Sample Type	Conc Level	Matrix	Conc A	Conc B	Conc C	Dil Factor	Samp Wt/Vol	Qualitative Method	Report Method
1	BFB	VOA	Tune Eval		Water				1.000	5.0	TICsearch10	
2	VOA STD. 5ng/ml	VOA	Init Calib		Water	25	250	250	1.000	5.0	TICsearch10	
3	VOA STD. 20ng/ml	VOA	Init Calib		Water	100	250	250	1.000	5.0	TICsearch10	
4	VOA STD. 50ng/ml	VOA	Init Calib		Water	250	250	250	1.000	5.0	TICsearch10	
5	VOA STD. 100ng/ml	VOA	Init Calib		Water	500	250	250	1.000	5.0	TICsearch10	
6	VOA STD. 200ng/ml	VOA	Init Calib		Water	1000	250	250	1.000	5.0	TICsearch10	
7	MethodBlank1	VOA	Meth Blank		Water		250	250	1.000	5.0	TICsearch10	
8	SpikeDup1	VOA	Spike		Water		250	250	1.000	5.0	TICsearch10	
9	LCS	VOA	Lab Control		Water		250	250	1.000	5.0	TICsearch10	LCS_VOA
10	MethodBlank2	VOA	Meth Blank		Water		250	250	1.000	5.0	TICsearch10	

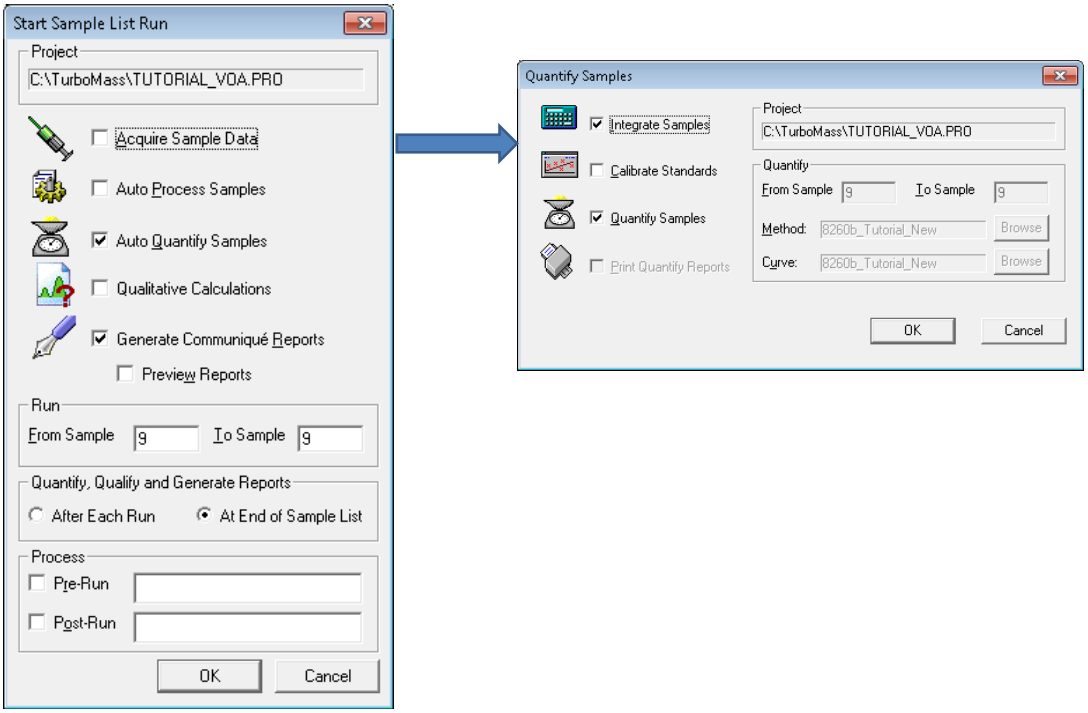
3. From the sample list window, **highlight the row** to be reported, then select the **Start Run** button.



4. Setup the Start Sample List Run window for Quantitative processing and to generate Communiqué reports.

5. Verify the **From Sample** and **To Sample** are correct.

6. Select **Integrate** (if samples were not previously processed), and **Quantify** Samples.



7. Select **OK** to generate the report.

Form1_CSV – Organics Analysis Data Sheet

- Reports the concentrations of Targets and Tentatively identified compounds (TICs) in a comma-separate format (CSV) file suitable for transferring data to LABWORKS LIMS.
- Reports Header and sample list information
- The "Q" field Sample displays the flags under the following conditions:
  - "U" flags are displayed when sample results are present below the MDL
  - "J" flags are displayed when Target samples results are present above or equal to the MDL and less than the RL (reporting threshold/reporting limit)
  - "J" flags are displayed when TIC results are identified
  - "E" flags are displayed when samples results are above the highest point in the calibration curve.
  - "N" flags are displayed for all TICs (Tentatively identified compounds)
  - No flag is displayed when sample results are present above or equal to the RL

Note: *Proper MDL and Reporting limit setup along with detailed calculations are located in* ***Appendix 1 Environmental Reporting Calculations*** *on page* ***169****.*

Below is an example of a Form1_CSV report:

	A	B	C	D
1	<//Start_Title//>			
2	COMBINED FORMS 1A/1B 1C/1D 1F/1G			
3	COMBINED VOLATILE AND SEMIVOLATILE ORGANICS ANALYSIS DATA SHEE			
4	COMBINED VOLATILE AND SEMIVOLATILE ORGANICS TENTATIVELY IDENTI			
5	<//End_Title//>			
6	<//Start_Misc_Info//>			
7	Sample Type:	Spike Dup		
8	Analysis:	VOA		
9	Submitter:	Forms		
10	Task:	Form1CSV		
11	File Text:	VOA		
12	Job:	PKI Tutorial		
13	User:	PerkinElmer		
14	Conditions:	123		
15	Surrogate Lot:	Surr_01		
16	ISTD Lot:	ISTD_01		
17	Vol Surrogate:	5		
18	Heated Purge:	N		
19	Analysis Timestamp:	October 04, 2005 04:47:54 PM Eastern D		
20	Project Directory:	C:\TurboMass\TUTORIAL_VOA.PRO		
21	<//End_Misc_Info//>			
22	<//Start_Form_1_Header//>			
23	EPA SAMPLE NO.	EPA_VOA_13		
24	Lab Name:	Tutorial		
25	Lab Code:	VOA_Testing		
26	Contract:	CONTRACT_VOA		
27	Case No.:	CASE_VOA		
28	SAS No.:	SAS_01		
29	SDG No.:	SDG_01		
30	Lab Sample ID:	123		
31	Lab File ID:	B10040514		
32	Date Received:	4-Oct-05		
33	Date Analyzed:	4-Oct-05		
34	Matrix: (soil/water)	Water		
35	Dilution Factor:	1		
36	Sample wt/vol:	5		

B10040514 122112 024935

	A	B	C	D	E	F	G
56	<//Start_Target_Compounds//>						
57	COMPOUND NAME	CAS NO.	R.T.	AREA	ADJ. CONC.	REV. FIT.	Q
58	Dichlorodifluoromethane	75-71-8	1.27	2887	0.18	224	U
59	Vinyl Chloride	75-01-4	1.53	392281	17.23	999	
60	Bromomethane	74-83-9	1.85	3433	0.22	8	U
61	Trichlorofluoromethane	75-69-4	2.19	2050	0.08	606	U
62	1,1-Dichloroethene	75-35-4	2.73	5692	0.31	911	U
63	Methyl Tert-butyl Ether	1634-04-4	3.8	2796	0.32	32	U
64	Methylene Chloride	75-09-2	3.35	12639	0	973	U
65	1,1,-Dichloroethane	75-34-3	4.42	148821	5.05	997	
66	2,2,-Dichloropropane	594-20-7	5.42	1082	0.06	140	U
67	Chloroform	67-66-3	5.91	258119	8.73	1000	
68	Benzene	71-43-2	6.74	115370	1.61	999	U
69	Trichloroethene	79-01-6	7.8	18847	1.09	998	U
70	Dibromomethane	79-95-3	8.32	751	0.07	807	U
71	1-Bromo-2-chloroethane	107-04-0	9.06	6615	0.29	965	U
72	Toluene	108-88-3	9.87	21659	0.52	998	U
73	trans-1,3-Dichloropropene	10061-02-6	10.29	2597	0.12	656	U
74	1,1,2-Trichloroethane	79-00-5	10.57	267571	16.72	1000	
75	Ethylbenzene	100-41-4	12.43	39144	0.66	993	U
76	p,m-Xylene	106-42-3/108	12.63	67872	1.52	1000	U
77	o-Xylene	95-47-6	13.29	33714	0.68	1000	U
78	Isopropylbenzene	98-82-8	13.95	9394	0.21	916	U
79	Bromobenzene	108-86-1	14.39	2692	0.18	580	U
80	n-Propylbenzene	103-65-1	14.68	13487	0.28	999	U
81	1,3,5-trimethylbenzene	108-67-8	15.01	12322	0.35	987	U
82	sec-Butylbenzene	135-98-8	15.96	10821	0.3	966	U
83	1,3-Dichlorobenzene	541-73-1	16.08	5459	0.25	994	U
84	1,4-Dichlorobenzene	106-46-7	16.24	6178	0.27	48	U
85	n-Butylbenzene	104-51-8	16.88	10562	0.41	945	U
86	1,2-Dichlorobenzene	95-50-1	16.8	5250	0.24	933	U
87	Hexachlorobutadiene	87-68-3	19.21	8742	9119.72	950	E
88	<//End_Target_Compounds//>						
89	<//Start_TIC_Table//>						
90	COMPOUND NAME	CAS NO.	R.T.	AREA	EST. CONC.	PROB.	Q
91	Carbon dioxide	124-38-9	1.14	13618653	110.9	46.8	JN
92	Butane, 2-methyl-	78-78-4	2.01	618285	5	78.2	JN
93	Ethane, 1,2-dichloro-	107-06-2	6.78	1242733	10.1	98.5	JN
94	Chlorobenzene-d5	3114-55-4	12.14	9665978	77.7	96.6	JN
95	<//End_TIC_Table//>						

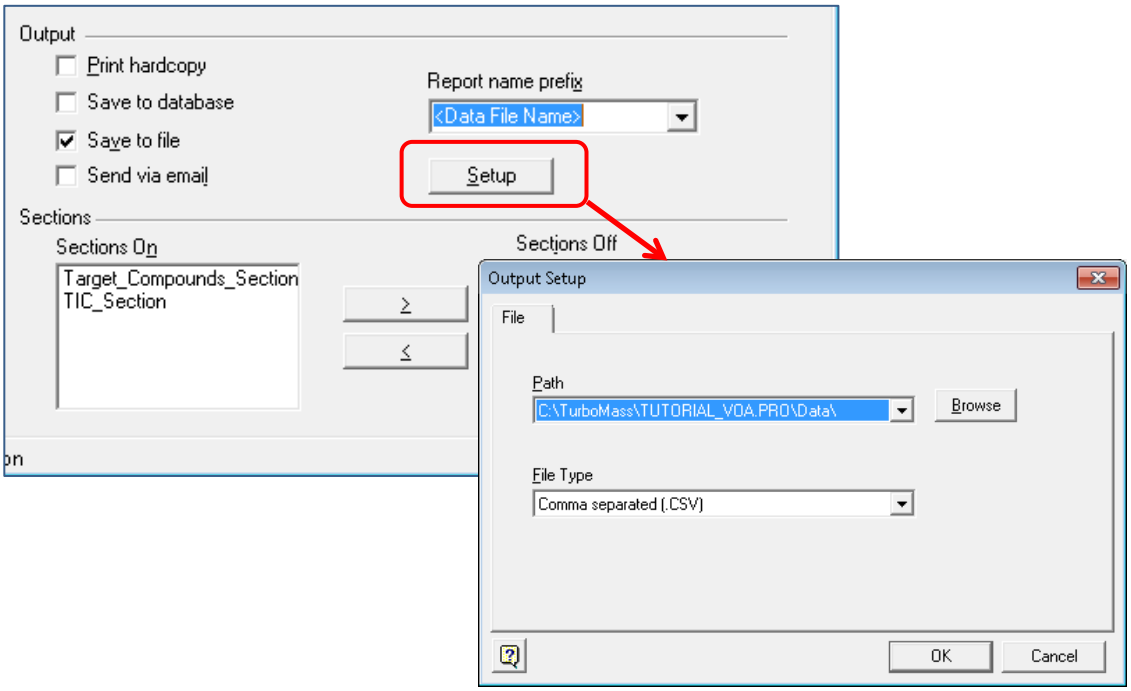
Report Method for Form1_CSV

A report method containing the Form1_CSV.tpl template must be created.

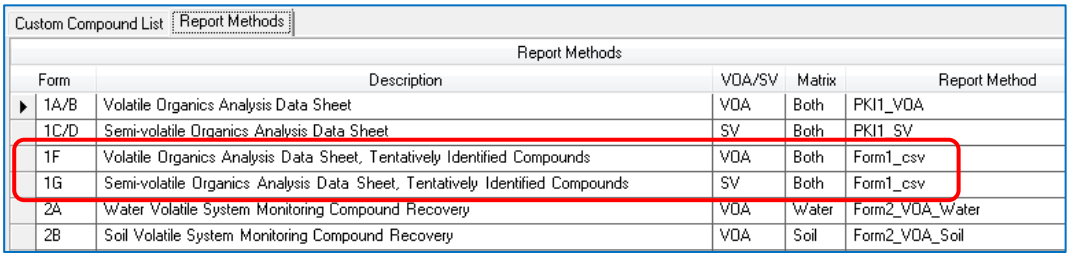
Note: *Refer to* ***Chapter 19 Report Method Editor of the TurboMass Software User's Guide*** *for information on how to create a report method.*

When creating the report method, be sure to set the following:

- Output set to "Save to File"
- Report name prefix is defined
- The path is defined where the file is to be saved
- The file type is selected as Comma Separated (CSV)



To generate a Form1_CSV report, the report method must be placed in **the Form 1F** row for **volatile** matrices and in the **Form 1G** row for **semi-volatile** fractions in the Submitter/Task list created.



Form	Description	VOA/SV	Matrix	Report Method
1A/B	Volatile Organics Analysis Data Sheet	VOA	Both	PKI1_VOA
1C/D	Semi-volatile Organics Analysis Data Sheet	SV	Both	PKI1_SV
1F	Volatile Organics Analysis Data Sheet, Tentatively Identified Compounds	VOA	Both	Form1_csv
1G	Semi-volatile Organics Analysis Data Sheet, Tentatively Identified Compounds	SV	Both	Form1_csv
2A	Water Volatile System Monitoring Compound Recovery	VOA	Water	Form2_VOA_Water
2B	Soil Volatile System Monitoring Compound Recovery	VOA	Soil	Form2_VOA_Soil

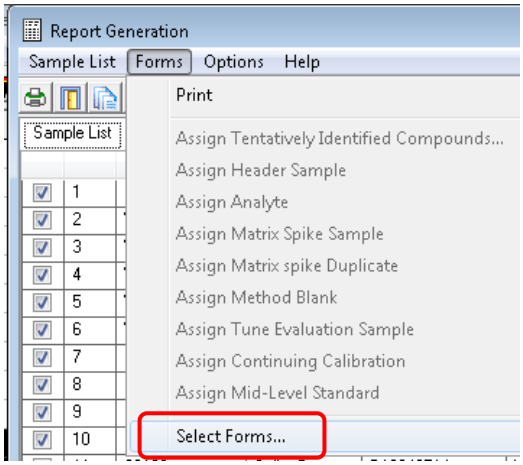
Generating a Form1_CSV

A form1_CSV contains information to produce ***both*** target compounds and tentatively identified compounds (TICs) on the report. Therefore, this form must follow setup procedures using both Form 1 types and Form 1 TIC reports.

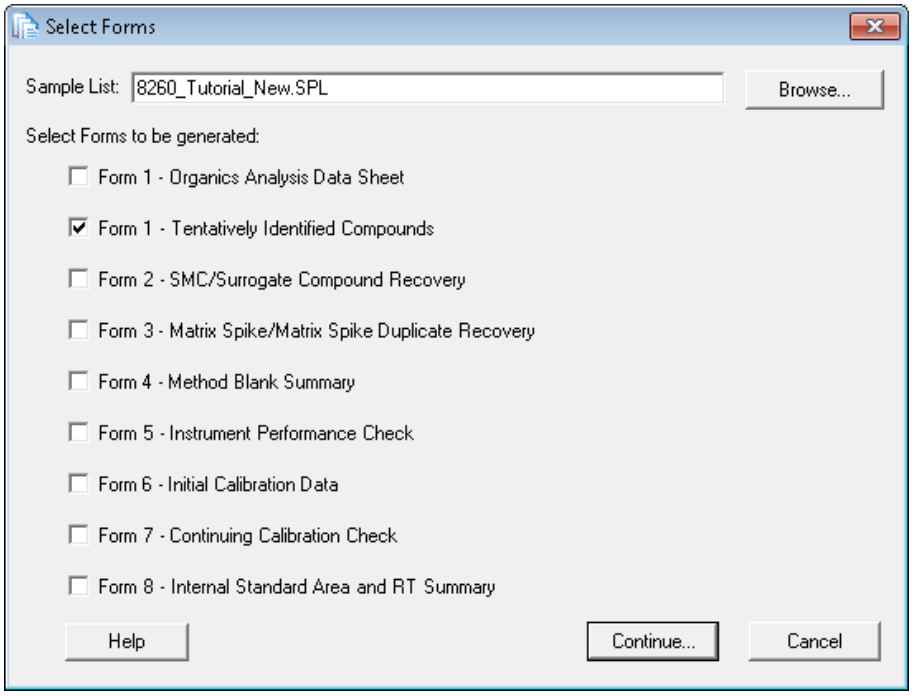
1. Refer to instructions for ***Form 1 – Organics Analysis Data Sheet*** on page ***101*** and follow in their entirety except for the last step of form generation.

2. Proceed to instructions for ***Form 1TIC – Tentatively Identified Compounds(TICs)*** on page ***109*** and follow in their entirety.

3. From the **Forms** menu in the **Report Generation** window, **Select Forms**



4. **Deselect the Form 1** and only have **Form1 TIC selected**, and then select **Continue**.

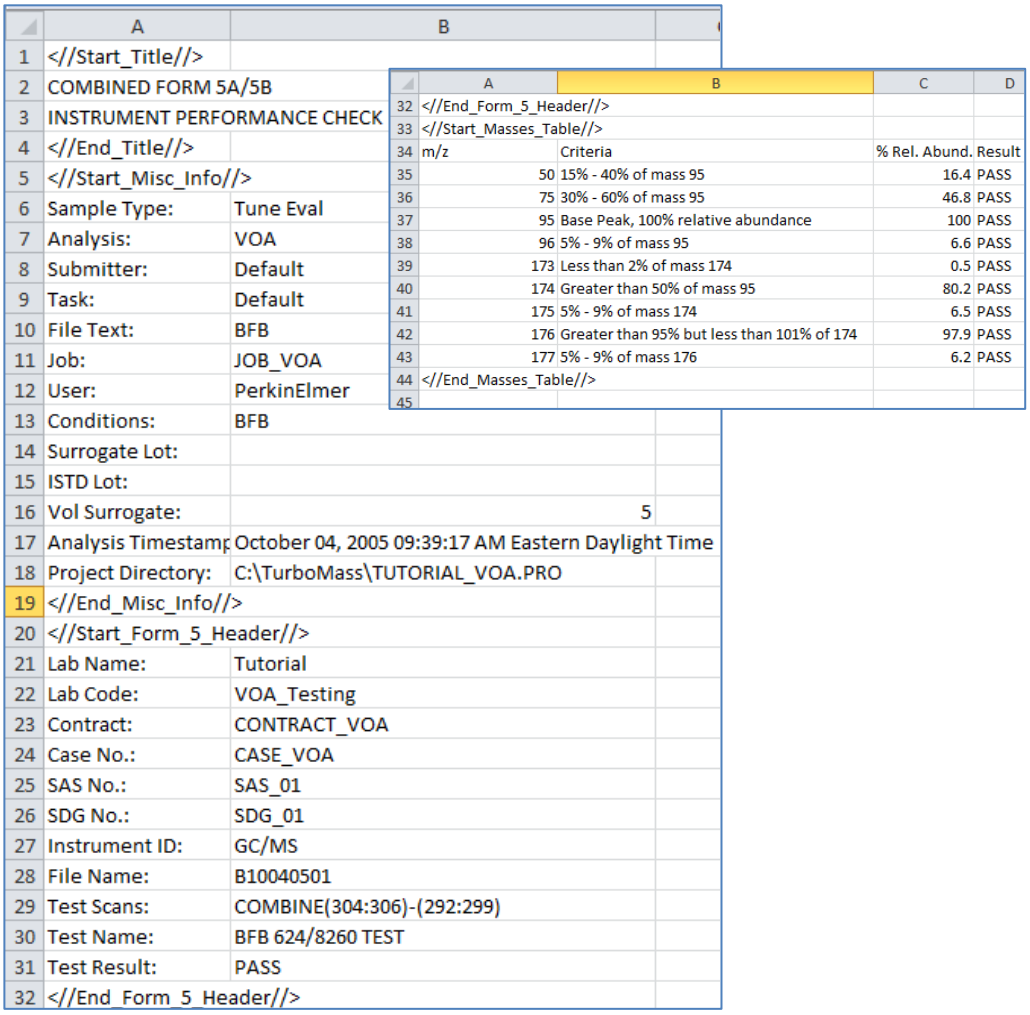


5. Generate the **Form1_CSV** by selecting **Print**.

Form5_CSV – Instrument Performance Check (CSV format)

This report displays the results and compliance of the tune evaluation sample in a comma-separate format (CSV) file.

Below is an example of Form5_CSV report:



	A	B
1	<//Start_Title//>	
2	COMBINED FORM 5A/5B	
3	INSTRUMENT PERFORMANCE CHECK	
4	<//End_Title//>	
5	<//Start_Misc_Info//>	
6	Sample Type:	Tune Eval
7	Analysis:	VOA
8	Submitter:	Default
9	Task:	Default
10	File Text:	BFB
11	Job:	JOB_VOA
12	User:	PerkinElmer
13	Conditions:	BFB
14	Surrogate Lot:	
15	ISTD Lot:	
16	Vol Surrogate:	5
17	Analysis Timestamp	October 04, 2005 09:39:17 AM Eastern Daylight Time
18	Project Directory:	C:\TurboMass\TUTORIAL_VOA.PRO
19	<//End_Misc_Info//>	
20	<//Start_Form_5_Header//>	
21	Lab Name:	Tutorial
22	Lab Code:	VOA_Testing
23	Contract:	CONTRACT_VOA
24	Case No.:	CASE_VOA
25	SAS No.:	SAS_01
26	SDG No.:	SDG_01
27	Instrument ID:	GC/MS
28	File Name:	B10040501
29	Test Scans:	COMBINE(304:306)-(292:299)
30	Test Name:	BFB 624/8260 TEST
31	Test Result:	PASS
32	<//End_Form_5_Header//>	

	A	B	C	D
32	<//End_Form_5_Header//>			
33	<//Start_Masses_Table//>			
34	m/z	Criteria	% Rel. Abund.	Result
35	50	15% - 40% of mass 95	16.4	PASS
36	75	30% - 60% of mass 95	46.8	PASS
37	95	Base Peak, 100% relative abundance	100	PASS
38	96	5% - 9% of mass 95	6.6	PASS
39	173	Less than 2% of mass 174	0.5	PASS
40	174	Greater than 50% of mass 95	80.2	PASS
41	175	5% - 9% of mass 174	6.5	PASS
42	176	Greater than 95% but less than 101% of 174	97.9	PASS
43	177	5% - 9% of mass 176	6.2	PASS
44	<//End_Masses_Table//>			
45				

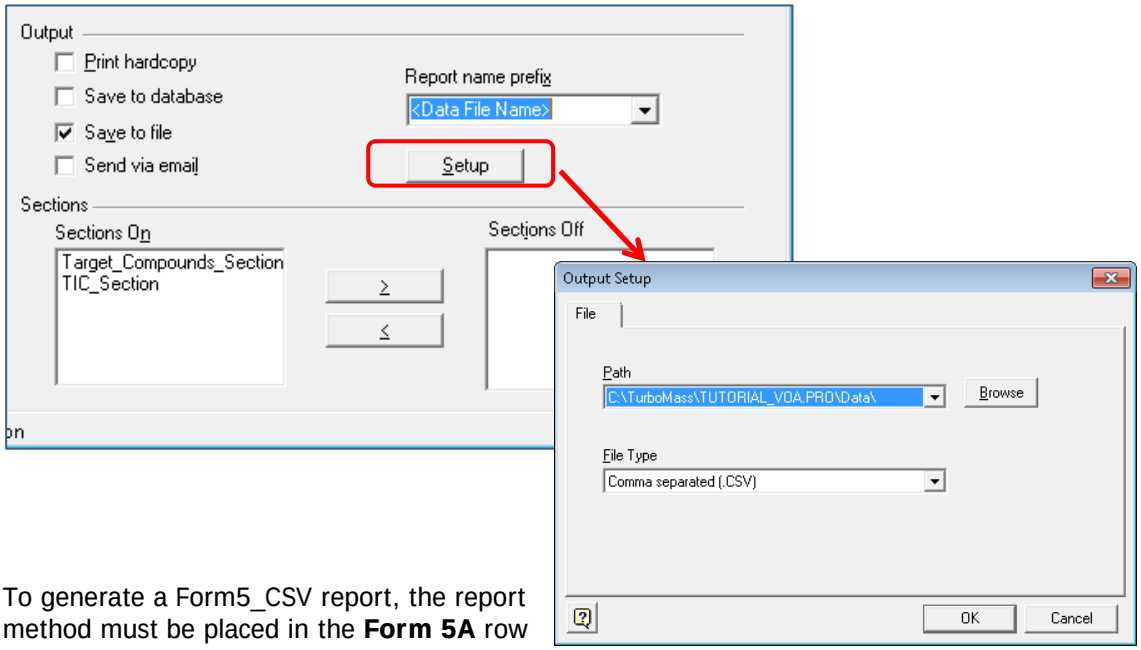
Report Method for Form5_CSV

A report method containing the **Form5_CSV.tpl** template **must** be created.

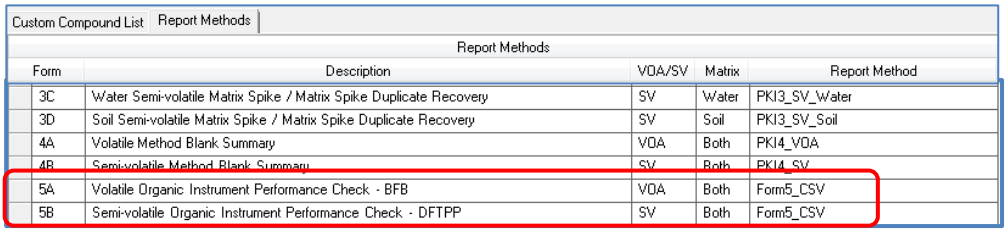
Note: *Refer to **Chapter 19 Report Method Editor of the TurboMass Software User's Guide** for information on how to create a report method.*

When creating the report method, be sure to set the following:

- Output set to "Save to File"
- Report name prefix is defined
- The path is defined where the file is to be saved
- The file type is selected as Comma Separated (CSV)



To generate a Form5_CSV report, the report method must be placed in the **Form 5A** row for **volatile** matrices and in the **Form 5B** row for **semi-volatile** fractions in the Submitter/Task list created.



Valid Sample Types for Form 5 Types

A **Form 5_CSV** can be generated using the following valid sample types only:

- Tune Eval – used to define the tune evaluation sample to report

Tune Evaluation Processing and Generation

1. Refer to instructions for ***Form 5 – Instrument Performance Check*** on page ***126*** and follow in its entirety.

Form6_CSV – Initial Calibration Data (CSV format)

- This report displays the initial calibration of the instrument based on a multi-point calibration in a comma-separate format (CSV) file.
- The average relative response factor (RRF) and the relative standard deviation (RSD is displayed.
- For compounds using a first order curve fit, the correlation coefficient (r^2) is reported. For compounds using a higher order curve fit, the coefficient of determination (CofD) is reported in the (r^2) column.

Below is an example of a Form6_CSV report:

	A	B	C
1	<//Start_Title//>		
2	COMBINED FORM 6A/6B/6C/6D		
3	VOLATILE ORGANICS INITIAL CALIBRATION		
4	<//End_Title//>		
5	<//Start_Misc_Info//>		
6	Sample Type:	Analyte	
7	Analysis:	VOA	
8	Submitter:	Default	
9	Task:	Default	
10	File Text:	VOA	
11	Job:	JOB_VOA	
12	User:	PerkinElmer	
13	Conditions:	Spike 50ug/L	
14	Surrogate Lot:	Surr_01	
15	ISTD Lot:	ISTD_01	
16	Vol Surrogate:	5	
17	Project Directory:	C:\TurboMass\TU	
18	<//End_Misc_Info//>		
19	<//Start_Form_6_Header//>		
20	Lab Name:	Tutorial	
21	Lab Code:	VOA_Testing	
22	Contract:	CONTRACT_VOA	
23	Case No.:	CASE_VOA	
24	SAS No.:	SAS_01	
25	SDG No.:	SDG_01	
26	Instrument ID:	GC/MS	
27	Heated Purge:	N	
28	GC Column:	Elite-VOA	
29	ID:	180 um	
30	Calibration Date(first)	Oct. 04, 05	
31	Calibration lime (first)	7:11:20 AM	
32	Calibration Date(last):	Oct. 04, 05	
33	Calibration lime (last)	9:59:37 AM	
34	<//End_Form_6_Header//>		
35	<//Start_Files_Table//>		
36	Level 1 =	B10040504	
37	Level 2 =	B10040505	
38	Level 3 =	B10040506	
39	Level 4 =	B10040507	
40	Level 5 =	B10040509	

	A	B	C	D	E	F	G	H	I	J	K
49	COMPOUND		Level1	Level2	Level3	Level4	Level5	AvgRRF	%RSD		r²
50	Dichlorodifluoromethane		0.375	0.377	0.374	0.351	0.327	0.361	6	*	
51	Vinyl Chloride		0.49	0.529	0.512	0.486	0.482	0.5	4	*	
52	Bromomethane		0.36	0.343	0.337	0.331	0.357	0.346	3.7	*	
53	Trichlorofluoromethane		0.568	0.59	0.568	0.546	0.553	0.565	3	*	
54	1,1-Dichloroethene		0.406	0.412	0.397	0.386	0.392	0.398	2.7	*	
55	Methyl Tert-butyl Ether		1.201	1.225	1.201	1.15	1.197				0.9995
56	Methylene Chloride		0.579	0.52	0.485	0.46	0.467				0.9998
57	1,1,-Dichloroethane	*	0.657	0.666	0.644	0.625	0.642	0.647	2.4	*	
58	2,2,-Dichloropropane		0.41	0.437	0.426	0.417	0.418	0.422	2.4	*	
59	Chloroform		0.649	0.669	0.651	0.634	0.642	0.649	2	*	
60	Benzene		0.987	1.019	0.985	0.965	0.985	0.988	2	*	
61	Trichloroethene		0.229	0.242	0.233	0.233	0.252	0.238	3.8	*	
62	Dibromomethane		0.139	0.146	0.144	0.141	0.143	0.143	1.8	*	
63	1-Bromo-2-chloroethane		0.311	0.324	0.321	0.316	0.327	0.32	2	*	
64	Toluene		0.566	0.594	0.578	0.565	0.578	0.576	2.1	*	
65	trans-1,3-Dichloropropene		0.262	0.286	0.293	0.304	0.339	0.297	9.5	*	
66	1,1,2-Trichloroethane		0.21	0.223	0.222	0.222	0.227	0.221	2.9	*	
67	Ethylbenzene		2.85	2.99	2.792	2.746	2.732	2.822	3.7	*	
68	p,m-Xylene		2.128	2.278	2.109	2.08	2.07	2.133	4	*	
69	o-Xylene		2.35	2.531	2.335	2.289	2.293	2.36	4.2	*	
70	Isopropylbenzene		2.086	2.255	2.07	2.039	2.028	2.096	4.4	*	
71	Bromobenzene		0.753	0.767	0.726	0.703	0.719	0.734	3.6	*	
72	n-Propylbenzene		2.245	2.446	2.255	2.206	2.197	2.27	4.5	*	
73	1,3,5-trimethylbenzene		1.655	1.816	1.703	1.663	1.672	1.702	3.9	*	
74	sec-Butylbenzene		1.741	1.839	1.719	1.706	1.699	1.741	3.3	*	
75	1,3-Dichlorobenzene		1.036	1.122	1.031	1.019	1.028	1.047	4	*	
76	1,4-Dichlorobenzene		1.096	1.159	1.084	1.067	1.104	1.102	3.2	*	
77	n-Butylbenzene		1.11	1.283	1.231	1.241	1.279	1.229	5.7	*	
78	1,2-Dichlorobenzene		0.988	1.094	1.017	1.004	1.029	1.026	4	*	
79	Hexachlorobutadiene		0.201	0.186	0.181	0.186	0.204	0.192	5.3	*	
80	<//End_Target_Compounds//>										
81	<//Start_SMC_Compounds//>										
82	SMC COMPOUND		Level1	Level2	Level3	Level4	Level5	AvgRRF	%RSD		r²
83	Dibromofluoromethane		0.407	0.41	0.411	0.412	0.417	0.411	0.9	*	
84	Toluene-d8		1.84	1.844	1.854	1.844	1.853	1.847	0.3	*	
85	Bromofluorobenzene		0.977	1.001	0.956	0.95	0.945	0.966	2.4	*	

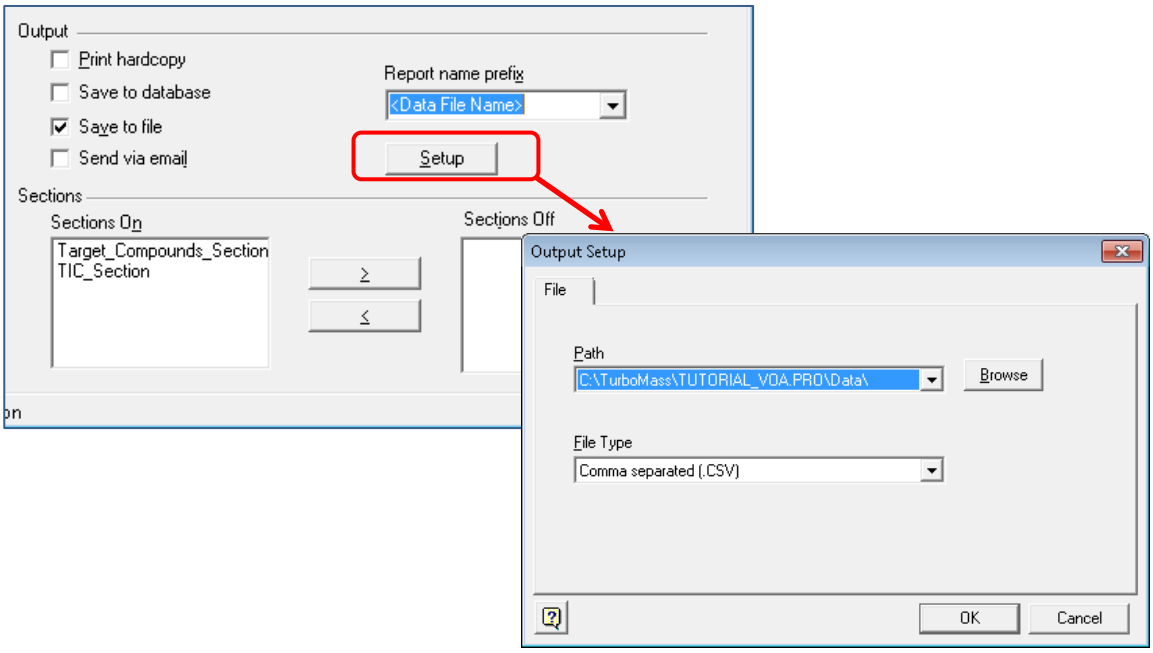
Report Method for Form6_CSV

A report method containing the Form6_CSV.tpl template must be created.

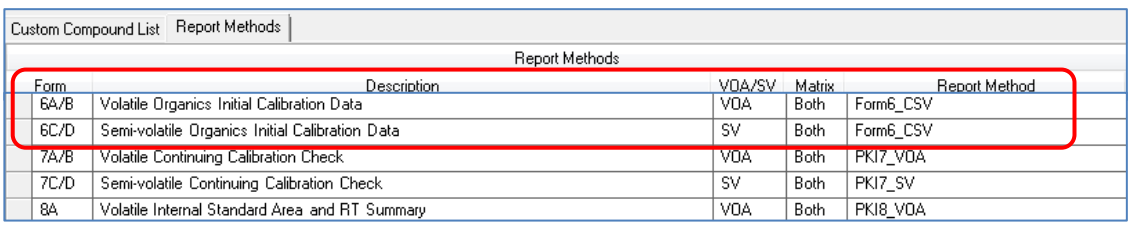
Note: *Refer to **Chapter 19 Report Method Editor of the TurboMass Software User's Guide** for how to create a report method.*

When creating the report method, be sure to set the following:

- Output set to "Save to File"
- Report name prefix is defined
- The path is defined where the file is to be saved
- The file type is selected as Comma Separated (CSV)



To generate a Form6_CSV report, the report method must be placed in the Form **6A/B** row for **volatile** matrices and in the Form **6C/D** row **for semi-volatile** fractions in the Submitter/Task list created.



Form	Description	VOA/SV	Matrix	Report Method
6A/B	Volatile Organics Initial Calibration Data	VOA	Both	Form6_CSV
6C/D	Semi-volatile Organics Initial Calibration Data	SV	Both	Form6_CSV
7A/B	Volatile Continuing Calibration Check	VOA	Both	PKI7_VOA
7C/D	Semi-volatile Continuing Calibration Check	SV	Both	PKI7_SV
8A	Volatile Internal Standard Area and RT Summary	VOA	Both	PKI8_VOA

Valid Sample Types for Form 6 types

A **Form 6_CSV** can be generated using the following valid sample types only:

- Init Calib – used to define initial calibration files.

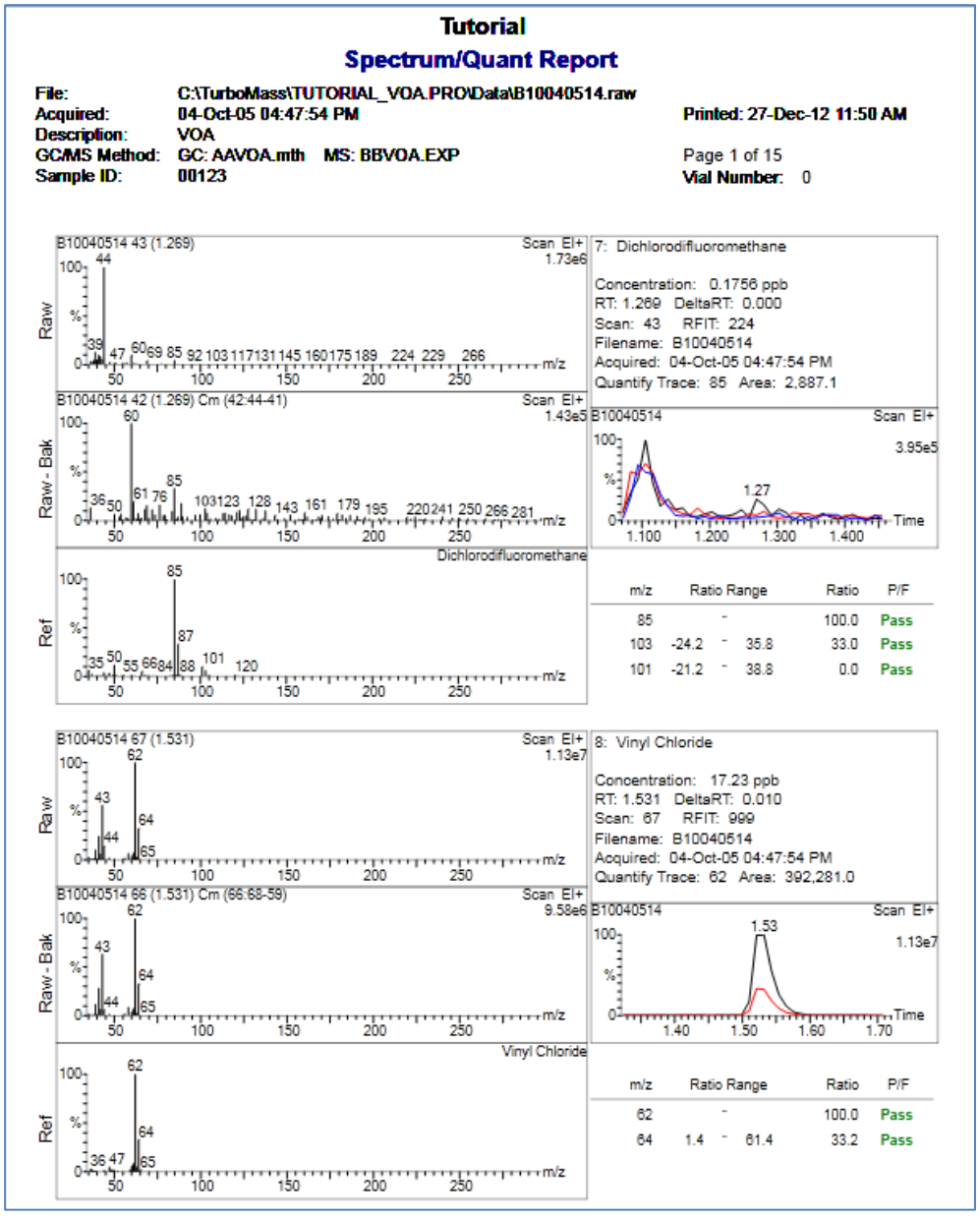
Initial Calibration Form6_csv Generation

1. Refer to instructions for ***Form 6 – Initial Calibration Data*** on page ***133*** and follow in its entirety.

SpectrumQuant Report_Env Template Form Generation

- Reports positive target compounds displaying the raw spectrum, background subtracted spectrum and reference spectrum.
- Quantitated results based on sample preparation information are displayed along with ion ratio's and the criteria.

Below is an example of a of SpectrumQuant Report_Env report:



Tutorial

Spectrum/Quant Report

File: C:\TurboMass\TUTORIAL_VOA.PRO\Data\B10040514.raw
Acquired: 04-Oct-05 04:47:54 PM
Printed: 27-Dec-12 11:50 AM
Description: VOA
GC/MS Method: GC: AAVOA.mth MS: BBVOA.EXP
Page 1 of 15
Sample ID: 00123
Vial Number: 0

B10040514 43 (1.269) Scan EI+ 1.73e6 — Raw

B10040514 42 (1.269) Cm (42:44-41) Scan EI+ 1.43e5 — Raw - Bak

Dichlorodifluoromethane — Ref

7: Dichlorodifluoromethane

Concentration: 0.1756 ppb
RT: 1.269 DeltaRT: 0.000
Scan: 43 RFIT: 224
Filename: B10040514
Acquired: 04-Oct-05 04:47:54 PM
Quantify Trace: 85 Area: 2,887.1

B10040514 Scan EI+ 3.95e5 — Time

m/z	Ratio Range		Ratio	P/F
85			100.0	Pass
103	-24.2	35.8	33.0	Pass
101	-21.2	38.8	0.0	Pass

B10040514 67 (1.531) Scan EI+ 1.13e7 — Raw

B10040514 66 (1.531) Cm (66:68-59) Scan EI+ 9.58e6 — Raw - Bak

Vinyl Chloride — Ref

8: Vinyl Chloride

Concentration: 17.23 ppb
RT: 1.531 DeltaRT: 0.010
Scan: 67 RFIT: 999
Filename: B10040514
Acquired: 04-Oct-05 04:47:54 PM
Quantify Trace: 62 Area: 392,281.0

B10040514 Scan EI+ 1.13e7 — Time

m/z	Ratio Range		Ratio	P/F
62			100.0	Pass
64	1.4	61.4	33.2	Pass

Report Method for Spectrum Quant Report_Env

A report method containing the **SpectrumQuant Report_Env.tpl** template must be created.

Note: *Refer to **Chapter 19 Report Method Editor of the TurboMass Software User's Guide** for information on how to create a report method.*

Note: *Refer to the section titled **Copy and Paste *.RME files to your new Project** on page **87** in this manual to obtain access to this report method.*

Valid Sample Types for SpectrumQuant Report_Env

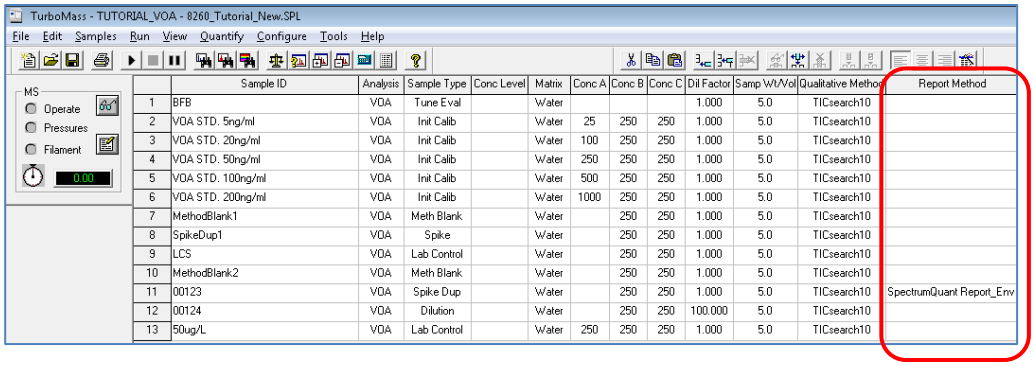
A SpectrumQuant Report_Env can be generated using the following valid sample types only:

- Analyte
- Analyte Dup
- Method Blank
- Spike
- Spike Dup
- Dilution
- Re-extract
- Lab Control

SpectrumQuant Report_Env Generation

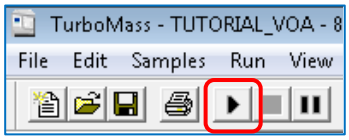
1. Verify Steps 1 through 5 have been followed (pages 91-96 of this tutorial).

2. Select the **Report Method** (SpectrumQuant Report_Env or whatever it has been named) on the **appropriate sample line.**



	Sample ID	Analysis	Sample Type	Conc Level	Matrix	Conc A	Conc B	Conc C	Dil Factor	Samp Wt/Vol	Qualitative Method	Report Method
1	BFB	VOA	Tune Eval		Water				1.000	5.0	TICsearch10	
2	VOA STD. 5ng/ml	VOA	Init Calib		Water	25	250	250	1.000	5.0	TICsearch10	
3	VOA STD. 20ng/ml	VOA	Init Calib		Water	100	250	250	1.000	5.0	TICsearch10	
4	VOA STD. 50ng/ml	VOA	Init Calib		Water	250	250	250	1.000	5.0	TICsearch10	
5	VOA STD. 100ng/ml	VOA	Init Calib		Water	500	250	250	1.000	5.0	TICsearch10	
6	VOA STD. 200ng/ml	VOA	Init Calib		Water	1000	250	250	1.000	5.0	TICsearch10	
7	MethodBlank1	VOA	Meth Blank		Water		250	250	1.000	5.0	TICsearch10	
8	SpikeDup1	VOA	Spike		Water		250	250	1.000	5.0	TICsearch10	
9	LCS	VOA	Lab Control		Water		250	250	1.000	5.0	TICsearch10	
10	MethodBlank2	VOA	Meth Blank		Water		250	250	1.000	5.0	TICsearch10	
11	00123	VOA	Spike Dup		Water		250	250	1.000	5.0	TICsearch10	SpectrumQuant Report_Env
12	00124	VOA	Dilution		Water		250	250	100.000	5.0	TICsearch10	
13	50ug/L	VOA	Lab Control		Water	250	250	250	1.000	5.0	TICsearch10	

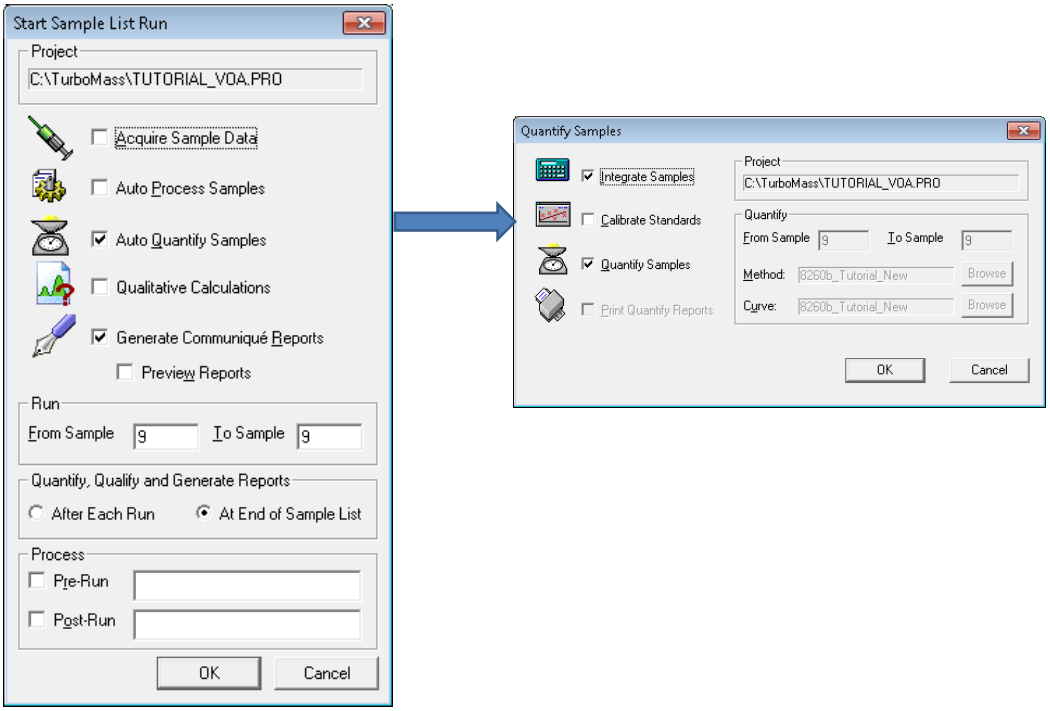
3. From the sample list window, **highlight the row** to be reported, then select the **Start Run** button.



4. Setup the **Start Sample List Run** window for Quantitative processing and to generate Communiqué reports.

5. Verify the **From Sample** and **To sample** are correct.

6. Select **Integrate** (if samples were not previously processed), and **Quantify** Samples and then select **OK** to generate the report.





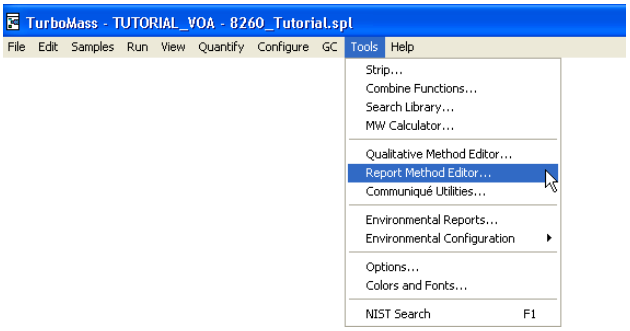
8 How to Save your Data to LIMS



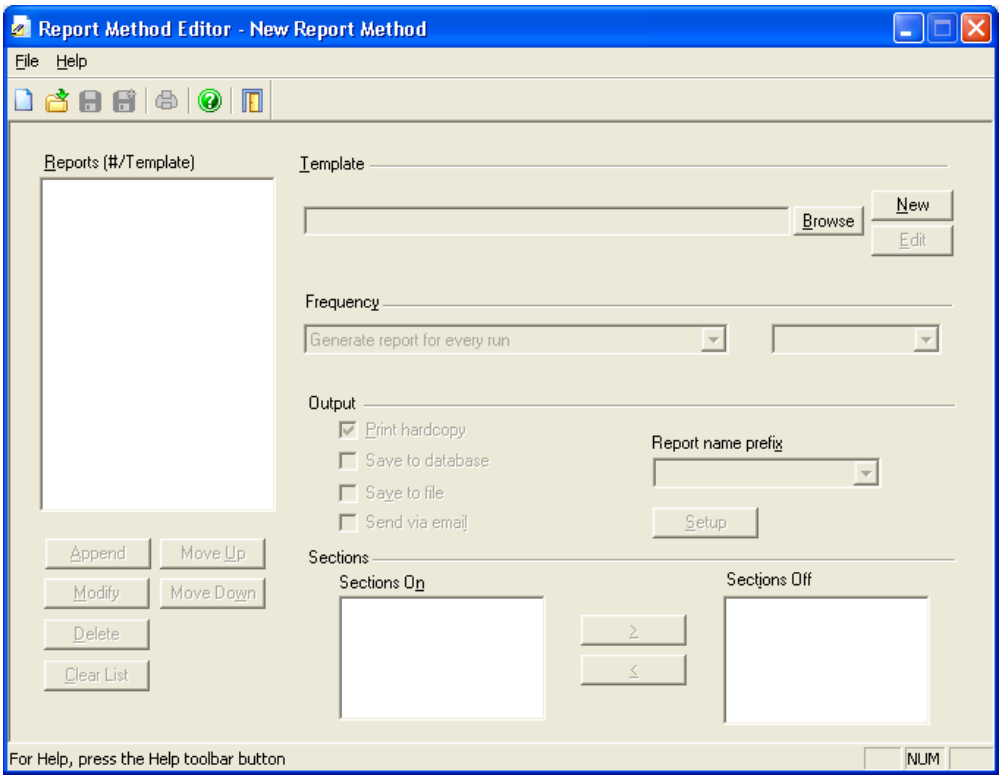
Exporting Your Environmental Data to LIMS

TurboMass provides the ability to save your data to LIMS for further analysis. The following steps summarize how to export your data to LIMS:

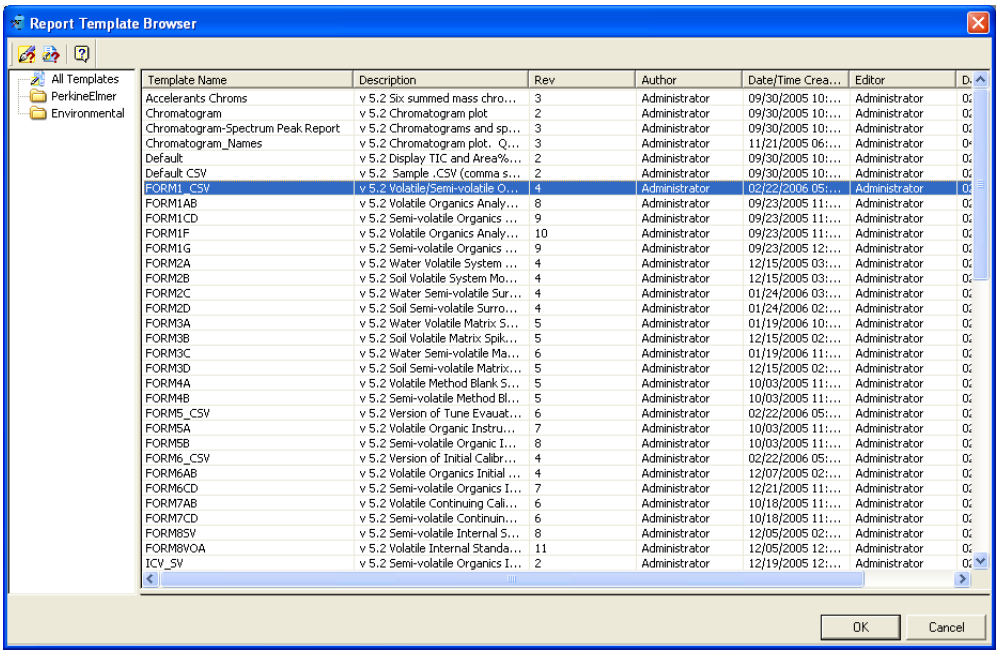
1. Select **Report Method Editor** from the **Tools** menu.



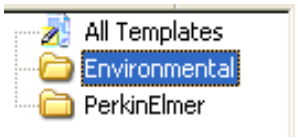
The **Report Method Editor** dialog appears:

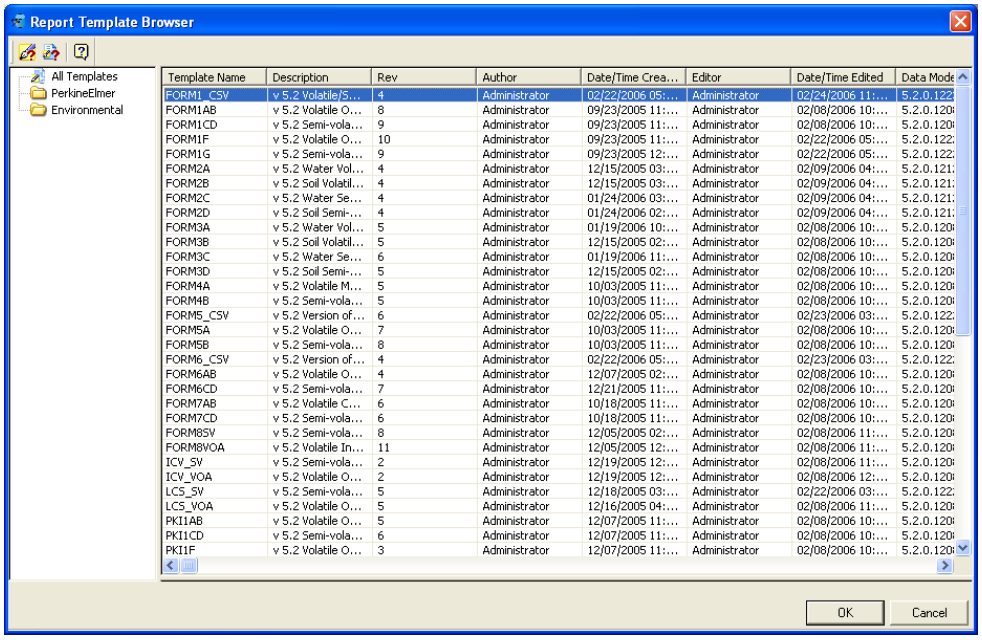


2. Click the **Browse** button to search for a Template.
   The **Report Template Browser** dialog appears with all templates displayed:

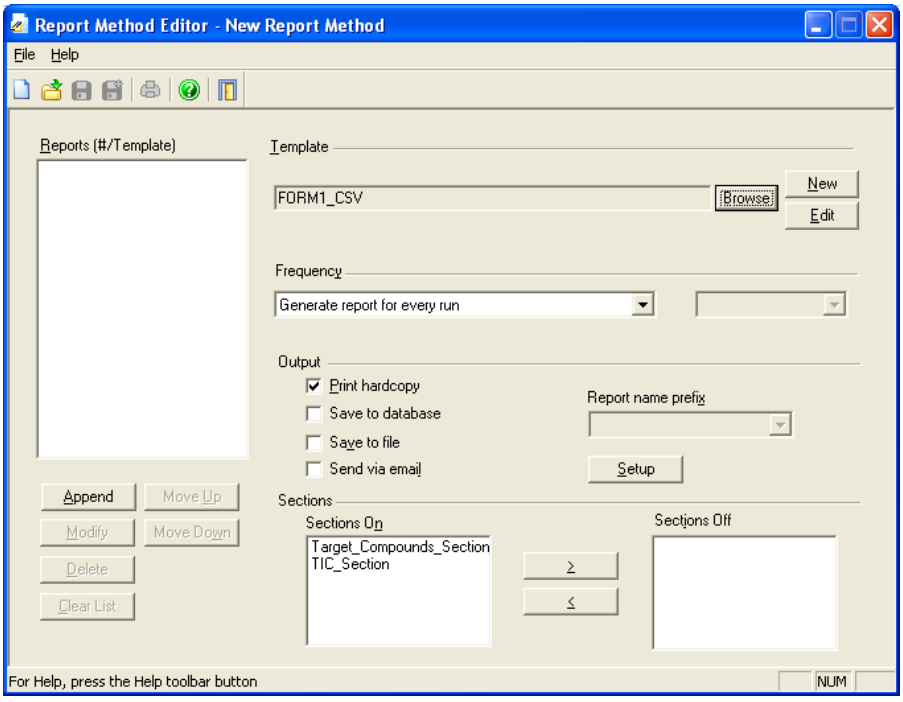


3. In the left pane, click the **Environmental** folder to display the Environmental templates.



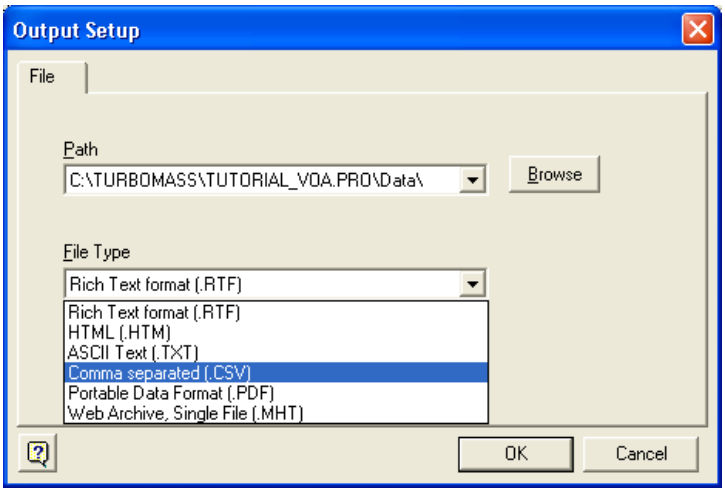


4. Select the **FORM1_CSV** template and click **OK**. It is now selected in the Report Method Editor.



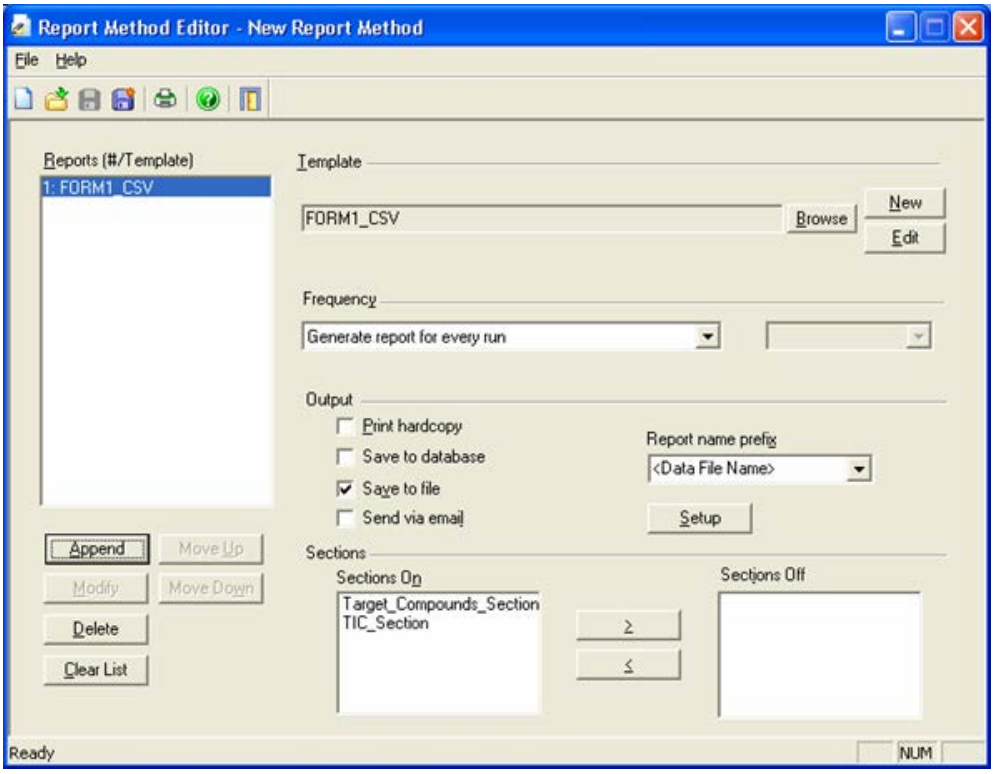
5. Under the Output section of the dialog, deselect **Print hardcopy** and select **Save to file**.

6. Click the **Setup** button and the following dialog appears:



Here you can select the **Path** where you want to send the output and the **File Type.** When you are outputting to LIMS, select **Comma separated (.CSV)** the click OK.

7. Click the **Append** button so that FORM1_CSV appears in the Reports (#/Template) list.



Appendix 1
Environmental Reporting Calculations



On-column Amount Calculations

TurboMass can fit several types of calibration curves which can be applied to environmental sample calculations. The following is an explanation on how TurboMass determines the on-column amounts (X_s) found in any sample type for 3 curve fits. The on-column amount (X_s) is then used in the appropriate environmental calculation dependent on Analysis type (Voa, Svoa), Conc Level (Blank, Low, Med), and Matrix type (Water or Soil).

Average Relative Response Factor (RRF)

X_s TurboMass amount of compound calculated on column in nanograms (ng).

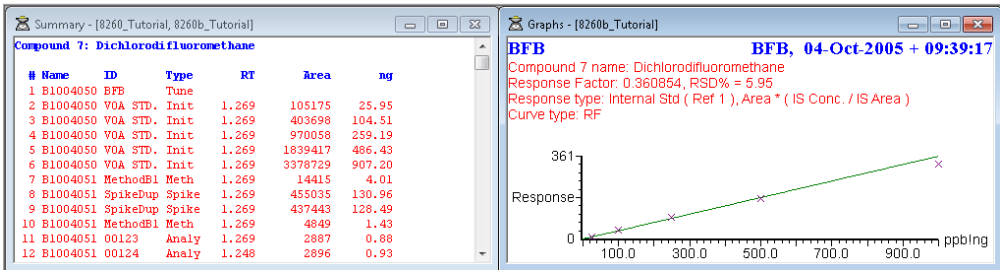
$$= \frac{(A_x)\ (I_s)}{(A_{is})\ (RRF)}$$

RRF The average relative response factor (RRF) of the target compound calculated from all levels that make up the calibration curve.

$$= \frac{(A_x)\ (I_s)}{(A_{is})\ (C_s)}$$

Where C_s Amount of the compound to be measured, in the calibration standard – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

Example Average Relative Response Factor Calculation from TurboMass Curve Display



With the following sample information from Sample list and Quantify> View Results:
File used from **VOA_TUTORIAL** project **B10040511** for **Dichlorodifluoromethane**.

Data from Quantify > View Results

Dichlorodifluoromethane EIC Area	= 455035	A_x
Pentafluorobenzene ISTD EIC Area	= 2407227	A_{is}
Pentafluorobenzene ISTD amount	= 250ng	I_s

Data from Quantify > View Results > Curve Display

RRF	= 0.360854	RRF

TurboMass Amount on-column calculation (ng)

TurboMass Amoung (X_s) = $\frac{(A_x)\ (I_s)}{(A_{is})\ (RRF)}$

= $\frac{(455035area)(250ng)}{(2407227area)(0.360854)}$

= 130.96ng

Linear Regression

The following equation demonstrates the Internal (relative) response type used in the calculation to determine the TurboMass amount of compound calculated on column in nanograms (ng) – defined as X_s :

$$X_s = \frac{Y - d}{c}$$

c = slope of the line
d = y-intercept
Y = Area of target corrected for Internal standard (ISTD) response

$$= \frac{(A_x)(I_s)}{(A_{is})}$$

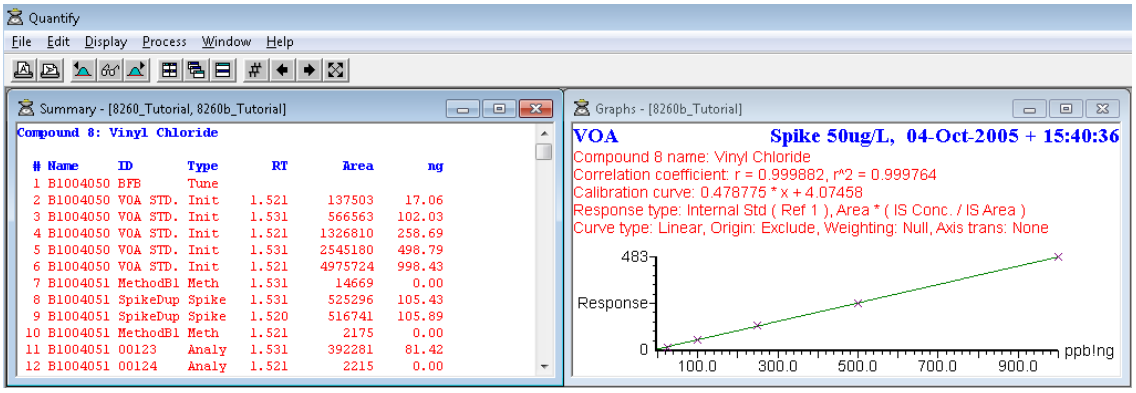
A_x Area[1] of the characteristic ion (Extracted Ion current profile – EICP) for the compound to be measured – from integration results.

A_{is} Area[1] of the characteristic ion (Extracted Ion current profile – EICP) for the internal standard – from integration results.

I_s Amount of internal standard added in nanograms (ng) – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

[1] "Area" is referenced here, since it is specified by the EPA methods, but the software will actually use whatever response mode (area or height) is specified in the Quantify Method. This applies to all the environmental calculations defined in this document.

Linear Regression – Example Calculation from TurboMass Curve Display



Example Linear Regression On-column Amount:

With the following sample information from Sample List and Quantify > View Results:
File used from **VOA_TUTORIAL** project **B10040511** for **vinyl chloride.**

Available from Quantify > View Results

Vinyl Chloride EIC Area	= 525296	A_x
Pentafluorobenzene ISTD EIC Area	= 2407227	A_{is}
Pentafluorobenzene ISTD amount	= 250ng	I_s

Available from Quantify > View Results > Curve Display

Slope	= 0.478775	c
y-intercept	= 4.07458	d

TurboMass Amount on-column calculation (ng)

$$\text{TurboMass Amoung } (X_s) = \frac{Y - d}{c}$$

$$= \frac{((525296\text{area} * 250\text{ng})/2407227\text{area})) - 4.07458}{0.478775}$$

= 105.43ng

Quadratic

The following demonstrates the Internal (relative) response type used in the calculation to determine the TurboMass amount of compound calculated on column in nanograms (ng) – defined as X_s (2 roots are derived for Xs as X_{s1} and X_{s2}, as explained below).

$$Y = ax^2 + bx + c$$

x = Sample amount in nanograms (ng)
Coefficients a, b, and c are known and calculated from the quadratic regression curve

Y_s = Area of target corrected for Internal standard (ISTD) response

$$Y_s = \frac{(A_x)(I_s)}{(A_{is})}$$

A_x Area[1] of the characteristic ion (Extracted Ion current profile – EICP) for the compound to be measured – from integration results.

A_{is} Area[1] of the characteristic ion (Extracted Ion current profile – EICP) for the internal standard – from integration results.

I_s Amount of internal standard added in nanograms (ng) – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

To determine the peak amount (x) for each given adjusted peak area (Y); the roots (2) of the quadratic equation are determined by:

$$\frac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}$$

Set Y = 0 and derive the two X roots (at the Y origin):

$$0 = ax^2 + bx + c$$

$$X_{s1} = \frac{-b + \sqrt{(b^2 - 4a(c - Y_s))}}{2a}$$

$$X_{s2} = \frac{-b - \sqrt{(b^2 - 4a(c - Y_s)}}{2a}$$

Negative values of X are excluded as it indicates a negative response for the sample amount, and are on the wrong side of the calibration curve.

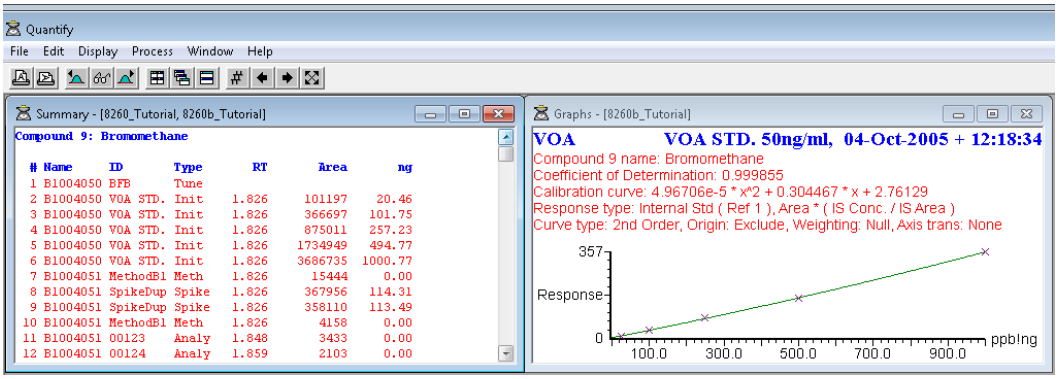
All other amounts are calculated by substituting the Y adjusted peak area and setting the equation to zero:

$$Y_s = ax^2 + bx + c$$

$$0 = ax^2 + bx + c - Y_s$$

[1] "Area" is referenced here, since it is specified by the EPA methods, but the software will actually use whatever response mode (area or height) is specified in the Quantify Method. This applies to all the environmental calculations defined in this document.

Example Quadratic Calculation from TurboMass curve display



Example Quadratic on-column amount:

With the following sample information from Sample list and Quantify/View Results:
File used from **VOA_TUTORIAL** project **B10040511** for **Bromomethane**.

<u>Available from Quantify\View Results</u>

Vinyl Chloride EIC Area	= 367956	A_x
Pentafluorobenzene ISTD EIC Area	= 2407227	A_{is}
Pentafluorobenzene ISTD amount	= 250ng	I_s

Y = (367956area)(250ng) / (2407227area)
= 38.21ng

<u>Available from Quantify\View Results\Curve Display</u>

Coefficient a	= 4.96706 e-5	a
Coefficient b	= 0.304467	b
Coefficient c	= 2.76129	c

TurboMass Amount on-column calculation (ng)

$0 = ax^2 + bx + c$

$$X_1 = \frac{-b + \sqrt{(b^2 - 4ac)}}{2a}$$

$$X_{s1} = \frac{-0.30447 + \sqrt{(0.30447^2 - (4(4.9671e\text{-}5))(2.7613\text{-}38.2137)}}{2(4.9671e\text{-}5)}$$

= 114.31 ng Positive result = detector response amount

$$X_2 = \frac{-b - \sqrt{(b^2 - 4ac)}}{2a}$$

$$X_{s2} = \frac{-0.30447 - \sqrt{(0.30447^2 - (4(4.9671e\text{-}5))(2.7613\text{-}38.2137)}}{2(4.9671e\text{-}5)}$$

= -6244 ng Negative result = below detectable range

Target Calculations

The calculation of compound concentration to be employed for a given sample will be determined by the analysis type (VOA or SV – taken from the 'Analysis' field in the Sample List), matrix type (water/soil – taken from the 'Matrix' field in the Sample List) and concentration level (low/medium – taken from the 'Level' field in the Sample List).

This value calculated from the equations below will be included in the data source as the item under the TargetCompounds() collection (i.e. it will not replace the standard TurboMass 'Concentration' value found in the Quantify view results.

Note: *The equations below are shown in two forms. Firstly, the form defined in the EPA methods and/or SOW. The TurboMass software will not perform these calculations exactly as defined in these equations but will instead derive the 'environmental concentrations' from the concentration value currently calculated and reported. The secondary calculations required to generate the required 'environmental concentration' value from the current TurboMass concentration are defined in the equations shown in parenthesis [ ] following the EPA version.*

Volatile Water Target

	Conditions	Analysis	Sample Type	Matrix	Conc A	Conc B	Conc C	Conc D	Dil Factor	Samp Wt/Vol
1	BFB	VOA	Tune Eval	Water					1.000	5.0
2	VOA STD. 5ug/L	VOA	Init Calib	Water	25	250	250	50	1.000	5.0
3	VOA STD. 20ug/L	VOA	Init Calib	Water	100	250	250	200	1.000	5.0
4	VOA STD. 50ug/L	VOA	Init Calib	Water	250	250	250	500	1.000	5.0
5	VOA STD. 100ug/L	VOA	Init Calib	Water	500	250	250	1000	1.000	5.0
6	VOA STD. 200ug/L	VOA	Init Calib	Water	1000	250	250	2000	1.000	5.0
7	BLANK	VOA	Meth Blank	Water		250	250		1.000	5.0
8	Sample 1	VOA	Analyte	Water		250	250		4.000	5.0

Concentration (ug/L) $= \frac{(A_x)\ (I_s)\ (Df)}{(A_{is})\ (RRF)\ (V_o)} = \frac{(X_s)\ (Df)}{(V_o)}$ (where X_s is TurboMass amount – ng[1])

A_x Area[2] of the characteristic ion (Extracted Ion current profile – EICP) for the compound to be measured – from integration results.

A_{is} Area[2] of the characteristic ion (Extracted Ion current profile – EICP) for the internal standard – from integration results.

I_s Amount of internal standard added in nanograms (ng) – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

$\overline{RRF}$ The average relative response factor (RRF) of target compound calculated from the calibration curve.

$$= \frac{(A_x)\ (I_s)}{(A_{is})\ (C_s)}$$

Where C_s Amount of the compound to be measured, in the calibration standard – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

V_o Volume of water purged in milliliters (mL) – derived from 'SampleWt/Vol' column in Sample List.

Df Dilution Factor – derived from 'Dil Factor – DILUTION FACTOR' from the Sample List.

X_s TurboMass amount of compound calculated on column in nanograms (ng).

$$= \frac{(A_x)\ (I_s)}{(A_{is})\ (\overline{RRF})}$$

[1] Xs is the target amount on column. This is calculated based on an amount (in ng) for the internal standard in the Sample List.

[2] "Area" is referenced here, since it is specified by the EPA methods, but the software will actually use whatever response mode (area or height) is specified in the Quantify Method. This applies to all the environmental calculations defined in this document.

Example Volatile Water Target:

With the following sample information from Sample List and Quantify > View Results:
File used from **VOA_TUTORIAL** project **B10040511**.

Available from Quantify\View Results		
1,1-dichloroethene EIC Area	= 407416	A_x
1,1-dichloroethene Avg RRF	= 0.39835	RRF
Pentafluorobenzene ISTD EIC Area	= 2407227	A_{is}

Available from Sample List columns		
Pentafluorobenzene ISTD Amount (ng)	= 250ng	I_s
Sample vol/wt	= 5.0mL	V_o
Dilution Factor	= 1	Df

$X_s = \frac{(407416)(250ng)}{(0.39835)(2407227)}$ = **106.22ng** on column of 1,1-Dichloroethene

Final concentration calculation

Concentration (ug/L) = $\frac{(A_x)\ (I_s)}{A_{is})\ (RRF)\ (V_o)}$ = $\frac{(X_s)\ (Df)}{(V_o)}$ (where X_s is TurboMass amount – ng)

= $\frac{(106.22ng)(1)}{(5.0mL)}$

= 21.24 ng/mL = 21.24 ug/L

Volatile Low Level Soil Target

	Conditions	Analysis	Sample Type	Matrix	Conc Level	Conc A	Conc B	Conc C	Conc D	Dil Factor	Samp Wt/Vol	Moisture
1	BFB	VOA	Tune Eval	Soil	Low					1.000	5.0	0.0
2	VOA STD. 5ug/L	VOA	Init Calib	Soil	Low	25	250	250	50	1.000	5.0	0.0
3	VOA STD. 20ug/L	VOA	Init Calib	Soil	Low	100	250	250	200	1.000	5.0	0.0
4	VOA STD. 50ug/L	VOA	Init Calib	Soil	Low	250	250	250	500	1.000	5.0	0.0
5	VOA STD. 100ug/L	VOA	Init Calib	Soil	Low	500	250	250	1000	1.000	5.0	0.0
6	VOA STD. 200ug/L	VOA	Init Calib	Soil	Low	1000	250	250	2000	1.000	5.0	0.0
7	BLANK	VOA	Meth Blank	Soil	Low		250	250		1.000	5.0	0.0
8	Sample 1	VOA	Analyte	Soil	Low		250	250		1.000	5.2	50.0

Concentration (ug/Kg) $= \frac{(A_x)(I_s)}{(A_{is})(\overline{RRF})(W_s)(D)} = \frac{(X_s)}{(W_s)(D)}$ (where X_s is TurboMass amount – ng[1])

A_x Area[2] of the characteristic ion (Extracted Ion current profile – EICP) for the compound to be measured – from integration results.

I_s Amount of internal standard added in nanograms (ng) – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

A_{is} Area[2] of the characteristic ion (Extracted Ion current profile – EICP) for the internal standard – from integration results.

$\overline{RRF}$ The average relative response factor (RRF) of target compound calculated from the calibration curve.

$$= \frac{(A_x)(I_s)}{(A_{is})(C_s)}$$

Where C_s Amount of the compound to be measured, in the calibration standard – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

W_s Weight of sample added to the purge tube in grams (g) – derived from 'SampleWt/Vol' column in Sample List.

D Adjustment for dry weight basis, calculated from 'Moisture' column in Sample List.

$$= \frac{100 - \%\ \text{moisture}}{100}$$

X_s TurboMass amount of compound calculated on column in nanograms (ng).

$$= \frac{(A_x)(I_s)}{(A_{is})(RRF)}$$

[1] Xs is the target amount on column. This is calculated based on an amount (in ng) for the internal standard in the Sample List.

[2] "Area" is referenced here, since it is specified by the EPA methods, but the software will actually use whatever response mode (area or height) is specified in the Quantify Method. This applies to all the environmental calculations defined in this document.

Example Volatile Low level Soil Target:

With the following sample information from Sample list and Quantify/View Results :
File used from **TUTORIAL_VOA project B10040511**.

Available from Quantify\View Results		
1,1-dichloroethene EIC Area	= 407416	A_x
1,1-dichloroethene Avg RRF	= 0.39835	RRF
Pentafluorobenzene ISTD EIC Area	= 2407227	A_{is}

Available from Sample List columns		
Pentafluorobenzene ISTD Amount (ng)	= 250ng	I_s
Sample vol/wt	= 5.2g	W_s
% moisture	= 50	D (see calculation below)

X_s = $\frac{(407416)(250ng)}{(0.39835)(2407227)}$ = **106.22ng** on column of 1,1-Dichloroethene

D = (100 – 50)/100
= 0.50

Final concentration calculation

Conc. (ug/Kg) = $\frac{(A_x)\ (I_s)}{(A_{is})\ (RRF)\ (W_s)(D)}$ = $\frac{(X_s)}{(W_s)(D)}$ (where X_s is TurboMass amount – ng)

= $\frac{(106.22ng)}{(5.2g)\ (0.50)}$

= 40.85 ng/g = 40.85 ug/Kg

Volatile Medium Level Soil Target

	Conditions	Analysis	Sample Type	Conc Level	Matrix	Conc A	Conc B	Conc C	Conc D	Dil Factor	VOA Moisture Calc	Moisture	Samp Wt/Vol	Aliqu Vol	Ext Vol
1	BFB	VOA	Tune Eval	Medium	Water					1.000		0.0	5.0	100.0	10000.0
2	VOA STD. 5ng/ml	VOA	Init Calib	Medium	Water	25	250	250	50	1.000		0.0	5.0	100.0	10000.0
3	VOA STD. 20ng/ml	VOA	Init Calib	Medium	Water	100	250	250	200	1.000		0.0	5.0	100.0	10000.0
4	VOA STD. 50ng/ml	VOA	Init Calib	Medium	Water	250	250	250	500	1.000		0.0	5.0	100.0	10000.0
5	VOA STD. 100ng/ml	VOA	Init Calib	Medium	Water	500	250	250	1000	1.000		0.0	5.0	100.0	10000.0
6	VOA STD. 200ng/ml	VOA	Init Calib	Medium	Water	1000	250	250	2000	1.000		0.0	5.0	100.0	10000.0
7	BLANK	VOA	Meth Blank	Medium	Water		250	250		1.000		0.0	5.0	100.0	10000.0

$$\text{Conc } (\mu g/kg) = \frac{(A_x)\,(I_s)\,(V_t^3)\,(Df)}{(A_{is})\,(RRF)\,(W_s)\,(V_a)\,(D)} = \frac{(X_s)\,(V_t)\,(Df)}{(W_s)\,(V_a)\,(D)}$$ (where X_s is TurboMass amount – ng[1])

A_x Area[2] of the characteristic ion (Extracted Ion current profile – EICP) for the compound to be measured – from integration results.

I_s Amount of internal standard added in nanograms (ng) – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

A_{is} Area[2] of the characteristic ion (Extracted Ion current profile – EICP) for the internal standard – from integration results.

RRF The average relative response factor (RRF) of target compound calculated from the calibration curve.

$$= \frac{(A_x)\,(I_s)}{(A_{is})\,(C_s)}$$

Where C_s Amount of the compound to be measured, in the calibration standard – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List)

V_t^3 The total volume of the methanol extract in microliters (uL). This value is applied in two different ways.

If **Additional Moisture** is set to **N** in the **VOA Moisture Calc Sample List** field , the Total volume of the methanol extract in microliters (uL) is used for V_t– derived directly from 'Ext Vol [SOIL_EXTRACT_VOLUME]' column in Sample List.

If **Additional Moisture is** set to **Y** in the **VOA Moisture Calc Sample List** field in the Sample List, the following additional calculation occurs and is used by the software for the V_t value in the concentration equation above.

$$= \left[\frac{(V_t)}{(1000uL/mL)} + \frac{(\%\ moisture)\,(Ws)}{100}\right] * \quad 1000uL/mL$$

[1] Xs is the target amount on column. This is calculated based on an amount (in ng) for the internal standard in the Sample List.

[2] "Area" is referenced here, since it is specified by the EPA methods, but the software will actually use whatever response mode (area or height) is specified in the Quantify Method. This applies to all the environmental calculations defined in this document.

[3] Identifies that there are two possible values for Vt dependent on the Additional Moisture column in the sample list

Va — Volume of the aliquot of the sample methanol extract (i.e., sample extract not including the methanol added to equal 100uL) in microliters (uL) added to reagent water for purging – derived from 'Aliqu Vol – SOIL_ALIQUOT_VOLUME' from the Sample List.

W_s — Weight of sample added to the purge tube in grams (g) – derived from 'SampleWt/Vol' column in Sample List.

D — Adjustment for dry weight basis, calculated from 'Moisture' column in Sample List.

$$= \frac{100 - \%\ \text{moisture}}{100}$$

Df — Dilution Factor – derived from 'Dil Factor – DILUTION FACTOR' from the Sample List.

X_s — TurboMass amount of compound calculated on column in nanograms (ng).

$$= \frac{(A_x)\ (I_s)}{(A_{is})\ (\text{RRF})}$$

Example Volatile Medium level Soil Target:

With the following sample information from Sample list and Quantify/View Results:
File used from **TUTORIAL_VOA** project **B10040511**.

Available from Quantify\View Results		
1,1-dichloroethene EIC Area	= 407416	A_x
1,1-dichloroethene Avg RRF	= 0.39835	RRF
Pentafluorobenzene ISTD EIC Area	= 2407227	A_{is}

Available from Sample List columns		
Pentafluorobenzene ISTD Amount (ng)	= 250ng	I_s
Sample vol/wt	= 5.0g	W_s
Ext Vol (Total volume of Methanol Extract)	= 10000uL	V_t (see calculation below)
Aliqu Vol (Soil Aliquot volume)	= 100uL	Va
% moisture	= 50	D (see calculation below)
Dilution Factor	= 1	Df

X_s = $\frac{(407416)(250ng)}{(0.39835)(2407227)}$ = **106.22ng** on column of 1,1-Dichloroethene

D = (100 – 50)/100
= 0.50

Vt with additional moisture set to "Y"

$$= \left[\frac{10000uL}{1000uL/mL} + \frac{(50)(5.0g)}{100} \right] * 1000uL/mL$$

= (10mL + 2.5) *1000uL/mL
Vt = 12500uL

Vt with additional moisture set to "N"

Vt = 10000uL

Concentration (ug/Kg) = $\frac{(A_x)\,(I_s)\,(V_t)^3\,(Df)}{(A_{is})\,(RRF)\,(W_s)\,(V_a)\,(D)}$ = $\frac{(X_s)\,(V_t)\,(Df)}{(W_s)\,(V_a)\,(D)}$ (where X_s is TurboMass amount – ng)

Final concentration with additional moisture calculation

= $\frac{(106.22ng)(12500uL)(1)}{(5.0g)(100uL)(0.50)}$

= 5311ng/g = 5311 ug/Kg

Final concentration without additional moisture calculation

= $\frac{(106.22ng)(10000uL)(1)}{(5.0g)(100uL)(0.50)}$

= 4249 ng/g = 4249ug/Kg

Semi-Volatile Water Target

	Conditions	Analysis	Sample Type	Matrix	Conc A	Conc B	Conc C	Conc D	Inject Volume	Dil Factor	GPC	Samp Wt/Vol	Conc Ext Vol
1	DFTPP	SV	Tune Eval	Water					1.000	1.000	N	1.0	1000.0
2	SV STD 80 ng BNs/200 ng acids	SV	Init Calib	Water	80	200	80	200	1.000	1.000	N	1.0	1000.0
3	SV STD 60 ng BNs/150 ng acids	SV	Init Calib	Water	60	150	60	150	1.000	1.000	N	1.0	1000.0
4	SV STD 40 ng BNs/100 ng acids	SV	Init Calib	Water	40	100	40	100	1.000	1.000	N	1.0	1000.0
5	SV STD 20 ng BNs/50 ng acids	SV	Init Calib	Water	20	50	20	50	1.000	1.000	N	1.0	1000.0
6	SV STD 8 ng BNs/20 ng acids	SV	Init Calib	Water	8	20	8	20	1.000	1.000	N	1.0	1000.0
7	CCS	SV	Cont Calib	Water	40	100	40	100	1.000	1.000	N	1.0	1000.0
8	MB	SV	Meth Blank	Water	25	50			1.000	1.000	N	1000.0	2000.0
9	Sample1	SV	Analyte	Water	25	50	25	50	1.000	25.000	Y	1000.0	2000.0

Conc. (ug/L) = $\frac{(A_x)\ (I_s)\ (V_t)(Df)(GPC)}{(A_{is})\ (RRF)\ (V_o)\ (V_i)} = \frac{(X_s)(V_t)(Df)(GPC)}{(V_o)(V_i)}$ (where X_s is TurboMass amount – ng[1])

A_x Area[2] of the characteristic ion (Extracted Ion current profile – EICP) for the compound to be measured – from integration results.

A_{is} Area[2] of the characteristic ion (Extracted Ion current profile – EICP) for the internal standard – from integration results.

I_s Amount of internal standard added in nanograms (ng) – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

RRF The average relative response factor (RRF) of target compound calculated from the calibration curve.

$= \frac{(A_x)\ (I_s)}{(A_{is})\ (C_s)}$

Where C_s Amount of the compound to be measured, in the calibration standard – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

V_o Volume of water extracted in milliliters (mL) – derived from 'SampleWt/Vol' column in Sample List.

V_i Volume of extract injected in microliters (uL) – derived from 'Inject Volume' column in Sample List.

V_t Volume of the concentrated extract in microliters (uL) – derived from 'Conc Ext Vol' column in Sample List.

GPC Gel Permeation Cleanup factor – derived from 'GPC' column setting in Sample list **Y** or **N**.
GPC = 1.0 if assigned N in GPC column in sample list
GPC = 2.0 if assigned Y in GPC column in sample list

[1] Xs is the target amount on column. This is calculated based on an amount (in ng) for the internal standard in the Sample List.

[2] "Area" is referenced here, since it is specified by the EPA methods, but the software will actually use whatever response mode (area or height) is specified in the Quantify Method. This applies to all the environmental calculations defined in this document.

Df — Dilution Factor – derived from 'Dil Factor – DILUTION FACTOR' from the Sample List.

X_s — TurboMass amount of compound calculated on column in nanograms (ng).

$$= \frac{(A_x)\,(I_s)}{(A_{is})\,(RRF)}$$

Example Semi-Volatile Water Target:

With the following sample information from Sample list and Quantify/View Results :
File used from **TUTORIAL_SVOA project 51006d06**.

Available from Quantify\View Results		
Pyrene EIC Area	= 4316281	A_x
Pyrene Avg RRF	= 1.2929	RRF
Chrysene-d12 ISTD EIC Area	= 2154951	A_{is}
Available from Sample List columns		
Chrysene-d12 ISTD Amount (ng)	= 10ng	I_s
Sample vol/wt	= 1000mL	V_o
Volume of extract injected (uL)	= 1uL	V_i
Volume of concentrated extract (uL)	= 2000uL	V_t
Dilution Factor	= 25	Df
GPC	= Y	GPC

$$\text{Concentration (ug/L)} = \frac{(A_x)\,(I_s)\,(V_t)(Df)(GPC)}{(A_{is})\,(RRF)\,(V_o)\,(V_i)} = \frac{(X_s)(V_t)(Df)(GPC)}{(V_o)(V_i)}$$ (where X_s is TurboMass amount – ng)

$$X_s = \frac{(4316281)(10ng)}{(1.2929)(2154951)} = \textbf{15.49 ng}$$ on column of Pyrene

Final concentration calculation

$$= \frac{(15.49ng)(2000uL)(25)(2)}{(1000mL)\,(1uL)}$$

= 1549 ng/mL = 1549 ug/L

Semi-Volatile Soil Target

	Conditions	Analysis	Sample Type	Matrix	Conc A	Conc B	Conc C	Conc D	Inject Volume	Dil Factor	GPC	Moisture	Samp Wt/Vol	Conc Ext Vol
1	DFTPP	SV	Tune Eval	Soil					1.000	1.000	N	0.0	1.0	1000.0
2	SV STD 80 ng BNs/20	SV	Init Calib	Soil	80	200	80	200	1.000	1.000	N	0.0	1.0	1000.0
3	SV STD 60 ng BNs/15	SV	Init Calib	Soil	60	150	60	150	1.000	1.000	N	0.0	1.0	1000.0
4	SV STD 40 ng BNs/10	SV	Init Calib	Soil	40	100	40	100	1.000	1.000	N	0.0	1.0	1000.0
5	SV STD 20 ng BNs/50	SV	Init Calib	Soil	20	50	20	50	1.000	1.000	N	0.0	1.0	1000.0
6	SV STD 8 ng BNs/20	SV	Init Calib	Soil	8	20	8	20	1.000	1.000	N	0.0	1.0	1000.0
7	CCS	SV	Cont Calib	Soil	40	100	40	100	1.000	1.000	N	0.0	1.0	1000.0
8	MB	SV	Meth Blank	Soil	25	50			1.000	1.000	N	0.0	15.0	2000.0
9	Sample1	SV	Analyte	Soil	25	50	25	50	1.000	25.000	Y	14.0	15.2	2000.0

Concentration (ug/Kg) = $\frac{(A_x)\ (I_s)\ (V_t)(Df)(GPC)}{(A_{is})\ (RRF)(V_i)(W_s)(D)} = \frac{(X_s)(V_t)(Df)(GPC)}{(V_i)\ (W_s)\ (D)}$ (where X_s is TurboMass amount – ng[1])

A_x Area[2] of the characteristic ion (Extracted Ion current profile – EICP) for the compound to be measured – from integration results.

A_{is} Area[2] of the characteristic ion (Extracted Ion current profile – EICP) for the internal standard – from integration results.

I_s Amount of internal standard added in nanograms (ng) – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

RRF The average relative response factor (RRF) of target compound calculated from the calibration curve.

$$= \frac{(A_x)\ (I_s)}{(A_{is})\ (C_s)}$$

Where C_s Amount of the compound to be measured, in the calibration standard – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

V_i Volume of extract injected in microliters (uL) – derived from 'Inject Volume' column in Sample List.

V_t Volume of the concentrated extract in microliters (uL) – derived from 'Conc Ext Vol' column in Sample List.

W_s Weight of sample extracted in grams (g) – derived from 'SampleWt/Vol' column in Sample List.

D Adjustment for dry weight basis, calculated from 'Moisture' column in Sample List.

$$= \frac{100 - \%\ \text{moisture}}{100}$$

[1] Xs is the target amount on column. This is calculated based on an amount (in ng) for the internal standard in the Sample List.

[2] "Area" is referenced here, since it is specified by the EPA methods, but the software will actually use whatever response mode (area or height) is specified in the Quantify Method. This applies to all the environmental calculations defined in this document.

GPC Gel Permeation Cleanup factor – derived from 'GPC' column setting in Sample list **Y** or **N**.

GPC = 1.0 if assigned N in GPC column in sample list
GPC = 2.0 if assigned Y in GPC column in sample list

Df Dilution Factor – derived from 'Dil Factor – DILUTION FACTOR' from the Sample List.

X_s TurboMass amount of compound calculated on column in nanograms (ng).

$$= \frac{(A_x)\,(I_s)}{(A_{is})\,(RRF)}$$

Example Semi-Volatile Soil Target:

With the following sample information from Sample list and Quantify/View Results :
File used from **TUTORIAL_SVOA project 51006d06**.

Available from Quantify\View Results		
Pyrene EIC Area	= 4316281	A_x
Pyrene Avg RRF	= 1.2929	RRF
Chrysene-d12 ISTD EIC Area	= 2154951	A_{is}
Available from Sample List columns		
Chrysene-d12 ISTD Amount (ng)	= 10ng	I_s
Sample vol/wt	= 15.2g	W_s
% Moisture	= 14.0	D (see calculation below)
Volume of extract injected (uL)	= 1uL	V_i
Volume of concentrated extract (uL)	= 2000uL	V_t
Dilution Factor	= 25	Df
GPC	= Y	GPC

$$X_s = \frac{(4316281)(10ng)}{(1.2929)(2154951)} = \textbf{15.49 ng} \text{ on column of Pyrene}$$

$$D = (100 - 14)/100 = 0.86$$

Final concentration calculation

$$\text{Concentration (ug/Kg)} = \frac{(A_x)\,(I_s)\,(V_t)(Df)(GPC)}{(A_{is})\,(RRF)(V_i)(W_s)(D)} = \frac{(X_s)(V_t)(Df)(GPC)}{(V_i)\,(W_s)\,(D)}$$

(where X_s is TurboMass amount – ng)

$$= \frac{(15.49ng)(2000uL)(25)(2)}{(1uL)\,(15.2g)\,(0.86)}$$

= 118497 ng/g = 118497 ug/Kg

Tentatively Identified Compounds Calculations

Volatile Water Tentatively Identified Compound

	Conditions	Analysis	Sample Type	Matrix	Conc A	Conc B	Conc C	Conc D	Dil Factor	Samp Wt/Vol	Qualitative Method
1	BFB	VOA	Tune Eval	Water					1.000	5.0	TICsearch10
2	VOA STD. 5ug/L	VOA	Init Calib	Water	25	250	250	50	1.000	5.0	TICsearch10
3	VOA STD. 20ug/L	VOA	Init Calib	Water	100	250	250	200	1.000	5.0	TICsearch10
4	VOA STD. 50ug/L	VOA	Init Calib	Water	250	250	250	500	1.000	5.0	TICsearch10
5	VOA STD. 100ug/L	VOA	Init Calib	Water	500	250	250	1000	1.000	5.0	TICsearch10
6	VOA STD. 200ug/L	VOA	Init Calib	Water	1000	250	250	2000	1.000	5.0	TICsearch10
7	BLANK	VOA	Meth Blank	Water		250	250		1.000	5.0	TICsearch10
8	Sample 1	VOA	Analyte	Water		250	250		4.000	5.0	TICsearch10

Estimated Concentration(ug/L) $= \frac{(A_T)\ (I_S)\ (Df)}{(A_{TI})\ (RRF)\ (V_o)} = \frac{(X_S)\ (Df)}{(V_o)}$ (where X_S is TurboMass amount – ng[1])

Where:

A_T Area of the total chromatogram (TIC) for the compound to be measured – from integrated results of the total ion chromatogram.

A_{TI} Area of the total ion chromatogram (TIC) for the internal standard – from integrated results of the total ion chromatogram.

I_s Amount of internal standard added in nanograms (ng) – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

RRF 1. The relative response factor value of a tentatively identified compound will always be one.

V_o Volume of water purged in milliliters (mL) – derived from 'SampleWt/Vol' column in Sample List.

Df Dilution Factor – derived from 'Dil Factor – DILUTION FACTOR' from the Sample List.

X_s TurboMass amount of tentatively identified compound calculated on column, in nanograms (ng).

$$= \frac{(A_T)\ (I_S)}{(A_{TI})\ (RRF)}$$

[1] Xs is the target amount on column. This is calculated based on an amount (in ng) for the internal standard in the Sample List.

Example Volatile Water Tentatively Identified Compound Calculation:

With the following sample information from Sample list and Chromatogram > View Results:
File used from **VOA_TUTORIAL** project **B10040511**.

Available from Chromatogram\View		
Peak TIC Area (peak 5.76)	= 18975064	A_T
Peak TIC RRF	= 1.00	RRF
Pentafluorobenzene ISTD TIC Area (peak 6.31)	= 6372218	A_{TI}
Available from Sample List columns		
Pentafluorobenzene ISTD Amount (ng)	= 250ng	I_s
Sample vol/wt	= 5.0mL	V_o
Dilution Factor	= 4	Df

$X_s = \frac{(18975064)(250ng)}{(1.00)(6372218)} = 744.45$**ng** on column of peak 5.76

Final concentration calculation

Estimated Concentration (ug/L) $= \frac{(A_T)\ (I_S)(Df)}{(A_{TI})\ (RRF)\ (V_o)} = \frac{(X_S)\ (Df)}{(V_o)}$ (where X_s is TurboMass amount – ng)

$= \frac{(744.45ng)(4)}{(5.0mL)}$

= 595.56 ng/mL = 595.56 ug/L

Volatile Low Level Soil Tentatively Identified Compound

	Conditions	Analysis	Sample Type	Matrix	Conc Level	Conc A	Conc B	Conc C	Conc D	Dil Factor	Samp Wt/Vol	Moisture	Qualitative Method
1	BFB	VOA	Tune Eval	Soil	Low					1.000	5.0	0.0	TICsearch10
2	VOA STD. 5ug/L	VOA	Init Calib	Soil	Low	25	250	250	50	1.000	5.0	0.0	TICsearch10
3	VOA STD. 20ug/L	VOA	Init Calib	Soil	Low	100	250	250	200	1.000	5.0	0.0	TICsearch10
4	VOA STD. 50ug/L	VOA	Init Calib	Soil	Low	250	250	250	500	1.000	5.0	0.0	TICsearch10
5	VOA STD. 100ug/L	VOA	Init Calib	Soil	Low	500	250	250	1000	1.000	5.0	0.0	TICsearch10
6	VOA STD. 200ug/L	VOA	Init Calib	Soil	Low	1000	250	250	2000	1.000	5.0	0.0	TICsearch10
7	BLANK	VOA	Meth Blank	Soil	Low		250	250		1.000	5.0	0.0	TICsearch10
8	Sample 1	VOA	Analyte	Soil	Low		250	250		1.000	5.2	50.0	TICsearch10

Estimated Conc. (ug/Kg) = $\frac{(A_T)\ (I_s)}{(A_{TI})\ (RRF)\ (W_s)(D)} = \frac{(X_s)}{(W_s)(D)}$ (where X_s is TurboMass amount – ng)

Where:

A_T Area of the total chromatogram (TIC) for the compound to be measured – from integrated results of the total ion chromatogram.

A_{TI} Area of the total ion chromatogram (TIC) for the internal standard – from integrated results of the total ion chromatogram.

I_s Amount of internal standard added in nanograms (ng) – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

RRF 1. The relative response factor value of a tentatively identified compound will always be one.

W_s Weight of sample added to the purge tube in grams (g) – derived from 'SampleWt/Vol' column in Sample List.

D Adjustment for dry weight basis, calculated from 'Moisture' column in Sample List.

$$= \frac{100 - \%\ \text{moisture}}{100}$$

X_s TurboMass amount of tentatively identified compound calculated on column, in nanograms (ng).

$$= \frac{(A_T)\ (I_s)}{(A_{TI})\ (RRF)}$$

Example Volatile Low level Soil Tentatively Identified Compound:

With the following sample information from Sample list and Chromatogram > View Results:
File used from **TUTORIAL_VOA** project **B10040511.**

Available from Chromatogram\View		
Peak TIC Area (peak 5.76)	= 18975064	A_T
Peak TIC RRF	= 1.00	RRF
Pentafluorobenzene ISTD TIC Area (peak 6.31)	= 6372218	A_{TI}

Available from Sample List columns		
Pentafluorobenzene ISTD Amount (ng)	= 250ng	I_s
Sample vol/wt	= 5.2g	W_s
% moisture	= 50	D (see calculation below)

X_s = $\frac{(18975064)(250ng)}{(1.00)(6372218)}$ = 744.45**ng** on column of peak 5.76

D = (100 – 50)/100
= 0.50

Final concentration calculation

Estimated Conc. (ug/Kg) = $\frac{(A_T)\ (I_s)}{(A_{TI})\ (RRF)\ (W_s)(D)}$ = $\frac{(X_s)}{(W_s)(D)}$ (where X_s is TurboMass amount – ng)

= $\frac{(744.45ng)(1)}{(5.2g)\ (0.50)}$

= 286.33 ng/g = 286.33 ug/Kg

Volatile Medium Level Soil Tentatively Identified Compound

	Conditions	Analysis	Sample Type	Conc Level	Matrix	Conc A	Conc B	Conc C	Conc D	Dil Factor	VOA Moisture Calc	Moisture	Samp Wt/Vol	Aliqu Vol	Ext Vol
1	BFB	VOA	Tune Eval	Medium	Water					1.000		0.0	5.0	100.0	10000.0
2	VOA STD. 5ng/ml	VOA	Init Calib	Medium	Water	25	250	250	50	1.000		0.0	5.0	100.0	10000.0
3	VOA STD. 20ng/ml	VOA	Init Calib	Medium	Water	100	250	250	200	1.000		0.0	5.0	100.0	10000.0
4	VOA STD. 50ng/ml	VOA	Init Calib	Medium	Water	250	250	250	500	1.000		0.0	5.0	100.0	10000.0
5	VOA STD. 100ng/ml	VOA	Init Calib	Medium	Water	500	250	250	1000	1.000		0.0	5.0	100.0	10000.0
6	VOA STD. 200ng/ml	VOA	Init Calib	Medium	Water	1000	250	250	2000	1.000		0.0	5.0	100.0	10000.0
7	BLANK	VOA	Meth Blank	Medium	Water		250	250		1.000		0.0	5.0	100.0	10000.0
8	Sample 1	VOA	Analyte	Medium	Water		250	250		1.000	Y	50.0	5.0	100.0	10000.0

Est. Conc. (ug/Kg) = $\frac{(A_T)\ (I_s)\ (V_t)^1\ (Df)}{(A_{TI})\ (RRF)\ (W_s)\ (V_a)\ (D)}$ = $\frac{(X_s)\ (V_t)\ (Df)}{(W_s)\ (V_a)\ (D)}$ (where X_s is TurboMass amount – ng)

Where:

A_T Area of the total chromatogram (TIC) for the compound to be measured – from integrated results of the total ion chromatogram.

A_{TI} Area of the total ion chromatogram (TIC) for the internal standard – from integrated results of the total ion chromatogram.

I_s Amount of internal standard added in nanograms (ng) – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

RRF 1. The relative response factor value of a tentatively identified compound will always be one.

V_t^1 The total volume of the methan.ol extract in microliters (uL). This value is applied in two different ways

If **Additional Moisture** is set to **"N"** in the **VOA Moisture Calc** field in the Sample List, the Total volume of the methanol extract in microliters (uL) is used for V_t– derived directly from 'Ext Vol [SOIL_EXTRACT_VOLUME]' column in Sample List.

If **Additional Moisture** is set to **"Y"** in the **VOA Moisture Calc** field in the Sample List, the following additional calculation occurs and is used by the software for the V_t value in the concentration equation above.

$$= \left[\frac{(V_t)}{(1000uL/mL)} + \frac{(\%\ moisture)\ (Ws)}{100} \right] * \quad 1000uL/mL$$

Va Volume of the aliquot of the sample methanol extract (i.e., sample extract not including the methanol added to equal 100uL) in microliters (uL) added to reagent water for purging – derived from 'Aliqu Vol – SOIL_ALIQUOT_VOLUME' from the Sample List.

W_s Weight of sample added to the purge tube in grams (g) – derived from 'SampleWt/Vol' column in Sample List.

[1] Identifies that there are two possible values for Vt dependent upon the Additional Moisture column in the Sample List.

D — Adjustment for dry weight basis, calculated from 'Moisture' column in Sample list.

$$= \frac{100 - \%\ \text{moisture}}{100}$$

Df — Dilution Factor – derived from 'Dil Factor – DILUTION FACTOR' from the sample list.

X_s — TurboMass amount of tentatively identified compound calculated on column, in nanograms (ng).

$$= \frac{(A_T)\ (I_s)}{(A_{TI})\ (RRF)}$$

Example Volatile Medium level Soil Calculation:

With the following sample information from Sample list and Chromatogram > View Results:
File used from **TUTORIAL_VOA** project **B10040511**.

Available from Chromatogram\View

Peak TIC Area (peak 5.76)	= 18975064	A_T
Peak TIC RRF	= 1.00	RRF
Pentafluorobenzene ISTD TIC Area (peak 6.31)	= 6372218	A_{TI}

Available from Sample List columns

Pentafluorobenzene ISTD Amount (ng)	= 250ng	I_s
Sample vol/wt	= 5.0g	W_s
Ext Vol (Total Vol. of Methanol Extract)	= 10000uL	V_t (see calculation below)
Aliqu Vol (Soil Aliquot Vol)	= 100uL	Va
% moisture	= 50	D (see calculation below)
Dilution Factor	= 1	Df

$X_s = \frac{(18975064)(250ng)}{(1.00)(6372218)} = 744.45$**ng** on column of peak 5.76

D = (100 – 50)/100
= 0.50

Vt with additional moisture set to "Y" in the VOA Moisture Calc field

$$= \left[\frac{10000uL}{1000uL/mL} + \frac{(50)(5.0g)}{100} \right] * 1000uL/mL$$

= (10mL + 2.5) *1000uL/mL
Vt = 12500uL

Vt with additional moisture set to "N" in the VOA Moisture Calc field:
Vt = 10000uL

Est. Conc. (ug/Kg) = $\frac{(A_T)(I_s)(V_t)^3(Df)}{(A_{TI})(RRF)(W_s)(V_a)(D)} = \frac{(X_s)(V_t)(Df)}{(W_s)(V_a)(D)}$ (where X_s is TurboMass amount – ng)

Final concentration with additional moisture calculation

$= \frac{(744.45ng)(12500uL)(1)}{(5.0g)(100uL)(0.50)}$

= 37222ng/g = 37222 ug/Kg

Final concentration without additional moisture calculation

$= \frac{(744.45ng)(10000uL)(1)}{(5.0g)(100uL)(0.50)}$

= 29778 ng/g = 29778 ug/Kg

Semi-Volatile Water Tentatively Identified Compound

	Conditions	Analysis	Sample Type	Matrix	Conc A	Conc B	Conc C	Conc D	Inject Volume	Dil Factor	GPC	Samp Wt/Vol	Conc Ext Vol
1	DFTPP	SV	Tune Eval	Water					1.000	1.000	N	1.0	1000.0
2	SV STD 80 ng BNs/200 ng acids	SV	Init Calib	Water	80	200	80	200	1.000	1.000	N	1.0	1000.0
3	SV STD 60 ng BNs/150 ng acids	SV	Init Calib	Water	60	150	60	150	1.000	1.000	N	1.0	1000.0
4	SV STD 40 ng BNs/100 ng acids	SV	Init Calib	Water	40	100	40	100	1.000	1.000	N	1.0	1000.0
5	SV STD 20 ng BNs/50 ng acids	SV	Init Calib	Water	20	50	20	50	1.000	1.000	N	1.0	1000.0
6	SV STD 8 ng BNs/20 ng acids	SV	Init Calib	Water	8	20	8	20	1.000	1.000	N	1.0	1000.0
7	CCS	SV	Cont Calib	Water	40	100	40	100	1.000	1.000	N	1.0	1000.0
8	MB	SV	Meth Blank	Water	25	50			1.000	1.000	N	1000.0	2000.0
9	Sample1	SV	Analyte	Water	25	50	25	50	1.000	25.000	Y	1000.0	2000.0

Est. Conc. (ug/L) = $\frac{(A_T)\ (I_s)\ (V_t)(Df)(GPC)}{A_{TI})\ (RRF)\ (V_o)\ (V_i)} = \frac{(X_s)(V_t)(Df)(GPC)}{(V_o)(V_i)}$ (where X_s is TurboMass amount – ng)

Where:

A_T — Area of the total chromatogram (TIC) for the compound to be measured – from integrated results of the total ion chromatogram.

A_{TI} — Area of the total ion chromatogram (TIC) for the internal standard – from integrated results of the total ion chromatogram.

I_s — Amount of internal standard added in nanograms (ng) – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

RRF — 1. The relative response factor value of a tentatively identified compound will always be one.

V_o — Volume of water extracted in milliliters (mL) – derived from 'SampleWt/Vol' column in Sample List.

V_i — Volume of extract injected in microliters (uL) – derived from 'Inject Volume' column in Sample List.

V_t — Volume of the concentrated extract in microliters (uL) – derived from 'Conc Ext Vol' column in Sample List.

GPC — Gel Permeation Cleanup factor – derived from 'GPC' column setting in Sample list **Y** or **N**.
GPC = 1.0 if assigned N in GPC column in sample list
GPC = 2.0 if assigned Y in GPC column in sample list

Df — Dilution Factor – derived from 'Dil Factor – DILUTION FACTOR' from the Sample List.

X_s — TurboMass amount of tentatively identified compound calculated on column, in nanograms (ng).

$= \frac{(A_T)\ (I_s)}{(A_{TI})\ (RRF)}$

Example Semi-Volatile Water Tentatively Identified Compound Calculation:

With the following sample information from Sample list and Chromatogram > View Results:
File used from **TUTORIAL_SVOA** project **51006d06**.

Available from Chromatogram View

Peak TIC Area (peak 7.11)	= 7687394	A_T
Peak TIC RRF	= 1.00	RRF
1,4-dichlorobenzene-d4 ISTD TIC Area (peak 6.56)	= 4687388	A_{TI}

Available from Sample List columns

Chrysene-d12 ISTD Amount (ng)	= 10ng	I_s
Sample vol/wt	= 1000mL	V_o
Volume of extract injected (uL)	= 1uL	V_i
Volume of concentrated extract (uL)	= 2000uL	V_t
Dilution Factor	= 25	Df
GPC	= Y	GPC

Est. Conc. (ug/L) $= \frac{(A_T)\,(I_s)\,(V_t)(Df)(GPC)}{(A_{TI})\,(RRF)\,(V_o)\,(V_i)} = \frac{(X_s)(V_t)(Df)(GPC)}{(V_o)(V_i)}$ (where X_s is TurboMass amount – ng)

$X_s = \frac{(7687394)(10ng)}{(1.00)(4687388)}$ = **16.40 ng** on column of Pyrene

Final concentration calculation

$= \frac{(16.40ng)(2000uL)(25)(2)}{(1000mL)\,(1uL)}$

= 1640 ng/mL = 1640 ug/L

Semi-Volatile Soil Tentatively Identified Compound

	Conditions	Analysis	Sample Type	Matrix	Conc A	Conc B	Conc C	Conc D	Inject Volume	Dil Factor	GPC	Moisture	Samp Wt/Vol	Conc Ext Vol
1	DFTPP	SV	Tune Eval	Soil					1.000	1.000	N	0.0	1.0	1000.0
2	SV STD 80 ng BNs/20	SV	Init Calib	Soil	80	200	80	200	1.000	1.000	N	0.0	1.0	1000.0
3	SV STD 60 ng BNs/15	SV	Init Calib	Soil	60	150	60	150	1.000	1.000	N	0.0	1.0	1000.0
4	SV STD 40 ng BNs/10	SV	Init Calib	Soil	40	100	40	100	1.000	1.000	N	0.0	1.0	1000.0
5	SV STD 20 ng BNs/50	SV	Init Calib	Soil	20	50	20	50	1.000	1.000	N	0.0	1.0	1000.0
6	SV STD 8 ng BNs/20	SV	Init Calib	Soil	8	20	8	20	1.000	1.000	N	0.0	1.0	1000.0
7	CCS	SV	Cont Calib	Soil	40	100	40	100	1.000	1.000	N	0.0	1.0	1000.0
8	MB	SV	Meth Blank	Soil	25	50			1.000	1.000	N	0.0	15.0	2000.0
9	Sample1	SV	Analyte	Soil	25	50	25	50	1.000	25.000	Y	14.0	15.2	2000.0

Est. Conc. (ug/Kg) = $\frac{(A_T)\ (I_s)\ (V_t)(Df)(GPC)}{(A_{TI})\ (RRF)(V_i)(W_s)(D)} = \frac{(X_s)(V_t)(Df)(GPC)}{(V_i)\ (W_s)\ (D)}$ (where X_s is TurboMass amount – ng)

Where:

A_T Area of the total chromatogram (TIC) for the compound to be measured – from integrated results of the total ion chromatogram.

A_{TI} Area of the total ion chromatogram (TIC) for the internal standard – from integrated results of the total ion chromatogram.

I_s Amount of internal standard added in nanograms (ng) – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

RRF 1. The relative response factor value of a tentatively identified compound will always be one.

V_i Volume of extract injected in microliters (uL) – derived from 'Inject Volume' column in Sample List.

V_t Volume of the concentrated extract in microliters (uL) – derived from 'Conc Ext Vol' column in Sample List.

W_s Weight of sample extracted in grams (g) – derived from 'SampleWt/Vol' column in Sample List.

D Adjustment for dry weight basis, calculated from 'Moisture' column in Sample List.

$$= \frac{100 - \%\ \text{moisture}}{100}$$

GPC Gel Permeation Cleanup factor – derived from 'GPC' column setting in Sample list **Y** or N**.**

GPC = 1.0 if assigned N in GPC column in sample list
GPC = 2.0 if assigned Y in GPC column in sample list

Df Dilution Factor – derived from 'Dil Factor – DILUTION FACTOR' from the sample list.

X_s TurboMass amount of tentatively identified compound calculated on column, in nanograms (ng).

$$= \frac{(A_T)\ (I_s)}{(A_{TI})\ (RRF)}$$

Example Semi-Volatile Soil Tentatively Identified Compound:

With the following sample information from Sample list or Chromatogram\View Results :
File used from **TUTORIAL_SVOA project 51006d06**.

Available from Chromatogram View

Peak TIC Area (peak 7.11)	= 7687394	A_T
Peak TIC RRF	= 1.00	RRF
1,4-dichlorobenzene-d4 ISTD TIC Area (peak 6.56)	= 4687388	A_{TI}

Available from Sample List columns

Chrysene-d12 ISTD Amount (ng)	= 10ng	I_s
Sample vol/wt	= 15.2g	W_s
% Moisture	= 14.0	D (see calculation below)
Volume of extract injected (uL)	= 1uL	V_i
Volume of concentrated extract (uL)	= 2000uL	V_t
Dilution Factor	= 25	Df
GPC	= Y	GPC

X_s = $\frac{(7687394)(10ng)}{(1.00)(4687388)}$ = **16.40 ng** on column of Pyrene

D = (100 – 14)/100
= 0.86

Final concentration calculation

Est. Conc (ug/Kg) = $\frac{(A_T)\ (I_s)\ (V_t)(Df)(GPC)}{(A_{TI})\ (RRF)(V_i)(W_s)(D)}$ = $\frac{(X_s)(V_t)(Df)(GPC)}{(V_i)\ (W_s)\ (D)}$ (where X_s is TurboMass amount – ng)

= $\frac{(16.40ng)(2000uL)(25)(2)}{(1uL)\ (15.2g)\ (0.86)}$

= 125458 ng/g = 125458 ug/Kg

Surrogate Recovery Calculations

Volatile Surrogate Recovery

This calculation applies to Volatile waters, Volatile low level soils, and Volatile medium level soils.

$$\text{Percent Recovery} = \left[\frac{X_s}{S_a}\right] * 100$$

Where:

X_s TurboMass amount of surrogate compound calculated on column in nanograms (ng).

$$= \frac{(A_x)\,(I_s)}{(A_{is})\,(RRF)}$$

S_a Amount of surrogate added in nanograms (ng) – derived from the environmental parameters section of the Quantify method.

Below is an example demonstrating how to calculate the value placed into the Surrogate amount field in the Quantify method. This value is based on the assumption that the surrogate is added to the purge tube with the aliquot of sample in it.

*= (Amount spiked into Sample in microliters (uL)) * (Concentration of surrogate solution in ug/mL)*

Concentration of surrogate solution in ug/mL = 125 ug/mL
Amount spiked into Sample in microliters (uL) = 2 uL

*= (2uL) * (125ug/mL)*
= 250ng

A_x Area[1] of the characteristic ion (Extracted Ion current profile – EICP) for the compound to be measured – from integration results.

A_{is} Area[1] of the characteristic ion (Extracted Ion current profile – EICP) for the internal standard – from integration results.

I_s Amount of internal standard added in nanograms (ng) – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

[1] Area" is referenced here, since it is specified by the EPA methods, but the software will actually use whatever response mode (area or height) is specified in the Quantify Method. This applies to all the environmental calculations defined in this document.

RRF The average relative response factor (RRF) of surrogate compound calculated from the calibration curve.

$$= \frac{(A_x)\ (I_s)}{(A_{is})\ (C_s)}$$

Where C_s Amount of the compound to be measured, in the calibration standard – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

Example Surrogate Recovery Calculation:

With the following sample information from Sample list, Quantify/View Results and Quantify Method:
File used from **VOA_TUTORIA**L project **B10040511**.

Available from Quantify\View Results		
Toluene-d8 surrogate EIC Area	= 4399797	A_x
Toluene-d8 Avg RRF	= 1.87411	RRF
Pentafluorobenzene ISTD EIC Area	= 2407227	A_{is}
Available from Sample List columns		
Pentafluorobenzene ISTD Amount (ng)	= 250ng	I_s
Available from Quantify Method		
Toluene-d8 Surrogate Amount (ng)	= 250ng	S_a

$$X_s = \frac{(4399797)(250ng)}{(1.87411)(2407227)} = \mathbf{243.82\ ng}$$ on column of Toluene-d8

Recovery Calculation

$$\%\ \text{Recovery} = \left[\frac{X_s}{S_a}\right] * 100$$

$$= \left[\frac{243.82ng}{250ng}\right] * 100$$

= 97.5% Recovery

Semi-Volatile Surrogate Recovery

$$\text{Percent Recovery} = \left[\frac{(X_s)\ (Df)\ (GPC)\ (V_t)}{(S_a)\ (V_i)} \right] * 100$$

Where:

X_s TurboMass amount of surrogate compound calculated on column in nanograms (ng).

$$= \frac{(A_x)\ (I_s)}{(A_{is})\ (RRF)}$$

S_a Amount of surrogate added in nanograms (ng) – derived from the environmental parameters section of the Quantify method.

Below is an example demonstrating how to calculate the value placed into the Surrogate amount field in the Quantify method. This value is based on the assumption that the surrogate is added to the sample at the time of extraction.

*= (Amount spiked into Sample in microliters (uL)) * (Concentration of surrogate solution in ug/mL)*

Concentration of surrogate solution in ug/mL = 100 ug/mL
Amount spiked into Sample in microliters (uL) = 500 uL

*= (500uL) * (100ug/mL)*
= 50000 ng

V_i Volume of extract injected in microliters (uL) – derived from 'Inject Volume' column in Sample List.

V_t Volume of the concentrated extract in microliters (uL) – derived from 'Conc Ext Vol' column in Sample List.

GPC Gel Permeation Cleanup factor – derived from 'GPC' column setting in Sample list **Y** or **N**.
GPC = 1.0 if assigned N in GPC column in sample list
GPC = 2.0 if assigned Y in GPC column in sample list

Df Dilution Factor – derived from 'Dil Factor – DILUTION FACTOR' from the sample list.

A_x Area[1] of the characteristic ion (Extracted Ion current profile – EICP) for the compound to be measured – from integration results.

A_{is} Area[2] of the characteristic ion (Extracted Ion current profile – EICP) for the internal standard – from integration results.

I_s Amount of internal standard added in nanograms (ng) – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

[1] Area" is referenced here, since it is specified by the EPA methods, but the software will actually use whatever response mode (area or height) is specified in the Quantify Method. This applies to all the environmental calculations defined in this document.

RRF The average relative response factor (RRF) of surrogate compound calculated from the calibration curve.

$$= \frac{(A_x)\ (I_s)}{(A_{is})\ (C_s)}$$

Where C_s Amount of the compound to be measured, in the calibration standard – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

Example Semi-Volatile Surrogate Recovery Calculation:

With the following sample information from Sample list, Quantify/View Results and Quantify Method:
File used from **SVOA_TUTORIAL** project **51006d06**.

Available from Quantify\View Results		
p-terphenyl-d14 surrogate EIC Area	= 2961600	A_x
p-terphenyl d-14 Avg RRF	= 0.85513	RRF
Chrysene-d12 ISTD EIC Area	= 2154951	A_{is}
Available from Sample List columns		
Chrysene-d12 ISTD Amount (ng)	= 10 ng	I_s
Volume of extract injected (uL)	= 1uL	V_i
Volume of concentrated extract (uL)	= 2000uL	V_t
Dilution Factor	= 2	Df
GPC	= N	GPC
Available from Quantify Method		
p-terphenyl-d14 Surrogate Amount (ng)	= 50000 ng	S_a

$$X_s = \frac{(2961600)(10ng)}{(0.85513)(2154951)} = \textbf{16.07 ng}$$ on column of p-terphenyl-d14

Recovery Calculation

$$\text{Percent Recovery} = \left[\frac{(X_s)\ (Df)\ (GPC)\ (V_t)}{(S_a)\ (V_i)}\right] * 100$$

$$= \left[\frac{(16.07ng)\ (2)\ (1)(2000uL)}{(50000ng)\ (1uL)}\right] * 100$$

= 128.56% Recovery

Spike Recovery Calculations

Volatile Spike Recovery

This calculation applies to Volatile waters, Volatile low level soils, and Volatile medium level soils.

$$\text{Percent Recovery} = \left[\frac{X_s - C_{us}}{S_a}\right] * 100$$

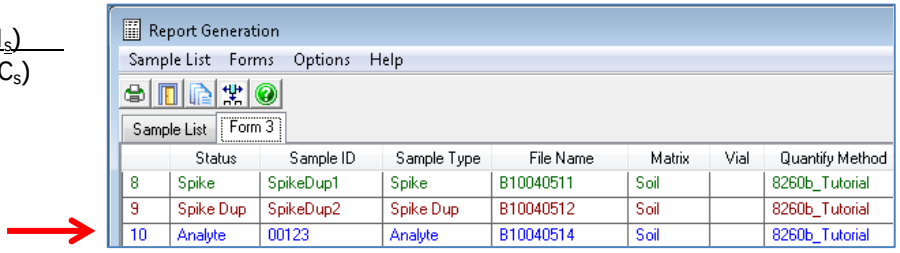
Where:

X_s TurboMass amount of spike compound calculated on column in nanograms (ng) for Spike or Spike Dup.

$$= \frac{(A_x)\,(I_s)}{(A_{is})\,(RRF)}$$

C_{us} TurboMass amount of spike compound calculated on column in nanograms (ng) for the unspiked sample – designated in the environmental form window as '**Analyte**'.

$$= \frac{(A_x)\,(I_s)}{(A_{is})\,(C_s)}$$



	Status	Sample ID	Sample Type	File Name	Matrix	Vial	Quantify Method
8	Spike	SpikeDup1	Spike	B10040511	Soil		8260b_Tutorial
9	Spike Dup	SpikeDup2	Spike Dup	B10040512	Soil		8260b_Tutorial
10	Analyte	00123	Analyte	B10040514	Soil		8260b_Tutorial

Note: If C_{us} is below the MDL specified in the quantify method, a value of zero is used.

A_x Area[1] of the characteristic ion (Extracted Ion current profile – EICP) for the compound to be measured – from integration results.

A_{is} Area[1] of the characteristic ion (Extracted Ion current profile – EICP) for the internal standard – from integration results.

I_s Amount of internal standard added in nanograms (ng) – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

RRF The average relative response factor (RRF) of spike compound calculated from the calibration curve.

$$= \frac{(A_x)\,(I_s)}{(A_{is})\,(C_s)}$$

Where C_s Amount of the compound to be measured, in the calibration standard – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

[1] Area" is referenced here, since it is specified by the EPA methods, but the software will actually use whatever response mode (area or height) is specified in the Quantify Method. This applies to all the environmental calculations defined in this document.

S_a Amount of spike added in nanograms (ng) – derived from the environmental parameters section of the Quantify method.

Below is an example demonstrating how to calculate the value placed into the Spike amount field in the Quantify method. This value is based on the assumption that the spike is added to the purge tube with the aliquot of sample in it.

*= (Amount spiked into Sample in microliters (uL)) * (Concentration of spike solution in ug/mL)*

Concentration of spike solution in ug/mL = 50 ug/mL
Amount spiked into Sample in microliters (uL) = 2 uL
*= (2uL) * (50ug/mL)*
= 100ng

Example Spike Recovery Calculation:

With the following sample information from Sample list, Quantify/View Results and Quantify method:
File used from **VOA_TUTORIAL** project **B10040511 Spike**, **B10040514 Analyte.**

Available from Quantify\View Results for B10040511 – designated as Spike

1,1-dichloroethene EIC Area	= 407416	A_x
1,1-dichloroethene Avg RRF	= 0.398355	RRF
Pentafluorobenzene ISTD EIC Area	= 2407227	A_{is}

Available from Quantify\View Results for B10040514 – designated as Analyte (unspiked sample)

1,1-dichloroethene surrogate EIC Area	= 5692	A_x
1,1-dichloroethene Avg RRF	= 0.398355	RRF
Pentafluorobenzene ISTD EIC Area	= 2277837	A_{is}

Available from Sample List columns

Pentafluorobenzene ISTD Amount (ng)	= 250ng	I_s

Available from Quantify Method

1,1-Dichloroethene Spike Amount (ng)	= 100ng	S_a

X_s = $\frac{(407416)(250ng)}{(0.398355)(2407227)}$ = **106.22 ng** on column of 1,1-Dichloroethene

U_s = $\frac{(5692)(250ng)}{(0.398355)(2277837)}$ = **1.568 ng** on column of 1,1-Dichloroethene

Recovery Calculation

Percent Recovery = $\left[\frac{X_s - C_{us}}{S_a}\right]$ * 100

= $\left[\frac{106.22ng - 1.568ng}{100ng}\right]$ * 100

= 104.65% Recovery

Semi-Volatile Spike Recovery

$$\text{Percent Recovery} = \left[\frac{(X_s - C_{us})\ (Df)\ (GPC)\ (V_t)}{(S_a)\ (V_i)} \right] * 100$$

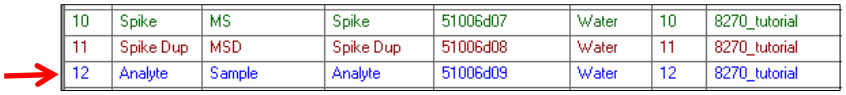
Where:

X_s TurboMass amount of spike compound calculated on column in nanograms (ng).

$$= \frac{(A_x)\ (I_s)}{(A_{is})\ (RRF)}$$

C_{us} TurboMass amount of spike compound calculated on column in nanograms (ng) for the unspiked sample – designated in the environmental form window as '**Analyte**'.

$$= \frac{(A_x)\ (I_s)}{(A_{is})\ (RRF)}$$



10	Spike	MS	Spike	51006d07	Water	10	8270_tutorial
11	Spike Dup	MSD	Spike Dup	51006d08	Water	11	8270_tutorial
12	Analyte	Sample	Analyte	51006d09	Water	12	8270_tutorial

Note: If C_{us} is below the MDL specified in the quantify method, a value of zero is used.

A_x Area[2] of the characteristic ion (Extracted Ion current profile – EICP) for the compound to be measured – from integration results.

A_{is} Area[2] of the characteristic ion (Extracted Ion Current Profile – EICP) for the internal standard – from integration results.

I_s Amount of internal standard added in nanograms (ng) – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

RRF The average relative response factor (RRF) of spike compound calculated from the calibration curve.

$$= \frac{(A_x)\ (I_s)}{(A_{is})\ (C_s)}$$

Where C_s Amount of the compound to be measured, in the calibration standard – derived from appropriate Concentration column of the Sample list (as defined in the Quantify Method and the Sample List).

S_a Amount of spike added in nanograms (ng) – derived from the environmental parameters section of the Quantify method.

Below is an example demonstrating how to calculate the value placed into the Spike amount field in the Quantify method. This value is based on the assumption that the surrogate is added to the sample at the time of extraction.

*= (Amount spiked into Sample in microliters (uL)) * (Concentration of spike solution in ug/mL)*

Concentration of spike solution in ug/mL = 100 ug/mL
Amount spiked into Sample in microliters (uL) = 500 uL
*= (500uL) * (100ug/mL)*
= 50000 ng

V_i Volume of extract injected in microliters (uL) – derived from 'Inject Volume' column in Sample List.

V_t Volume of the concentrated extract in microliters (uL) – derived from 'Conc Ext Vol' column in Sample List.

GPC Gel Permeation Cleanup factor – derived from 'GPC' column setting in Sample List **Y** or **N**.
GPC = 1.0 if assigned N in GPC column in sample list.
GPC = 2.0 if assigned Y in GPC column in sample list.

Df Dilution Factor – derived from 'Dil Factor – DILUTION FACTOR' from the Sample List.

Example Semi-Volatile Spike Recovery Calculation:

With the following sample information from Sample list, Quantify/View Results, and Quantify Method:

File used from **SVOA_TUTORIAL** project **51006d07 Spike**, **51006d09 Unspiked Sample**.

Available from Quantify\View Results for 51006d07 – designated as Spike

Pyrene EIC Area	= 4339724	A_x
Pyrene Avg RRF	= 1.2929	RRF
Chrysene-d12 ISTD EIC Area	= 1408276	A_{is}

Available from Quantify\View Results for 51006d09 – designated as Analyte (unspiked sample)

Pyrene EIC Area	= 9129	A_x
Pyrene Avg RRF	= 1.2929	RRF
Chrysene-d12 ISTD EIC Area	= 1310746	A_{is}

Available from Sample List columns

Chrysene-d12 ISTD Amount (ng)	= 10 ng	I_s
Volume of extract injected (uL)	= 1uL	V_i
Volume of concentrated extract (uL)	= 2000uL	V_t
Dilution Factor	= 1	Df
GPC	= N	GPC

Available from Quantify Method

Pyrene Spike Amount (ng)	= 50000 ng	S_a

$X_s = \frac{(4339724)(10ng)}{(1.2929)(1408276)}$ = **23.83 ng** on column of p-terphenyl-d14

$C_{us} = \frac{(9129)(10ng)}{(1.2929)(1310746)}$ = **0.0539 ng** on column of p-terphenyl-d14

Recovery Calculation

$$\text{Percent Recovery} = \left[\frac{(X_s - C_{us})\,(Df)\,(GPC)\,(V_t)}{(S_a)\,(V_i)}\right] * 100$$

$$= \left[\frac{(23.83 - 0.0539)\,(1)\,(1)(2000uL)}{(50000ng)\,(1uL)}\right] * 100$$

= 95.14 % Recovery

Concentrations for Spike Added Calculations

The value displayed on the Form 3 in the Spike Added column.

Spike Added Volatile Water Concentration

$$\text{Concentration (ug/L)} = \frac{(S_a)\ (Df)}{(V_o)}$$

Where:

S_a Amount of spike compound - derived from Quantify method for spike amount in nanograms (ng).

Amount of spike added in nanograms (ng) – derived from the environmental parameters section of the Quantify method.

Below is an example demonstrating how to calculate the value placed into the Spike amount field in the Quantify method. This value is based on the assumption that the spike is added to the purge tube with the aliquot of sample in it.

*= (Amount spiked into Sample in microliters (uL)) * (Concentration of spike solution in ug/mL)*

Concentration of spike solution in ug/mL = 50 ug/mL
Amount spiked into Sample in microliters (uL) = 2 uL
*= (2uL) * (50ug/mL)*
= 100ng

V_o Volume of water purged in milliliters (mL) – derived from 'SampleWt/Vol' column in Sample List.

Df Dilution Factor – derived from 'Dil Factor – DILUTION FACTOR' from the Sample List.

Example Volatile Water Calculation:

With the following sample information from Sample list and Quantify Method:
File used from **VOA_TUTORIAL** project **B10040511**.

Available from Quantify Method

1,1-Dichloroethene Spike Amount (ng)	= 250ng	S_a

Available from Sample List columns

Sample vol/wt	= 5.0mL	V_o
Dilution Factor	= 1	Df

Spike Added concentration calculation

$$\text{Concentration (ug/L)} = \frac{(S_s)\ (Df)}{(V_o)}$$ (where X_s is TurboMass amount – ng)

$$= \frac{(250ng)(1)}{(5.0mL)}$$

= 50 ng/mL = 50 ug/L

Spike Added Volatile Low Level Soil Concentration

The value displayed on the Form 3 in the Spike Added column

$$\text{Concentration (ug/Kg)} = \frac{(S_a)}{(W_s)(D)}$$

Where:

S_a Amount of spike compound - derived from Quantify method for spike amount in nanograms (ng).

Below is an example demonstrating how to calculate the value placed into the Spike amount field in the Quantify method. This value is based on the assumption that the spike is added to the purge tube with the aliquot of sample in it.

*= (Amount spiked into Sample in microliters (uL)) * (Concentration of spike solution in ug/mL).*

Concentration of spike solution in ug/mL = 50 ug/mL
Amount spiked into Sample in microliters (uL) = 2 uL
*= (2uL) * (50ug/mL)*
= 100ng

W_s Weight of sample added to the purge tube in grams (g) – derived from 'SampleWt/Vol' column in Sample List.

D Adjustment for dry weight basis, calculated from 'Moisture' column in Sample List.

$$= \frac{100 - \%\ \text{moisture}}{100}$$

Example Volatile Low Level Soil Calculation:

With the following sample information from Sample list and Quantify Method:
File used from **TUTORIAL_VOA** project **B10040511**.

Available from Quantify Method

1,1-Dichloroethene Spike Amount (ng)	= 250ng	S_a

Available from Sample List columns

Sample vol/wt	= 5.2g	W_s
% moisture	= 50	D (see calculation below)

D = (100 – 50)/100
= 0.50

Spike Added concentration calculation

$$\text{Concentration (ug/Kg)} = \frac{(S_s)}{(W_s)(D)}$$

$$= \frac{(250\text{ng})}{(5.2\text{g})\ (0.50)}$$

= 96.15 ng/g = 96.15 ug/Kg

Spike Added Volatile Medium Level Soil Concentration

The value displayed on the Form 3 in the Spike Added column

$$\text{Concentration (ug/Kg)} = \frac{(S_a)\ (V_t)\ (Df)}{(W_s)\ (V_a)\ (D)}$$

Where:

S_a Amount of spike compound - derived from Quantify method for spike amount in nanograms (ng).

Below is an example demonstrating how to calculate the value placed into the Spike amount field in the Quantify method. This value is based on the assumption that the spike is added to the purge tube with the aliquot of sample in it.

*= (Amount spiked into Sample in microliters (uL)) * (Concentration of spike solution in ug/mL).*

Concentration of spike solution in ug/mL = 50 ug/mL
Amount spiked into Sample in microliters (uL) = 2 uL
*= (2uL) * (50ug/mL)*
= 100ng

V_t[1] The total volume of the methanol extract in microliters (uL). This value is applied in two different ways.

If **Additional Moisture** iis set to "**N**" in the **VOA Moisture Calc** field in the Sample List, the Total volume of the methanol extract in microliters (uL) is used for V_t– derived directly from 'Ext Vol [SOIL_EXTRACT_VOLUME]' column in Sample List.

If **Additional Moisture** button is set to "**Y**" in the **VOA Moisture Calc** field in the Sample List, the following additional calculation occurs and is used by the software for the V_t value in the concentration equation above.

$$= \left[\frac{(V_t)}{(1000uL/mL)} + \frac{(\%\ \text{moisture})\ (Ws)}{100} \right] * \ 1000uL/mL$$

Va Volume of the aliquot of the sample methanol extract (i.e., sample extract not including the methanol added to equal 100uL) in microliters (uL) added to reagent water for purging – derived from 'Aliqu Vol – SOIL_ALIQUOT_VOLUME' from the Sample List.

W_s Weight of sample added to the purge tube in grams (g) – derived from 'SampleWt/Vol' column in Sample List

D Adjustment for dry weight basis, calculated from 'Moisture' column in Sample List.

$$= \frac{100 - \%\ \text{moisture}}{100}$$

Df Dilution Factor – derived from 'Dil Factor – DILUTION FACTOR' from the Sample List.

[1] Identifies that there are two possible values for Vt dependent on the Additional Moisture column in the Sample List.

Example Volatile Medium Level Soil Calculation:

With the following sample information from Sample list and Quantify Method:
File used from **TUTORIAL_VOA** project **B10040511**.

Available from Quantify Method

1,1-Dichloroethene Spike Amount (ng)	= 250ng	S_a

Available from Sample List columns

Sample vol/wt	= 5.0g	W_s
Ext Vol (Total volume of Methanol Extract)	= 10000uL	V_t (see calculation below)
Aliqu Vol (Soil Aliquot volume)	= 100uL	Va
% moisture	= 50	D (see calculation below)
Dilution Factor	= 1	Df

D = (100 – 50)/100
= 0.50

Vt with additional moisture set to "Y" in the VOA Moisture Calc field

$$= \left[\frac{10000uL}{1000uL/mL} + \frac{(50)(5.0g)}{100} \right] * 1000uL/mL$$

= (10mL + 2.5) *1000uL/mL
Vt = 12500uL

Vt with additional moisture set to "N" in the VOA Moisture Calc field:
Vt = 10000uL

Spike Added Concentration (ug/Kg) = $\frac{(X_s)\ (V_t)\ (Df)}{(W_s)\ (V_a)\ (D)}$

Concentration with additional moisture calculation

= $\frac{(250ng)(12500uL)(1)}{(5.0g)(100uL)(0.50)}$

= 12500 ng/g = 12500 ug/Kg

Concentration without additional moisture calculation

= $\frac{(250ng)(10000uL)(1)}{(5.0g)(100uL)(0.50)}$

= 10000 ng/g = 10000 ug/Kg

Spike Added Semi-Volatile Water Concentration

The value displayed on the Form 3 in the Spike Added column

$$\text{Concentration (ug/L)} = \frac{(S_a)}{(V_o)}$$

Where:

S_a Amount of spike compound - derived from Quantify method for spike amount in nanograms (ng).

Below is an example demonstrating how to calculate the value placed into the Spike amount field in the Quantify method. This value is based on the assumption that the surrogate is added to the sample at the time of extraction.

*= (Amount spiked into Sample in microliters (uL)) * (Concentration of spike solution in ug/mL).*

Concentration of spike solution in ug/mL = 100 ug/mL
Amount spiked into Sample in microliters (uL) = 500 uL
*= (500uL) * (100ug/mL)*
= 50000 ng

V_o Volume of water extracted in milliliters (mL) – derived from 'SampleWt/Vol' column in Sample List.

Example Semi-Volatile Water Calculation

With the following sample information from Sample list and Quantify Method:
File used from **TUTORIAL_SVOA** project **51006d06**.

Available from Quantify Method
Pyrene Spike Amount (ng) = 50000ng S_a

Available from Sample List columns
Sample vol/wt = 1000mL V_o

$$\text{Concentration (ug/L)} = \frac{(S_a)}{(V_o)}$$

Spike Added concentration calculation

$$= \frac{(50000\text{ng})}{(1000\text{mL})}$$

= 50 ng/mL = 50 ug/L

Spike Added Semi-Volatile Soil Concentration

The value displayed on the Form 3 in the Spike Added column

$$\text{Concentration (ug/Kg)} = \frac{(S_a)}{(W_s)\ (D)}$$

Where:

S_a Amount of spike compound - derived from Quantify method for spike amount in nanograms (ng).

Below is an example demonstrating how to calculate the value placed into the Spike amount field in the Quantify method. This value is based on the assumption that the surrogate is added to the sample at the time of extraction.

*= (Amount spiked into Sample in microliters (uL)) * (Concentration of spike solution in ug/mL).*

Concentration of spike solution in ug/mL = 100 ug/mL
Amount spiked into Sample in microliters (uL) = 500 uL
*= (500uL) * (100ug/mL)*
= 50000 ng

W_s Weight of sample extracted in grams (g) – derived from 'SampleWt/Vol' column in Sample List.

D Adjustment for dry weight basis, calculated from 'Moisture' column in Sample List.

$$= \frac{100 - \%\ \text{moisture}}{100}$$

Example Semi-Volatile Soil Calculation

With the following sample information from Sample list and Quantify Method:
File used from **TUTORIAL_SVOA** project **51006d06**.

Available from Quantify Method

Pyrene Spike Amount (ng)	= 50000ng	S_a

Available from Sample List columns

Sample vol/wt	= 15.2g	W_s
% Moisture	= 14.0	D (see calculation below)

D = (100 – 14)/100
= 0.86

Concentration calculation

$$\text{Concentration (ug/Kg)} = \frac{(S_s)}{(W_s)\,(D)}$$

$$= \frac{(50000\text{ng})}{(15.2\text{g})\,(0.86)}$$

= 3824 ng/g = 3824 ug/Kg

Relative Percent Difference (RPD) Spike Calculations

RPD by Concentration

This calculation applies to Volatile and Soil Analysis types for all matrices.

$$RPD = \left[\frac{| C1 - C2 |}{((C1 + C2)/2)} \right] * 100$$

C1 The calculated final concentration in the Spike.
C2 The calculated final concentration in the Spike Duplicate .

Example RPD by Concentration :

With the following sample information from the Form 1 or Form 3:

C1 = 21.24 ug/L
C2 = 19.80 ug/L

$$RPD = \left[\frac{| C1 - C2 |}{((C1 + C2)/2)} \right] * 100$$

$$= \frac{| 21.24 - 19.80 |}{((21.24 + 19.80/2)} * 100$$

$$= 7.0 \%$$

RPD by Recovery

This calculation applies to Volatile and Soil Analysis types for all matrices.

$$RPD = \left[\frac{|R1 - R2|}{((R1 + R2)/2)} \right] * 100$$

R1 The recovery of the Spike.
R2 The recovery of the Spike Duplicate.

Example RPD by Recovery :

With the following sample information from the Form 3:

R1 = 100.2%
R2 = 93.0%

$$RPD = \left[\frac{|R1 - R2|}{((R1 + R2)/2)} \right] * 100$$

$$= \frac{|100.2 - 93|}{((100.2 + 93)/2)} * 100$$

$$= 7.5\ \%$$

Reporting Limit Calculations (RL or MDL)

Volatile Water Reporting Limit using RL or MDL

$$\text{Concentration (ug/L)} = \frac{\text{(RL or MDL) (Df)}}{(V_o)}$$

Where:

RL	On-column Reporting limit in nanograms (ng) – derived from Submitter\Task Custom compound list in the 'Water Limit' column or from the Reporting Threshold field in the Quantify Method. This value is typically determined by the lowest level that is used in the initial calibration curve.
MDL	On-column Method detection limit in nanograms (ng) – derived from the Quantify Method in the Environmental Parameters section in the 'MDL Water' field. This value is typically determined statistically by analyzing 7 replicate samples, however it can be any user determined value.
V_o	Volume of water purged in milliliters (mL) – derived from 'SampleWt/Vol' column in Sample List.
Df	Dilution Factor – derived from 'Dil Factor – DILUTION FACTOR' from the Sample List.

Example Volatile RL or MDL Calculation:

With the following sample information from Sample list, Submitter\Task or Quantify Method:

Available from Submitter\Task		
On column reporting Limit for 1,1-dichloroethene	= 2.5 ng	RL
Available from Quantify Method		
On column MDL for 1,1-dichloroethene	= 0.234 ng	MDL
Available from Sample List columns		
Sample vol/wt	= 5.0mL	V_o
Dilution Factor	= 1	Df

Final Reporting limit calculation by RL

$$\text{RL/MDL Concentration (ug/L)} = \frac{\text{(RL) (Df)}}{(V_o)}$$

$$= \frac{\text{(2.5 ng)(1)}}{\text{(5.0mL)}}$$

= 0.50 ng/mL = 0.50 ug/L

Final Reporting limit calculation by MDL

$$\text{RL/MDL Concentration (ug/L)} = \frac{\text{(MDL) (Df)}}{(V_o)}$$

$$= \frac{\text{(0.234 ng)(1)}}{\text{(5.0mL)}}$$

= 0.0468 ng/mL = 0.0468 ug/L

Volatile Low Level Soil Reporting Limit (RL or MDL)

$$\text{Concentration (ug/Kg)} = \frac{\text{(RL or MDL)}}{(W_s)(D)}$$

Where:

RL — On-column Reporting limit in nanograms (ng) – derived from Submitter\Task Custom compound list in the 'Water Limit' column or from the Reporting Threshold in the Quantify Method. This value is typically determined by the lowest level that is used in the initial calibration curve.

MDL — Method detection limit in nanograms (ng) – derived from the Quantify Method in the Environmental Parameters section in the 'MDL Water' field. This value is typically determined statistically by analyzing 7 replicate samples, however it can be any user determined value.

W_s — Weight of sample added to the purge tube in grams (g) – derived from 'SampleWt/Vol' column in Sample List.

D — Adjustment for dry weight basis, calculated from 'Moisture' column in Sample List.

$$= \frac{100 - \%\ \text{moisture}}{100}$$

Example Volatile Low Level Soil RL/MDL Calculation:

With the following sample information from Sample list, Submitter\Task or Quantify Method:

Available from Submitter\Task
On column reporting Limit for 1,1-dichloroethene = 25 ng RL

Available from Quantify Method
On column MDL for 1,1-dichloroethene = 0.345 ng MDL

Available from Sample List columns
Sample vol/wt = 5.2g W_s
% moisture = 50 D (see calculation below)

D = (100 – 50)/100
= 0.50

Final Reporting limit calculation by RL

$$\text{RL/MDLConcentration (ug/Kg)} = \frac{\text{(RL)}}{(W_s)(D)}$$

$$= \frac{(25\ \text{ng})}{(5.0\text{g})(0.50)}$$

= 10 ng/g = 10 ug/Kg

Final Reporting limit calculation by MDL

$$\text{RL/MDL Concentration (ug/Kg)} = \frac{\text{(MDL)}}{(W_s)(D)}$$

$$= \frac{(0.345\ \text{ng})}{(5.0\text{g})(0.50)}$$

= 0.138 ng/g = 0.138 ug/Kg

Volatile Medium Level Soil Reporting Limit (RL or MDL)

$$\text{Concentration (ug/Kg)} = \frac{(\text{RL or MDL})\ (V_t^3)\ (Df)}{(W_s)\ (V_a)\ (D)}$$

Where:

RL — On-column Reporting limit in nanograms (ng) – derived from Submitter\Task Custom compound list in the 'Water Limit' column or from the Reporting Threshold field in the Quantify Method. This value is typically determined by the lowest level that is used in the initial calibration curve.

MDL — Method detection limit in nanograms (ng) – derived from the Quantify Method in the Environmental Parameters section in the 'MDL Water' field. This value is typically determined statistically by analyzing 7 replicate samples, however it can be any user determined value.

V_t[1] — The total volume of the methanol extract in microliters (uL). This value is applied in two different ways.

If **Additional Moisture** is set to "**N**" in the **VOA Moisture Calc** field in the Sample List, the Total volume of the methanol extract in microliters (uL) is used for V_t– derived directly from 'Ext Vol [SOIL_EXTRACT_VOLUME]' column in Sample List.

If **Additional Moisture** is set to **"Y"** in the **VOA Moisture Calc** field in the Sample List, the following additional calculation occurs and is used by the software for the V_t value in the concentration equation above.

$$= \left[\frac{(V_t)}{(1000\text{uL/mL})} + \frac{(\%\ \text{moisture})\ (Ws)}{100} \right] * \quad 1000\text{uL/mL}$$

Va — Volume of the aliquot of the sample methanol extract (i.e., sample extract not including the methanol added to equal 100uL) in microliters (uL) added to reagent water for purging – derived from 'Aliqu Vol – SOIL_ALIQUOT_VOLUME' from the Sample List.

W_s — Weight of sample added to the purge tube in grams (g) – derived from 'SampleWt/Vol' column in Sample List.

D — Adjustment for dry weight basis, calculated from 'Moisture' column in Sample List.

$$= \frac{100 - \%\ \text{moisture}}{100}$$

Df — Dilution Factor – derived from 'Dil Factor – DILUTION FACTOR' from the Sample List.

[1] Identifies that there are two possible values for Vt dependent on the Additional Moisture column in the Sample List.

Example Volatile Medium Level Soil Calculation:

With the following sample information from Sample list, Submitter\Task or Quantify Method:

Available from Submitter\Task
On column reporting Limit for 1,1-dichloroethene = 2.5 ng RL

Available from Quantify Method
On column MDL for 1,1-dichloroethene = 0.234 ng MDL

Available from Sample List columns

Sample vol/wt	= 5.0g	W_s
Ext Vol (Total volume of Methanol Extract)	= 10000uL	V_t (see calculation below)
Aliqu Vol (Soil Aliquot volume)	= 100uL	Va
% moisture	= 50	D (see calculation below)
Dilution Factor	= 1	Df

D = (100 – 50)/100
= 0.50

Vt with additional moisture is set to “Y” in the VOA Moisture Calc Field:

$$= \left[\frac{10000uL}{1000uL/mL} + \frac{(50)(5.0g)}{100} \right] * 1000uL/mL$$

= (10mL + 2.5) *1000uL/mL
Vt = 12500uL

Vt with additional moisture set to “N” in the VOA Moisture Calc field:
Vt = 10000uL

$$\text{RL/MDL Concentration (ug/Kg)} = \frac{(\text{RL or MDL})(V_t^3)(Df)}{(W_s)(V_a)(D)}$$

Reporting limit(RL) with additional moisture Calculation

$$= \frac{(2.5ng)(12500uL)(1)}{(5.0g)(100uL)(0.50)}$$

= 125ng/g = 125 ug/Kg

Reporting limit(RL) without additional moisture calculation

$$= \frac{(2.5ng)(10000uL)(1)}{(5.0g)(100uL)(0.50)}$$

= 100 ng/g = 100 ug/Kg

Reporting limit(MDL) with additional moisture

Calculation

$$= \frac{(0.234ng)(12500uL)(1)}{(5.0g)(100uL)(0.50)}$$

= 11.7ng/g = 11.7 ug/Kg

Reporting limit(MDL) without additional moisture calculation

$$= \frac{(0.234ng)(10000uL)(1)}{(5.0g)(100uL)(0.50)}$$

= 9.36 ng/g = 9.36 ug/Kg

Semi-Volatile Water Reporting Limit (RL or MDL)

$$\text{Concentration (ug/L)} = \frac{(\text{RL or MDL})(V_t)(Df)(GPC)}{(V_o)(V_i)}$$

Where:

RL	On-column Reporting limit in nanograms (ng) – derived from Submitter\Task Custom compound list in the 'Water Limit' column or from the Reporting Threshold field in the Quantify Method. This value is typically determined by the lowest level that is used in the initial calibration curve.
MDL	Method detection limit in nanograms (ng) – derived from the Quantify Method in the Environmental Parameters section in the 'MDL Water' field. This value is typically determined statistically by analyzing 7 replicate samples, however it can be any user determined value.
V_o	Volume of water extracted in milliliters (mL) – derived from 'SampleWt/Vol' column in Sample List.
V_i	Volume of extract injected in microliters (uL) – derived from 'Inject Volume' column in Sample List.
V_t	Volume of the concentrated extract in microliters (uL) – derived from 'Conc Ext Vol' column in Sample List.
GPC	Gel Permeation Cleanup factor – derived from 'GPC' column setting in Sample list **Y** or **N** GPC = 1.0 if assigned N in GPC column in Sample List. GPC = 2.0 if assigned Y in GPC column in Sample List.
Df	Dilution Factor – derived from 'Dil Factor – DILUTION FACTOR' from the Sample List.

Example Semi-Volatile Water Calculation:

With the following sample information from Sample list, Submitter\Task or Quantify Method:

Available from Submitter\Task		
On column reporting Limit for Pyrene	= 4.0 ng	RL
Available from Quantify Method		
On column MDL for Pyrene	= 0.567 ng	MDL
Available from Sample List columns		
Sample vol/wt	= 1000mL	V_o
Volume of extract injected (uL)	= 1uL	V_i
Volume of concentrated extract (uL)	= 2000uL	V_t
Dilution Factor	= 25	Df
GPC	= Y	GPC

$$\text{RL/MDL Concentration (ug/L)} = \frac{(\text{RL or MDL})(V_t)(Df)(GPC)}{(V_o)(V_i)}$$

Reporting Limit (RL) calculation

$$= \frac{(4.0ng)(2000uL)(25)(2)}{(1000mL)\ (1uL)}$$

= 400 ng/mL = 400 ug/L

Reporting Limit (MDL) calculation

$$= \frac{(0.567ng)(2000uL)(25)(2)}{(1000mL)\ (1uL)}$$

= 56.7 ng/mL = 56.7 ug/L

Semi-Volatile Soil Reporting Limit (RL or MDL)

$$\text{RL/MDL Concentration (ug/Kg)} = \frac{(\text{RL or MDL})(V_t)(Df)(GPC)}{(V_i)\ (W_s)\ (D)}$$

Where:

RL	On-column Reporting limit in nanograms (ng) – derived from Submitter\Task Custom compound list in the 'Water Limit' column or from the Reporting Threshold field in the Quantify Method. This value is typically determined by the lowest level that is used in the initial calibration curve.
MDL	Method detection limit in nanograms (ng) – derived from the Quantify Method in the Environmental Parameters section in the 'MDL Water' field. This value is typically determined statistically by analyzing 7 replicate samples, however it can be any user determined value.
V_i	Volume of extract injected i.n microliters (uL) – derived from 'Inject Volume' column in Sample List
V_t	Volume of the concentrated extract in microliters (uL) – derived from 'Conc Ext Vol' column in Sample List.
W_s	Weight of sample extracted in grams (g) – derived from 'SampleWt/Vol' column in Sample List.
D	Adjustment for dry weight basis, calculated from 'Moisture' column in Sample List. $= \frac{100 - \%\ \text{moisture}}{100}$
GPC	Gel Permeation Cleanup factor – derived from 'GPC' column setting in Sample list **Y** or **N**. GPC = 1.0 if assigned N in GPC column in Sample List. GPC = 2.0 if assigned Y in GPC column in Sample List.
Df	Dilution Factor – derived from 'Dil Factor – DILUTION FACTOR' from the Sample List.

Example Semi-Volatile Soil Reporting Limit (RL or MDL) Calculation:

With the following sample information from Sample list, Submitter/Task or Quantify Method:

Available from Submitter\Task		
On column reporting Limit for Pyrene	= 4.0 ng	RL
Available from Quantify Method		
On column MDL for Pyrene	= 1.23 ng	MDL
Available from Sample List columns		
Sample vol/wt	= 15.2g	W_s
% Moisture	= 14.0	D (see calculation below)
Volume of extract injected (uL)	= 1uL	V_i
Volume of concentrated extract (uL)	= 2000uL	V_t
Dilution Factor	= 25	Df
GPC	= Y	GPC

D = (100 – 14)/100
= 0.86

$$\text{RL/MDL Concentration (ug/Kg)} = \frac{(\text{RL or MDL})(V_t)(Df)(GPC)}{(V_i)\,(W_s)\,(D)}$$

Reporting Limit (RL) calculation

$$= \frac{(4.0\text{ng})(2000\text{uL})(25)(2)}{(1\text{uL})\,(15.2\text{g})\,(0.86)}$$

= 30600 ng/g = 30600 ug/Kg

Reporting Limit (MDL) calculation

$$= \frac{(1.23\text{ng})(2000\text{uL})(25)(2)}{(1\text{uL})\,(15.2\text{g})\,(0.86)}$$

= 9409 ng/mL = 9409 ug/L

Appendix 2
Error Messages and Warnings



About the Error Messages and Warnings

The list of samples in Environmental Reports is checked when it is opened and whenever the rows are added or the state of any row is changed. All selected rows of the Sample List are examined to identify errors or warnings. If any errors are identified then the Sample List title of the tab is displayed in red and all Form tabs are disabled. General errors and warnings are displayed in the message pane located on the bottom of the window.

Error Messages

An error is defined as a condition that will prevent the generation of a coherent data set for reporting. The Print command remains disabled while such errors exist. The following checks are made:

- All sample rows must reference the same Analysis type (VOA, SV or QA/QC). If more than one Analysis type is present (empty Analysis fields will be ignored for this test) an error message will be displayed: General Error Sample list contains mixed Analysis types Rows, a, b, x-z
- All Sample List rows selected for processing, other than Tune Eval types, must reference the same Calibration file. If more than one Calibration file is referenced an error message will be displayed: General Error Sample List contains more than one Calibration file Rows a, b, xz
- If no Calibration file is included in the Sample List rows selected for processing, then an error message will be displayed: General Error No Calibration file is defined Rows az
- All Sample List rows selected for processing, other than Tune Eval types, must reference the same Quantify Method. If more than one Quantify Method is referenced an error message will be displayed: General Error Sample List contains more than one Quantify Method file Rows a, b, xz
- If no selected rows reference environmental sample types (e.g., all set to Standard) an error message will be displayed: General Error Sample List contains no environmental sample types
- All Sample List rows selected for processing that specify a matrix type must specify the same type, or an error message will be displayed: General Error Sample List contains mixed matrix types Rows a, b, xz
- All Sample List rows selected for processing that specify a concentration level (low/med) must specify the same level, or an error message will be displayed: General Error Sample List contains mixed matrix concentration levels Rows a, b, xz
- A Sample List row selected for processing cannot be reassigned as different sample types on different Form tabs, with the exception of Tune Eval (Form 5) and Cont Calib (Form 7) [or Form 8], which is valid. For example assigning the same row as Meth Blank for Form 4 and as Tune Eval for Form 5 would cause an error message to be displayed: General Error Sample List contains invalid multiple sample type assignments Rows a, b, xz

- If the calibration file identified in the Sample List does not exist in the Project CurveDB directory then an error message will be displayed: General Error Calibration file cannot be found Rows az

- Selected rows, other than Tune Eval or Calib types, must contain values for Analysis and Matrix. General Error Sample List contains sample rows with blank Analysis or Matrix column Rows az

- If Matrix is Soil, then the Conc Level column cannot be blank. General Error Sample List contains Soil sample with blank Conc Level column Rows az

Note: *For items 1 to 5, the display of row numbers exhibiting the error is determined in one of the two following ways:*

1. The first Calibration file, Quantify Method, etc located (ignoring any referenced by the Tune Eval sample) will be taken as the correct one and any row that references a different one will be flagged as in error.

2. The Calibration file, Quantify Method, etc that is referenced most frequently in the selected rows will be taken as the correct one and any row that references a different one will be flagged as in error.

Contiguous groups of rows will be displayed in the form xz, while isolated rows will displayed separated by commas.

General errors must be cleared before the report generation process can proceed. Although general errors can be cleared by unchecking rows this may leave a Sample List that cannot be used to generate the required forms (i.e., Form specific errors may be generated in the process). In this event you may have to leave this environment and edit the Sample List or create a new one before generating reports.

Warnings

A warning is defined as a condition that does not meet the strict rules of CLP reporting but may be valid in the context of "CLP–like" reporting. Warnings do not prevent access to the Form tabs nor do they prevent reports being generated, although there is no guarantee the results will be complete or entirely valid. The following checks are made:

- If Sample List rows selected for processing specify a sample type other than one of those listed (plus Analyte), a warning message will be displayed: General Warning Sample List contains nonenvironmental sample types Rows a, b, xz

Note: *The display of row numbers exhibiting the warning should be determined as described in the previous section.*

- A Sample List row selected for processing should not be reassigned as different sample types on different Form tabs, with the exception of Tune Eval (Form 5) and Cont Calib (Form 7) [or Form 8], which is valid. For example assigning the same row as Meth Blank for Form 4 and as Tune Eval for Form 5 would cause a warning message to be displayed: General Warning Sample List contains invalid multiple sample type assignments Rows a, b, xz

- If the injection time of any rows selected for processing is more than 12 hours after the injection time of the tune evaluation sample.

Form Specific Checks

The Sample List is checked when it is opened and whenever the state of any row is changed. All selected rows of the Sample List will be examined to identify errors or warnings. If any errors related to a specific Form are identified then the title of that Form tab will be displayed in red. The error will be displayed in the message pane on the Form tab as well as on the Sample List tab. Form-specific warnings will also be displayed in both places.

Form–specific errors/warnings follow General errors/warning in the message pane on the Sample List. A Form–specific error is defined as a condition that will prevent the generation of that Form. The Print command will remain disabled while such errors exist.

A Form–specific warning is defined as a condition that does not meet the strict rules of CLP reporting but may be valid in the context of "**CLP–like**" reporting. Warnings do not prevent reports being generated, although there is no guarantee the results will be complete or entirely valid.

Form 1

The Sample List is checked for the existence of sample types reported on Form 1.

Errors:
If no rows selected for processing are of sample type other than Tune Eval, Init Calib or Cont Calib then an error message will be displayed: Form 1 Error Sample List contains no sample types reported on Form 1

Form 1 TIC

The Sample List is checked for the existence of sample types reported on Form 1 TIC.
Errors:
If any rows selected for processing (of sample type other than Tune Eval, Init Calib or Cont Calib) have no associated qualitative results saved, then an error message will be displayed: Form 1 TIC Error No qualitative data found for some selected samples Rows a, b, x–z
If no rows selected for processing are of sample type other than Tune Eval, Init Calib or Cont Calib then an error message will be displayed: Form 1 TIC Error Sample List contains no sample types reported on Form 1 TIC
Either of these errors will prevent the TIC assignment process being performed.

Warnings:
If any rows selected for processing (other than Tune Eval, Init Calib or Cont Calib rows) still have the Pending status then an error message will be displayed: Form 1 TIC Warning Samples with incomplete TIC selection will not be reported Rows a, b, x–z

Form 2

The Sample List is checked for the existence of sample types reported on Form 2. The first Analyte (or Analyte Dup) row found in the selected rows will be marked as the source of the header information for Form 2. You can change this assignment if necessary.

Errors:
If no rows selected for processing are of sample type other than Tune Eval, Init Calib or Cont Calib then an error message will be displayed: Form 2 Error Sample List contains no sample types reported on Form 2

Warnings:
No warnings specific to Form 2 have been identified.

Form 3

Generation of Form 3 requires three (and only three) files: An Analyte sample, a Matrix Spike sample (prepared by spiking the analyte) and a Matrix Spike (a second spiked sample). The default rows are identified as follows:

Matrix Spike Duplicate:

- Look for the last Spike Dup sample in the list. If no Spike Dup can be located a warning condition exists (see below).

Matrix Spike:

- If a Spike Dup was located and its Sample ID ends with MSD, look for the Spike sample with the same root Sample ID but ending with MS. If this is not located, look for the first Spike sample preceding the Spike Dup. If one is not found look for the first Spike sample following the Spike Dup. If no Spike can be located an error condition exists (see below).

- If a Spike Dup was located and its Sample ID did not end with MSD look for the first Spike sample preceding the Spike Dup. If one is not found look for the first Spike sample following the Spike Dup. If no Spike can be located an error condition exists (see below).

- If no Spike Dup was located, look for the last Spike sample in the Sample List. If no Spike can be located an error condition exists (see below).

Analyte:

- If a Spike Dup was located and its Sample ID ends with MSD, look for the Analyte sample with the Sample ID formed by removing the MSD (i.e. same root, no suffix). If this is not located, look for the first Analyte (or Analyte Dup) sample preceding the Spike Dup. If one is not found look for the first Analyte (or Analyte Dup)sample following the Spike Dup. If no Analyte (or Analyte Dup) can be located an error condition exists (see below).

- If a Spike Dup was located and its Sample ID did not end with MSD look for the first Analyte (or Analyte Dup) sample preceding the Spike Dup. If one is not found look for the first Analyte sample following the Spike Dup. If no Analyte (or Analyte Dup) can be located an error condition exists (see below).

- If no Spike Dup was located, look for the first Analyte (or Analyte Dup) sample preceding the Spike in the Sample List. If one is not found, look for the first Analyte (or Analyte Dup) following the Spike. If no Analyte (or Analyte Dup) can be located an error condition exists (see below).

Errors:

- If no Spike sample was located in the rows selected for processing (according to the rules given above), then an error message will be displayed: Form 3 Error No matrix spike sample identified

- If no Analyte (or Analyte Dup) sample was located in the rows selected for processing (according to the rules given above), then an error message will be displayed: Form 3 Error No Unspiked sample identified

Warnings:

If no Spike Dup sample was located in the rows selected for processing (according to the rules given above), then a warning message will be displayed: Form 3 Warning No matrix spike duplicate sample identified – form will be incomplete. This is a valid condition, since Spike Dup samples are not always run.

Form 4

Primary data for Form 4 will be taken from a Meth Blank sample with summary information from samples types other than Tune Eval, Init Calib or Cont Calib. The default Meth Blank to be used as the primary data set will be the last Meth Blank sample row selected in the Sample List.

Errors:

If no Meth Blank sample was located in the rows selected for processing (according to the rules given above), then an error message will be displayed: Form 4 Error No method blank sample identified.

Warnings:

If no rows selected for processing (excluding the Meth Blank row) are of sample type other than Tune Eval, Init Calib or Cont Calib then a warning message will be displayed: Form 4 Warning Sample List contains no sample types reported in Form 4 summary.

Form 5

Primary data for Form 5 will be taken from a Tune Eval sample with summary information from other samples types (other than Tune Eval). The default Tune Eval to be used as the primary data set will be the last Tune Eval sample row selected in the Sample List.

Errors:

If no Tune Eval sample was located in the rows selected for processing (according to the rules given above), then an error message will be displayed: Form 5 Error No tune evaluation sample identified.

Warnings:
If no rows selected for processing are of sample type other than Tune Eval then a warning message will be displayed: Form 5 Warning Sample List contains no sample types reported in Form 5 summary. If the Analysis type is Volatiles and the specified Tune Eval file references DFTPP (or Analysis is Semivolatiles and Tune Eval references BFB) then a warning message will be displayed: Form 5 Warning Tune evaluation sample references the wrong compound for the Analysis type.

Form 6

The data for Form 6 will be taken from the Calibration file referenced by most of the rows in the Sample List. The general checks will have ensured that only one Calibration file is referenced. This Calibration file can be taken from any sample row other than the Tune Eval (which is ignored).

The first Analyte (or Analyte Dup) row found in the selected rows of the Sample List will be marked as the source of the header information for Form 6. The file name of this sample will be displayed on the Form 6 tab). The user can change this assignment if necessary.

Errors:
No errors specific to Form 6 have been identified.

Warnings:
No warnings specific to Form 6 have been identified.

Form 7

Primary data for Form 7 will be taken from a Cont Calib sample. The default Cont Calib will be the last Cont Calib sample row selected in the Sample List.

Errors:
If no Cont Calib sample was located in the rows selected for processing (according to the rules given above), then an error message will be displayed: Form 7 Error No continuing calibration sample identified

Warnings:
No warnings specific to Form 7 have been identified.

Form 8

Primary data for Form 8 will be taken from a Cont Calib sample (which may be the Mid-Level of an Initial Calibration with summary information from other samples types (other than Tune Eval, Cont Calib or Init Calib types). The default Cont Calib to be used as the primary data set will be determined as follows:

- If the selected sample rows contain Cont Calib rows then the primary data set will be last Cont Calib sample row selected in the Sample List.

- If the selected sample rows do not contain a Cont Calib row but contain one or Init Calib rows then the middle row of this set will be marked (in the Status column) as the Cont Calib. If an even number of Init Calib rows are selected then the closest row past the mid–point will be flagged as the Cont Calib. If only one Init Calib row exists then that will be chosen.

Errors:
If no Cont Calib sample was located in the rows selected for processing (according to the rules given above), then an error message will be displayed: Form 8 Error No continuing calibration sample identified.

Warnings:
If no rows selected for processing are of sample type other than Tune Eval, Cont Calib or Init Calib then a warning message will be displayed: Form 8 Warning Sample List contains no sample types reported in Form 8 summary.

Index

A

Analytes, 20
Assign Method Blank, 125

C

Calibration, chromatographic
  library, 20
Conventions used, 14
Custom Compound List, 55

D

Decimal Places in Sample List, 72
Display chromatogram, 50

E

Environmental Parameters, setting, 35
Environmental Reporting, 83
EPA Qualifiers, 107, 114
EPA Test, 129
External standard, 19, 20
External standard method, 19

F

Flags, 107, 114
Format File, 24

I

Injection
  size variations, 19
Interactive Data Review, 79
Internal standard
  defined, 20
Internal standard method, 19
Introduction to TurboMass, 10

M

Manual Peak Integration, 79
Maximum in Blank, 41
MDL, 42, 107

P

Process Quantification Results, 75

Q

Qualifiers, 107, 114
Quantification
  creating the method, 22
  Sample List, 65
Quantify
  List by Compound, 78
  List by Sample, 78
  processing samples, 76
  view results, 78
Quantitation, 20

R

Report Method Data Files, 54
Report Methods, defining, 53
Reporting Threshold, 33
Response Factor Curve, 20

S

Sample Injection Information, 68
Sample List
  creation, 65
Sample List Information Fields, 91
Sample List Wizard, 66
Sample Prep Information, 69
Select Forms, 98
Spike, 39
Surrogate, 38

T

Target, 37
Target compound, 20
Tentatively Identified Compounds, 112
Touch Screen, 15



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